

Egyptian ERP System

Comprehensive User Manual and Reference Guide

Complete Documentation for Customer Support Chatbot

(RAG Implementation - Retrieval Augmented Generation)

Version 2025.1 - Enterprise Edition

Release Date: January 2025

Document Version: 1.0

Egyptian ERP Solutions

Cairo, Arab Republic of Egypt

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Regulatory Compliance Statement

This software and documentation are designed to assist organizations in complying with Egyptian regulatory requirements including:

- Egyptian Tax Authority (ETA) E-Invoicing Regulations
- Value Added Tax (VAT) Law No. 67 of 2016
- Social Insurance Law and GOSI Requirements
- Egyptian Labor Law No. 12 of 2003 (as amended by Law No. 14 of 2025)
- Commercial Registry Regulations
- Central Bank of Egypt Reporting Requirements

Important: While this system is designed to facilitate regulatory compliance, users are responsible for ensuring their specific implementations meet all applicable legal

requirements. Consult with qualified legal and accounting professionals for advice on your specific situation.

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Preface

About This Manual

Welcome to the Comprehensive User Manual for the Egyptian ERP System. This manual has been meticulously crafted to serve as the definitive reference guide for all users of the Egyptian ERP System, from system administrators and power users to end users seeking to understand specific functions and processes. The manual is designed to be both a learning resource for new users and a comprehensive reference for experienced practitioners.

This documentation serves a dual purpose: first, as a traditional user manual providing step-by-step guidance for using the system; and second, as the knowledge base for the AI-powered Customer Support Chatbot that assists users through the RAG (Retrieval Augmented Generation) implementation. Every concept, procedure, and configuration option documented here is accessible to the chatbot to provide accurate, contextual assistance.

Who Should Read This Manual

This manual is intended for a diverse audience including:

System Administrators

IT professionals responsible for configuring, maintaining, and supporting the Egyptian ERP System will find comprehensive guidance on system setup, security configuration, user management, integration, and troubleshooting. Chapters 26, 27, 28, and 31 are particularly relevant for administrators.

Finance and Accounting Professionals

Accountants, financial controllers, CFOs, and finance team members will find detailed coverage of all financial modules including General Ledger, Accounts Payable, Accounts Receivable, Fixed Assets, Cost Accounting, and Cash Management. Special attention is given to Egyptian regulatory compliance including ETA E-Invoicing, VAT, and tax reporting requirements.

Human Resources and Payroll Professionals

HR managers, payroll administrators, and compensation specialists will find comprehensive coverage of employee management, attendance tracking, leave management, and payroll processing with full compliance to Egyptian Labor Law and GOSI requirements.

Operations and Supply Chain Professionals

Warehouse managers, procurement officers, production planners, and operations managers will find detailed guidance on Inventory Management, Purchasing, Sales, Manufacturing, Quality Management, and Maintenance modules.

End Users

All employees who interact with the system for daily operations will find user-friendly explanations of common tasks, navigation guidance, and troubleshooting tips.

Implementation Partners

Consultants and implementation partners will find the Implementation Methodology chapter (Chapter 32) invaluable for planning and executing successful ERP deployments.

How This Manual Is Organized

The manual is organized into seven major parts, each focusing on a specific aspect of the Egyptian ERP System:

Part I: Foundation and Getting Started provides essential background information about the system, its architecture, and basic navigation. New users should start here.

Part II: Financial Management covers all financial modules in depth, with special emphasis on Egyptian regulatory compliance.

Part III: Human Capital Management addresses HR and payroll functions with complete coverage of Egyptian labor law and social insurance requirements.

Part IV: Supply Chain and Operations provides comprehensive guidance on inventory, procurement, sales, manufacturing, quality, and maintenance.

Part V: Business Operations covers project management, CRM, and business intelligence.

Part VI: Technical and Administration addresses system integration, mobile access, AI capabilities, and administration.

Part VII: Support and Implementation provides troubleshooting guidance, training resources, and implementation methodology.

Document Conventions

This manual uses the following conventions to help you navigate the content:

Note: Indicates important information that provides additional context or helpful tips.

Warning: Alerts you to potential issues or actions that could cause problems if not handled correctly.

Tip: Provides helpful suggestions or shortcuts to improve efficiency.

Important: Highlights critical information that requires special attention.

Navigation Menu References

Navigation paths are shown using the arrow notation: **Module** → **Submenu** → **Function**. For example, "Navigate to Financial → General Ledger → Journal Entries" indicates you should click on Financial in the main menu, then General Ledger in the submenu, and finally Journal Entries.

Keyboard Shortcuts

Keyboard shortcuts are shown in the format **Ctrl+K** or **Alt+F4**. These shortcuts can significantly improve your efficiency when using the system.

Version Information

This manual corresponds to Egyptian ERP System Version 2025.1, released in January 2025. Features and functionality described in this manual may differ from earlier or later versions. Always refer to the release notes for your specific version for any version-specific information.

Feedback and Support

We welcome your feedback on this manual. If you find errors, omissions, or have suggestions for improvement, please contact our documentation team through the support portal. Your feedback helps us continuously improve our documentation.

Acknowledgments

This manual was developed by the Egyptian ERP Solutions documentation team in collaboration with subject matter experts from finance, HR, operations, and technical teams. We extend our gratitude to all contributors who helped make this comprehensive resource possible.

Chapter 1: Introduction to the Egyptian ERP System

1.1 Overview of the Egyptian ERP System

The Egyptian ERP System represents a comprehensive, integrated enterprise resource planning solution specifically designed to meet the unique business and regulatory requirements of organizations operating in the Arab Republic of Egypt. This state-of-the-art system combines international best practices in ERP design with deep localization for Egyptian business processes, tax regulations, labor laws, and reporting requirements. The system has been developed with a thorough understanding of the Egyptian business environment, incorporating features that address the specific challenges faced by companies operating in this dynamic and evolving market.

From the bustling commercial districts of Cairo and Alexandria to the industrial zones of the Suez Canal Economic Zone and the New Administrative Capital, the Egyptian ERP System provides the tools necessary for businesses to thrive in Egypt's rapidly evolving economic landscape. The system supports businesses of all sizes, from small and medium enterprises to large multinational corporations with complex operations spanning multiple industries and geographic locations.

1.1.1 System Philosophy and Design Principles

The Egyptian ERP System has been built on several core design principles that guide its architecture, functionality, and user experience. These principles ensure the system delivers maximum value to Egyptian businesses while maintaining the flexibility to adapt to changing requirements.

Integration: The system provides seamless connectivity across all business functions, eliminating data silos and ensuring information flows freely between departments. When a sales order is created, inventory is automatically updated, production is scheduled if needed, and financial transactions are generated without manual intervention. This integration reduces errors, improves efficiency, and provides real-time visibility into business operations.

Compliance-First Design: Every feature is designed with Egyptian regulatory requirements in mind. From the ETA E-Invoicing requirements to GOSI social insurance calculations, from VAT compliance to labor law adherence, the system ensures your business remains compliant with all applicable regulations. The compliance framework is regularly updated to reflect changes in Egyptian laws and regulations.

User-Centric Interface: The modern, intuitive interface reduces training time and increases user adoption. Context-sensitive help, guided workflows, and consistent design patterns make the system accessible to users of all technical backgrounds. The interface is available in both Arabic and English, supporting Egypt's bilingual business environment.

Scalability: Whether you are a startup with ten employees or a multinational corporation with thousands of users, the system scales to meet your needs. The cloud-native architecture allows you to start small and grow without system limitations, adding users, companies, and functionality as your business expands.

Security: Multi-layered security protects your business data from unauthorized access, cyber threats, and data breaches. Role-based access control ensures users only see and access the information relevant to their responsibilities. All data is encrypted in transit and at rest, with comprehensive audit trails tracking every system interaction.

1.1.2 Historical Context and Evolution

The Egyptian ERP System has evolved in response to significant changes in Egypt's business and regulatory environment. Understanding this evolution helps appreciate the system's current capabilities and future direction.

The Digital Transformation Era (2016-2019): The initial development of the Egyptian ERP System began during Egypt's digital transformation initiative. The government recognized that modernizing business systems was essential for economic growth and competitiveness. Early versions focused on core financial management with basic Egyptian localization.

The E-Invoicing Revolution (2020-2021): The introduction of the Egyptian Tax Authority's E-Invoicing system in 2020 marked a turning point in digital transformation for Egyptian businesses. The Egyptian ERP System was among the first ERP solutions to achieve full compliance with ETA requirements, helping hundreds of companies transition from paper-based invoicing to electronic submission. This experience shaped the system's approach to regulatory compliance.

The Comprehensive ERP Era (2022-2024): Building on the success of the financial modules, the system expanded to include comprehensive HR, payroll, inventory, manufacturing, and business intelligence capabilities. Each module was developed with the same attention to Egyptian regulatory requirements and local business practices.

The AI-Powered Future (2025 and Beyond): The current version introduces artificial intelligence and machine learning capabilities to enhance decision-making, automate routine tasks, and provide predictive insights. The integrated chatbot assistant uses this manual as its knowledge base to provide contextual help to users.

1.1.3 Market Position and Adoption

The Egyptian ERP System has been adopted by organizations across various industries, establishing itself as a leading ERP solution in the Egyptian market. The system's deep understanding of local requirements combined with international best practices has made it the preferred choice for Egyptian businesses seeking a modern, compliant ERP solution.

Industry Coverage: The system serves diverse industries including manufacturing (textiles, food processing, chemicals, pharmaceuticals), trading (import/export, distribution, retail), services (consulting, healthcare, education), construction and real estate, and government and public sector organizations.

Geographic Reach: While primarily serving organizations within Egypt, the system supports multi-country operations for businesses with regional presence. The multi-currency, multi-language, and multi-company capabilities enable seamless management of operations across borders.

Company Size: The system serves businesses of all sizes, from SMEs with simple requirements to large enterprises with complex, multi-site operations. The modular architecture allows organizations to implement only the modules they need, expanding as requirements grow.

1.2 System Architecture

The Egyptian ERP System employs a modern, cloud-native architecture built on microservices principles. This architecture provides the flexibility, scalability, and reliability required by modern businesses while maintaining the security and compliance standards demanded by Egyptian regulations.

1.2.1 Technical Architecture Overview

The microservices architecture divides the system into small, independent services that communicate through well-defined APIs. Each service is responsible for a specific business function and can be developed, deployed, and scaled independently.

Benefits of Microservices Architecture:

- **Modularity:** Each business function is implemented as an independent service, making the system easier to understand, develop, and maintain.
- **Scalability:** Individual services can be scaled based on demand. High-traffic services like sales order processing can have more resources allocated without affecting other services.
- **Resilience:** Failure in one service does not cascade to others. If the inventory service experiences issues, financial transactions can continue processing.
- **Technology Flexibility:** Each service can use the technology stack best suited to its requirements. The ETA integration service uses technologies optimized for government portal communication, while the reporting service uses technologies optimized for data analytics.
- **Continuous Deployment:** Services can be updated independently, enabling faster delivery of new features and bug fixes without system-wide downtime.

1.2.2 Architecture Layers

The system consists of four main layers, each with specific responsibilities and capabilities:

Presentation Layer: The presentation layer provides user interfaces across multiple channels. The web application offers the full functionality of the system through modern browsers. The mobile application extends key functions to iOS and Android devices, enabling field staff to access and update information on the go. The API layer enables integration with third-party applications and custom interfaces.

Application Layer: The application layer contains the business logic microservices. Each module (General Ledger, Inventory, HR, etc.) is implemented as one or more microservices. These services handle business rules, data validation, workflow processing, and integration with external systems. The services communicate through asynchronous messaging for loose coupling and improved reliability.

Data Layer: The data layer manages all data storage and retrieval. Transactional databases store operational data with ACID compliance for data integrity. Analytical databases and data warehouses support reporting and business intelligence with optimized query performance. Document storage handles attachments, scanned documents, and generated reports. Cache layers improve performance for frequently accessed data.

Infrastructure Layer: The infrastructure layer provides the foundation for the system. Cloud infrastructure from leading providers ensures high availability and global reach. Container orchestration manages the deployment and scaling of microservices. API gateways handle authentication, rate limiting, and request routing. Monitoring and logging provide visibility into system health and performance.

1.2.3 Security Architecture

Security is built into every layer of the architecture, not added as an afterthought. The security architecture addresses authentication, authorization, data protection, and audit requirements.

Authentication: Users authenticate through multiple methods including username/password, multi-factor authentication (MFA), single sign-on (SSO), and API keys for system integration. Password policies enforce strong credentials with regular rotation requirements.

Authorization: Role-based access control (RBAC) defines what each user can see and do. Roles are assigned based on job functions, with granular permissions at the module, function, and data level. Data restrictions can limit access by company, department, cost center, or other dimensions.

Data Protection: All data is encrypted in transit using TLS 1.3 and at rest using AES-256 encryption. Database connections are secured and audited. Sensitive data like passwords and API keys are hashed using industry-standard algorithms.

Audit and Compliance: Comprehensive audit trails record all system activities including user logins, data modifications, and administrative actions. Audit logs are tamper-proof and retained according to regulatory requirements.

1.3 Core Modules Overview

The Egyptian ERP System includes comprehensive modules covering all aspects of business operations. Each module is designed to work seamlessly with others while providing deep

functionality in its specific domain.

1.3.1 Financial Management Modules

The financial management modules form the backbone of the system, ensuring accurate financial records, regulatory compliance, and timely reporting.

General Ledger (GL): The General Ledger is the central repository of all financial transactions. It supports multi-company, multi-currency, and multi-dimensional accounting. The chart of accounts can be configured to match your organization's structure, with support for Egyptian accounting standards. Period-end closing processes are automated with comprehensive checklists and validation.

Accounts Payable (AP): The AP module manages all vendor-related financial transactions from invoice receipt through payment processing. It supports three-way matching with purchase orders and goods receipts, automated payment proposals, and comprehensive vendor analysis. Integration with the ETA portal ensures compliance with e-invoicing requirements.

Accounts Receivable (AR): The AR module handles customer invoicing, receipt processing, and collections management. Sales invoices are automatically submitted to the ETA portal with digital signatures and QR codes. Credit management features help control customer credit exposure and reduce bad debt.

Cash and Bank Management: This module provides comprehensive control over cash positions, bank accounts, and treasury operations. Bank reconciliation is streamlined through automatic statement import and intelligent matching. Cash flow forecasting helps manage liquidity and plan funding requirements.

Fixed Assets: The Fixed Assets module tracks all capital assets from acquisition through disposal. It supports multiple depreciation methods including Egyptian tax depreciation rates. Asset transfers, revaluations, and disposals are processed with automatic GL postings.

Cost Accounting: Cost Accounting provides detailed cost tracking and analysis to support management decision-making. Cost centers, profit centers, and internal orders enable multi-dimensional cost analysis. Product costing supports standard, actual, and activity-based costing methods.

1.3.2 Supply Chain Management Modules

The supply chain modules manage the flow of materials and products from suppliers through production to customers.

Inventory Management: Real-time visibility into stock levels across multiple warehouses and locations. Support for multiple valuation methods (FIFO, LIFO, Average, Standard), batch/serial tracking, and cycle counting. Automated reorder point calculations help maintain optimal inventory levels.

Purchasing: The complete procurement lifecycle from requisition to payment. Purchase requisitions with approval workflows, purchase orders with vendor management, goods receipt with quality inspection, and invoice matching for payment authorization.

Sales Management: The complete sales cycle from quotation to cash collection. Customer management with credit control, sales quotations and orders, delivery management, invoicing with ETA integration, and receipt processing.

1.3.3 Manufacturing Modules

The manufacturing modules support production planning and control for make-to-stock, make-to-order, and assemble-to-order environments.

Manufacturing Management: Bill of Materials (BOM) management with version control, routing definition with work center capacity planning, production order management with material issuance and output recording, and Material Requirements Planning (MRP) for automated planning.

Quality Management: Quality planning with inspection plans, receiving inspection, in-process inspection, and final inspection. Non-conformance management with Corrective and Preventive Actions (CAPA).

Maintenance Management: Preventive maintenance scheduling, work order management, spare parts inventory, and equipment history tracking.

1.3.4 Human Capital Management Modules

The HCM modules manage the complete employee lifecycle from hiring through termination.

Human Resources: Employee master with comprehensive personal and employment information, organizational structure management, attendance tracking with biometric integration, leave management with Egyptian labor law compliance, and employee self-service portal.

Payroll: Complete payroll processing with Egyptian regulatory compliance. Earnings and deductions management, social insurance (GOSI) calculations and reporting, income tax calculation per Egyptian tax brackets, and bank transfer file generation.

1.4 Egyptian Regulatory Compliance

The Egyptian ERP System ensures full compliance with Egyptian regulations through built-in features, automated calculations, and integrated reporting. Understanding these compliance features is essential for proper system configuration and use.

1.4.1 Egyptian Tax Authority (ETA) E-Invoicing

The ETA E-Invoicing system is a mandatory requirement for all B2B and B2G transactions in Egypt. The Egyptian ERP System provides complete integration with the ETA portal, automating the submission process and ensuring compliance.

Key Requirements:

- **UUID Generation:** Each invoice must have a unique identifier (UUID) in the format specified by ETA.
- **Digital Signature:** Invoices must be digitally signed using an ETA-approved certificate to ensure authenticity and integrity.
- **QR Code:** Each invoice must include a QR code containing key invoice data for verification purposes.
- **JSON Format:** Invoices must be submitted in the specific JSON format defined by ETA.
- **Real-time Submission:** Invoices must be submitted to the ETA portal at the time of issuance.

System Integration: When a sales invoice is created in the system, it automatically undergoes the ETA submission process. The system validates the invoice data against ETA requirements, generates the UUID, creates the JSON format, applies the digital signature, submits to the ETA portal, and receives the ETA ID and QR code. The invoice can only be printed after successful ETA submission.

1.4.2 Value Added Tax (VAT)

Egypt applies a standard VAT rate of 14% on most goods and services. The Egyptian ERP System provides comprehensive VAT management to ensure accurate calculation, reporting, and compliance.

VAT Configuration: The system supports multiple tax codes including VAT14 (Standard 14%), VAT0 (Zero-rated 0%), and EXEMPT (Exempt). Tax accounts are automatically maintained for input VAT (on purchases), output VAT (on sales), and VAT payable/receivable.

VAT Calculation: VAT is automatically calculated on transactions based on the tax code assigned to items, customers, and vendors. The formula used is: $\text{VAT Amount} = (\text{Subtotal} - \text{Discount}) \times \text{Tax Rate}$. The system handles complex scenarios including partial payments, returns, and adjustments.

VAT Return: The monthly VAT return is automatically prepared based on transaction data. The return includes total sales (taxable and exempt), output VAT, total purchases, input VAT, and net VAT payable or receivable. The return must be filed within 30 days of month end.

1.4.3 Social Insurance (GOSI)

The General Organization for Social Insurance (GOSI) manages Egypt's social insurance system. Employers must calculate and remit social insurance contributions for all employees.

Contribution Rates (2025):

Insurance Type	Employer Share	Employee Share	Total
Old Age, Disability, and Death	15%	11%	26%
Work Injury	2%	0%	2%
Unemployment	1%	1%	2%
Total	18%	12%	30%

Maximum Insurance Salary: The maximum salary subject to social insurance is EGP 8,400 per month (2025). Contributions are calculated on the lower of actual salary or the maximum.

GOSI Reporting: Monthly reports must be submitted to GOSI with employee details, salary information, and contribution calculations. The system generates the required reports in the format specified by GOSI.

1.4.4 Labor Law Compliance

The Egyptian ERP System ensures compliance with Egyptian Labor Law No. 12 of 2003 as amended by Law No. 14 of 2025. The HR and Payroll modules incorporate all relevant labor law provisions.

Working Hours: Normal working hours are 8 hours per day or 48 hours per week. Overtime is compensated at 135% of the hourly rate for regular overtime, 200% for rest days, and 200% for official holidays.

Leave Entitlements: Annual leave is 21 days for employees with less than 10 years of service, increasing to 30 days after 10 years. Sick leave is up to 180 days per year at varying pay rates. Maternity leave is 90 days with full pay for female employees.

End of Service Benefits: Employees are entitled to end of service benefits calculated based on years of service and final salary. The system automatically calculates these benefits according to the formulas specified in labor law.

1.5 Getting Help and Support

The Egyptian ERP System provides multiple channels for getting help and support. Whether you need quick answers to simple questions or in-depth assistance with complex issues, support resources are readily available.

1.5.1 In-System Help

The system includes comprehensive help resources accessible directly from the user interface:

Field-Level Tooltips: Hover over any field label to see a brief description of the field's purpose and any relevant guidelines.

Page Help: Click the help icon on any page to access detailed documentation for that specific function.

Process Guides: Step-by-step guides walk you through complex processes with screenshots and detailed instructions.

Searchable Documentation: The complete user manual is searchable from within the system, allowing you to quickly find information on any topic.

1.5.2 Customer Support

Our customer support team is available through multiple channels:

Help Desk: Submit support tickets through the customer portal for tracking and resolution. Tickets are assigned priority levels based on impact and urgency.

Live Chat: Get immediate assistance through live chat during business hours for quick questions and guidance.

Phone Support: Call our support hotline for urgent issues requiring immediate attention.

Email Support: Send detailed inquiries to support@egypterp.com for non-urgent issues.

1.5.3 AI-Powered Chatbot

The integrated AI chatbot uses this manual as its knowledge base to provide contextual assistance:

Natural Language Queries: Ask questions in natural language, and the chatbot will find relevant information from this manual.

Contextual Help: The chatbot understands your current context in the system and provides relevant guidance.

Process Guidance: Get step-by-step guidance for completing tasks and processes.

Troubleshooting: The chatbot can help diagnose and resolve common issues.

1.5.4 Training Resources

Comprehensive training resources help users develop system expertise:

Online Courses: Self-paced e-learning courses cover all modules from basic to advanced topics.

Video Tutorials: Short video demonstrations show how to complete common tasks.

User Guides: Module-specific guides provide detailed reference information.

Certification Programs: Formal certification validates user competencies and enhances professional credentials.

1.5.5 Community and Knowledge Base

Connect with other users and access shared knowledge:

Community Forum: Exchange ideas, ask questions, and share best practices with other Egyptian ERP System users.

Knowledge Base: Searchable repository of articles, FAQs, and solutions to common issues.

Release Notes: Stay informed about new features, enhancements, and bug fixes in each release.

Tip: Bookmark the customer portal and save the support hotline number for quick access when you need assistance.

Chapter 2: Getting Started with the System

2.1 System Access

The Egyptian ERP System provides multiple access methods to accommodate different user needs and work environments. Understanding these access options ensures you can use the system effectively from any location.

2.1.1 Accessing the Web Application

The primary method of accessing the Egyptian ERP System is through the web application. The web interface provides full functionality and can be accessed from any device with a modern web browser and internet connection.

Supported Browsers: The system is optimized for the following browsers and versions:

- **Google Chrome:** Version 90 or later (Recommended)
- **Mozilla Firefox:** Version 88 or later
- **Microsoft Edge:** Version 90 or later (Chromium-based)
- **Apple Safari:** Version 14 or later

Warning: Internet Explorer is not supported. Users attempting to access the system with Internet Explorer will be redirected to a page suggesting an upgrade to a supported browser.

Access Procedure:

To access the web application:

1. Open your web browser.

2. Enter the system URL provided by your system administrator (typically <https://yourcompany.egypterp.com>).
3. The login page will load, displaying your company logo and the login form.
4. Proceed to the login procedure described in Section 2.1.2.

Browser Settings: For optimal performance, ensure your browser is configured as follows:

- JavaScript must be enabled
- Cookies must be enabled for the system domain
- Pop-up blockers should be disabled or configured to allow pop-ups from the system domain
- Browser zoom should be set to 100% for proper display

2.1.2 Login Procedure

The login process verifies your identity and grants access to the system based on your user credentials and assigned permissions.

To log in to the system:

1. Navigate to the system login page.
2. Enter your Username in the Username field. This is typically your employee ID or email address.
3. Enter your Password in the Password field. The password is masked for security.
4. If your organization uses Multi-Factor Authentication (MFA), you will be prompted to complete the second verification step after entering your password.
5. Click the **Login** button or press Enter.
6. Upon successful authentication, you will be directed to your personalized dashboard.

Multi-Factor Authentication: If MFA is enabled for your account, you will need to complete an additional verification step after entering your password:

- **SMS:** A verification code will be sent to your registered mobile number. Enter this code to complete login.
- **Email:** A verification code will be sent to your registered email address.
- **Authenticator App:** Open your authenticator app (Google Authenticator, Microsoft Authenticator, etc.) and enter the current code.
- **Hardware Token:** Enter the code displayed on your hardware security token.

Note: If you enter incorrect credentials multiple times, your account may be temporarily locked for security. Contact your system administrator to unlock your account.

2.1.3 Password Management

Proper password management is essential for system security. The Egyptian ERP System enforces password policies to ensure strong credentials.

Password Requirements: Your password must meet the following criteria:

- Minimum length: 8 characters (recommended: 12+)
- At least one uppercase letter (A-Z)
- At least one lowercase letter (a-z)
- At least one number (0-9)
- At least one special character (!@#\$%^&*)
- Cannot be the same as your last 5 passwords
- Cannot contain your username or common words

To change your password:

1. Click your profile icon in the top-right corner of the screen.
2. Select **Change Password** from the dropdown menu.

3. Enter your Current Password.
4. Enter your New Password.
5. Confirm your New Password by entering it again.
6. Click **Change Password** to save.

Forgot Password: If you forget your password, click the **Forgot Password** link on the login page. Enter your username or email address, and a password reset link will be sent to your registered email. Follow the link to create a new password.

2.1.4 Session Management

Understanding session management helps you work efficiently and securely.

Session Timeout: For security, your session will automatically timeout after a period of inactivity (typically 30 minutes). You will receive a warning before timeout occurs, giving you the opportunity to extend your session.

Concurrent Sessions: Depending on your organization's policy, you may be allowed to have multiple concurrent sessions (e.g., on different devices) or may be limited to a single session at a time.

Logout: Always log out when you finish using the system, especially on shared computers. Click your profile icon and select **Logout**, or simply close the browser window.

2.2 User Interface Overview

The Egyptian ERP System features a modern, intuitive user interface designed for efficiency and ease of use. Understanding the interface layout helps you navigate the system effectively.

2.2.1 Main Navigation

The navigation menu provides access to all system functions organized hierarchically by module groups, modules, and functions.

Navigation Structure:

- **Module Groups:** High-level categories (Financial, HR, Inventory, etc.)

- **Modules:** Specific functional areas within each group
- **Functions:** Individual transactions, reports, and setup options

Navigation Menu Types:

- **Sidebar Menu:** The default navigation displays as a collapsible sidebar on the left side of the screen. Click the hamburger icon to collapse or expand the menu.
- **Mega Menu:** Some configurations use a mega menu that displays all options in a grid layout when hovering over module groups.
- **Search Navigation:** Press Ctrl+K to open the quick search and find any function by typing its name.

2.2.2 Dashboard Layout

The dashboard is your personalized home screen, displaying information relevant to your role and responsibilities.

Dashboard Components:

- **KPI Cards:** Key performance indicators displayed as cards with current values, trends, and comparisons.
- **Charts and Graphs:** Visual representations of data trends, comparisons, and distributions.
- **Lists:** Recent transactions, pending approvals, alerts, and notifications.
- **Quick Actions:** Shortcuts to frequently used functions.
- **Calendar:** Important dates, deadlines, and events.

Dashboard Customization: You can customize your dashboard by adding, removing, or rearranging widgets. Click the **Customize Dashboard** button to enter customization mode.

2.2.3 Common UI Elements

The system uses consistent UI elements throughout for familiarity and ease of use.

Header Bar: The top bar contains:

- Company logo and name
- Global search (Ctrl+K)

- Notifications bell
- Help icon
- User profile menu

Content Area: The main content area displays:

- Breadcrumb navigation showing your current location
- Page title and action buttons
- Form fields, tables, or other content

Footer: The footer displays:

- System version information
- Copyright notice
- Quick links to support resources

2.3 Basic Navigation

Mastering basic navigation techniques will significantly improve your efficiency when using the system.

2.3.1 Using the Search Function

The global search function allows you to quickly find transactions, master data, and functions without navigating through menus.

To use the search function:

1. Press **Ctrl+K** or click the search icon in the header bar.
2. Type your search query (at least 3 characters).
3. Results are displayed in categories: Functions, Transactions, Masters, Reports.
4. Click on a result to navigate directly to that item.

Search Tips:

- Use partial matches: Searching for "inv" will find "Invoice", "Inventory", etc.
- Search by code or name: Enter a customer code, invoice number, or name.
- Recent searches: Your recent searches are saved for quick access.

2.3.2 Using Filters

Filters help you find specific records in lists and reports.

Quick Filters: Common filters are displayed as buttons above lists. Click a filter to apply it instantly.

Advanced Filters: Click the **Filters** button to open the advanced filter panel:

- Select filter fields from the dropdown
- Choose comparison operators (equals, contains, greater than, etc.)
- Enter filter values
- Click **Apply** to filter the list
- Click **Save Filter** to save the filter for future use

Date Range Filters: Date fields support range selection:

- Today, Yesterday, This Week, This Month, This Year
- Last 7 Days, Last 30 Days, Last 90 Days
- Custom range with start and end dates

2.3.3 Sorting and Pagination

Lists display data in a tabular format with sorting and pagination controls.

Sorting: Click any column header to sort by that column. Click again to reverse the sort order. A small arrow indicates the current sort direction.

Pagination: When a list contains many records, pagination controls appear at the bottom:

- Page size selector: Choose how many records to display per page (10, 25, 50, 100)
- Page navigation: First, Previous, Next, Last buttons
- Page number input: Jump directly to a specific page
- Record count: Display showing "Showing 1-25 of 1,234 records"

Exporting: Click the export button to download the current list view:

- **Excel:** Download as .xlsx file for further analysis
- **PDF:** Download as formatted PDF document
- **CSV:** Download as comma-separated values for import into other systems

2.4 Personalization

The system allows you to personalize various aspects to match your preferences and work style.

2.4.1 Customizing Your Dashboard

Your dashboard can be customized to display the information most relevant to you.

To customize your dashboard:

1. Click the **Customize** button on your dashboard.
2. To add a widget, click **Add Widget** and select from available widgets.
3. To remove a widget, click the X icon on the widget header.
4. To move a widget, drag and drop it to the desired position.
5. To configure a widget, click the gear icon and adjust settings.
6. Click **Save** to apply your changes.

2.4.2 Setting Preferences

Personal preferences control various aspects of the user interface.

To set your preferences:

1. Click your profile icon and select **Preferences**.

2. Configure the following options:

Language: Select your preferred language (Arabic or English). This affects all system text and reports.

Date Format: Choose your preferred date format:

- DD/MM/YYYY (default for Egypt)
- MM/DD/YYYY
- YYYY-MM-DD

Number Format: Configure decimal separator and thousands separator:

- 1,234.56 (comma thousands, point decimal)
- 1.234,56 (point thousands, comma decimal)

Time Zone: Select your local time zone (default: Cairo, Egypt).

Notifications: Choose how you want to receive notifications:

- In-app notifications
- Email notifications
- SMS notifications (if enabled)

2.4.3 Favorites and Shortcuts

Favorites provide quick access to frequently used functions.

To add a function to favorites:

1. Navigate to the function you want to favorite.
2. Click the star icon in the page header.
3. The function is added to your favorites list.

Accessing Favorites: Your favorites appear in the sidebar under the **Favorites** section. You can organize favorites into folders for better management.

Keyboard Shortcuts: The system supports various keyboard shortcuts for power users:

Shortcut	Action
Ctrl+K	Open global search
Ctrl+S	Save current record
Ctrl+N	Create new record
Ctrl+F	Find in page
Escape	Close modal / Cancel
Alt+Left Arrow	Go back
Alt+Right Arrow	Go forward

2.5 Getting Help

Multiple help resources are available throughout the system to assist you when needed.

2.5.1 In-Context Help

In-context help provides immediate assistance without leaving your current task.

Field Tooltips: Hover over any field label to see a brief description of the field's purpose, data type, and any validation rules.

Help Icons: Click the question mark icon on any page to access detailed help documentation for that specific function.

Process Guides: Complex processes include step-by-step guides accessible from the help menu. These guides walk you through the entire process with explanations at each step.

2.5.2 Chatbot Assistance

The AI-powered chatbot provides instant answers to your questions using this manual as its knowledge base.

To use the chatbot:

1. Click the chatbot icon in the bottom-right corner of the screen.
2. Type your question in natural language.
3. The chatbot will search this manual and provide relevant answers.
4. You can ask follow-up questions for clarification.

Example Questions:

- "How do I create a journal entry?"
- "What is the VAT rate in Egypt?"
- "How do I process payroll?"
- "Why can't I post this invoice?"

2.6 Security Best Practices

Following security best practices protects your account and the organization's data.

2.6.1 Protecting Your Account

Password Security:

- Never share your password with anyone, including IT support
- Use a strong, unique password not used for other systems
- Change your password regularly (every 90 days recommended)
- Never write down your password or store it in unsecured locations

Session Security:

- Always log out when leaving your workstation
- Never leave your computer unlocked and unattended
- Be cautious of shoulder surfing when entering passwords
- Report suspicious activity immediately

2.6.2 Recognizing Phishing Attempts

Phishing attempts may try to steal your credentials through fake emails or websites.

Warning: Egyptian ERP Solutions will never ask you to provide your password via email. Always access the system by typing the URL directly into your browser, not by clicking links in emails.

Signs of Phishing:

- Emails with urgent or threatening language
- Requests to verify account information
- Links that don't match the displayed text
- Misspellings or poor grammar
- Suspicious sender email addresses

2.6.3 Reporting Security Issues

If you suspect a security issue, report it immediately:

- Contact your system administrator
- Call the IT security hotline
- Email security@yourcompany.com

Tip: When in doubt, verify! If something seems suspicious, contact IT support to confirm before taking action.

Chapter 3: User Management and Security

3.1 Introduction to Security

The Egyptian ERP System implements a comprehensive security framework designed to protect sensitive business data, ensure regulatory compliance, and maintain the integrity of business operations. Understanding this security framework is essential for system administrators, security officers, and all users who interact with the system.

Security in the Egyptian ERP System is built on the principle of defense in depth, with multiple layers of protection working together to safeguard your data. From authentication and authorization to encryption and audit trails, every aspect of security has been carefully designed and implemented.

3.1.1 Security Framework Overview

The security framework encompasses several key components that work together to protect the system:

Authentication: Verifies the identity of users attempting to access the system. The system supports multiple authentication methods including username/password, multi-factor authentication (MFA), and single sign-on (SSO) integration.

Authorization: Determines what authenticated users are allowed to do within the system. Role-based access control (RBAC) ensures users only have access to the functions and data necessary for their job responsibilities.

Encryption: Protects data both in transit and at rest. All communications between users and the system use TLS 1.3 encryption, while data at rest is protected using AES-256 encryption.

Audit Trails: Comprehensive logging of all system activities enables monitoring, compliance reporting, and forensic investigation when needed.

Data Segregation: Ensures data from different companies or business units remains properly isolated while allowing authorized cross-company transactions where needed.

3.1.2 Security Responsibilities

Security is a shared responsibility involving multiple roles:

System Administrators: Responsible for configuring security settings, managing user accounts, monitoring security logs, and responding to security incidents.

Security Officers: Define security policies, conduct security assessments, manage compliance requirements, and oversee the security program.

End Users: Responsible for protecting their credentials, following security policies, reporting suspicious activity, and using the system in accordance with security guidelines.

Management: Sets security priorities, allocates resources for security initiatives, and ensures security requirements are met.

3.2 User Management

User management is the foundation of system security. Proper user management ensures that only authorized individuals have access to the system and that their access is appropriate for their role.

3.2.1 Creating Users

Creating a new user account involves several steps to ensure proper setup and security configuration.

To create a new user:

1. Navigate to **Administration** → **Security** → **Users**.
2. Click the **Create New** button.
3. Enter the following information:

Basic Information:

- **User ID:** Unique identifier for the user (typically employee ID)

- **Full Name:** User's full name as it should appear in the system
- **Email Address:** Valid email address for notifications and password recovery
- **Phone Number:** Mobile number for SMS notifications and MFA
- **Department:** User's department for organizational reporting
- **Employee ID:** Link to employee record if applicable

Security Settings:

- **Password:** Initial password (user will be forced to change on first login)
- **Force Password Change:** Require password change on first login (recommended)
- **MFA Required:** Enable multi-factor authentication for this user
- **Account Expiry:** Set expiration date for temporary accounts

Role Assignment:

- Select one or more roles to assign to the user
- Roles determine what functions the user can access
- Multiple roles can be combined for complex access requirements

4. Click **Save** to create the user account.

5. The system will send a welcome email to the user with login instructions.

3.2.2 Managing Users

Ongoing user management ensures accounts remain secure and appropriate as roles change.

Activating/Deactivating Users:

To deactivate a user:

1. Navigate to **Administration** → **Security** → **Users**.
2. Search for and select the user.
3. Click **Edit**.
4. Set **Status** to **Inactive**.

5. Click Save.

Note: Deactivating a user immediately prevents login but preserves all audit history and transaction records associated with the user.

Resetting Passwords:**To reset a user's password:**

1. Navigate to **Administration** → **Security** → **Users**.
2. Search for and select the user.
3. Click **Reset Password**.
4. Enter a temporary password or generate a random one.
5. The user will receive an email with the temporary password and must change it on next login.

Updating User Information: User information can be updated at any time to reflect changes in department, contact details, or other attributes. Navigate to the user record and click Edit to make changes.

3.2.3 User Self-Service

Users can manage certain aspects of their own accounts through the profile menu:

- Change password
- Update contact information
- Configure notification preferences
- Set language and regional preferences
- View login history

3.3 Role-Based Access Control (RBAC)

Role-based access control is the primary mechanism for managing user permissions in the Egyptian ERP System. Roles define sets of permissions that can be assigned to users based on their job functions.

3.3.1 Understanding Roles

A role is a collection of permissions that define what actions a user can perform and what data they can access. Roles provide several benefits:

- **Simplified Administration:** Assign permissions to roles once, then assign roles to multiple users
- **Consistency:** Users with the same job function have the same access
- **Auditability:** Easy to review who has access to what
- **Scalability:** New users can be quickly assigned appropriate access

3.3.2 Standard Roles

The system includes pre-configured standard roles for common job functions:

Role	Description	Typical Access
System Administrator	Full system access	All modules and functions
Finance Manager	Financial management	All financial modules, reports
Accountant	Day-to-day accounting	GL, AP, AR, limited reporting
HR Manager	Human resources management	HR and Payroll modules
Warehouse Manager	Inventory management	Inventory, Purchasing
Sales Representative	Sales operations	Sales, Customer inquiries
End User	Basic system access	Self-service, limited transactions

3.3.3 Creating Custom Roles

When standard roles don't meet your requirements, you can create custom roles with specific permissions.

To create a custom role:

1. Navigate to **Administration** → **Security** → **Roles**.
2. Click **Create New**.
3. Enter Role Code (unique identifier).
4. Enter Role Name (descriptive name).
5. Enter Description (purpose of the role).
6. Select Parent Role if this role extends another role.
7. Assign permissions for each module and function.
8. Set data restrictions (company, department, cost center).
9. Click **Save**.

Permission Types:

- **View:** Can view records but not modify
- **Create:** Can create new records
- **Edit:** Can modify existing records
- **Delete:** Can delete records
- **Post:** Can post/approve transactions
- **Report:** Can generate reports
- **Export:** Can export data

3.3.4 Data Restrictions

Data restrictions provide additional control over what data users can see and modify:

Company Restriction: Limit user access to specific companies in a multi-company environment.

Department Restriction: Restrict access to data belonging to specific departments.

Cost Center Restriction: Limit access to transactions for specific cost centers.

Amount Limit: Set maximum transaction amounts that users can process without additional approval.

3.4 Authentication and Security Settings

System-wide authentication and security settings are configured by administrators to enforce organizational security policies.

3.4.1 Password Policy

The password policy defines requirements for user passwords to ensure adequate strength.

To configure password policy:

1. Navigate to **Administration** → **Security** → **Password Policy**.
2. Configure the following settings:

Password Requirements:

- **Minimum Length:** Minimum number of characters (recommended: 8-12)
- **Require Uppercase:** At least one uppercase letter required
- **Require Lowercase:** At least one lowercase letter required
- **Require Numbers:** At least one numeric digit required
- **Require Special Characters:** At least one special character required

Password Expiration:

- **Expiration Period:** Number of days before password must be changed (0 = never)
- **Warning Period:** Days before expiration to start warning users

- **Password History:** Number of previous passwords to prevent reuse

Account Lockout:

- **Failed Attempts:** Number of failed login attempts before lockout
- **Lockout Duration:** Minutes before automatic unlock (0 = require admin unlock)

3.4.2 Multi-Factor Authentication (MFA)

MFA adds an extra layer of security by requiring a second verification factor in addition to the password.

MFA Methods:

- **SMS:** Verification code sent via text message
- **Email:** Verification code sent to registered email
- **Authenticator App:** Time-based one-time passwords (TOTP)
- **Hardware Token:** Physical security token

To enable MFA for a user:

1. Navigate to **Administration** → **Security** → **Users**.
2. Select the user.
3. Click **Edit**.
4. Set **MFA Required** to **Yes**.
5. Select the MFA method.
6. Click **Save**.
7. The user will be prompted to set up MFA on next login.

3.4.3 Session Security

Session security settings control how user sessions are managed:

Session Timeout: Number of minutes of inactivity before automatic logout (recommended: 30 minutes).

Concurrent Sessions: Maximum number of simultaneous sessions allowed per user:

- Single session: User can only be logged in from one device at a time
- Multiple sessions: User can be logged in from multiple devices

IP Restrictions: Limit access to specific IP addresses or ranges. Useful for restricting access to office locations.

Device Registration: Require users to register devices before accessing from new locations. Provides additional security for remote access.

3.5 Audit Trails

Comprehensive audit trails record all system activities for security monitoring, compliance reporting, and forensic investigation.

3.5.1 Transaction Logging

Every significant action in the system is logged with the following information:

Field	Description
Timestamp	Date and time of the action
User ID	User who performed the action
Action Type	Create, Update, Delete, View, Login, Logout, etc.
Module	System module where action occurred
Function	Specific function or transaction
Record ID	Identifier of affected record
Before Values	Values before change (for updates)
After Values	Values after change

IP Address	IP address of the user
Session ID	Session identifier for correlation

3.5.2 Audit Reports

Several audit reports are available for monitoring and compliance:

User Activity Report: Shows all activities performed by a specific user or group of users within a date range. Useful for investigating suspicious activity or reviewing user actions.

Transaction History Report: Shows the complete history of changes to a specific record. Useful for understanding how data has evolved over time.

Login/Logout Report: Tracks user login and logout activity including failed login attempts. Useful for monitoring access patterns and detecting unauthorized access attempts.

Failed Access Attempts Report: Shows all failed login attempts with details. Useful for identifying potential security threats.

Data Modification Report: Shows all create, update, and delete operations. Useful for compliance auditing and change tracking.

To generate an audit report:

1. Navigate to **Administration** → **Security** → **Audit Reports**.
2. Select the report type.
3. Set filter criteria (date range, users, modules, etc.).
4. Click **Generate**.
5. Review the report on screen or export to Excel/PDF.

3.5.3 Audit Log Retention

Audit logs are retained according to configurable retention policies:

- **Online Logs:** Recent logs kept in the database for quick access (default: 90 days)

- **Archived Logs:** Older logs moved to archive storage (default: 7 years)
- **Compliance Retention:** Some logs may need to be retained longer for regulatory compliance

Important: Audit logs are tamper-proof and cannot be modified or deleted through the user interface. This ensures the integrity of audit records for compliance and forensic purposes.

3.6 Security Best Practices

Following security best practices helps maintain a secure system environment:

3.6.1 User Account Management

- Review user accounts quarterly and deactivate unused accounts
- Remove access immediately when employees leave the organization
- Review role assignments annually to ensure appropriate access
- Implement the principle of least privilege - users should have minimum necessary access
- Use shared accounts only when absolutely necessary and monitor closely

3.6.2 Password Management

- Enforce strong password policies
- Require regular password changes
- Prevent password reuse
- Consider implementing password managers for users
- Never store passwords in plain text or shared documents

3.6.3 Monitoring and Response

- Review audit logs regularly for suspicious activity

- Set up alerts for critical security events
- Have an incident response plan for security breaches
- Conduct periodic security assessments
- Keep the system updated with security patches

Chapter 4: Company Setup and General Ledger

4.1 Introduction to Company Setup

Proper company setup is the foundation of successful ERP implementation. The company master defines your organization's structure, fiscal calendar, and basic configuration that affects all modules. Taking time to configure company settings correctly at the beginning will save significant effort later and ensure accurate financial reporting and regulatory compliance.

The company setup process involves defining organizational information, configuring the fiscal year structure, setting up the chart of accounts, and establishing basic accounting policies. This chapter guides you through each step of the company setup process and explains how to maintain these settings as your organization evolves.

4.1.1 Pre-Implementation Planning

Before beginning company setup, gather all required information to ensure accurate configuration:

Legal Information:

- Company legal name (English and Arabic)
- Commercial registration number
- Tax registration number (from ETA)
- Social insurance number (from GOSI)
- Legal form (LLC, JSC, etc.)
- Date of establishment

Contact Information:

- Head office address
- Phone and fax numbers

- Email address
- Website

Fiscal Information:

- Fiscal year start and end dates
- Accounting period structure (monthly, 4-week periods)
- Base currency (typically EGP for Egyptian companies)
- Additional currencies if applicable

4.2 Company Master Setup

The company master is the central record that defines your organization in the system. All other configuration builds upon this foundation.

4.2.1 Basic Company Information

To set up company master:

1. Navigate to **Administration** → **Company Setup** → **Company Master**.
2. Click **Create New** or **Edit** to modify existing.
3. Enter the following information:

Company Code: Unique identifier for the company (2-10 characters, alphanumeric). This code appears on all transactions and reports.

Company Name: Legal name of the company in English. This appears on official documents, invoices, and reports.

Company Name (Arabic): Legal name in Arabic. Required for official Egyptian documents and ETA submissions.

Commercial Registration Number: Your company's commercial registration number from the Ministry of Trade and Industry.

Tax Registration Number: Your ETA tax registration number. This is critical for E-Invoicing compliance.

Social Insurance Number: Your GOSI registration number for social insurance reporting.

4.2.2 Address Information

Complete address information is required for official documents and regulatory reporting:

- **Street Address:** Building number, street name, district
- **City:** City name
- **Governorate:** Select from dropdown of Egyptian governorates
- **Postal Code:** Egyptian postal code
- **Country:** Egypt (default)

4.2.3 Contact Information

Contact details for business communications:

- **Phone Number:** Primary contact number with country code (+20 for Egypt)
- **Fax Number:** If applicable
- **Email Address:** Official company email
- **Website:** Company website URL

4.2.4 Fiscal Year Setup

The fiscal year configuration determines your accounting calendar and reporting periods.

Fiscal Year Start: The first day of your fiscal year. Common configurations:

- January 1 (calendar year)
- July 1 (Egyptian fiscal year)
- Other dates based on business requirements

Fiscal Year End: Automatically calculated based on start date and period structure.

Number of Periods:

- **12 Periods:** Standard monthly accounting periods
- **13 Periods:** Four-week periods for more consistent period lengths

Period Type:

- **Calendar Months:** January through December
- **4-4-5 Weeks:** Common retail calendar with consistent week counts
- **Custom:** Define your own period structure

4.2.5 Period Status Management

Each accounting period has a status that controls transaction processing:

Status	Description	Transactions Allowed
Open	Period is active for transaction entry	Yes
Closed	Period is closed, no transactions allowed	No
Pending Close	Closing process in progress	Limited
Future	Period not yet opened	No

To open/close periods:

1. Navigate to **Financial** → **General Ledger** → **Period Management**.
2. Select the period to modify.
3. Click **Open** or **Close** as needed.
4. Enter justification for the change.
5. Click **Confirm**.

Warning: Closing a period prevents any further transactions in that period. Ensure all entries are complete before closing. Reopening a closed period requires special

authorization and should be done only when necessary.

4.3 Chart of Accounts

The Chart of Accounts is the foundation of financial reporting. It defines all accounts used to record financial transactions and must be carefully planned to meet current and future reporting needs.

4.3.1 Chart of Accounts Structure

A well-designed chart of accounts follows a logical structure that facilitates reporting and analysis. For Egyptian companies, the typical structure aligns with Egyptian accounting standards:

Account Range	Category	Balance Type
1xxxx	Assets	Debit
2xxxx	Liabilities	Credit
3xxxx	Equity	Credit
4xxxx	Revenue	Credit
5xxxx	Cost of Sales	Debit
6xxxx	Operating Expenses	Debit
7xxxx	Other Income/Expenses	Varies

Account Number Format: The system supports flexible account numbering with segments for different dimensions:

- **Simple:** 11000 (Cash)
- **With Cost Center:** 11000-100 (Cash - Head Office)
- **With Project:** 11000-100-PRJ001 (Cash - Head Office - Project 001)

4.3.2 Creating Accounts

To create a new account:

1. Navigate to **Financial** → **General Ledger** → **Chart of Accounts**.
2. Click **Create New**.
3. Enter the following information:

Account Code: Unique account number following your chart of accounts structure.

Account Name: Descriptive name for the account (English).

Account Name (Arabic): Arabic name for official documents.

Account Type: Select from:

- Asset
- Liability
- Equity
- Revenue
- Expense

Account Category: Further classification for reporting:

- Current Asset
- Fixed Asset
- Current Liability
- Long-term Liability
- Direct Revenue
- Indirect Revenue
- Operating Expense
- Administrative Expense

Parent Account: For sub-accounts, select the parent account. This creates a hierarchical structure for summary reporting.

4.3.3 Account Properties

Additional properties control how accounts behave:

Balance Type: Determines normal balance (Debit or Credit). Affects balance display and reconciliation.

Currency: Base currency for the account (typically EGP). Foreign currency accounts track balances in both local and foreign currency.

Cash Flow Category: Classification for cash flow statement reporting:

- Operating Activities
- Investing Activities
- Financing Activities

Budget Control: Enable to track actual vs. budgeted amounts for this account.

Blocked: Prevent new postings to this account while preserving historical data.

4.4 General Ledger Operations

The General Ledger is the central repository of all financial transactions. Understanding GL operations is essential for accurate financial reporting.

4.4.1 Journal Entries

Journal entries record financial transactions in the general ledger. Each entry consists of a header and multiple line items that must balance (total debits equal total credits).

To create a journal entry:

1. Navigate to **Financial** → **General Ledger** → **Journal Entries** → **Create New**.
2. Enter header information:

Journal Entry Header:

- **Entry Number:** Automatically assigned or manually entered based on configuration
- **Entry Date:** Date of the transaction (must be in an open period)
- **Posting Date:** Date when entry affects account balances (may differ from entry date)
- **Reference:** External reference number (e.g., check number, invoice number)
- **Description:** Brief description of the transaction
- **Journal Type:** Category of journal entry (General, Adjustment, Closing, etc.)

3. Add line items:**Journal Entry Lines:**

- **Line Number:** Sequential line identifier
- **Account:** GL account to debit or credit
- **Description:** Line-specific description (optional)
- **Debit Amount:** Amount to debit (or leave blank for credit)
- **Credit Amount:** Amount to credit (or leave blank for debit)
- **Cost Center:** Cost center for dimensional reporting
- **Project:** Project code if applicable
- **Reference:** Additional reference information

4. Verify that Total Debits equals Total Credits.**5. Click **Save** to save as draft or **Post** to post immediately.****Example: Recording Office Rent Expense****Header:**

- Entry Date: 01/01/2025
- Description: January 2025 Office Rent
- Reference: INV-2025-001

Lines:

Account	Description	Debit	Credit
61010 - Rent Expense	Office Rent	10,000.00	
21010 - Accounts Payable	Rent Payable		10,000.00

4.4.2 Posting Journal Entries

Posting updates account balances and makes entries part of the official financial records.

To post a journal entry:

1. Review the entry for accuracy.
2. Click **Post**.
3. System validates the entry (period open, accounts valid, debits = credits).
4. If validation passes, the entry is posted and account balances are updated.
5. Posted entries cannot be modified; corrections require reversal entries.

Important: Once posted, journal entries cannot be deleted or modified. This ensures audit integrity. If corrections are needed, create a reversing entry and then post the correct entry.

4.4.3 Reversing Entries

When a posted entry contains errors, use reversal to correct it:

To reverse a journal entry:

1. Navigate to **Financial** → **General Ledger** → **Journal Entries**.
2. Find and open the entry to reverse.
3. Click **Reverse**.
4. Enter reversal date (typically current date).
5. Enter reason for reversal.
6. Click **Confirm**.
7. System creates a reversal entry with opposite debits/credits.
8. Post the reversal entry.
9. Create and post the correct entry if needed.

4.5 Financial Reporting

The General Ledger provides standard financial reports essential for business management and regulatory compliance.

4.5.1 Trial Balance

The Trial Balance shows all account balances at a specific point in time. It is the fundamental report used to verify the accuracy of accounting records.

Trial Balance Contents:

- **Account Code and Name:** Each account in the chart of accounts
- **Opening Balance:** Balance at the start of the period
- **Period Debits:** Total debits during the period
- **Period Credits:** Total credits during the period
- **Closing Balance:** Balance at the end of the period

To generate a Trial Balance:

1. Navigate to **Financial** → **General Ledger** → **Reports** → **Trial Balance**.
2. Select the reporting period.
3. Choose detail level (summary or detailed).
4. Select additional options (include zero balances, show cost centers, etc.).
5. Click **Generate**.
6. Review on screen or export to Excel/PDF.

Note: A properly balanced system will show equal total debits and credits on the Trial Balance. Any imbalance indicates an error that must be investigated and corrected.

4.5.2 Income Statement

The Income Statement (Profit and Loss Statement) shows financial performance over a period of time.

Income Statement Structure:

- **Revenue:** Sales and other income
- **Less: Cost of Sales:** Direct costs of goods/services sold
- = **Gross Profit:** Revenue minus cost of sales
- **Less: Operating Expenses:** Selling, administrative, and general expenses
- = **Operating Income:** Profit from operations
- **Plus/Less: Other Income/Expenses:** Non-operating items
- = **Net Income:** Bottom line profit or loss

4.5.3 Balance Sheet

The Balance Sheet shows the financial position at a specific point in time.

Balance Sheet Structure:

- **Assets:** Current Assets and Non-Current Assets

- **Liabilities:** Current Liabilities and Non-Current Liabilities
- **Equity:** Share Capital, Retained Earnings, Reserves

Note: The fundamental accounting equation must always balance: $\text{Assets} = \text{Liabilities} + \text{Equity}$

4.5.4 Cash Flow Statement

The Cash Flow Statement shows how cash moved during a period, categorized by activity type.

Cash Flow Categories:

- **Operating Activities:** Cash from normal business operations
- **Investing Activities:** Cash from asset purchases/sales
- **Financing Activities:** Cash from loans, equity, dividends

The system generates the Cash Flow Statement using the indirect method, starting with net income and adjusting for non-cash items and changes in working capital.

Chapter 5: Accounts Payable Management

5.1 Introduction to Accounts Payable

The Accounts Payable (AP) module manages all vendor-related financial transactions from invoice receipt through payment processing. Effective AP management ensures timely payments to vendors, accurate financial records, and optimal cash flow management. The module integrates seamlessly with Purchasing, General Ledger, and Cash Management to provide a complete view of payables.

In the Egyptian business context, the AP module also ensures compliance with tax regulations including proper VAT tracking and ETA E-Invoicing requirements for vendor invoices. Understanding these compliance aspects is essential for proper AP operations.

5.1.1 AP Process Overview

The accounts payable process follows a structured workflow:

1. **Vendor Setup:** Create and maintain vendor master records with all required information
2. **Invoice Receipt:** Receive and record vendor invoices
3. **Invoice Verification:** Validate invoice accuracy and match to supporting documents
4. **Approval:** Route invoices for appropriate authorization
5. **Payment Processing:** Generate and execute payments
6. **Reconciliation:** Reconcile vendor statements and resolve discrepancies

5.1.2 Key Concepts

Three-Way Matching: The process of verifying that purchase order, goods receipt, and vendor invoice all agree before payment. This control prevents payment for goods not received or at incorrect prices.

Payment Terms: Conditions specified by vendors for payment, including due date, early payment discounts, and late payment penalties. Common terms include Net 30 (payment due in 30 days) and 2/10 Net 30 (2% discount if paid within 10 days, otherwise net amount due in 30 days).

Aging: Categorization of payables by how long they have been outstanding. Aging reports help identify overdue invoices and manage cash flow.

5.2 Vendor Master

The vendor master contains all information about your vendors. Accurate vendor setup is essential for proper invoice processing, payment execution, and regulatory compliance.

5.2.1 Creating Vendor Records

To create a new vendor:

1. Navigate to **Accounts Payable** → **Vendors** → **Create New**.
2. Enter Basic Information:

Basic Information:

- **Vendor Code:** Unique identifier (alphanumeric, typically 5-10 characters)
- **Vendor Name:** Legal name as it appears on invoices and contracts
- **Vendor Name (Arabic):** Arabic name for official documents
- **Vendor Type:** Local, Foreign, Government, or Employee
- **Category:** Classification for reporting (Raw Materials, Services, Utilities, etc.)
- **Status:** Active, On Hold, or Blocked

Tax Information:

- **Tax Registration Number:** Vendor's ETA registration number (required for B2B)
- **Commercial Registration Number:** Vendor's commercial registration
- **VAT Treatment:** Standard (14%), Zero-rated, or Exempt

Payment Information:

- **Payment Terms:** Select from predefined terms or create custom
- **Credit Limit:** Maximum outstanding balance allowed
- **Currency:** Default currency for transactions
- **Payment Method:** Preferred payment method (Bank Transfer, Check, Cash)

Bank Information:

- **Bank Name:** Vendor's bank
- **Account Number:** Vendor's bank account number
- **IBAN:** International Bank Account Number
- **SWIFT Code:** Bank identifier for international transfers

Contact Information:

- **Address:** Complete mailing address
- **Contact Person:** Primary contact name
- **Phone/Email:** Contact details

5.2.2 Vendor Classification

Vendors can be classified by status to control transaction processing:

Status	Description	Transactions Allowed
Active	Normal vendor, all transactions allowed	All
On Hold	Temporary suspension, requires review	None until reviewed
Blocked	Permanently blocked, no transactions	None
Provisional	New vendor pending approval	Limited

5.2.3 Vendor Evaluation

The system supports vendor performance tracking to help manage vendor relationships:

Performance Metrics:

- **Quality Rating:** Based on quality inspection results
- **Delivery Performance:** On-time delivery percentage
- **Price Competitiveness:** Comparison to market prices
- **Service Level:** Responsiveness and support quality

5.3 Invoice Processing

Invoice processing is the core AP function. Proper invoice handling ensures accurate records, timely payments, and good vendor relationships.

5.3.1 Invoice Entry

To enter a vendor invoice:

1. Navigate to **Accounts Payable** → **Invoices** → **Create New**.
2. Enter Header Information:

Invoice Header:

- **Vendor:** Select from vendor master
- **Invoice Number:** Vendor's invoice number (must be unique per vendor)
- **Invoice Date:** Date on vendor's invoice
- **Posting Date:** Date to record in system
- **Due Date:** Payment due date (calculated from payment terms)
- **Reference:** Purchase order number or other reference

3. Enter Line Items:**Invoice Lines:**

- **Line Type:** Item, Expense, or GL Account
- **Account/Item:** GL account or inventory item

- **Description:** Description of goods/services
- **Quantity:** Number of units
- **Unit Price:** Price per unit
- **Line Total:** Calculated as Quantity × Unit Price
- **Tax Code:** VAT treatment for the line
- **Tax Amount:** Calculated VAT amount

4. Review invoice totals and tax calculations.
5. Click **Save** to save as draft or **Submit** for approval.

5.3.2 Three-Way Matching

Three-way matching verifies invoice accuracy by comparing three documents:

1. **Purchase Order (PO):** What was ordered (quantity, price)
2. **Goods Receipt Note (GRN):** What was received (quantity, condition)
3. **Vendor Invoice:** What was billed (quantity, price)

Matching Process:

- System automatically matches invoice to PO and GRN based on PO number
- Quantities must match within tolerance (configurable, default 0%)
- Prices must match PO prices unless variance is approved
- If all match, invoice is approved for payment
- If discrepancies exist, invoice is flagged for review

To handle matching discrepancies:

1. Review the discrepancy report showing differences between PO, GRN, and Invoice.
2. Investigate the cause (short shipment, price change, data entry error).
3. Take appropriate action:

- **Accept Variance:** If within approved tolerance, approve with comment

- **Request Credit Note:** If overcharged, request vendor credit
- **Reject Invoice:** Return invoice to vendor for correction
- **Adjust PO:** If PO was incorrect, process amendment

5.3.3 Invoice Approval

Invoices require approval before payment based on your organization's approval matrix:

Approval Levels (Example):

Invoice Amount	Approver
Up to EGP 5,000	Department Manager
EGP 5,001 - 25,000	Finance Manager
EGP 25,001 - 100,000	CFO
Above EGP 100,000	CEO + CFO

To approve an invoice:

1. Navigate to **Accounts Payable** → **Invoices** → **Pending Approval**.
2. Review invoice details and supporting documents.
3. Click **Approve** to approve or **Reject** to return with comments.
4. Approved invoices proceed to payment processing.

5.4 Payment Processing

Payment processing executes vendor payments efficiently while maintaining proper controls.

5.4.1 Payment Proposal

The payment proposal identifies invoices to pay based on due dates, cash availability, and vendor priorities.

To generate a payment proposal:

1. Navigate to **Accounts Payable** → **Payments** → **Payment Proposal**.
2. Set selection criteria:

Selection Criteria:

- **Payment Date:** Date to execute payments
- **Due Date Range:** Include invoices due through this date
- **Vendor Selection:** All vendors or specific vendors
- **Payment Method:** Bank transfer, check, cash
- **Currency:** Filter by currency if applicable

3. Click **Generate Proposal**.

4. Review proposed payments:

Payment Proposal Review:

- Review list of invoices selected for payment
- Verify amounts and due dates
- Check available cash/bank balance
- Remove any invoices that should not be paid
- Add any urgent invoices not in the proposal

5. Click **Create Payment Batch** to proceed.

5.4.2 Payment Methods

The system supports multiple payment methods:

Bank Transfer:

- Most common for vendor payments

- Electronic file generated for bank upload
- Supports local and international transfers
- Payment advice automatically emailed to vendors

Check:

- Printed checks for vendors who prefer them
- Check printing integrated with system
- Check register tracks all issued checks
- Voided checks properly recorded

Cash:

- For small payments or vendors without bank accounts
- Requires additional authorization
- Proper documentation and receipts required

5.4.3 Payment Execution

To execute payments:

1. Review payment batch and verify all details.
2. For bank transfers:

- Generate payment file in bank format
- Upload to online banking or submit to bank
- Confirm successful transmission

3. For checks:

- Load check stock in printer
- Print checks from system
- Obtain required signatures
- Mail or deliver to vendors

4. Record payment in system:

- Update invoice status to Paid
- Record payment reference (check number, transfer reference)
- Post to General Ledger

5.5 Reporting

AP reports provide visibility into payables and support decision-making.

5.5.1 AP Aging Report

The AP Aging Report categorizes outstanding payables by age:

Category	Description
Current	Not yet due
1-30 Days	Overdue up to 30 days
31-60 Days	Overdue 31-60 days
61-90 Days	Overdue 61-90 days
Over 90 Days	Overdue more than 90 days

Using the Aging Report:

- Identify overdue invoices requiring attention
- Manage cash flow by prioritizing payments
- Negotiate payment terms with vendors
- Identify disputes requiring resolution

5.5.2 Vendor Balance Report

Shows the balance for each vendor including:

- Opening balance

- Invoices during period
- Payments during period
- Credit notes/adjustments
- Closing balance

5.5.3 Payment History Report

Tracks all payments made to vendors:

- Payment date and amount
- Payment method and reference
- Invoices paid
- Discounts taken

5.6 Period-End Procedures

Monthly closing procedures ensure accurate AP balances:

AP month-end checklist:

1. Enter all invoices received during the month
2. Process all approved payments
3. Reconcile vendor statements
4. Resolve any discrepancies
5. Review AP aging for unusual items
6. Generate AP reconciliation report
7. Verify GL balance matches AP subledger

Chapter 6: Accounts Receivable Management

6.1 Introduction to Accounts Receivable

The Accounts Receivable (AR) module manages all customer-related financial transactions from invoicing through cash collection. Effective AR management is critical for maintaining healthy cash flow, minimizing bad debt, and ensuring strong customer relationships. The module integrates with Sales, General Ledger, and the ETA E-Invoicing portal to provide comprehensive receivables management.

In Egypt, the AR module must comply with ETA E-Invoicing requirements, ensuring all B2B and B2G invoices are properly submitted to the Egyptian Tax Authority with digital signatures and QR codes.

6.1.1 AR Process Overview

The accounts receivable process encompasses:

1. **Customer Setup:** Create and maintain customer master records
2. **Sales Invoicing:** Create and submit customer invoices
3. **ETA Submission:** Submit invoices to Egyptian Tax Authority
4. **Receipt Processing:** Record and apply customer payments
5. **Credit Management:** Monitor and control customer credit exposure
6. **Collections:** Follow up on overdue accounts

6.2 Customer Master

The customer master contains all information about your customers. Accurate customer setup is essential for proper invoicing, credit management, and regulatory compliance.

6.2.1 Creating Customer Records

To create a new customer:

1. Navigate to **Accounts Receivable** → **Customers** → **Create New**.
2. Enter Basic Information:

Basic Information:

- **Customer Code:** Unique identifier (alphanumeric)
- **Customer Name:** Legal name as it should appear on invoices
- **Customer Name (Arabic):** Arabic name for official documents
- **Customer Type:** Individual, Company, Government
- **Category:** Classification for reporting (Retail, Wholesale, Export, etc.)
- **National ID/Tax ID:** Required for B2B customers

Tax Information:

- **Tax Registration Number:** Customer's ETA registration (mandatory for B2B)
- **VAT Treatment:** Standard (14%), Zero-rated, or Exempt
- **E-Invoice Required:** Yes for B2B/B2G customers

Credit Information:

- **Credit Limit:** Maximum outstanding balance allowed
- **Credit Period:** Payment terms (days)
- **Payment Terms:** Net 30, Net 60, etc.
- **Credit Hold:** Block new orders if exceeded

Contact Information:

- **Address:** Billing and shipping addresses
- **Contact Person:** Primary contact
- **Phone/Email:** Contact details

6.2.2 Customer Classification

Customers can be classified by status:

Status	Description
Active	Normal customer, all transactions allowed
On Hold	Credit hold, no new orders
Blocked	Permanently blocked
Prospect	Potential customer, no transactions yet

6.3 Sales Invoicing

Sales invoices document revenue and create customer receivables. In Egypt, B2B and B2G invoices must be submitted to the ETA portal.

6.3.1 Creating Invoices

To create a sales invoice:

1. Navigate to **Accounts Receivable** → **Invoices** → **Create New**.
2. Enter Header Information:

Invoice Header:

- **Customer:** Select from customer master
- **Invoice Date:** Date of invoice
- **Due Date:** Payment due date (from customer terms)
- **Reference:** Sales order or delivery note number
- **Delivery Address:** Where goods were delivered

3. Add Line Items:

Invoice Lines:

- **Item:** Product or service code
- **Description:** Item description
- **Quantity:** Number of units
- **Unit Price:** Price per unit
- **Discount:** Line discount percentage or amount
- **Tax Code:** VAT treatment

4. Review totals and taxes.

5. Click **Save** to save or **Submit** to proceed to ETA submission.

6.3.2 ETA E-Invoice Submission

For B2B and B2G customers, invoices must be submitted to the ETA portal:

ETA submission process:

1. System validates invoice data against ETA requirements
2. Generates UUID (Unique Identifier)
3. Creates JSON format per ETA specification
4. Applies digital signature using company certificate
5. Submits to ETA portal via API
6. Receives ETA ID and QR code
7. Updates invoice with ETA reference
8. Invoice can now be printed with QR code

Note: Invoices for B2B/B2G customers cannot be printed or sent to customers until successfully submitted to ETA and validated.

6.4 Receipt Processing

Receipts record customer payments and apply them to outstanding invoices.

6.4.1 Recording Receipts

To record a customer receipt:

1. Navigate to **Accounts Receivable** → **Receipts** → **Create New**.
2. Enter Receipt Header:

Receipt Header:

- **Customer:** Select customer
- **Receipt Date:** Date payment received
- **Receipt Number:** Auto-generated or manual
- **Payment Method:** Cash, Check, Bank Transfer, Credit Card
- **Reference:** Check number, transfer reference, etc.
- **Amount:** Total receipt amount
- **Bank Account:** Account where deposited

3. Apply to Invoices:

Application Options:

- **Auto-Apply:** System applies to oldest invoices first
- **Manual Apply:** Select specific invoices to apply
- **On Account:** Record as unapplied credit

4. Click **Post** to record the receipt.

6.4.2 Receipt Types

Cash Receipts: Record cash payments received at office or retail locations.

Check Receipts: Record customer checks with check number and bank details.

Bank Transfer Receipts: Record wire transfers and electronic payments.

Credit Card Receipts: Record credit card payments with authorization codes.

6.5 Credit Management

Credit management monitors and controls customer credit exposure to minimize bad debt risk.

6.5.1 Credit Control

Credit Limit Monitoring:

- System tracks each customer's outstanding balance
- Warns when approaching credit limit
- Blocks new orders when limit exceeded

Credit Review Process:

1. Review customer payment history
2. Analyze aging and collection patterns
3. Assess financial stability (for large limits)
4. Set or adjust credit limit
5. Document approval for significant increases

6.5.2 Collections

Collections follow up on overdue accounts to secure payment:

Collection Activities:

- **Reminder Letters:** Automated emails at 5, 15, 30 days overdue
- **Phone Calls:** Personal follow-up for significant amounts
- **Collection Letters:** Formal demand letters
- **Payment Plans:** Negotiate installment arrangements

- **Legal Action:** Final recourse for uncollectible amounts

To record collection activities:

1. Navigate to **Accounts Receivable** → **Collections**.
2. Select the customer.
3. Click **Add Activity**.
4. Enter activity type, date, and notes.
5. Record any payment promises made.
6. Set follow-up date if needed.

6.6 Reporting

6.6.1 AR Aging Report

The AR Aging Report shows customer balances by age:

Category	Description
Current	Not yet due
1-30 Days	Overdue up to 30 days
31-60 Days	Overdue 31-60 days
61-90 Days	Overdue 61-90 days
Over 90 Days	Overdue more than 90 days

6.6.2 Customer Statement

Shows all transactions for a customer period:

- Opening balance

- Invoices
- Receipts
- Credit notes
- Closing balance

6.6.3 Collection Effectiveness Report

Measures collection performance:

- Days Sales Outstanding (DSO)
- Collection percentage by aging bucket
- Bad debt ratio

Chapter 7: ETA E-Invoicing Integration

7.1 Introduction to ETA E-Invoicing

The Egyptian Tax Authority (ETA) introduced mandatory E-Invoicing for all B2B and B2G transactions to improve tax collection, combat tax evasion, and modernize Egypt's tax administration. The Egyptian ERP System provides comprehensive integration with the ETA portal, automating the submission process and ensuring full compliance.

7.1.1 E-Invoicing Overview

E-Invoicing requires all qualifying transactions to be submitted electronically to the ETA portal before the invoice can be issued to the customer. The system generates invoices in the ETA-specified format, applies digital signatures, and handles all communication with the ETA portal.

7.1.2 Who Must Comply

E-Invoicing is mandatory for:

- All B2B (Business-to-Business) transactions
- All B2G (Business-to-Government) transactions
- All exporters (regardless of customer type)

B2C (Business-to-Consumer) transactions are currently exempt but may be included in future phases.

7.2 E-Invoice Requirements

7.2.1 Mandatory Fields

Field	Description	Example
UUID	Unique invoice identifier	ABC123456789

Invoice Number	Internal invoice number	INV-2025-0001
Invoice Date	Date of issuance	2025-01-15
Seller Tax ID	Company tax registration	123456789
Buyer Tax ID	Customer tax registration	987654321
Item Description	Description of goods/services	Product A
Quantity	Number of units	10
Unit Price	Price per unit	100.00
Total Amount	Line total before tax	1000.00
VAT Amount	Tax amount	140.00
Total with VAT	Final invoice amount	1140.00

7.2.2 Digital Signature

Each invoice must be digitally signed using an ETA-approved certificate. The signature ensures:

- Authenticity - invoice came from the claimed sender
- Integrity - invoice has not been altered
- Non-repudiation - sender cannot deny issuing the invoice

7.2.3 QR Code

The QR code contains key invoice data that can be verified using ETA's mobile application. The code includes:

- Seller and buyer tax IDs
- Invoice date and total amount
- VAT amount
- Digital signature hash

7.3 System Configuration

7.3.1 ETA Portal Registration

Before using E-Invoicing, your company must register with the ETA:

Registration steps:

1. Visit the ETA portal (<https://eta.gov.eg>)
2. Register using your tax ID
3. Complete company verification
4. Obtain API credentials
5. Purchase digital certificate from approved provider
6. Configure certificate in ERP system

7.3.2 ERP Configuration

To configure ETA integration:

1. Navigate to **Administration** → **Integration** → **ETA Configuration**.
2. Enter API Settings:
 - **API Endpoint:** ETA portal URL
 - **API Key:** Your ETA API key
 - **Client ID:** ETA client identifier
 - **Client Secret:** ETA client secret
3. Upload Digital Certificate:
 - Certificate file (PFX format)

- Certificate password
- Test certificate validation

4. Configure Submission Mode:

- **Real-time:** Submit immediately on invoice creation
- **Batch:** Submit multiple invoices together

7.4 Invoice Submission Process

7.4.1 Submission Flow

When creating a B2B invoice:

1. Create invoice in ERP system
2. System validates data completeness
3. Generates UUID per ETA specification
4. Creates JSON format invoice
5. Applies digital signature
6. Submits to ETA portal via API
7. Receives submission response
8. Stores ETA ID and QR code
9. Invoice ready for printing

7.4.2 Submission Status

Status	Description	Action Required
Pending	Waiting to be submitted	None
Submitted	Sent to ETA, awaiting response	Monitor

Valid	Accepted by ETA	None
Invalid	Rejected by ETA	Correct and resubmit
Cancelled	Cancelled after submission	Issue credit note

7.5 Error Handling

7.5.1 Common Errors

Error	Cause	Resolution
Invalid Tax ID	Customer tax ID incorrect	Verify and correct customer master
Invalid Signature	Certificate issue	Check certificate validity
Duplicate UUID	UUID already exists	System regenerates automatically
Schema Validation	Data format error	Check field formats
Portal Unavailable	ETA portal down	Retry later

7.6 Reporting

7.6.1 ETA Reports

Submission Status Report: Shows all invoices submitted to ETA with their current status.

Rejected Invoices Report: Lists invoices rejected by ETA with error details.

ETA Reconciliation: Compares system invoices to ETA portal records.

Chapter 8: VAT Management

8.1 Introduction to VAT in Egypt

Value Added Tax (VAT) is a consumption tax applied at each stage of the supply chain in Egypt. The standard VAT rate is 14%, with certain goods and services subject to zero-rating or exemption. The Egyptian ERP System provides comprehensive VAT management to ensure accurate calculation, reporting, and compliance.

8.1.1 VAT Overview

VAT is charged on:

- Sale of goods and services in Egypt
- Import of goods
- Receipt of certain services from abroad

Businesses collect VAT on sales (Output VAT) and reclaim VAT on purchases (Input VAT). The difference is remitted to the tax authority or claimed as a refund.

8.1.2 VAT Rates

Type	Rate	Applies To
Standard	14%	Most goods and services
Zero-rated	0%	Exports, certain food items, medical supplies
Exempt	N/A	Financial services, education, healthcare

8.2 VAT Configuration

8.2.1 Tax Codes

Tax codes define how VAT is calculated and reported:

Code	Description	Rate
VAT14	Standard VAT	14%
VAT0	Zero-rated	0%
EXEMPT	Exempt	N/A

8.2.2 Tax Accounts

VAT transactions post to specific GL accounts:

- **Input VAT (22010):** VAT on purchases (asset)
- **Output VAT (42010):** VAT on sales (liability)
- **VAT Payable (22020):** Net VAT due to ETA
- **VAT Receivable (12020):** Net VAT claimable from ETA

8.3 VAT Transactions

8.3.1 Sales VAT

VAT on sales is automatically calculated:

Sales VAT Calculation

Subtotal: EGP 1,000.00

Discount: EGP 100.00

Taxable Amount: EGP 900.00

VAT (14%): EGP 126.00

Total: EGP 1,026.00

8.3.2 Purchase VAT

VAT on purchases can be reclaimed as input VAT:

Recording purchase VAT:

1. Enter vendor invoice with VAT amount
2. System posts to Input VAT account
3. VAT is tracked for return preparation
4. Reclaim on monthly VAT return

8.4 VAT Return

8.4.1 Return Preparation

The monthly VAT return includes:

Section	Description
Total Sales	Taxable and exempt sales
Output VAT	VAT collected on sales
Total Purchases	Taxable and exempt purchases
Input VAT	VAT paid on purchases
Net VAT	Output VAT minus Input VAT

To generate VAT return:

1. Navigate to **Financial** → **VAT** → **VAT Return**.
2. Select reporting period.

3. Review auto-populated figures.
4. Make any necessary adjustments.
5. Generate return in ETA format.
6. Submit through ETA portal.

8.4.2 Filing Deadline

VAT returns must be filed within 30 days of month end. Late filing penalties apply.

8.5 Special VAT Scenarios

8.5.1 Reverse Charge

For imported services, the buyer accounts for VAT under reverse charge mechanism. The system automatically handles reverse charge transactions.

8.5.2 Withholding VAT

Certain transactions require the buyer to withhold VAT and remit directly to ETA. The system tracks withholding VAT separately.

8.5.3 Export VAT

Exports are zero-rated (0% VAT). Proper documentation is required to support zero-rating.

Chapter 9: Cost Accounting and Management

9.1 Introduction to Cost Accounting

Cost Accounting provides detailed cost tracking and analysis to support management decision-making. Unlike financial accounting which focuses on external reporting, cost accounting provides internal insights into where and how costs are incurred, enabling better control and optimization.

9.1.1 Cost Accounting Overview

The Cost Accounting module enables:

- Multi-dimensional cost tracking by cost center, project, and product
- Cost allocation and distribution
- Product costing for inventory valuation and pricing
- Variance analysis for performance measurement
- Profitability analysis by product, customer, and segment

9.2 Cost Center Accounting

9.2.1 Cost Center Structure

Cost centers represent organizational units that incur costs:

Cost Center Type	Examples
Department	Sales, Marketing, Production, HR
Function	Assembly, Quality Control, Maintenance

Location Cairo Plant, Alexandria Warehouse

Project Project Alpha, Implementation

9.2.2 Creating Cost Centers

To create a cost center:

1. Navigate to **Financial** → **Cost Accounting** → **Cost Centers**.
2. Click **Create New**.
3. Enter cost center code and name.
4. Select cost center category.
5. Assign responsible person.
6. Set validity period.
7. Click **Save**.

9.2.3 Cost Allocation

Indirect costs are allocated to cost centers using defined bases:

Cost Type	Allocation Base
Rent	Floor area occupied
Utilities	Meter readings or headcount
IT Services	Number of users
HR Services	Headcount

9.3 Cost Elements

9.3.1 Primary Cost Elements

Primary cost elements correspond to GL expense accounts:

- **Materials:** Raw materials, components, packaging
- **Labor:** Direct and indirect labor costs
- **Overhead:** Indirect manufacturing costs

9.3.2 Secondary Cost Elements

Secondary cost elements track internal allocations:

- Cost center allocations
- Activity type costs
- Internal service charges

9.4 Product Costing

9.4.1 Standard Costing

Standard costs are predetermined costs used for planning and control:

Setting standard costs:

1. Define standard material costs
2. Set standard labor rates
3. Establish overhead rates
4. Calculate standard product costs
5. Review and update periodically

9.4.2 Actual Costing

Actual costs reflect real expenditures:

- Material costs from purchase invoices
- Labor costs from payroll
- Overhead from actual expenses

9.4.3 Variance Analysis

Variances highlight differences between standard and actual costs:

Variance Type	Formula	Cause
Material Price	$(\text{Actual Price} - \text{Standard Price}) \times \text{Actual Quantity}$	Price changes, purchasing efficiency
Material Usage	$(\text{Actual Quantity} - \text{Standard Quantity}) \times \text{Standard Price}$	Waste, efficiency, quality issues
Labor Rate	$(\text{Actual Rate} - \text{Standard Rate}) \times \text{Actual Hours}$	Wage changes, overtime
Labor Efficiency	$(\text{Actual Hours} - \text{Standard Hours}) \times \text{Standard Rate}$	Productivity, training

9.5 Reporting

9.5.1 Cost Center Reports

Show costs by cost center with comparisons to budget and prior periods.

9.5.2 Product Cost Reports

Detail product costs with breakdown by material, labor, and overhead.

9.5.3 Variance Reports

Highlight significant variances requiring management attention.

Chapter 10: Fixed Assets Management

10.1 Introduction to Fixed Assets

The Fixed Assets module tracks all capital assets from acquisition through disposal. Proper fixed asset management ensures accurate financial reporting, tax compliance, and optimal asset utilization.

10.1.1 Fixed Assets Overview

Fixed assets are long-term tangible assets used in business operations:

- Buildings and land
- Machinery and equipment
- Vehicles
- Furniture and fixtures
- Computer equipment

10.2 Asset Master

10.2.1 Asset Classification

Assets are classified by type with associated depreciation rates:

Asset Class	Tax Depreciation Rate	Typical Useful Life
Buildings	5%	20 years
Machinery	15%	7 years
Vehicles	25%	4 years

Furniture	10%	10 years
Computers	33%	3 years

10.2.2 Creating Asset Master

To create an asset master:

1. Navigate to **Financial** → **Fixed Assets** → **Asset Master** → **Create New**.
2. Enter asset code and description.
3. Select asset class.
4. Enter acquisition date and cost.
5. Select depreciation method.
6. Enter useful life.
7. Specify location and department.
8. Click **Save**.

10.3 Depreciation

10.3.1 Depreciation Methods

Method	Description	Best For
Straight Line	Equal amount each period	Most assets
Declining Balance	Percentage of book value	Technology assets
Tax Depreciation	Per Egyptian tax rates	Tax reporting

10.3.2 Depreciation Calculation

Straight Line Formula:

Annual Depreciation = (Cost - Salvage Value) / Useful Life

Depreciation Example

Asset Cost: EGP 100,000

Salvage Value: EGP 10,000

Useful Life: 5 years

Annual Depreciation = (100,000 - 10,000) / 5 = EGP 18,000

10.3.3 Depreciation Run**To run monthly depreciation:**

1. Navigate to **Financial** → **Fixed Assets** → **Depreciation Run**.
2. Select period.
3. Preview depreciation amounts.
4. Click **Post** to create GL entries.

10.4 Asset Transactions

10.4.1 Asset Acquisition

Record new asset purchases:

- Create asset master
- Record acquisition cost
- Post to GL
- Start depreciation from acquisition date

10.4.2 Asset Transfer

Process asset movements:

- Inter-location transfers
- Department transfers
- Cost center changes

10.4.3 Asset Disposal

To dispose of an asset:

1. Navigate to **Financial** → **Fixed Assets** → **Disposal**.
2. Select asset to dispose.
3. Enter disposal date.
4. Enter disposal value (sale price or zero).
5. System calculates gain/loss.
6. Post disposal entry.

10.5 Reporting

10.5.1 Asset Register

Complete listing of all assets with key information.

10.5.2 Depreciation Schedule

Shows depreciation for each asset by period.

10.5.3 Asset Movement Report

Tracks all asset transactions and changes.

Chapter 11: Cash and Bank Management

11.1 Introduction to Cash Management

The Cash and Bank Management module provides comprehensive control over all cash-related transactions, ensuring accurate tracking of cash positions and efficient treasury operations.

11.1.1 Cash Management Overview

The module manages:

- Bank accounts and cash on hand
- Payments and receipts
- Bank reconciliation
- Cash flow forecasting
- Check management

11.2 Bank Account Setup

11.2.1 Bank Master

To set up a bank:

1. Navigate to **Cash Management** → **Setup** → **Bank Master**.
2. Enter bank code and name.
3. Enter SWIFT code.
4. Enter address and contact information.

11.2.2 Bank Account Creation

To create a bank account:

1. Navigate to **Cash Management** → **Setup** → **Bank Accounts**.
2. Select bank.
3. Enter account number and IBAN.
4. Select account type (Current, Savings).
5. Select currency.
6. Map to GL account.
7. Enter opening balance.

11.3 Payment Processing

11.3.1 Payment Methods

Method	Use Case	Processing
Bank Transfer	Vendor payments	Electronic file to bank
Check	Traditional payments	Print and sign
Cash	Small payments	Physical handling
Credit Card	Employee expenses	Card processing

11.3.2 Check Management

Complete check lifecycle management:

- Checkbook setup and tracking

- Check printing with security features
- Void check handling
- Stop payment processing

11.4 Bank Reconciliation

11.4.1 Importing Bank Statements

To import bank statement:

1. Download statement from bank.
2. Navigate to **Cash Management** → **Reconciliation** → **Import**.
3. Select bank account.
4. Upload statement file.
5. Map file fields to system fields.
6. Validate and import.

11.4.2 Reconciliation Process

To reconcile bank statement:

1. Navigate to **Cash Management** → **Reconciliation**.
2. Select bank account and statement.
3. System suggests matches based on amount and reference.
4. Review and confirm matches.
5. Investigate unmatched items.
6. Create adjustments for bank charges/interest.

7. Complete reconciliation when balanced.

11.4.3 Common Differences

Difference	Cause	Resolution
Outstanding checks	Checks issued not yet cleared	Verify check status
Deposits in transit	Deposits not yet credited	Confirm deposit date
Bank charges	Fees not recorded	Record bank charge
Interest earned	Interest not recorded	Record interest income

11.5 Cash Flow Management

11.5.1 Cash Flow Forecasting

Predict future cash positions:

- Projected receipts from customers
- Projected payments to vendors
- Recurring payments (salaries, rent)
- Cash position prediction

11.5.2 Cash Position Report

Real-time view of cash across all accounts.

Chapter 12: Human Resources Management

12.1 Introduction to HR Management

The Human Resources Management module manages the complete employee lifecycle from hiring through termination. The module ensures compliance with Egyptian Labor Law No. 12 of 2003 as amended by Law No. 14 of 2025, and provides comprehensive tools for personnel administration, organizational management, attendance tracking, and employee self-service.

12.1.1 HR Module Overview

The HR module covers:

- Organizational structure management
- Employee master data
- Attendance and time management
- Leave management
- Employee self-service
- Document management
- HR reporting and analytics

12.2 Organizational Structure

12.2.1 Organization Units

The organizational hierarchy typically includes:

Level	Description	Example
Company	Legal entity	Egyptian ERP Solutions

Division	Business division	Manufacturing Division
Department	Functional area	Finance Department
Section	Sub-department	Accounts Payable Section

12.2.2 Job Structure

Jobs are defined independently of employees:

- **Job Code:** Unique identifier
- **Job Title:** Position name
- **Job Grade:** Level in hierarchy
- **Salary Band:** Approved salary range
- **Job Description:** Responsibilities and requirements

12.3 Employee Master

12.3.1 Personal Information

To create an employee record:

1. Navigate to **HR** → **Employees** → **Create New**.
2. Enter personal information:

- **Full Name (English):** As per national ID
- **Full Name (Arabic):** For official documents
- **National ID:** 14-digit Egyptian national ID
- **Date of Birth:** From national ID
- **Nationality:** Country of citizenship
- **Gender:** Male/Female
- **Marital Status:** Single, Married, Divorced, Widowed

- **Military Status:** Completed, Exempted, Postponed, NA

12.3.2 Employment Information

- **Employee ID:** Company employee number
- **Joining Date:** First day of employment
- **Department:** Assigned department
- **Job Title:** Current position
- **Job Grade:** Grade level
- **Reporting Manager:** Direct supervisor
- **Employment Type:** Permanent, Contract, Temporary
- **Employment Status:** Active, On Leave, Suspended, Terminated

12.3.3 Contact Information

- Address (current residence)
- Phone numbers (mobile, home)
- Email address
- Emergency contact

12.4 Attendance Management

12.4.1 Shift Management

Define work schedules:

Shift Type	Hours	Common Use
Day Shift	08:00 - 17:00	Office staff
Night Shift	20:00 - 05:00	Production
Morning Shift	07:00 - 15:00	Early operations

Afternoon Shift

15:00 - 23:00

Extended coverage

12.4.2 Time Recording

Multiple time recording options:

- **Biometric Integration:** Fingerprint or facial recognition
- **Mobile Check-in:** GPS-enabled mobile app
- **Web Check-in:** Browser-based for remote workers
- **Manual Entry:** Supervisor-approved corrections

12.4.3 Overtime Calculation

Per Egyptian Labor Law:

Overtime Type	Rate	Calculation
Regular Overtime	135%	Hourly rate × 1.35
Rest Day	200%	Hourly rate × 2.0
Official Holiday	200%	Hourly rate × 2.0

Note: Maximum overtime is 2 hours per day per Egyptian Labor Law.

12.5 Leave Management

12.5.1 Leave Types

Leave Type	Entitlement	Conditions
Annual Leave	21-30 days/year	Based on years of service
Sick Leave	Up to 180 days/year	Medical certificate required

Casual Leave	6 days/year	No prior notice required
Maternity Leave	90 days	Female employees only
Paternity Leave	3 days	Within 1 month of birth
Hajj Leave	30 days	Once during employment

12.5.2 Leave Request Process

Leave workflow:

1. Employee submits leave request
2. System checks leave balance
3. Manager receives notification
4. Manager approves or rejects
5. Employee receives notification
6. Balance updated if approved

12.6 HR Reporting

12.6.1 Standard Reports

- Employee list with contact details
- Organization chart
- Attendance summary
- Leave balance report
- Headcount analysis

Chapter 13: Payroll Management

13.1 Introduction to Payroll

The Payroll module handles all aspects of salary calculation, ensuring accurate and timely payment to employees while maintaining full compliance with Egyptian regulatory requirements including GOSI contributions and income tax.

13.1.1 Payroll Overview

The payroll process includes:

- Earnings calculation (basic salary, allowances, overtime)
- Deductions calculation (social insurance, income tax, loans)
- GOSI contribution calculation and reporting
- Income tax calculation per Egyptian tax law
- Payslip generation
- Bank transfer file generation
- GL posting

13.2 Payroll Structure

13.2.1 Earnings

Earning Type	Description	Taxable
Basic Salary	Fixed monthly salary	Yes
Housing Allowance	Allowance for accommodation	Yes
Transportation	Commuting allowance	Yes
Food Allowance	Meal allowance	Yes

Overtime	Extra hours payment	Yes
Bonus	Performance or annual bonus	Yes

13.2.2 Deductions

Deduction Type	Description
Social Insurance	Employee GOSI contribution
Income Tax	Salary tax per tax brackets
Loan Repayment	Employee loan installments
Health Insurance	Private health insurance premium
Other Deductions	Any authorized deductions

13.3 Social Insurance (GOSI)

13.3.1 Contribution Rates (2025)

Insurance Type	Employer	Employee	Total
Old Age/Disability/Death	15%	11%	26%
Work Injury	2%	0%	2%
Unemployment	1%	1%	2%
Total	18%	12%	30%

13.3.2 Maximum Insurance Salary

The maximum salary subject to social insurance is EGP 8,400 per month (2025). Contributions are calculated on the lower of actual salary or the maximum.

GOSI Calculation Example

Employee Salary: EGP 10,000

Maximum Insurance Salary: EGP 8,400

Employee Contribution: $8,400 \times 12\% = \text{EGP } 1,008$

Employer Contribution: $8,400 \times 18\% = \text{EGP } 1,512$

13.4 Income Tax

13.4.1 Tax Brackets (2025)

Annual Income (EGP)	Tax Rate
0 - 30,000	0%
30,001 - 45,000	10%
45,001 - 60,000	15%
60,001 - 200,000	20%
200,001 - 400,000	22.5%
Above 400,000	25%

13.4.2 Tax Calculation

Tax Calculation Example

Annual	Salary:	EGP		120,000
First	30,000:	0%	=	0
30,001-45,000	(15,000):	10%	=	1,500
45,001-60,000	(15,000):	15%	=	2,250
60,001-120,000	(60,000):	20%	=	12,000

Total Annual Tax: EGP 15,750

Monthly Tax: EGP 1,312.50

13.5 Payroll Processing

13.5.1 Monthly Process

Monthly payroll process:

1. Collect attendance data
2. Calculate overtime
3. Process variable payments (bonuses, commissions)
4. Calculate deductions (GOSI, tax, loans)
5. Generate payslips
6. Generate bank transfer file
7. Post to GL

13.5.2 Payroll Journal Entries

Automatic GL posting:

Account	Debit	Credit
Salary Expense	XXX	
Social Insurance Expense	XXX	
Social Insurance Payable		XXX
Tax Payable		XXX

Net Salary Payable

XXX

13.6 Payroll Reporting

- Payslip (individual employee)
- Payroll summary
- GOSI monthly report (Form 1)
- Income tax report
- Bank transfer report

Chapter 14: Inventory Management

14.1 Introduction to Inventory

The Inventory Management module provides real-time visibility into stock levels, tracks inventory movements, and supports multiple valuation methods. Effective inventory management ensures optimal stock levels, reduces carrying costs, and prevents stockouts.

14.1.1 Inventory Overview

The module manages warehouses, items, transactions, valuation, and physical counts. Integration with Purchasing, Sales, and Manufacturing ensures accurate inventory tracking across all operations.

14.2 Warehouse Setup

14.2.1 Warehouse Master

Configure warehouses with code, name, type (main, distribution, retail), location, and manager assignment.

14.2.2 Location Structure

Define storage locations using Zone → Aisle → Rack → Shelf → Bin hierarchy for precise item tracking.

14.3 Item Master

14.3.1 Item Categories

Classify items as Raw Materials, Work in Progress, Finished Goods, Trading Goods, or Consumables.

14.3.2 Item Settings

Configure valuation method (FIFO, LIFO, Average, Standard), tracking method (None, Batch, Serial), reorder point, and safety stock.

14.4 Inventory Transactions

14.4.1 Goods Receipt

Create GRN, enter received quantities, perform quality check, and update inventory.

14.4.2 Goods Issue

Process production issues, sales issues, and transfer issues with proper documentation.

14.4.3 Stock Transfer

Handle inter-warehouse transfers with complete tracking and documentation.

14.5 Inventory Valuation

14.5.1 Valuation Methods

Method	Description
FIFO	First In, First Out - oldest stock used first
LIFO	Last In, First Out - newest stock used first
Weighted Average	Average cost of all units
Standard Cost	Predetermined standard cost

14.6 Physical Count

14.6.1 Cycle Counting

ABC classification with count frequency: A items (high value) counted monthly, B items quarterly, C items annually.

14.6.2 Annual Stock Take

Freeze inventory, conduct physical count, reconcile differences, and post adjustments.

Chapter 15: Purchasing and Procurement

15.1 Introduction to Purchasing

The Purchasing module manages the complete procurement lifecycle from requisition to payment authorization.

15.2 Vendor Management

15.2.1 Vendor Master

Maintain vendor information including code, name, tax registration, payment terms, and credit limit.

15.2.2 Vendor Evaluation

Track quality rating, delivery performance, price competitiveness, and service level.

15.3 Purchase Requisition

15.3.1 Creating Requisitions

Enter item requirements, specify quantity and date, justify need, and submit for approval.

15.3.2 Approval Workflow

Route requisitions based on amount: Department Manager (up to EGP 5,000), Finance Manager (up to EGP 25,000), CFO (above EGP 25,000).

15.4 Purchase Orders

15.4.1 PO Creation

Select vendor, add line items, enter prices, calculate totals, and send to vendor.

15.4.2 PO Management

Track status, manage amendments, and monitor delivery.

15.5 Goods Receipt

15.5.1 Receiving Process

Verify against PO, count quantities, inspect quality, and record receipt.

15.5.2 Three-Way Match

Match PO, GRN, and Invoice before payment authorization.

15.6 Reporting

Generate Open PO Report, Vendor Performance, Purchase Analysis, and Spend by Category.

Chapter 16: Sales Management

16.1 Introduction to Sales

The Sales Management module handles the complete sales cycle from inquiry to cash collection.

16.2 Customer Management

16.2.1 Customer Master

Maintain customer code, name, tax registration number, credit limit, and payment terms.

16.2.2 Credit Control

Monitor credit limit, aging analysis, and place credit holds when necessary.

16.3 Sales Quotations

16.3.1 Creating Quotations

Select customer, add items, apply pricing, and set validity period.

16.3.2 Pricing and Discounts

Apply list prices, customer-specific prices, volume discounts, and promotional prices.

16.4 Sales Orders

16.4.1 Order Creation

Create order header, add line items, check availability, and reserve inventory.

16.4.2 Order Management

Track order status, manage amendments, and handle backorders.

16.5 Delivery and Invoicing

16.5.1 Delivery Notes

Create delivery notes from sales orders with complete tracking.

16.5.2 Sales Invoicing

Create invoice from delivery, submit to ETA, and print with QR code.

16.6 Sales Returns

16.6.1 Return Process

Process return authorization, goods receipt, and credit note.

16.7 Reporting

Generate Sales by Customer, Sales by Product, Sales Pipeline, and Revenue Analysis.

Chapter 17: Manufacturing Management

17.1 Introduction to Manufacturing

The Manufacturing module provides production planning and control for make-to-stock, make-to-order, and assemble-to-order environments.

17.2 Bill of Materials

17.2.1 BOM Structure

Define parent item, component items, quantities, and scrap factors.

17.2.2 BOM Versions

Manage multiple BOM versions with effective dates for engineering changes.

17.3 Routing

17.3.1 Work Centers

Define work center code, capacity, and cost rates.

17.3.2 Operations

Specify operation sequence, setup time, run time, and work center assignment.

17.4 Production Orders

17.4.1 Order Creation

Create order header, explode BOM, reserve materials, and schedule operations.

17.4.2 Order Execution

Issue materials, record labor, report output, and record scrap.

17.5 MRP

17.5.1 MRP Inputs

Demand (forecasts, orders), inventory levels, BOM data, and lead times.

17.5.2 MRP Output

Planned orders, purchase requisitions, and action messages.

17.6 Reporting

Generate Production Schedule, WIP Report, Efficiency Analysis, and Scrap Report.

Chapter 18: Quality Management

18.1 Introduction to Quality

The Quality Management module ensures product quality through planning, inspection, and non-conformance management.

18.2 Quality Planning

18.2.1 Inspection Plans

Define inspection type, quality characteristics, specification limits, and sampling plan.

18.2.2 Sampling

AQL-based sampling per ISO 2859 standards.

18.3 Quality Inspection

18.3.1 Inspection Types

Receiving inspection, in-process inspection, and final inspection.

18.3.2 Inspection Process

Create inspection lot, record results, and make decision.

18.3.3 Inspection Decisions

Accept, Reject, Rework, or Accept with deviation.

18.4 Non-Conformance

18.4.1 NCR Creation

Document description, quantity, and severity.

18.4.2 Disposition

Use as is, Rework, Scrap, or Return to vendor.

18.5 CAPA

18.5.1 CAPA Process

Problem identification, root cause analysis, corrective action, preventive action, and verification.

Chapter 19: Maintenance Management

19.1 Introduction to Maintenance

The Maintenance module manages asset maintenance through preventive, corrective, predictive, and shutdown maintenance.

19.2 Asset Master

19.2.1 Equipment Master

Track equipment code, description, category, location, acquisition date, and cost.

19.2.2 Asset Hierarchy

Functional location → Equipment → Sub-equipment structure.

19.3 Preventive Maintenance

19.3.1 Maintenance Plans

Define frequency, activities, labor requirements, and spare parts.

19.3.2 Scheduling

Generate maintenance schedule based on plans.

19.4 Work Orders

19.4.1 Work Order Types

Preventive, Corrective, and Project work orders.

19.4.2 Work Order Process

Create work order, plan resources, execute work, and complete and close.

19.5 Spare Parts

19.5.1 Spare Parts Management

Manage spare parts master, inventory tracking, and reorder planning.

Chapter 20: Project Management

20.1 Introduction to Project Management

The Project Management module provides project planning and control for internal projects, customer projects, construction projects, and implementation projects.

20.2 Project Structure

20.2.1 WBS

Work Breakdown Structure for project organization.

20.2.2 Project Master

Define project code, name, manager, start date, end date, and budget.

20.3 Project Planning

20.3.1 Activity Planning

Define activities, durations, dependencies, and resources.

20.3.2 Cost Planning

Plan labor costs, material costs, external costs, and overhead.

20.4 Project Execution

20.4.1 Time Recording

Timesheets for project labor tracking.

20.4.2 Progress Tracking

Track percent complete, earned value, and milestones.

20.5 Project Control

20.5.1 Earned Value

Calculate PV, EV, AC, CPI, and SPI for project performance measurement.

20.5.2 Reporting

Generate project status, cost report, and variance analysis.

Chapter 21: Customer Relationship Management

21.1 Introduction to CRM

The CRM module manages customer interactions and relationships including lead management, opportunity management, activity management, campaign management, and service management.

21.2 Lead Management

21.2.1 Lead Process

Lead capture, lead qualification, lead scoring, and lead assignment.

21.3 Opportunity Management

21.3.1 Opportunity Process

Opportunity creation, sales stages, pipeline tracking, and forecasting.

21.4 Activity Management

21.4.1 Activity Types

Calls, meetings, tasks, and emails.

21.5 Service Management

21.5.1 Service Process

Ticket management, SLA tracking, and knowledge base.

Chapter 22: Business Intelligence and Reporting

22.1 Introduction to BI

The BI module provides analytics and reporting capabilities including data warehouse, ETL processes, data models, dashboards, reports, and KPIs.

22.2 Dashboards

22.2.1 Dashboard Types

Executive dashboard, financial dashboard, sales dashboard, and operations dashboard.

22.3 Reports

22.3.1 Report Types

Financial reports, operational reports, and custom reports.

22.4 KPIs

22.4.1 KPI Categories

Financial KPIs, operational KPIs, and custom KPIs.

Chapter 23: System Integration

23.1 Introduction to Integration

The system provides comprehensive integration capabilities with government portals, banking systems, e-commerce, third-party applications, and IoT devices.

23.2 ETA Integration

23.2.1 ETA Integration Details

API connection, JSON format invoices, digital signature, and QR code generation.

23.3 Bank Integration

23.3.1 Bank Integration Features

Statement import, payment file generation, balance inquiry, and transaction status.

23.4 API Framework

23.4.1 API Details

REST endpoints, JSON request/response, OAuth 2.0 authentication, and rate limiting.

23.5 Webhooks

23.5.1 Webhook Configuration

Define endpoint URL, select events to subscribe, set retry policy, and monitor delivery.

Chapter 24: Mobile Application

24.1 Introduction to Mobile ERP

The mobile app extends ERP functionality to iOS, Android, and Progressive Web App (PWA) platforms.

24.2 Mobile Features

24.2.1 Feature Categories

Sales features, service features, manager features, and employee features.

24.3 Offline Capability

24.3.1 Offline Features

Data synchronization, offline data entry, queue for upload, and conflict resolution.

24.4 Security

24.4.1 Mobile Security

Biometric authentication, device registration, remote wipe, and data encryption.

Chapter 25: Artificial Intelligence and Machine Learning

25.1 Introduction to AI in ERP

AI capabilities enhance decision-making and automate processes including predictive analytics, natural language processing, intelligent automation, anomaly detection, and recommendation engines.

25.2 Predictive Analytics

25.2.1 Prediction Types

Demand forecasting, cash flow prediction, and customer churn prediction.

25.3 Intelligent Automation

25.3.1 Automation Types

Invoice processing automation and email classification.

25.4 NLP

25.4.1 NLP Features

Chatbot integration and sentiment analysis.

Chapter 26: System Administration

26.1 Introduction to System Administration

System administration manages configuration, users, security, and system health.

26.2 User Management

26.2.1 User Administration

Creating users, managing users, and role assignment.

26.3 Security

26.3.1 Security Configuration

Password policy, MFA, and session management.

26.4 Configuration

26.4.1 System Configuration

Company settings and module configuration.

26.5 Audit

26.5.1 Audit Functions

Audit trail and system logs.

26.6 Backup

26.6.1 Backup Management

Backup strategy and recovery procedures.

Chapter 27: Data Management

27.1 Introduction to Data Management

The Data Management module manages data import, export, migration, and quality.

27.2 Data Import

27.2.1 Import Process

Import templates, field mapping, validation rules, and sample files.

27.3 Data Export

27.3.1 Export Options

Excel, CSV, PDF, XML, and JSON formats with scheduled exports.

27.4 Data Migration

27.4.1 Migration Process

Migration planning, migration approach, data cleansing, and execution.

27.5 Data Quality

27.5.1 Quality Management

Data validation, data cleansing, duplicate detection, and standardization.

Chapter 28: Audit and Compliance

28.1 Introduction to Audit and Compliance

The Audit and Compliance module provides tools for internal controls and regulatory compliance.

28.2 Internal Controls

28.2.1 Control Framework

Segregation of duties and approval workflows.

28.3 Audit Trail

28.3.1 Audit Logging

Transaction logging and audit reports.

28.4 Regulatory Compliance

28.4.1 Compliance Areas

Tax compliance, GOSI compliance, and labor law compliance.

Chapter 29: Troubleshooting and Support

29.1 Introduction to Troubleshooting

This chapter provides guidance on identifying and resolving common issues.

29.2 Common Issues and Solutions

29.2.1 Login Issues

Cannot log in due to incorrect credentials, locked account, or expired password.

29.2.2 Performance Issues

Slow response due to network issues, server load, or large queries.

29.2.3 Data Issues

Cannot save records due to missing fields, validation errors, or duplicate keys.

29.3 Support Channels

29.3.1 Support Options

Help desk, knowledge base, and community forum.

29.4 System Health

29.4.1 Health Monitoring

Health dashboard and alerts.

Chapter 30: Training and User Adoption

30.1 Introduction to Training

Comprehensive training ensures users can effectively use the system.

30.2 Training Program

30.2.1 Training Types

Administrator training, power user training, end user training, and executive training.

30.2.2 Training Methods

Instructor-led, self-paced e-learning, hands-on workshops, video tutorials, and documentation.

30.3 Training Materials

30.3.1 Material Types

User guides, video library, quick reference cards, and process flowcharts.

30.4 User Adoption

30.4.1 Adoption Strategy

Change management, communication plan, and adoption metrics.

30.5 Certification

30.5.1 Certification Program

End User, Power User, Administrator, and Consultant certification levels.

Chapter 31: Advanced Configuration

31.1 Introduction to Advanced Configuration

Advanced configuration allows customization to meet specific business requirements.

31.2 Custom Fields

31.2.1 Field Configuration

Field types include text, number, date, dropdown, checkbox, multi-select, and lookup.

31.3 Custom Workflows

31.3.1 Workflow Design

Define states, transitions, conditions, and actions.

31.4 Custom Reports

31.4.1 Report Builder

Select data source, design layout, add calculations, configure filters, and schedule delivery.

31.5 Integration Configuration

31.5.1 Integration Setup

API configuration and webhook configuration.

Chapter 32: Implementation Methodology

32.1 Introduction to Implementation

A structured approach ensures successful ERP implementation.

32.2 Implementation Phases

32.2.1 Phase 1: Discovery

Business requirements gathering, current state analysis, future state design, gap analysis, and project planning.

32.2.2 Phase 2: Design

Solution design, process mapping, system configuration design, integration design, and report design.

32.2.3 Phase 3: Build

System configuration, customization development, integration development, report development, and unit testing.

32.2.4 Phase 4: Test

System testing, integration testing, UAT, performance testing, and security testing.

32.2.5 Phase 5: Deploy

Data migration, user training, go-live preparation, cutover, and hypercare support.

32.2.6 Phase 6: Optimize

Post-go-live support, issue resolution, performance optimization, enhancements, and continuous improvement.

32.3 Data Migration

32.3.1 Migration Strategy

Big bang, phased, or parallel running approaches.

32.3.2 Data Cleansing

Data quality assessment, duplicate removal, and standardization.

32.4 Change Management

32.4.1 Change Management Activities

Communication plan and training strategy.

Appendix A: Complete API Reference

A.1 Introduction to APIs

The Egyptian ERP System provides a comprehensive REST API for integration with external systems. This appendix documents all available endpoints, request/response formats, and authentication methods.

A.2 Authentication

A.2.1 OAuth 2.0

All API requests require OAuth 2.0 authentication. Obtain an access token using client credentials flow.

```
POST /oauth/token Content-Type: application/x-www-form-urlencoded
grant_type=client_credentials &client_id=YOUR_CLIENT_ID
&client_secret=YOUR_CLIENT_SECRET &scope=api_access
```

A.2.2 Using Access Token

Include the access token in the Authorization header of all API requests:

```
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
```

A.3 Customer API

A.3.1 Get Customers

Retrieve a list of customers with pagination and filtering.


```
GET /api/v1/customers?page=1&page_size=50&search=abc
```

Response:

```
{ "data": [ { "customer_code": "CUST001", "customer_name": "ABC Company",
"tax_registration": "123456789", "credit_limit": 100000, "balance": 25000 } ],
"pagination": { "page": 1, "page_size": 50, "total_records": 150 } }
```

A.3.2 Create Customer

Create a new customer record.

```
POST /api/v1/customers Content-Type: application/json { "customer_code": "CUST002",
"customer_name": "XYZ Trading", "customer_name_ar": "إكس واي زد للتجارة",
"tax_registration": "987654321", "address": "123 Main Street, Cairo", "credit_limit":
50000, "payment_terms": "NET30" }
```

A.4 Invoice API

A.4.1 Create Invoice

Create a new sales invoice.

```
POST /api/v1/invoices Content-Type: application/json { "customer_code": "CUST001",
"invoice_date": "2025-01-15", "due_date": "2025-02-15", "reference": "S0-2025-001",
"lines": [ { "item_code": "PROD001", "description": "Product A", "quantity": 10,
"unit_price": 100, "tax_code": "VAT14" } ] }
```

A.4.2 Get Invoice

Retrieve invoice details including ETA submission status.

```
GET /api/v1/invoices/INV-2025-001
```

A.5 Inventory API

A.5.1 Get Stock Balance

Retrieve current stock balance for an item.

```
GET /api/v1/inventory/balance?item_code=PROD001&warehouse=MAIN
```

A.5.2 Create Goods Receipt

Record goods receipt from vendor.

```
POST /api/v1/inventory/goods-receipt Content-Type: application/json { "grn_number":  
"GRN-2025-001", "grn_date": "2025-01-15", "vendor_code": "VEND001", "po_reference":  
"PO-2025-001", "lines": [ { "item_code": "RAW001", "quantity": 100, "unit_price": 50,  
"warehouse": "MAIN" } ] }
```

A.6 Payroll API

A.6.1 Get Employees

Retrieve employee list.

```
GET /api/v1/hr/employees?department=FIN&status=ACTIVE
```

A.6.2 Process Payroll

Trigger payroll processing for a period.

```
POST /api/v1/payroll/process Content-Type: application/json { "period": "2025-01",  
"company_code": "COMP001" }
```

A.7 Error Codes

Code	Description	Resolution
400	Bad Request	Check request parameters
401	Unauthorized	Check authentication token
403	Forbidden	Check API permissions
404	Not Found	Verify resource exists
409	Conflict	Resource already exists
422	Validation Error	Check field validations
429	Rate Limited	Reduce request frequency
500	Server Error	Contact support

A.8 Rate Limiting

API requests are rate limited to ensure fair usage:

- Standard: 1000 requests per hour
- Premium: 10000 requests per hour

Appendix B: Regulatory Forms Reference

B.1 Egyptian Tax Authority Forms

B.1.1 VAT Forms

- VAT Return Form (Monthly)
- VAT Registration Form
- VAT Exemption Application
- VAT Refund Request

B.1.2 Income Tax Forms

- Salary Tax Form 1 (Monthly)
- Annual Tax Return (Form 2)
- Withholding Tax Form
- Tax Clearance Certificate Request

B.2 Social Insurance Forms

B.2.1 GOSI Monthly Forms

Form	Purpose	Frequency
Form 1	Monthly Contributions	Monthly
Form 2	New Employees Registration	As needed
Form 3	Terminated Employees	As needed

Form 4	Salary Changes	As needed
Form 6	Annual Reconciliation	Annual

B.3 Labor Office Forms

- Work Permit Application
- Employment Contract Registration
- Leave Notification Form
- Termination Notification Form
- End of Service Certificate

B.4 Commercial Registry Forms

- Commercial Registration Application
- Company Data Update Form
- Branch Registration
- Activity Addition/Removal

Appendix C: Configuration Templates

C.1 Chart of Accounts Template

Sample chart of accounts for a manufacturing company:

Account Code	Account Name	Type
11000	Cash on Hand	Asset
11100	Bank Accounts	Asset
12000	Accounts Receivable	Asset
13000	Inventory	Asset
14000	Fixed Assets	Asset
21000	Accounts Payable	Liability
22000	VAT Payable	Liability
23000	Social Insurance Payable	Liability
31000	Share Capital	Equity
32000	Retained Earnings	Equity
41000	Sales Revenue	Revenue
51000	Cost of Goods Sold	Expense
61000	Salaries and Wages	Expense
62000	Rent Expense	Expense

C.2 Approval Workflow Template

Sample approval matrix for purchase orders:

Amount Range (EGP)	Approver Level 1	Approver Level 2
Up to 5,000	Department Manager	-
5,001 - 25,000	Department Manager	Finance Manager
25,001 - 100,000	Finance Manager	CFO
Above 100,000	CFO	CEO

C.3 Role Template

Sample role definition for Sales Manager:

Module	View	Create	Edit	Delete	Approve
Sales	Yes	Yes	Yes	No	Yes
Customers	Yes	Yes	Yes	No	No
Reports	Yes	Yes	Yes	No	-
Inventory	Yes	No	No	No	No

Appendix D: Detailed Case Studies

D.1 Manufacturing Company Implementation

D.1.1 Company Background

ABC Manufacturing is a mid-sized Egyptian company producing consumer goods with 200 employees, annual revenue of EGP 150 million, and operations across two facilities.

D.1.2 Challenges

- Disconnected systems for inventory, production, and finance
- Manual ETA E-Invoicing compliance
- Inaccurate product costing
- Delayed financial reporting

D.1.3 Solution

Implemented Egyptian ERP System with Manufacturing, Inventory, Quality, and Financial modules over 6 months.

D.1.4 Results

- 50% reduction in month-end closing time
- Automated ETA compliance
- Accurate real-time product costing
- Integrated operations visibility

D.2 Trading Company Implementation

D.2.1 Company Background

XYZ Trading imports and distributes electronics with 50 employees and annual revenue of EGP 80 million.

D.2.2 Challenges

- Complex import documentation
- Multi-currency transactions
- Customer credit management
- VAT compliance on imports

D.2.3 Solution

Implemented Purchasing, Sales, Inventory, and Financial modules with multi-currency support.

D.2.4 Results

- Streamlined import processes
- Automated multi-currency accounting
- Improved collections by 30%
- Accurate VAT reporting

Appendix E: Comprehensive Glossary

E.1 ERP Terminology

Term	Definition
AP	Accounts Payable - money owed to vendors
AR	Accounts Receivable - money owed by customers
API	Application Programming Interface
BOM	Bill of Materials - list of components
CAPA	Corrective and Preventive Action
CRM	Customer Relationship Management
ETA	Egyptian Tax Authority
GOSI	General Organization for Social Insurance
GL	General Ledger - central accounting record
GRN	Goods Receipt Note
KPI	Key Performance Indicator
MRP	Material Requirements Planning
NCR	Non-Conformance Report
PO	Purchase Order
RAG	Retrieval Augmented Generation
RBAC	Role-Based Access Control
SLA	Service Level Agreement

SO	Sales Order
UUID	Universally Unique Identifier
VAT	Value Added Tax
WBS	Work Breakdown Structure
WIP	Work In Progress

E.2 Egyptian Regulatory Terms

Term	Definition
Commercial Registration	Official business registration document
Tax Card	Taxpayer identification document
E-Invoice	Electronically submitted tax invoice
Digital Signature	Electronic signature for document authenticity
QR Code	Quick Response code for invoice verification
Social Insurance	Mandatory employee benefits program
End of Service	Employee termination benefits

E.3 Accounting Terms

Term	Definition
Accrual	Revenue/expense recognized when earned/incurred
Depreciation	Allocation of asset cost over useful life
Fiscal Year	12-month accounting period

Journal Entry	Record of financial transaction
Reconciliation	Process of matching records
Trial Balance	List of all account balances
Variance	Difference between expected and actual

Appendix F: Complete Index

F.1 Index by Topic

Accounting and Finance

- General Ledger - Chapter 4
- Accounts Payable - Chapter 5
- Accounts Receivable - Chapter 6
- VAT Management - Chapter 8
- Cost Accounting - Chapter 9
- Fixed Assets - Chapter 10
- Cash Management - Chapter 11

Human Resources

- HR Management - Chapter 12
- Payroll - Chapter 13

Inventory and Manufacturing

- Inventory Management - Chapter 14
- Purchasing - Chapter 15
- Sales - Chapter 16
- Manufacturing - Chapter 17
- Quality Management - Chapter 18
- Maintenance - Chapter 19

Business Operations

- Project Management - Chapter 20

- CRM - Chapter 21
- Business Intelligence - Chapter 22

Technical

- System Integration - Chapter 23
- Mobile App - Chapter 24
- AI and ML - Chapter 25
- System Administration - Chapter 26
- Data Management - Chapter 27
- Audit and Compliance - Chapter 28

Support

- Troubleshooting - Chapter 29
- Training - Chapter 30
- Advanced Configuration - Chapter 31
- Implementation - Chapter 32

Regulatory Compliance

- ETA E-Invoicing - Chapter 7
- VAT - Chapter 8
- GOSI - Chapter 13
- Labor Law - Chapter 12

F.2 Index by Module

Module	Primary Chapter	Related Chapters
General Ledger	4	5, 6, 8, 9, 10, 11
Accounts Payable	5	4, 8, 11, 15

Accounts Receivable	6	4, 7, 8, 16
HR	12	13
Payroll	13	12
Inventory	14	15, 16, 17
Purchasing	15	5, 14
Sales	16	6, 7, 14
Manufacturing	17	14, 18, 19

1.6 System Requirements

1.6.1 Hardware Requirements

The Egyptian ERP System is a cloud-based solution that requires minimal local hardware. However, understanding the requirements ensures optimal performance.

End User Devices:

Component	Minimum	Recommended
Processor	Dual Core 2.0 GHz	Quad Core 2.5 GHz
RAM	4 GB	8 GB
Storage	10 GB free	50 GB free
Display	1366x768	1920x1080
Internet	2 Mbps	10 Mbps

1.6.2 Software Requirements

Operating Systems: Windows 10/11, macOS 10.15+, Linux (Ubuntu 18.04+)

Browsers: Chrome 90+, Firefox 88+, Edge 90+, Safari 14+

Additional Software: PDF reader, Microsoft Office or equivalent, Anti-virus software

1.7 Deployment Options

1.7.1 Cloud Deployment

The standard deployment model uses cloud infrastructure with the following benefits:

- No local server hardware required
- Automatic updates and maintenance
- Scalable resources based on demand
- Built-in disaster recovery
- 24/7 monitoring and support

1.7.2 On-Premise Deployment

For organizations requiring local deployment:

- Full control over infrastructure
- Custom security configurations
- Integration with local systems
- Requires dedicated IT resources

1.8 Data Security and Privacy

1.8.1 Data Encryption

All data is protected using industry-standard encryption:

- **In Transit:** TLS 1.3 encryption for all data transmission
- **At Rest:** AES-256 encryption for database storage
- **Backups:** Encrypted backup storage with geographic redundancy

1.8.2 Data Privacy Compliance

The system complies with Egyptian data protection regulations and international standards:

- Personal data minimization
- Consent management
- Right to access and deletion
- Data breach notification procedures

1.9 System Updates and Maintenance

1.9.1 Release Cycle

The Egyptian ERP System follows a regular release cycle:

- **Major Releases:** Annually with new features and enhancements
- **Minor Releases:** Quarterly with improvements and fixes
- **Patches:** As needed for critical issues

1.9.2 Maintenance Windows

Scheduled maintenance is performed during low-usage periods with advance notification.

1.10 Training and Certification

1.10.1 Training Programs

Comprehensive training programs are available:

- **End User Training:** Basic system usage (2 days)
- **Power User Training:** Advanced features and reporting (3 days)
- **Administrator Training:** System configuration and management (5 days)
- **Developer Training:** Customization and integration (10 days)

1.10.2 Certification Levels

Level	Requirements	Valid For
Certified End User	Complete training + exam	2 years
Certified Professional	Advanced training + project	2 years
Certified Administrator	Admin training + practical	2 years
Certified Consultant	All training + experience	2 years

Extended Troubleshooting Guide

T.1 Login and Access Issues

T.1.1 Cannot Log In - Invalid Credentials

Symptom: Error message "Invalid username or password"

Possible Causes:

- Incorrect username or password
- Caps Lock is on
- Account is locked due to failed attempts
- Password has expired
- Account is deactivated

Solutions:

1. Verify username is correct (check for typos)
2. Check Caps Lock is off
3. Try typing password in Notepad to verify
4. If account locked, contact administrator to unlock
5. If password expired, use Forgot Password link

T.1.2 Session Timeout Issues

Symptom: "Your session has expired" message appears frequently

Solutions:

1. Save work frequently

2. Contact administrator to extend session timeout
3. Check for browser extensions that may interfere
4. Clear browser cache and cookies

T.1.3 MFA Not Working

Symptom: Cannot complete multi-factor authentication

Solutions:

1. Verify device time is synchronized
2. For SMS: Check phone signal and message inbox
3. For Authenticator: Re-sync the app
4. Contact administrator to reset MFA

T.2 Performance Issues

T.2.1 Slow Page Loading

Symptom: Pages take long time to load

Possible Causes:

- Slow internet connection
- Large data queries
- Browser cache issues
- Server load

Solutions:

1. Check internet speed (minimum 2 Mbps recommended)
2. Clear browser cache

3. Use filters to limit data displayed
4. Try different browser
5. Contact support if issue persists

T.2.2 Report Generation Slow

Symptom: Reports take long time to generate

Solutions:

1. Use date range filters to limit data
2. Schedule reports to run during off-hours
3. Use summary reports instead of detailed
4. Export to Excel for complex analysis

T.3 Data Entry Issues

T.3.1 Cannot Save Record

Symptom: Save button disabled or error on save

Possible Causes:

- Required fields not filled
- Validation errors
- Duplicate key violation
- Period is closed

Solutions:

1. Check for red-highlighted required fields
2. Review error messages

3. Verify unique fields are not duplicates
4. Check accounting period status

T.3.2 Data Not Appearing in Reports

Symptom: Recently entered data not showing in reports

Solutions:

1. Verify data was saved successfully
2. Check report date range includes entry date
3. Refresh the report
4. Check if data needs to be posted (not just saved)
5. Wait for ETL process if using BI reports

T.4 ETA E-Invoicing Issues

T.4.1 Invoice Rejected by ETA

Symptom: Invoice submission failed with error

Common Errors:

Error Code	Cause	Solution
INV001	Invalid seller tax ID	Verify company tax registration
INV002	Invalid buyer tax ID	Verify customer tax registration
INV003	Invalid date format	Check system date format settings
INV004	Duplicate invoice number	Use unique invoice number

INV005

Invalid signature

Check digital certificate

T.4.2 Certificate Issues

Symptom: Digital signature validation fails

Solutions:

1. Verify certificate has not expired
2. Check certificate is properly uploaded
3. Regenerate certificate if corrupted
4. Contact certificate provider for support

T.5 Integration Issues

T.5.1 Bank Integration Failed

Symptom: Cannot import bank statement

Solutions:

1. Verify file format matches bank specification
2. Check file encoding (UTF-8 recommended)
3. Verify date format matches system settings
4. Check for special characters in data

T.5.2 API Errors

Symptom: API calls returning errors

Solutions:

1. Verify API credentials are valid
2. Check rate limit not exceeded
3. Validate request format
4. Check API endpoint URL
5. Review API documentation for changes

T.6 Module-Specific Issues

T.6.1 Payroll Calculation Errors

Symptom: Incorrect salary calculations

Solutions:

1. Verify employee master data is complete
2. Check attendance data is imported
3. Verify GOSI rates are current
4. Check tax brackets for current year
5. Review earning/deduction formulas

T.6.2 Inventory Discrepancies

Symptom: Physical count differs from system

Solutions:

1. Check for unprocessed transactions
2. Verify all receipts and issues are recorded
3. Check for negative inventory settings

4. Review unit of measure conversions
5. Investigate possible data entry errors

T.6.3 Financial Imbalance

Symptom: Trial Balance does not balance

Solutions:

1. Check for unposted journal entries
2. Verify all entries have equal debits/credits
3. Check for foreign exchange rounding differences
4. Review period-end adjustments
5. Contact support for data integrity check

Frequently Asked Questions (FAQ)

General Questions

Q: What is the Egyptian ERP System?

The Egyptian ERP System is a comprehensive enterprise resource planning solution designed specifically for businesses operating in Egypt. It includes modules for finance, HR, inventory, manufacturing, and more, with full compliance to Egyptian regulatory requirements.

Q: Is the system cloud-based or on-premise?

The system is primarily offered as a cloud-based SaaS solution, but on-premise deployment is available for organizations with specific requirements.

Q: What are the system requirements?

Users need a modern web browser (Chrome, Firefox, Edge, Safari) and internet connection. No local installation is required for the cloud version.

Q: How is data security ensured?

Data is protected with TLS 1.3 encryption in transit, AES-256 encryption at rest, role-based access control, and comprehensive audit trails.

Financial Management

Q: Does the system support multi-currency?

Yes, the system supports multiple currencies with automatic exchange rate updates and conversion calculations.

Q: Can I customize the chart of accounts?

Yes, the chart of accounts is fully customizable to match your organization's structure and reporting needs.

Q: How does period closing work?

Periods can be opened or closed individually. Once closed, no transactions can be posted to that period without reopening it.

Q: Can I reverse a posted journal entry?

Yes, posted entries can be reversed by creating a reversal entry. The original entry remains visible for audit purposes.

Q: Does the system handle year-end closing?

Yes, the system provides a comprehensive year-end closing checklist and automated processes for closing the fiscal year.

ETA E-Invoicing

Q: Is ETA E-Invoicing mandatory?

Yes, E-Invoicing is mandatory for all B2B and B2G transactions in Egypt.

Q: How do I register for ETA E-Invoicing?

You must register on the ETA portal (eta.gov.eg) using your tax ID, obtain API credentials, and purchase a digital certificate from an approved provider.

Q: What happens if an invoice is rejected by ETA?

The system will display the error code and description. You must correct the issue and resubmit the invoice.

Q: Can I print an invoice before ETA submission?

No, B2B and B2G invoices must be successfully submitted to ETA and validated before they can be printed or sent to customers.

Q: What is the QR code for?

The QR code contains key invoice data and can be scanned using ETA's mobile app to verify the invoice's authenticity.

VAT

Q: What is the standard VAT rate in Egypt?

The standard VAT rate is 14%. Some goods and services are zero-rated (0%) or exempt.

Q: How often must VAT returns be filed?

VAT returns must be filed monthly within 30 days of the month end.

Q: Can I claim VAT on all purchases?

VAT can be claimed on business-related purchases. Some expenses like entertainment may not be reclaimable.

Q: How does the system calculate VAT?

VAT is automatically calculated based on the tax code assigned to items, customers, and vendors.

Payroll and HR

Q: What are the current GOSI contribution rates?

Employer contributes 18% (15% old age + 2% work injury + 1% unemployment), employee contributes 12% (11% old age + 1% unemployment).

Q: What is the maximum salary for GOSI calculation?

The maximum salary subject to social insurance is EGP 8,400 per month (2025).

Q: How is income tax calculated?

Income tax is calculated using progressive tax brackets ranging from 0% to 25% based on annual income.

Q: Can the system handle multiple pay structures?

Yes, the system supports multiple payroll structures for different employee categories.

Q: How is overtime calculated?

Overtime is calculated at 135% for regular overtime, 200% for rest days and holidays.

Inventory

Q: What valuation methods are supported?

The system supports FIFO, LIFO, Weighted Average, and Standard Cost valuation methods.

Q: Can I track items by batch or serial number?

Yes, the system supports batch tracking and serial number tracking for items requiring traceability.

Q: How does reorder point work?

When inventory falls below the reorder point, the system can automatically generate purchase requisitions.

Q: Can I have multiple warehouses?

Yes, the system supports multiple warehouses and locations with inter-warehouse transfers.

Purchasing and Sales

Q: What is three-way matching?

Three-way matching verifies that the purchase order, goods receipt, and vendor invoice all agree before payment is authorized.

Q: Can I set approval workflows?

Yes, approval workflows can be configured based on amount, department, or document type.

Q: How does credit control work?

The system monitors customer credit limits and can block new orders when limits are exceeded.

System Administration

Q: How do I create a new user?

Navigate to Administration → Security → Users → Create New and fill in the required information.

Q: What are roles and permissions?

Roles define sets of permissions that determine what users can see and do in the system.

Q: How do I back up my data?

Data backups are handled automatically for cloud deployments. On-premise deployments require configured backup procedures.

Q: Can I customize fields and forms?

Yes, the system supports custom fields, custom workflows, and custom reports.

Integration

Q: Does the system have an API?

Yes, the system provides a comprehensive REST API for integration with other systems.

Q: Can I import data from Excel?

Yes, data can be imported from Excel using the import templates provided.

Q: Does the system integrate with banks?

Yes, the system can import bank statements and generate payment files for bank integration.

Support

Q: How do I get support?

Support is available through the help desk, live chat, email, and phone.

Q: What are support hours?

Standard support is available Sunday-Thursday, 9 AM - 5 PM Egypt time. 24/7 support is available for critical issues.

Q: Is training available?

Yes, comprehensive training programs are available including online courses, instructor-led training, and certification programs.

Best Practices Guide

BP.1 General Best Practices

BP.1.1 Data Entry Best Practices

- Always enter data in real-time rather than batching
- Double-check all amounts before saving
- Use copy function for similar transactions
- Save work frequently to prevent loss
- Use comments to document unusual transactions
- Verify master data (customers, vendors, items) before using

BP.1.2 Security Best Practices

- Never share your password with anyone
- Use strong, unique passwords
- Change passwords regularly (every 90 days)
- Log out when leaving your workstation
- Lock your computer when away
- Report suspicious activity immediately
- Use MFA whenever available

BP.1.3 Report Generation Best Practices

- Use filters to limit data and improve performance
- Schedule large reports to run during off-hours
- Save frequently used filter combinations
- Export to Excel for complex analysis
- Verify report parameters before generating

BP.2 Financial Management Best Practices

BP.2.1 Chart of Accounts Best Practices

- Plan the chart of accounts carefully before implementation
- Use a logical numbering scheme
- Leave gaps in numbering for future accounts
- Document the purpose of each account
- Review and clean up unused accounts annually
- Align with Egyptian accounting standards

BP.2.2 Journal Entry Best Practices

- Always use descriptive references
- Include supporting document numbers
- Attach supporting documents when possible
- Review entries before posting
- Use reversing entries for accruals
- Never delete posted entries - reverse them

BP.2.3 Period Closing Best Practices

- Follow the period-end checklist
- Complete all entries before closing
- Review trial balance for unusual balances
- Reconcile all control accounts
- Document any adjustments made
- Close periods promptly after month-end

BP.2.4 Bank Reconciliation Best Practices

- Reconcile bank accounts monthly
- Import statements promptly
- Investigate unmatched items immediately
- Document reasons for adjustments
- Review old unmatched items regularly
- Segregate reconciliation duties

BP.3 ETA E-Invoicing Best Practices

BP.3.1 Customer Master Best Practices

- Verify tax registration numbers before creating customers
- Keep customer master data up to date
- Mark B2B customers correctly for E-Invoicing
- Validate customer information regularly

BP.3.2 Invoice Processing Best Practices

- Submit invoices to ETA immediately upon creation
- Monitor submission status regularly
- Address rejected invoices promptly
- Keep digital certificates up to date
- Test ETA connection periodically

BP.4 Payroll Best Practices

BP.4.1 Employee Master Best Practices

- Verify national ID numbers for accuracy
- Keep contact information current
- Document all salary changes

- Review employee data quarterly

BP.4.2 Payroll Processing Best Practices

- Process payroll on a consistent schedule
- Verify attendance data before processing
- Review payroll summary before posting
- Keep backup of payroll data
- Submit GOSI reports on time

BP.5 Inventory Best Practices

BP.5.1 Item Master Best Practices

- Use consistent item numbering
- Include clear descriptions
- Set appropriate reorder points
- Classify items using ABC analysis
- Review and update costs regularly

BP.5.2 Inventory Transaction Best Practices

- Record transactions immediately
- Use barcode scanning when possible
- Verify quantities before confirming
- Document any discrepancies
- Perform regular cycle counts

BP.5.3 Physical Count Best Practices

- Plan counts in advance
- Count during low-activity periods

- Use count teams of two people
- Investigate variances immediately
- Document count procedures

BP.6 Purchasing Best Practices

BP.6.1 Vendor Management Best Practices

- Evaluate vendors regularly
- Maintain accurate vendor information
- Document vendor performance
- Negotiate payment terms strategically

BP.6.2 Purchase Order Best Practices

- Always use formal purchase orders
- Include all required terms and conditions
- Obtain proper approvals
- Track PO status regularly
- Follow up on overdue deliveries

BP.7 Sales Best Practices

BP.7.1 Customer Management Best Practices

- Perform credit checks on new customers
- Set appropriate credit limits
- Monitor aging regularly
- Follow up on overdue accounts
- Document all customer communications

BP.7.2 Sales Order Best Practices

- Verify item availability before confirming
- Confirm pricing with customers
- Document special requirements
- Track order status proactively
- Communicate delays promptly

BP.8 System Administration Best Practices

BP.8.1 User Management Best Practices

- Follow principle of least privilege
- Review user access quarterly
- Deactivate users promptly upon termination
- Document role assignments
- Use standard roles when possible

BP.8.2 Security Best Practices

- Enforce strong password policies
- Enable MFA for all users
- Review audit logs regularly
- Keep system updated
- Test backup recovery periodically

BP.8.3 Data Management Best Practices

- Cleanse data before import
- Validate imported data
- Archive old data regularly

- Maintain data quality
- Document data procedures

Regulatory Compliance Guide

RC.1 Egyptian Tax Authority (ETA) Compliance

RC.1.1 E-Invoicing Requirements

All B2B and B2G transactions must comply with ETA E-Invoicing regulations:

Requirement	Description	System Feature
UUID Generation	Unique identifier per invoice	Auto-generated
Digital Signature	ETA-approved certificate	Integrated signing
QR Code	Verification code on invoice	Auto-generated
JSON Format	ETA-specified format	Auto-formatted
Real-time Submission	Submit at time of issuance	Automatic submission

RC.1.2 ETA Portal Registration

Registration Steps:

1. Visit <https://eta.gov.eg>
2. Register using company tax ID
3. Submit required documents
4. Obtain API credentials
5. Purchase digital certificate
6. Configure in ERP system

RC.1.3 ETA Submission Process

1. Create invoice in ERP system
2. System validates data format
3. Generate UUID per ETA spec
4. Create JSON document
5. Apply digital signature
6. Submit to ETA portal
7. Receive submission response
8. Store ETA ID and QR code
9. Print invoice with QR code

RC.2 Value Added Tax (VAT) Compliance

RC.2.1 VAT Rates and Categories

Category	Rate	Examples
Standard Rate	14%	Most goods and services
Zero-Rated	0%	Exports, essential food
Exempt	N/A	Financial services, education

RC.2.2 VAT Registration

Businesses must register for VAT if annual turnover exceeds EGP 500,000.

RC.2.3 VAT Return Filing

- Monthly returns required
- Due within 30 days of month end

- File electronically through ETA portal
- Pay any VAT due

RC.2.4 Input VAT Recovery

Businesses can recover VAT paid on:

- Business purchases
- Imports for business use
- Capital assets

RC.3 Social Insurance (GOSI) Compliance

RC.3.1 Contribution Rates (2025)

Type	Employer	Employee	Total
Old Age/Disability/Death	15%	11%	26%
Work Injury	2%	0%	2%
Unemployment	1%	1%	2%
Total	18%	12%	30%

RC.3.2 Maximum Insurance Salary

Maximum salary subject to contributions: EGP 8,400 per month (2025)

RC.3.3 Monthly Reporting

- Submit Form 1 monthly
- Report new employees on Form 2
- Report terminations on Form 3
- Report salary changes on Form 4

RC.3.4 Annual Reconciliation

Submit Form 6 annually with complete employee and salary information.

RC.4 Labor Law Compliance

RC.4.1 Working Hours

Type	Hours	Rate
Normal	8 hours/day, 48 hours/week	100%
Regular Overtime	Up to 2 hours/day	135%
Rest Day	Any hours	200%
Official Holiday	Any hours	200%

RC.4.2 Leave Entitlements

Leave Type	Entitlement	Conditions
Annual Leave	21-30 days	Based on service years
Sick Leave	Up to 180 days	Medical certificate
Maternity Leave	90 days	Female employees
Paternity Leave	3 days	Within 1 month of birth
Hajj Leave	30 days	Once during employment

RC.4.3 End of Service Benefits

Employees are entitled to end of service benefits calculated as:

- First 5 years: 1/2 month per year
- After 5 years: 1 month per year
- Based on last basic salary

RC.5 Commercial Registration

RC.5.1 Registration Requirements

- Company incorporation documents
- Tax registration
- Bank account
- Legal address

RC.5.2 Annual Renewal

Commercial registration must be renewed annually with updated company information.

RC.6 Data Protection

RC.6.1 Employee Data Protection

- Obtain consent for data collection
- Limit access to authorized personnel
- Secure storage of personal data
- Allow employee access to their data
- Proper disposal of old records

RC.6.2 Customer/Vendor Data Protection

- Collect only necessary data
- Secure data transmission
- Implement access controls

- Have data breach response plan

RC.7 Compliance Checklist

RC.7.1 Monthly Compliance

- Submit ETA E-Invoices
- File VAT return
- Submit GOSI Form 1
- Process payroll with deductions

RC.7.2 Quarterly Compliance

- Review and update registrations
- Verify tax compliance
- Update employee records

RC.7.3 Annual Compliance

- Submit annual tax return
- Submit GOSI Form 6
- Renew commercial registration
- Review all compliance procedures

Workflow Documentation

WF.1 Purchase to Pay Workflow

WF.1.1 Process Overview

1. Department identifies need for goods/services
2. Create purchase requisition
3. Obtain approval based on amount
4. Procurement creates purchase order
5. Vendor delivers goods/services
6. Warehouse receives goods (GRN)
7. Accounts payable receives invoice
8. Three-way match verification
9. Invoice approval
10. Payment processing

WF.1.2 Detailed Steps

Step 1: Create Purchase Requisition

1. Navigate to Purchasing → Requisitions → Create New
2. Select requesting department
3. Add required items/services
4. Enter quantities and estimated prices
5. Specify required date
6. Add justification/comments

7. Submit for approval

Step 2: Approve Requisition

1. Approver receives notification
2. Review requisition details
3. Check budget availability
4. Approve or reject with comments

Step 3: Create Purchase Order

1. Procurement converts approved requisition to PO
2. Select vendor
3. Confirm pricing and terms
4. Add delivery instructions
5. Send PO to vendor

WF.2 Order to Cash Workflow

WF.2.1 Process Overview

1. Customer inquiry/quotation
2. Create sales quotation
3. Customer approves quotation
4. Create sales order
5. Credit check
6. Reserve inventory

7. Pick and pack goods
8. Create delivery note
9. Ship goods
10. Create invoice
11. Submit to ETA (B2B)
12. Send invoice to customer
13. Receive payment
14. Apply payment to invoice

WF.2.2 Detailed Steps

Step 1: Create Sales Quotation

1. Navigate to Sales → Quotations → Create New
2. Select customer
3. Add items with quantities
4. Apply pricing and discounts
5. Set quotation validity
6. Send to customer

Step 2: Create Sales Order

1. Convert approved quotation to sales order
2. Verify customer credit limit
3. Check inventory availability
4. Reserve inventory
5. Confirm delivery date

Step 3: Create Invoice

1. Create invoice from delivery
2. Verify invoice details
3. Submit to ETA (B2B)
4. Print with QR code
5. Send to customer

WF.3 Payroll Workflow

WF.3.1 Process Overview

1. Collect attendance data
2. Import attendance to system
3. Calculate overtime
4. Process variable payments
5. Calculate deductions
6. Calculate GOSI contributions
7. Calculate income tax
8. Generate payslips
9. Review and approve payroll
10. Post to GL
11. Generate bank transfer file
12. Submit GOSI report

WF.4 Period End Closing Workflow

WF.4.1 Monthly Closing

Month-End Checklist:

1. Complete all transaction entry
2. Post all journal entries
3. Reconcile bank accounts
4. Reconcile AP subledger to GL
5. Reconcile AR subledger to GL
6. Reconcile inventory to GL
7. Run depreciation
8. Process payroll
9. Review trial balance
10. Generate financial reports
11. Close accounting period

WF.5 Inventory Management Workflow

WF.5.1 Goods Receipt

1. Receive PO notification
2. Prepare receiving area
3. Verify delivery against PO
4. Inspect goods for damage
5. Count and verify quantities
6. Create GRN in system
7. Update inventory
8. Move goods to storage

WF.5.2 Goods Issue

1. Receive issue request
2. Verify authorization
3. Pick items from storage
4. Verify quantities
5. Create goods issue in system
6. Update inventory
7. Deliver to requestor

WF.5.3 Stock Transfer

1. Create transfer order
2. Pick items from source location
3. Ship to destination
4. Receive at destination
5. Confirm transfer completion

WF.5.4 Physical Count

1. Plan count schedule
2. Freeze inventory movements
3. Print count sheets
4. Perform physical count
5. Enter count results
6. Review variances
7. Investigate significant variances
8. Post adjustments
9. Unfreeze inventory

WF.6 Manufacturing Workflow

WF.6.1 Production Order Process

1. Create production order
2. Reserve raw materials
3. Issue materials to production
4. Record labor hours
5. Record machine hours
6. Report finished goods
7. Record scrap/waste
8. Complete production order
9. Calculate production cost

WF.7 Fixed Assets Workflow

WF.7.1 Asset Acquisition

1. Receive asset
2. Inspect and verify
3. Create asset master
4. Assign asset tag
5. Record in location
6. Post acquisition entry

WF.7.2 Asset Depreciation

1. Review asset register
2. Run depreciation calculation
3. Review depreciation amounts

4. Post depreciation entry
5. Update asset values

WF.7.3 Asset Disposal

1. Approve disposal
2. Remove asset from location
3. Create disposal entry
4. Calculate gain/loss
5. Post disposal to GL
6. Remove from asset register

Extended General Ledger Guide

GL.1 Advanced Journal Entry Topics

GL.1.1 Recurring Journal Entries

Recurring journal entries automate repetitive transactions such as monthly rent, depreciation, and accruals.

To create a recurring journal entry:

1. Navigate to Financial → General Ledger → Recurring Entries → Create New
2. Enter template name and description
3. Select frequency (Daily, Weekly, Monthly, Quarterly, Annually)
4. Set start date and end date (if applicable)
5. Enter journal entry lines
6. Set posting date option (fixed or relative)
7. Save template

Example: Monthly Rent Accrual

Template Name: Monthly Rent Accrual

Frequency: Monthly

Posting Date: Last day of month

Lines:

Account	Debit	Credit

61010 - Rent Expense

10,000

21010 - Rent Payable

10,000

GL.1.2 Reversing Journal Entries

Reversing entries automatically reverse on the first day of the next period, commonly used for accruals.

To create a reversing entry:

1. Create journal entry as normal
2. Check "Auto Reverse" option
3. Select reversal date
4. Post the entry
5. System will create reversal entry automatically

GL.1.3 Journal Entry Templates

Templates speed up data entry for common transactions:

- Save frequently used journal entries as templates
- Share templates across users
- Modify amounts while keeping account structure

GL.2 Advanced Reporting

GL.2.1 Comparative Reports

Compare financial data across multiple periods:

- Current period vs prior period
- Current year vs prior year
- Actual vs budget
- Multiple year trend analysis

GL.2.2 Consolidated Reporting

For multi-company organizations:

- Consolidate financial statements
- Eliminate intercompany transactions
- Report in functional or reporting currency
- Partial ownership handling

GL.3 Period-End Procedures

GL.3.1 Month-End Checklist

Task	Responsible	Deadline
Complete all transaction entry	All Users	Month + 2 days
Post all journal entries	Accountant	Month + 3 days
Reconcile bank accounts	Cashier	Month + 5 days
Reconcile AP subledger	AP Clerk	Month + 5 days
Reconcile AR subledger	AR Clerk	Month + 5 days
Reconcile inventory	Warehouse	Month + 5 days
Run depreciation	Accountant	Month + 5 days
Review trial balance	Accountant	Month + 7 days

Generate financial statements	Accountant	Month + 7 days
Management review	CFO	Month + 10 days
Close period	Controller	Month + 10 days

GL.3.2 Year-End Procedures

Additional year-end tasks:

- Physical inventory count
- Fixed assets verification
- Accrual review and adjustment
- Prepaid expense review
- Bad debt provision
- Depreciation review
- Tax provision calculation
- Closing entries
- Financial statement preparation
- Audit preparation

GL.4 Multi-Currency Accounting

GL.4.1 Currency Setup

Configure multiple currencies:

- Base currency (EGP)
- Transaction currencies (USD, EUR, GBP)
- Exchange rate sources
- Rate update frequency

GL.4.2 Foreign Currency Transactions

Recording foreign currency transactions:

- Enter amount in foreign currency
- System converts using exchange rate
- Track exchange rate differences
- Realized vs unrealized gains/losses

GL.4.3 Exchange Rate Revaluation

To run exchange rate revaluation:

1. Navigate to Financial → General Ledger → Revaluation
2. Select revaluation date
3. Select currency to revalue
4. Preview revaluation amounts
5. Post revaluation entry

GL.5 Budget Management

GL.5.1 Budget Setup

Create annual budgets:

- Define budget accounts
- Set budget amounts by period
- Allocate to cost centers
- Set budget versions

GL.5.2 Budget Control

Monitor actual vs budget:

- Warning when approaching budget
- Block transactions exceeding budget
- Budget transfer between accounts
- Budget revision workflow

GL.5.3 Budget Reports

Generate budget variance reports:

- Budget vs actual by account
- Budget vs actual by cost center
- Variance analysis
- Trend analysis

Extended Accounts Payable Guide

AP.1 Advanced Invoice Processing

AP.1.1 Recurring Invoices

Handle regular vendor payments like utilities, rent, and subscriptions:

- Set up recurring invoice templates
- Automatic generation at scheduled intervals
- Variable amount handling
- Approval workflow integration

AP.1.2 Debit Memos

Process vendor returns and adjustments:

To create a debit memo:

1. Navigate to AP → Debit Memos → Create New
2. Select vendor
3. Reference original invoice
4. Enter reason for debit
5. Enter amount
6. Apply to open invoice or keep as credit

AP.1.3 Withholding Tax

Handle tax withholding on vendor payments:

Payment Type	Withholding Rate
Professional Services	5%
Contracting Services	1.5%
Rent	10%
Dividends	10%
Interest	20%

AP.2 Advanced Payment Processing

AP.2.1 Payment Scheduling

Optimize cash flow through payment scheduling:

- Early payment discount capture
- Cash flow forecasting
- Payment prioritization
- Vendor payment terms optimization

AP.3 Vendor Management

AP.3.1 Vendor Onboarding

Vendor onboarding checklist:

1. Collect vendor information form
2. Verify tax registration
3. Check commercial registration

- 4. Obtain bank details
- 5. Verify bank account
- 6. Set credit limit
- 7. Assign vendor category
- 8. Create in system

AP.3.2 Vendor Performance Tracking

Monitor vendor performance metrics:

Metric	Measurement	Target
On-Time Delivery	% delivered on promised date	>95%
Quality Rating	Defect rate	<1%
Price Competitiveness	vs market average	Within 5%
Invoice Accuracy	% error-free invoices	>98%

AP.4 Period-End Procedures

AP.4.1 Month-End Checklist

- Enter all received invoices
- Process all approved payments
- Reconcile vendor statements
- Resolve discrepancies
- Review AP aging
- Verify GL reconciliation
- Accrue unbilled expenses

Extended Accounts Receivable Guide

AR.1 Advanced Invoicing

AR.1.1 Recurring Invoices

Automate regular customer billing:

- Monthly service fees
- Subscription billing
- Lease payments
- Installment billing

AR.1.2 Credit Notes

Process customer returns and adjustments:

To create a credit note:

1. Navigate to AR → Credit Notes → Create New
2. Select customer
3. Reference original invoice
4. Enter reason for credit
5. Enter credit amount
6. Apply to open invoice or keep as credit

AR.2 Credit Management

AR.2.1 Credit Evaluation

Assess customer creditworthiness:

Factor	Assessment
Financial Statements	Analyze liquidity and profitability
Payment History	Review past payment behavior
Credit References	Check with other suppliers
Industry Risk	Consider industry conditions

AR.3 Collections Management

AR.3.1 Collection Strategy

Aging	Action
Current	Send statement
1-15 days	Reminder email
16-30 days	Follow-up call
31-45 days	Formal demand letter
46-60 days	Management involvement
60+ days	Collection agency or legal

AR.4 Period-End Procedures

AR.4.1 Month-End Checklist

- Ensure all invoices are created
- Post all receipts

- Apply all payments to invoices
- Review AR aging
- Follow up on overdue accounts
- Reconcile AR to GL

Extended Inventory Management Guide

INV.1 Advanced Item Management

INV.1.1 Item Classification

ABC Analysis for inventory control:

Class	% of Items	% of Value	Control Level
A	20%	80%	Tight control, frequent review
B	30%	15%	Moderate control
C	50%	5%	Simple control

INV.1.2 Item Attributes

Track additional item characteristics:

- Brand/Manufacturer
- Model/Specification
- Country of origin
- Shelf life/Expiry date
- Storage requirements
- Hazardous material classification

INV.2 Inventory Planning

INV.2.1 Reorder Point Calculation

Formula: $\text{Reorder Point} = (\text{Average Daily Usage} \times \text{Lead Time}) + \text{Safety Stock}$

Reorder Point Example

Average Daily Usage: 10 units

Lead Time: 7 days

Safety Stock: 20 units

Reorder Point = $(10 \times 7) + 20 = 90$ units

INV.2.2 Economic Order Quantity (EOQ)

Formula: $\text{EOQ} = \sqrt{(2 \times \text{Annual Demand} \times \text{Order Cost}) / \text{Holding Cost per Unit}}$

INV.2.3 Safety Stock

Factors affecting safety stock levels:

- Demand variability
- Lead time variability
- Desired service level
- Stockout costs

INV.3 Advanced Transactions

INV.3.1 Assembly/Kitting

Combine multiple items into a kit:

- Define BOM for assembly
- Issue component items
- Receive finished assembly
- Track assembly costs

INV.3.2 Disassembly

Break down items into components:

- Issue assembly item
- Receive component items
- Record any loss/waste

INV.4 Inventory Valuation

INV.4.1 FIFO Method

First In, First Out - oldest inventory used first:

- Matches physical flow for perishables
- Ending inventory reflects current costs
- COGS reflects older costs

INV.4.2 Weighted Average Method

Average cost recalculated after each receipt:

- Smooths out price fluctuations
- Simple to calculate
- Used for homogeneous products

INV.5 Physical Inventory

INV.5.1 Cycle Counting

Count frequencies by ABC class:

Class	Count Frequency
A	Monthly
B	Quarterly

C

Annually

INV.5.2 Count Variance Analysis

Investigate count differences:

- Data entry errors
- Transaction timing differences
- Theft or damage
- Unit of measure errors
- System errors

INV.6 Reporting

INV.6.1 Inventory Turnover

Formula: $\text{Inventory Turnover} = \text{Cost of Goods Sold} / \text{Average Inventory}$

INV.6.2 Days Inventory Outstanding

Formula: $\text{DIO} = \text{Average Inventory} / \text{Cost of Goods Sold} \times 365$

Extended Payroll Guide

PAY.1 Advanced Earnings

PAY.1.1 Earning Types

Earning	Description	Tax Treatment
Basic Salary	Fixed monthly pay	Taxable
Housing Allowance	Accommodation support	Taxable
Transportation	Commuting allowance	Taxable
Meal Allowance	Food subsidy	Taxable
Mobile Allowance	Phone reimbursement	Taxable
Shift Differential	Extra for night/weekend	Taxable
Bonus	Performance/annual	Taxable
Commission	Sales commission	Taxable

PAY.1.2 Overtime Calculation

Hourly Rate = Basic Salary / (Standard Hours per Month)

Standard Hours: 8 hours/day × 6 days × 4 weeks = 192 hours

Overtime Calculation

Basic	Salary:	EGP	5,000
Hourly	Rate:	5,000 / 192 =	EGP 26.04

Regular Overtime (10 hours): $10 \times 26.04 \times 1.35 = \text{EGP } 351.54$

Holiday Overtime (5 hours): $5 \times 26.04 \times 2.0 = \text{EGP } 260.40$

PAY.2 Advanced Deductions

PAY.2.1 Deduction Types

Deduction	Description	Calculation
Social Insurance	GOSI employee share	12% of capped salary
Income Tax	Salary tax	Per tax brackets
Loan Repayment	Employee loan	As per agreement
Health Insurance	Private medical	Fixed or %
Life Insurance	Insurance premium	Fixed amount
Union Dues	Union membership	Fixed amount

PAY.3 GOSI Detailed Calculation

PAY.3.1 Contribution Breakdown

Type	Employer %	Employee %	Purpose
Old Age/Disability/Death	15%	11%	Pension benefits
Work Injury	2%	0%	Workplace accidents
Unemployment	1%	1%	Unemployment benefits

PAY.3.2 Maximum Insurance Amount

Maximum Salary: EGP 8,400 per month (2025)

Contributions calculated on lesser of actual salary or maximum

PAY.4 Income Tax Detailed

PAY.4.1 Tax Brackets (2025)

Annual Income (EGP)	Rate	Tax Amount
0 - 30,000	0%	0
30,001 - 45,000	10%	1,500
45,001 - 60,000	15%	2,250
60,001 - 200,000	20%	28,000
200,001 - 400,000	22.5%	45,000
Above 400,000	25%	Variable

PAY.4.2 Tax Exemptions

Personal exemption: EGP 15,000 annually

Social insurance contributions are deductible

PAY.5 Payroll Processing

PAY.5.1 Monthly Checklist

- Import attendance data
- Calculate overtime
- Process new hires
- Process terminations

- Update salary changes
- Process variable payments
- Calculate deductions
- Generate payslips
- Review and approve
- Post to GL
- Generate bank file
- Submit GOSI report

PAY.6 End of Service Benefits

PAY.6.1 Calculation Formula

For first 5 years: 1/2 month per year of service

After 5 years: 1 month per year of service

Based on last basic salary

End of Service Example

Years of Service: 8

Last Basic Salary: EGP 6,000

First 5 years: $5 \times 0.5 \times 6,000 = \text{EGP } 15,000$

Next 3 years: $3 \times 1 \times 6,000 = \text{EGP } 18,000$

Total End of Service: EGP 33,000

Extended Human Resources Guide

HR.1 Employee Lifecycle

HR.1.1 Hiring Process

New Hire Process:

1. Receive approved requisition
2. Post job advertisement
3. Screen applications
4. Conduct interviews
5. Make job offer
6. Complete background check
7. Prepare employment contract
8. Collect required documents
9. Create employee record
10. Conduct orientation

HR.1.2 Required Documents

- National ID copy
- Birth certificate
- Educational certificates
- Military status document
- Medical certificate
- Previous employment certificates

- Tax registration (if applicable)
- Bank account details
- Emergency contact information
- Passport photos

HR.2 Leave Management

HR.2.1 Leave Entitlements by Service

Years of Service	Annual Leave Days
Less than 10 years	21 days
10 years or more	30 days

HR.2.2 Sick Leave Provisions

Period	Payment	Requirements
First 90 days	100%	Medical certificate
Next 90 days	75%	Medical committee approval

HR.2.3 Leave Request Workflow

Leave Request Process:

1. Employee submits request
2. System checks balance
3. Manager receives notification
4. Manager approves/rejects

5. Employee notified
6. Balance updated

HR.3 Attendance Management

HR.3.1 Shift Types

Shift	Hours	Break
Day	08:00 - 17:00	1 hour
Morning	07:00 - 15:00	30 min
Afternoon	15:00 - 23:00	30 min
Night	23:00 - 07:00	30 min

HR.3.2 Overtime Rules

Maximum overtime: 2 hours per day

Weekly maximum: 12 hours

HR.4 Performance Management

HR.4.1 Performance Review Cycle

- Goal setting (annual)
- Mid-year review
- Year-end evaluation
- Performance rating
- Development planning

HR.4.2 Performance Ratings

Rating	Description	Distribution
5 - Exceptional	Exceeds all expectations	5%
4 - Exceeds	Exceeds most expectations	20%
3 - Meets	Meets expectations	60%
2 - Needs Improvement	Below expectations	10%
1 - Unsatisfactory	Significantly below	5%

HR.5 Termination Process

HR.5.1 Termination Types

Type	Notice Period	End of Service
Resignation	2-3 months	Yes
Termination (Company)	2-3 months	Yes
End of Contract	As per contract	Yes
Mutual Agreement	Negotiated	Negotiated

HR.5.2 Termination Checklist

- Receive resignation/termination notice
- Calculate notice period
- Process handover
- Collect company assets
- Disable system access

- Calculate final settlement
- Process end of service benefits
- Prepare experience certificate
- Update employee status

Extended Manufacturing Guide

MFG.1 Production Planning

MFG.1.1 Production Types

Type	Description	Use Case
Make to Stock	Produce to inventory	Standard products
Make to Order	Produce after order	Custom products
Assemble to Order	Assemble after order	Configured products
Engineer to Order	Design and build	Complex projects

MFG.1.2 Capacity Planning

Factors to consider:

- Available work center hours
- Machine capacity
- Labor availability
- Tooling availability
- Material availability

MFG.2 Bill of Materials

MFG.2.1 BOM Levels

- Level 0: Finished product
- Level 1: Major sub-assemblies

- Level 2: Components
- Level 3: Raw materials

MFG.2.2 BOM Types

Type	Description
Engineering BOM	As designed
Manufacturing BOM	As produced
Service BOM	For maintenance
Sales BOM	For quotations

MFG.3 Production Order Execution

MFG.3.1 Order Status

Status	Description
Created	Order created, not released
Released	Ready for production
In Progress	Production started
Partially Complete	Some output reported
Complete	Production finished
Closed	Order closed

MFG.3.2 Material Issuance

To issue materials:

1. Create material requisition
2. Pick materials from warehouse
3. Issue to production order
4. Update inventory

MFG.4 Quality in Manufacturing

MFG.4.1 Inspection Points

- Incoming material inspection
- In-process inspection
- Final inspection
- Outgoing quality check

MFG.4.2 Quality Metrics

Metric	Formula	Target
First Pass Yield	$(\text{Units Passed} / \text{Total}) \times 100$	>95%
Scrap Rate	$(\text{Scrap} / \text{Total Produced}) \times 100$	<2%
Rework Rate	$(\text{Rework} / \text{Total}) \times 100$	<3%

MFG.5 Costing

MFG.5.1 Cost Components

- Direct Materials

- Direct Labor
- Manufacturing Overhead

MFG.5.2 Costing Methods

Method	Description
Standard Costing	Predetermined costs
Actual Costing	Actual costs incurred
Absorption Costing	All manufacturing costs

Extended Reporting and Business Intelligence Guide

BI.1 Report Types

BI.1.1 Financial Reports

Report	Description	Frequency
Trial Balance	All account balances	Monthly
Income Statement	Revenue and expenses	Monthly
Balance Sheet	Assets, liabilities, equity	Monthly
Cash Flow	Cash movements	Monthly
AP Aging	Payables by age	Weekly
AR Aging	Receivables by age	Weekly

BI.1.2 Operational Reports

Report	Description	Frequency
Inventory Status	Stock levels	Daily
Production Schedule	Planned production	Daily
Sales Pipeline	Opportunity tracking	Weekly
Purchase Orders	Open POs	Weekly
Work Orders	Production status	Daily

BI.2 Dashboard Components

BI.2.1 KPI Cards

Key metrics at a glance:

- Total Revenue
- Total Expenses
- Net Income
- Cash Balance
- AR Balance
- AP Balance
- Inventory Value

BI.2.2 Chart Types

Chart Type	Best For
Line Chart	Trends over time
Bar Chart	Comparing categories
Pie Chart	Showing proportions
Area Chart	Cumulative totals
Gauge	Performance vs target
Table	Detailed data

BI.3 Financial KPIs

BI.3.1 Profitability Ratios

Ratio	Formula	Interpretation
Gross Margin	Gross Profit / Revenue	Pricing power
Operating Margin	Operating Income / Revenue	Operating efficiency
Net Margin	Net Income / Revenue	Overall profitability
ROA	Net Income / Total Assets	Asset efficiency
ROE	Net Income / Equity	Shareholder return

BI.3.2 Liquidity Ratios

Ratio	Formula	Healthy Range
Current Ratio	Current Assets / Current Liabilities	1.5 - 2.0
Quick Ratio	(Current Assets - Inventory) / Current Liabilities	1.0 - 1.5
Cash Ratio	Cash / Current Liabilities	>0.5

BI.3.3 Efficiency Ratios

Ratio	Formula	Interpretation
Inventory Turnover	COGS / Average Inventory	Higher is better
AR Turnover	Revenue / Average AR	Higher is better
AP Turnover	COGS / Average AP	Manage payment terms
Asset Turnover	Revenue / Total Assets	Asset efficiency

BI.4 Custom Reports

BI.4.1 Report Builder

Creating Custom Reports:

- 1. Select data source
- 2. Choose fields to display
- 3. Add filters
- 4. Add calculations
- 5. Design layout
- 6. Save and schedule

BI.5 Data Export

BI.5.1 Export Formats

Format	Use Case
Excel	Data analysis
PDF	Distribution
CSV	Data import
XML	System integration
JSON	API integration

Extended System Administration Guide

SA.1 User Management

SA.1.1 User Types

Type	Access Level	Typical Users
System Administrator	Full access	IT Team
Power User	Module admin	Department Heads
Standard User	Limited access	Staff
Read-Only User	View only	Auditors

SA.1.2 User Lifecycle

User Management Process:

1. Receive user request
2. Verify authorization
3. Create user account
4. Assign roles
5. Set password
6. Notify user
7. Monitor usage

8. Deactivate when needed

SA.2 Security Configuration

SA.2.1 Password Policy

Setting	Recommended
Minimum Length	8 characters
Complexity	Upper, lower, number, special
Expiration	90 days
History	Remember last 5
Lockout	After 5 failed attempts

SA.2.2 Session Settings

Setting	Recommended
Session Timeout	30 minutes
Concurrent Sessions	1 per user
Idle Warning	5 minutes before timeout

SA.3 Backup and Recovery

SA.3.1 Backup Strategy

Backup Type	Frequency	Retention
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Full Backup	Weekly	4 weeks
Incremental	Daily	7 days
Transaction Log	Every 15 min	24 hours

SA.3.2 Recovery Procedures

Disaster Recovery Steps:

- 1. Assess damage
- 2. Restore from latest backup
- 3. Apply transaction logs
- 4. Verify data integrity
- 5. Notify users
- 6. Document incident

SA.4 System Monitoring

SA.4.1 Performance Metrics

Metric	Warning Threshold	Critical Threshold
CPU Usage	70%	90%
Memory Usage	80%	95%
Disk Space	80%	95%
Response Time	3 seconds	5 seconds

SA.4.2 Alert Configuration

Set up alerts for:

- System performance issues
- Failed login attempts
- Backup failures
- Disk space warnings
- License expiration

SA.5 Maintenance Tasks

SA.5.1 Daily Tasks

- Check system health dashboard
- Review error logs
- Verify backup completion
- Monitor user activity

SA.5.2 Weekly Tasks

- Review security logs
- Check disk space
- Update system statistics
- Review performance metrics

SA.5.3 Monthly Tasks

- User access review
- Security patch assessment
- Capacity planning review
- Disaster recovery test

Extended Purchasing Guide

PUR.1 Procurement Methods

PUR.1.1 Purchase Types

Type	Description	Use Case
Standard Purchase	Regular procurement	Normal operations
Emergency Purchase	Urgent requirement	Critical shortage
Blanket Order	Multiple deliveries	Regular supplies
Framework Agreement	Agreed terms	Strategic vendors

PUR.1.2 Sourcing Methods

Method	Description	When to Use
Single Source	One approved vendor	Specialized items
Multiple Source	Several vendors	Commodity items
Competitive Bidding	RFQ process	High value
Auction	Reverse auction	Standard items

PUR.2 Purchase Requisition

PUR.2.1 Requisition Types

- Standard - Normal procurement

- Urgent - Expedited processing
- Project - Specific project needs
- Stock - Inventory replenishment

PUR.2.2 Approval Matrix

Amount (EGP)	Approver
Up to 5,000	Department Manager
5,001 - 25,000	Finance Manager
25,001 - 100,000	CFO
Above 100,000	CEO + CFO

PUR.3 Purchase Orders

PUR.3.1 PO Terms and Conditions

- Payment terms
- Delivery terms
- Quality requirements
- Inspection requirements
- Penalty clauses
- Force majeure

PUR.3.2 PO Amendments

Amending a Purchase Order:

1. Open original PO

- 2. Create amendment
- 3. Specify changes
- 4. Get approval if needed
- 5. Send to vendor

PUR.4 Vendor Management

PUR.4.1 Vendor Evaluation Criteria

Criteria	Weight	Measurement
Quality	30%	Defect rate
Delivery	25%	On-time %
Price	25%	Competitiveness
Service	20%	Responsiveness

PUR.4.2 Vendor Classification

Class	Description	Action
A	Strategic partner	Long-term relationship
B	Preferred vendor	Regular business
C	Approved vendor	Occasional use
D	Provisional	Monitor closely

PUR.5 Procurement Analytics

PUR.5.1 Key Metrics

Metric	Formula	Target
Purchase Order Cycle Time	PO Date to Receipt Date	<7 days
Cost Savings	Target Price - Actual Price	>5%
Spend Under Management	Managed Spend / Total Spend	>80%
Vendor Consolidation	Number of Vendors	Optimize

Extended Sales Guide

SAL.1 Sales Process

SAL.1.1 Sales Stages

Stage	Description	Probability
Prospecting	Initial contact	10%
Qualification	Needs identified	25%
Proposal	Quote submitted	50%
Negotiation	Terms discussed	75%
Closing	Order received	100%

SAL.1.2 Sales Team Roles

Role	Responsibilities
Sales Rep	Customer contact, quotes
Sales Manager	Team management, key accounts
Sales Director	Strategy, major deals

SAL.2 Customer Management

SAL.2.1 Customer Segmentation

Segment	Criteria	Strategy
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Key Accounts	High value, strategic	Dedicated rep
Growth Accounts	High potential	Develop relationship
Core Accounts	Steady business	Maintain service
Small Accounts	Low value	Self-service

SAL.2.2 Credit Management

Credit Evaluation Process:

1. Request credit application
2. Check trade references
3. Review financial statements
4. Set credit limit
5. Monitor usage

SAL.3 Pricing and Discounts

SAL.3.1 Pricing Strategies

Strategy	Description	When to Use
List Price	Standard price	Standard orders
Volume Discount	Quantity breaks	Large orders
Customer Specific	Negotiated price	Key accounts
Promotional	Temporary reduction	Promotions

SAL.3.2 Discount Authorization

Discount %	Approver
Up to 5%	Sales Rep
5% - 10%	Sales Manager
10% - 15%	Sales Director
Above 15%	Management

SAL.4 Sales Analytics

SAL.4.1 Key Metrics

Metric	Formula	Target
Revenue	Total sales value	Growth 10%
Win Rate	Won / Total Opportunities	>30%
Average Deal Size	Revenue / Orders	Growth 5%
Sales Cycle	Average days to close	Reduce 10%

SAL.4.2 Sales Forecasting

Methods for sales forecasting:

- Pipeline weighted by probability
- Historical trend analysis
- Market-based forecasting
- Bottom-up rep forecasts

Extended ETA E-Invoicing Guide

ETA.1 E-Invoice Format

ETA.1.1 JSON Structure

The ETA requires invoices in specific JSON format:

```
{ "issuer": { "name": "Company Name", "taxId": "123456789", "address": { "street": "123 Main St", "city": "Cairo", "country": "EG" } }, "receiver": { "name": "Customer Name", "taxId": "987654321", "address": { "street": "456 Street", "city": "Alexandria", "country": "EG" } }, "document": { "uuid": "ABC123456789", "number": "INV-2025-001", "date": "2025-01-15", "total": 1140.00, "totalTax": 140.00 }, "items": [ { "description": "Product A", "quantity": 10, "unitPrice": 100.00, "total": 1000.00, "tax": 140.00 } ] }
```

ETA.1.2 Mandatory Fields

Field	Description	Format
UUID	Unique identifier	String, 20 chars
Invoice Number	Internal number	String, max 50
Invoice Date	Issue date	YYYY-MM-DD
Seller Tax ID	Company TRN	9 digits
Buyer Tax ID	Customer TRN	9 digits
Total Amount	With tax	Decimal
Tax Amount	VAT amount	Decimal

ETA.2 Digital Signature

ETA.2.1 Certificate Requirements

- Must be from ETA-approved provider
- RSA 2048-bit or higher
- SHA-256 hashing
- Valid for 1-3 years

ETA.2.2 Signing Process

Digital Signature Steps:

1. Generate invoice JSON
2. Create hash of JSON
3. Sign hash with private key
4. Attach signature to invoice
5. Verify with public key

ETA.3 QR Code

ETA.3.1 QR Code Contents

The QR code contains:

- Seller tax ID
- Buyer tax ID
- Invoice date
- Total amount
- Tax amount
- Signature hash

ETA.3.2 QR Code Verification

Customers can verify invoices using:

- ETA mobile application
- ETA website
- Third-party verification tools

ETA.4 Submission Process

ETA.4.1 API Endpoints

Endpoint	Method	Purpose
/api/v1/invoices	POST	Submit invoice
/api/v1/invoices/{id}	GET	Get status
/api/v1/invoices/{id}	DELETE	Cancel invoice

ETA.4.2 Response Codes

Code	Description	Action
200	Success	Continue
400	Invalid data	Correct and retry
401	Unauthorized	Check credentials
429	Rate limited	Wait and retry
500	Server error	Contact support

ETA.5 Error Handling

ETA.5.1 Common Errors

Error	Cause	Solution
Invalid Tax ID	Wrong format	Verify 9 digits
Invalid Date	Future date	Use current date
Duplicate UUID	Already exists	Regenerate UUID
Invalid Signature	Certificate issue	Check certificate

ETA.5.2 Retry Strategy

- Immediate retry for network errors
- Exponential backoff for rate limits
- Manual review for validation errors
- Contact support for persistent issues

ETA.6 Compliance Monitoring

ETA.6.1 Submission Monitoring

Track submission status:

- Pending - Not yet submitted
- Submitted - Sent to ETA
- Valid - Accepted by ETA
- Invalid - Rejected by ETA
- Cancelled - Cancelled after submission

ETA.6.2 Reporting

Generate compliance reports:

- Submission status report
- Rejected invoices report
- Pending submissions report
- Monthly submission summary

Extended VAT Guide

VAT.1 VAT Overview

VAT.1.1 VAT Scope

VAT applies to:

- Sale of goods in Egypt
- Provision of services in Egypt
- Import of goods
- Receipt of services from abroad

VAT.1.2 VAT Rates

Type	Rate	Applies To
Standard	14%	Most goods/services
Zero-rated	0%	Exports, essential goods
Exempt	N/A	Financial, education, health

VAT.2 VAT Registration

VAT.2.1 Registration Threshold

Mandatory registration if:

- Annual turnover exceeds EGP 500,000
- Importer of goods
- Provider of taxable services

VAT.2.2 Registration Process

VAT Registration Steps:

1. Complete registration form
2. Submit commercial registration
3. Submit tax ID
4. Provide bank details
5. Submit to ETA
6. Receive VAT certificate

VAT.3 VAT Transactions

VAT.3.1 Output VAT (Sales)

VAT charged on sales:

Output VAT Calculation

Subtotal: EGP 1,000

VAT (14%): EGP 140

Total Invoice: EGP 1,140

VAT.3.2 Input VAT (Purchases)

VAT paid on purchases:

- Can be reclaimed against output VAT
- Must have valid tax invoice
- Must be for business purposes

VAT.3.3 Reverse Charge

For imported services:

- Buyer accounts for VAT
- Both input and output VAT recorded
- Net effect is zero

VAT.4 VAT Return

VAT.4.1 Return Contents

Section	Description
Total Sales	Taxable and exempt sales
Output VAT	VAT on sales
Total Purchases	Taxable purchases
Input VAT	VAT on purchases
Net VAT	Output - Input

VAT.4.2 Filing Requirements

- Monthly filing mandatory
- Due within 30 days of month end
- File electronically via ETA portal
- Pay any VAT due

VAT.5 Special Scenarios

VAT.5.1 Export VAT

Exports are zero-rated (0% VAT):

- Must have export documentation
- Customs declaration required
- Input VAT can still be reclaimed

VAT.5.2 Bad Debt Relief

VAT on uncollectible debts:

- Can claim relief after 1 year
- Must document collection efforts
- Adjust in VAT return

VAT.5.3 Credit Notes

Adjusting VAT for returns:

- Issue credit note with VAT adjustment
- Include in next VAT return
- Link to original invoice

VAT.6 VAT Audit

VAT.6.1 Audit Preparation

- Maintain complete records
- Reconcile VAT to GL
- Document special transactions
- Prepare supporting documents

VAT.6.2 Common Audit Issues

Issue	Prevention
-------	------------

Missing invoices	Proper filing system
Incorrect rates	Regular rate verification
Late filing	Calendar reminders
Math errors	System validation

Extended Quality Management Guide

QM.1 Quality Standards

QM.1.1 ISO Standards

Standard	Description	Focus Area
ISO 9001	Quality Management	General quality
ISO 14001	Environmental Management	Environment
ISO 45001	Occupational Health & Safety	Workplace safety
ISO 22000	Food Safety	Food industry

QM.1.2 Quality Policy

Elements of a quality policy:

- Commitment to quality
- Customer focus
- Continuous improvement
- Compliance with requirements
- Management commitment

QM.2 Inspection Planning

QM.2.1 Inspection Types

Type	When	What to Check
------	------	---------------

Incoming	Material receipt	Quantity, quality, documents
In-Process	During production	Process parameters
Final	Before shipment	Finished product
Outgoing	Before delivery	Packaging, labeling

QM.2.2 Sampling Plans

AQL (Acceptable Quality Level) sampling:

AQL	Defect Rate	Use Case
0.1	0.1%	Critical defects
1.0	1%	Major defects
2.5	2.5%	Minor defects
4.0	4%	Cosmetic defects

QM.3 Quality Control Tools

QM.3.1 Statistical Process Control

Control chart types:

- X-bar chart - Process mean
- R chart - Process range
- p chart - Proportion defective
- c chart - Count of defects

QM.3.2 Seven Basic Tools

1. Cause and Effect Diagram (Fishbone)

2. Check Sheet
3. Control Chart
4. Histogram
5. Pareto Chart
6. Scatter Diagram
7. Stratification

QM.4 Non-Conformance Management

QM.4.1 NCR Types

Type	Description	Example
Critical	Safety/Regulatory issue	Failed safety test
Major	Significant defect	Function failure
Minor	Cosmetic issue	Scratch on surface

QM.4.2 Disposition Options

- Use As Is - Accept with deviation
- Rework - Correct the defect
- Repair - Fix for use
- Scrap - Dispose of item
- Return to Vendor - Reject and return

QM.5 CAPA Process

QM.5.1 Problem Solving Steps

8D Problem Solving:**D1.** Form team**D2.** Describe problem**D3.** Implement containment**D4.** Identify root cause**D5.** Develop corrective action**D6.** Implement corrective action**D7.** Prevent recurrence**D8.** Recognize team**QM.5.2 Root Cause Analysis**

Methods for identifying root causes:

- 5 Whys - Ask why 5 times
- Fishbone Diagram - Categorize causes
- FMEA - Failure Mode Effects Analysis
- Fault Tree Analysis

QM.6 Quality Metrics**QM.6.1 Key Quality Indicators**

Metric	Formula	Target
First Pass Yield	$(\text{Passed} / \text{Total}) \times 100$	>95%
Defect Rate	$(\text{Defects} / \text{Total}) \times 100$	<2%

Customer Complaints	Count per month	Decreasing
Scrap Rate	$(\text{Scrap} / \text{Input}) \times 100$	<3%

Extended Maintenance Management Guide

MT.1 Maintenance Types

MT.1.1 Maintenance Strategies

Type	Description	When to Use
Preventive	Scheduled maintenance	Critical equipment
Predictive	Condition-based	Monitored equipment
Corrective	Repair after failure	Non-critical items
Shutdown	Planned outage	Annual maintenance

MT.1.2 Maintenance Categories

- Mechanical - Motors, pumps, conveyors
- Electrical - Panels, wiring, controls
- Instrument - Sensors, meters, gauges
- Civil - Buildings, structures
- Utility - HVAC, compressed air

MT.2 Preventive Maintenance

MT.2.1 PM Schedule Types

Type	Trigger	Example
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Time-based	Calendar	Monthly inspection
Usage-based	Operating hours	Every 500 hours
Event-based	Condition	Oil analysis

MT.2.2 PM Task List

Typical preventive maintenance tasks:

- Visual inspection
- Lubrication
- Filter replacement
- Belt tension check
- Alignment check
- Vibration analysis
- Temperature check
- Electrical testing

MT.3 Work Order Management

MT.3.1 Work Order Types

Type	Description	Priority
Emergency	Breakdown	Immediate
Urgent	High risk	Within 24 hours
Routine	Normal work	As scheduled
Project	Improvement	Planned

MT.3.2 Work Order Status

- Created - Work order entered
- Planned - Resources assigned
- Scheduled - Date assigned
- In Progress - Work started
- Waiting Parts - Awaiting materials
- Complete - Work finished
- Closed - Approved and archived

MT.4 Spare Parts Management

MT.4.1 Spare Parts Classification

Class	Description	Stock Level
Critical	Shutdown risk	Always in stock
Essential	Important	Maintain stock
Desirable	Convenient	Order as needed
Non-critical	Long lead time OK	No stock

MT.4.2 Spare Parts Planning

Factors for spare parts inventory:

- Equipment criticality
- Failure history
- Lead time
- Cost
- Storage space

MT.5 Equipment History

MT.5.1 Equipment Master Data

- Equipment code and description
- Manufacturer and model
- Serial number
- Installation date
- Location
- Cost and depreciation
- Warranty information

MT.5.2 Maintenance History

Track for each equipment:

- All work orders
- Parts used
- Labor hours
- Costs incurred
- Failures and breakdowns
- Modifications

MT.6 Maintenance Metrics

MT.6.1 Key Performance Indicators

Metric	Formula	Target
MTBF	Operating Hours / Failures	Maximize
MTTR	Repair Time / Failures	Minimize
Availability	Uptime / Total Time	>95%

Planned Maintenance %	Planned / Total	>80%
Maintenance Cost	Cost / Asset Value	<5%

MT.6.2 OEE (Overall Equipment Effectiveness)

OEE = Availability × Performance × Quality

World-class OEE: 85%

Extended Project Management Guide

PM.1 Project Types

PM.1.1 Project Categories

Type	Description	Examples
Internal	Company projects	IT upgrade, renovation
Customer	External projects	Construction, consulting
Investment	Capital projects	New facility, equipment
Research	R&D projects	Product development

PM.1.2 Project Phases

1. Initiation - Project charter, stakeholders
2. Planning - Scope, schedule, budget
3. Execution - Do the work
4. Monitoring - Track progress
5. Closing - Handover, lessons learned

PM.2 Work Breakdown Structure

PM.2.1 WBS Levels

- Level 0: Project
- Level 1: Major deliverables
- Level 2: Work packages
- Level 3: Activities

- Level 4: Tasks

PM.2.2 WBS Example

ERP Implementation WBS

- 1.0 ERP Implementation
 - 1.1 Project Management
 - 1.2 Discovery
 - 1.3 Design
 - 1.4 Build
 - 1.5 Testing
 - 1.6 Training
 - 1.7 Deployment
 - 1.8 Support

PM.3 Project Planning

PM.3.1 Activity Sequencing

Dependency	Description	Example
Finish-Start	B starts after A finishes	Design then Build
Start-Start	B starts when A starts	Parallel tasks
Finish-Finish	B finishes when A finishes	Concurrent end

PM.3.2 Duration Estimation

Methods for estimating activity duration:

- Analogous - Based on similar projects
- Parametric - Statistical relationship

- Three-point - Optimistic, pessimistic, most likely
- Expert judgment - Based on experience

PM.4 Resource Management

PM.4.1 Resource Types

Type	Examples	Unit
Labor	Engineers, technicians	Hours
Equipment	Machinery, tools	Hours
Material	Raw materials	Quantity
Cost	Expenses	Currency

PM.4.2 Resource Leveling

Techniques to balance resource demand:

- Adjust activity timing
- Split activities
- Assign alternative resources
- Reduce scope

PM.5 Earned Value Management

PM.5.1 EVM Terms

Term	Formula	Description
PV	Planned Value	Budgeted cost of work scheduled

EV	Earned Value	Budgeted cost of work performed
AC	Actual Cost	Actual cost of work performed
SV	EV - PV	Schedule Variance
CV	EV - AC	Cost Variance
SPI	EV / PV	Schedule Performance Index
CPI	EV / AC	Cost Performance Index

PM.5.2 EVM Example

EVM Calculation

Project Budget: EGP 100,000

Planned % Complete: 50%

Actual % Complete: 40%

Actual Cost: EGP 45,000

$PV = 100,000 \times 50\% = \text{EGP } 50,000$

$EV = 100,000 \times 40\% = \text{EGP } 40,000$

$AC = \text{EGP } 45,000$

$SV = 40,000 - 50,000 = -10,000$ (Behind schedule)

$CV = 40,000 - 45,000 = -5,000$ (Over budget)

$SPI = 40,000 / 50,000 = 0.8$

$CPI = 40,000 / 45,000 = 0.89$

PM.6 Project Reporting

PM.6.1 Status Report Contents

- Executive summary
- Key accomplishments

- Upcoming milestones
- Issues and risks
- Budget status
- Schedule status
- Action items

PM.6.2 Project Dashboard

Key metrics to display:

- % Complete
- Budget consumed
- Schedule variance
- Open issues
- Risk status

Extended Customer Relationship Management Guide

CRM.1 Customer Lifecycle

CRM.1.1 Customer Journey

- 1. Awareness - Customer learns about company
- 2. Interest - Customer shows interest
- 3. Evaluation - Customer compares options
- 4. Purchase - Customer buys
- 5. Retention - Customer stays
- 6. Advocacy - Customer recommends

CRM.1.2 Touch Points

Channel	Purpose
Website	Information, self-service
Email	Communication, marketing
Phone	Support, sales
Social Media	Engagement, brand
In-Person	Meetings, events

CRM.2 Lead Management

CRM.2.1 Lead Sources

Source	Description
Website	Inquiry forms, chat
Referral	Existing customer
Trade Show	Event leads
Cold Call	Outbound prospecting
Social Media	LinkedIn, etc.

CRM.2.2 Lead Scoring

Factors for lead scoring:

Factor	Points
Job title (Decision maker)	+20
Company size	+10 to +30
Budget confirmed	+25
Timeline (3 months)	+20
Email opened	+5
Website visited	+10

CRM.3 Opportunity Management

CRM.3.1 Sales Stages

Stage	Probability	Activities
-------	-------------	------------

Prospecting	10%	Initial contact
Qualification	25%	Needs analysis
Proposal	50%	Quote submitted
Negotiation	75%	Terms discussion
Closing	90%	Final agreement
Won	100%	Order received

CRM.3.2 Opportunity Forecasting

Weighted pipeline value:

Forecast = Sum of (Opportunity Value × Probability)

CRM.4 Activity Management

CRM.4.1 Activity Types

Activity	Purpose
Call	Direct communication
Meeting	In-person discussion
Email	Written communication
Task	Follow-up action
Note	Record information

CRM.4.2 Activity Tracking

Benefits of activity tracking:

- Complete customer history

- Performance measurement
- Handover between team members
- Compliance documentation

CRM.5 Service Management

CRM.5.1 Ticket Management

Priority	Response Time	Resolution Time
Critical	1 hour	4 hours
High	4 hours	1 day
Medium	1 day	3 days
Low	3 days	1 week

CRM.5.2 Service Metrics

Metric	Formula	Target
First Response Time	Time to first response	Per SLA
Resolution Time	Time to resolve	Per SLA
First Contact Resolution	Resolved on first contact	>70%
Customer Satisfaction	Satisfaction score	>4/5

CRM.6 Marketing Integration

CRM.6.1 Campaign Management

- Email campaigns

- Event campaigns
- Social media campaigns
- Webinar campaigns

CRM.6.2 Campaign Metrics

Metric	Description
Open Rate	Emails opened / Emails sent
Click Rate	Clicks / Emails sent
Conversion Rate	Conversions / Clicks
ROI	$(\text{Revenue} - \text{Cost}) / \text{Cost}$

Extended System Integration Guide

INT.1 Integration Architecture

INT.1.1 Integration Patterns

Pattern	Description	Use Case
Point-to-Point	Direct connection	Simple integrations
Hub and Spoke	Central integration	Multiple systems
Bus	Message bus	Enterprise integration
API Gateway	Unified API access	External access

INT.1.2 Integration Methods

Method	Description	Best For
API	REST/SOAP calls	Real-time data
File Transfer	Batch file exchange	Large data volumes
Database	Direct DB access	Reporting
Message Queue	Asynchronous messaging	Event-driven

INT.2 ETA Integration

INT.2.1 ETA API Endpoints

Endpoint	Method	Description
----------	--------	-------------

/api/v1/invoices	POST	Submit invoice
/api/v1/invoices/{id}	GET	Get invoice status
/api/v1/invoices/{id}	DELETE	Cancel invoice
/api/v1/receipts	POST	Submit receipt

INT.2.2 ETA Authentication

```
POST /oauth/token Content-Type: application/x-www-form-urlencoded
grant_type=client_credentials &client_id=YOUR_CLIENT_ID
&client_secret=YOUR_CLIENT_SECRET
```

INT.3 Bank Integration

INT.3.1 Bank Statement Import

Supported formats:

- MT940 - SWIFT statement
- CSV - Comma separated
- Excel - Microsoft format
- XML - Structured format
- JSON - API format

INT.3.2 Payment File Generation

Supported payment formats:

- SEPA - European payments
- ACH - US payments
- Local formats - Country specific
- Custom formats - Configurable

INT.4 API Framework

INT.4.1 API Authentication

OAuth 2.0 flow:

1. Request access token
2. Include token in API calls
3. Token expires after set time
4. Refresh token when needed

INT.4.2 Rate Limiting

Plan	Requests/Hour	Concurrent
Standard	1,000	10
Premium	10,000	50
Enterprise	Unlimited	100

INT.4.3 API Best Practices

- Use pagination for large datasets
- Implement retry logic
- Handle errors gracefully
- Cache responses when appropriate
- Monitor API usage

INT.5 Webhooks

INT.5.1 Webhook Events

Event	Description
invoice.created	New invoice created
invoice.paid	Invoice paid
order.shipped	Order shipped
payment.received	Payment received
customer.created	New customer added

INT.5.2 Webhook Configuration

Setting up Webhooks:

1. Navigate to Integration → Webhooks
2. Click Create New
3. Enter endpoint URL
4. Select events to subscribe
5. Set retry policy
6. Save configuration

INT.6 Third-Party Integrations

INT.6.1 Common Integrations

System	Integration Type	Data
BI Tools	API/Database	Reporting data
E-commerce	API	Orders, customers

Shipping	API	Tracking, rates
CRM	API	Customer data
HR Systems	File/API	Employee data

INT.6.2 Integration Testing

- Test in development environment
- Use test data
- Verify data mapping
- Test error scenarios
- Performance testing

Extended Mobile Application Guide

MOB.1 Mobile Overview

MOB.1.1 Supported Platforms

Platform	Version	Features
iOS	14+	Full functionality
Android	10+	Full functionality
PWA	Modern browsers	Core functionality

MOB.1.2 Mobile Features

- Sales - Create orders, check prices
- Service - Manage tickets, update status
- Inventory - Check stock, receive goods
- Approvals - Approve requests on the go
- Dashboard - View KPIs
- Notifications - Real-time alerts

MOB.2 Mobile Sales

MOB.2.1 Creating Orders

Mobile Order Entry:

1. Open Sales app

2. Select customer
3. Scan item barcode or search
4. Enter quantity
5. Apply discounts
6. Review and submit

MOB.2.2 Offline Capability

Work without internet:

- Cache customer data
- Store product catalog
- Queue orders for sync
- Auto-sync when connected

MOB.3 Mobile Service

MOB.3.1 Ticket Management

- View assigned tickets
- Update ticket status
- Add notes and comments
- Upload photos
- Record time spent

MOB.3.2 Field Service

Features for field technicians:

- GPS navigation to customer
- Check-in/check-out

- Parts request
- Signature capture
- Invoice generation

MOB.4 Mobile Inventory

MOB.4.1 Stock Inquiry

- Scan barcode to check stock
- View item details
- Check multiple warehouses
- View transaction history

MOB.4.2 Goods Receipt

Mobile GRN Process:

1. Select PO
2. Scan received items
3. Enter quantities
4. Take photos if needed
5. Submit receipt

MOB.5 Mobile Approvals

MOB.5.1 Approval Types

- Purchase requisitions
- Purchase orders
- Invoices

- Expense claims
- Leave requests

MOB.5.2 Approval Process

Mobile Approval:

1. Receive notification
2. Review request details
3. View attachments
4. Approve or reject
5. Add comments if needed

MOB.6 Mobile Security

MOB.6.1 Authentication

- Username/password
- Biometric (fingerprint/face)
- PIN code
- Device registration

MOB.6.2 Data Protection

- Data encryption at rest
- Secure communication (TLS)
- Remote wipe capability
- Session timeout

MOB.6.3 App Security

- App store distribution only
- Code obfuscation
- Anti-tampering
- Jailbreak detection

Extended AI and Machine Learning Guide

AI.1 AI Overview

AI.1.1 AI Capabilities

Capability	Description	Application
Predictive Analytics	Forecast future outcomes	Demand, cash flow
NLP	Understand text	Chatbot, sentiment
Computer Vision	Analyze images	Quality inspection
Anomaly Detection	Identify outliers	Fraud detection
Recommendation	Suggest actions	Pricing, products

AI.1.2 AI Architecture

- Data collection and preparation
- Model training
- Model deployment
- Inference engine
- Feedback loop

AI.2 Predictive Analytics

AI.2.1 Demand Forecasting

Factors considered:

- Historical sales data
- Seasonal patterns
- Market trends
- Promotional calendar
- Economic indicators

AI.2.2 Cash Flow Prediction

Inputs for cash flow forecasting:

- AR aging
- AP aging
- Recurring payments
- Historical patterns
- Seasonal variations

AI.2.3 Customer Churn Prediction

Indicators of churn risk:

- Declining purchase frequency
- Decreasing order values
- Late payments
- Support ticket volume
- Engagement metrics

AI.3 Natural Language Processing

AI.3.1 Chatbot Integration

Chatbot capabilities:

- Answer FAQs
- Guide users through processes

- Provide contextual help
- Escalate to human support

AI.3.2 Sentiment Analysis

Analyze customer communications:

- Email sentiment
- Support ticket tone
- Social media mentions
- Survey responses

AI.4 Intelligent Automation

AI.4.1 Invoice Processing

Automated invoice handling:

- Extract data from PDF
- Match to PO
- Route for approval
- Post to AP

AI.4.2 Email Classification

Automatically categorize emails:

- Customer inquiry
- Vendor communication
- Internal request
- Spam

AI.5 Anomaly Detection

AI.5.1 Fraud Detection

Suspicious patterns identified:

- Unusual transaction amounts
- Off-hours activity
- Unusual vendor/customer
- Duplicate payments
- Round amounts

AI.5.2 Quality Anomalies

Detect quality issues:

- Out-of-spec measurements
- Process deviations
- Unusual defect patterns
- Supplier quality issues

AI.6 Recommendation Engine

AI.6.1 Product Recommendations

Suggest products based on:

- Purchase history
- Browsing behavior
- Similar customers
- Seasonal trends

AI.6.2 Pricing Recommendations

Optimize pricing based on:

- Demand patterns
- Competitor pricing
- Inventory levels
- Customer segment

AI.7 AI Governance

AI.7.1 Model Monitoring

- Track prediction accuracy
- Monitor data drift
- Detect model degradation
- Retrain when needed

AI.7.2 Ethical AI

- Fairness in predictions
- Transparency in decisions
- Privacy protection
- Human oversight

Extended Data Management Guide

DM.1 Data Governance

DM.1.1 Data Ownership

Data Type	Owner	Responsibility
Master Data	Data Steward	Accuracy, completeness
Transactional	System	Integrity, audit trail
Reference	Department	Maintenance, updates
Reporting	BI Team	Accuracy, timeliness

DM.1.2 Data Quality Dimensions

Dimension	Description	Metric
Accuracy	Correctness	Error rate
Completeness	Missing values	% complete
Consistency	Across systems	Match rate
Timeliness	Up-to-date	Age of data
Validity	Format compliance	% valid

DM.2 Data Import

DM.2.1 Import Templates

Available templates for:

- Customers
- Vendors
- Items
- Chart of Accounts
- Employees
- Opening Balances
- Price Lists

DM.2.2 Import Process

Data Import Steps:

1. Download template
2. Fill with data
3. Validate format
4. Upload file
5. Review validation results
6. Fix errors
7. Import data

DM.2.3 Import Validation Rules

- Required fields must be populated
- Data types must match
- Codes must be unique
- References must exist
- Dates must be valid

DM.3 Data Export

DM.3.1 Export Formats

Format	Extension	Best For
Excel	.xlsx	Analysis, reporting
CSV	.csv	Data exchange
PDF	.pdf	Distribution
XML	.xml	System integration
JSON	.json	API integration

DM.3.2 Scheduled Exports

Automate regular exports:

- Daily sales report
- Weekly inventory status
- Monthly financial reports
- Custom schedules

DM.4 Data Migration

DM.4.1 Migration Strategy

Strategy	Description	Best For
Big Bang	All at once	Small companies
Phased	Module by module	Medium complexity

Parallel	Run both systems	High risk
----------	------------------	-----------

DM.4.2 Migration Process

Data Migration Steps:

1. Data assessment
2. Data cleansing
3. Mapping design
4. Extract data
5. Transform data
6. Load data
7. Validate results
8. Reconcile balances

DM.5 Data Archiving

DM.5.1 Archiving Policy

Data Type	Retention	Archive After
Transactions	7 years	2 years
Audit Logs	10 years	1 year
Reports	5 years	1 year
Attachments	7 years	3 years

DM.5.2 Archive Process

- Identify data to archive
- Export to archive storage
- Verify archive integrity
- Remove from production
- Document archive location

DM.6 Data Security

DM.6.1 Access Control

- Role-based access
- Data classification
- Field-level security
- Row-level security

DM.6.2 Data Masking

Protect sensitive data:

- Credit card numbers
- National IDs
- Bank account numbers
- Personal information

Extended Audit and Compliance Guide

AUD.1 Internal Controls

AUD.1.1 Control Framework

Control Type	Description	Example
Preventive	Stop errors before occur	Approval workflows
Detective	Find errors after occur	Reconciliation
Corrective	Fix errors found	Reversal entries

AUD.1.2 Segregation of Duties

Incompatible duties to separate:

- Authorization vs. Recording
- Custody vs. Recording
- Authorization vs. Custody

AUD.1.3 Approval Workflows

Transaction	Level 1	Level 2	Level 3
Journal Entry	Accountant	Manager	Controller
Payment	AP Clerk	Manager	CFO
PO	Buyer	Manager	Director

AUD.2 Audit Trail

AUD.2.1 Audit Log Contents

Field	Description
Timestamp	Date and time of action
User	Who performed action
Action	Create, Update, Delete, View
Object	What was affected
Before	Values before change
After	Values after change
IP Address	Source of action

AUD.2.2 Audit Reports

- User Activity Report
- Transaction History
- Login/Logout Report
- Failed Access Attempts
- Data Modification Report

AUD.3 Regulatory Compliance

AUD.3.1 Tax Compliance

- VAT return filing
- E-invoice submission

- Tax payment
- Tax record retention

AUD.3.2 Labor Compliance

- Working hours
- Overtime payment
- Leave entitlements
- Social insurance

AUD.3.3 Financial Compliance

- Accounting standards
- Financial reporting
- Audit requirements
- Record retention

AUD.4 Compliance Monitoring

AUD.4.1 Compliance Dashboard

Track compliance status:

- Upcoming deadlines
- Overdue items
- Compliance score
- Risk indicators

AUD.4.2 Compliance Calendar

Requirement	Frequency	Deadline
VAT Return	Monthly	Month + 30 days

GOSI Report	Monthly	Month + 15 days
Income Tax	Monthly	Month + 30 days
Annual Audit	Annual	Year + 6 months

AUD.5 Risk Management

AUD.5.1 Risk Assessment

Risk	Likelihood	Impact	Mitigation
Data loss	Low	High	Backup
Unauthorized access	Medium	High	Security
System failure	Low	High	Redundancy

AUD.5.2 Risk Mitigation

- Regular backups
- Access controls
- Disaster recovery plan
- Business continuity

Extended Training Guide

TRN.1 Training Program

TRN.1.1 Training Types

Type	Description	Duration
Administrator	System setup, security	5 days
Power User	Advanced features	3 days
End User	Basic operations	2 days
Executive	Dashboards, reports	1 day

TRN.1.2 Training Methods

Method	Description	Best For
Instructor-led	Classroom training	Complex topics
E-learning	Self-paced online	Flexibility
Hands-on	Practice exercises	Skill building
Video	Recorded tutorials	Reference

TRN.2 Training Curriculum

TRN.2.1 Financial Module Training

- General Ledger basics

- Journal entry processing
- Period-end closing
- Financial reporting
- Budget management

TRN.2.2 Inventory Module Training

- Item master setup
- Goods receipt
- Goods issue
- Stock transfers
- Physical counting

TRN.2.3 HR/Payroll Training

- Employee master
- Attendance management
- Leave processing
- Payroll calculation
- Statutory reporting

TRN.3 Certification Program

TRN.3.1 Certification Levels

Level	Requirements	Valid For
End User	Training + Exam	2 years
Professional	Advanced + Project	2 years
Administrator	Admin + Practical	2 years

Consultant

All + Experience

2 years

TRN.3.2 Exam Format

- Multiple choice questions
- Practical exercises
- Pass mark: 70%
- Time limit: 2 hours

TRN.4 Training Materials

TRN.4.1 User Guides

- Getting Started Guide
- Module User Guides
- Quick Reference Cards
- Process Flowcharts

TRN.4.2 Video Library

- Feature demonstrations
- Process walkthroughs
- Tips and tricks
- Troubleshooting

TRN.5 User Adoption

TRN.5.1 Adoption Strategy

- Executive sponsorship
- Change champions
- Communication plan

- Incentive programs
- Feedback mechanisms

TRN.5.2 Adoption Metrics

Metric	Target
User Login Rate	>90%
Feature Usage	>80%
Training Completion	100%
Support Tickets	Decreasing

Extended Implementation Guide

IMP.1 Implementation Phases

IMP.1.1 Phase Overview

Phase	Duration	Key Activities
Discovery	2-4 weeks	Requirements, current state
Design	3-6 weeks	Solution design, process mapping
Build	6-12 weeks	Configuration, development
Test	4-6 weeks	Testing, UAT
Deploy	2-4 weeks	Data migration, go-live
Optimize	Ongoing	Support, enhancements

IMP.2 Discovery Phase

IMP.2.1 Current State Analysis

- Document existing processes
- Map data flows
- Identify pain points
- Catalog integrations
- Assess data quality

IMP.2.2 Requirements Gathering

Requirement Type	Description
Functional	What system should do
Technical	Performance, security
Integration	External system connections
Reporting	Information needs
Compliance	Regulatory requirements

IMP.3 Design Phase

IMP.3.1 Solution Design

- Process design
- Data model design
- Integration architecture
- Security model
- Reporting framework

IMP.3.2 Fit-Gap Analysis

Type	Action	Example
Fit	Use standard	Standard GL
Gap	Configuration	Custom fields
Gap	Customization	Custom workflow
Gap	Workaround	Process change

IMP.4 Build Phase

IMP.4.1 Configuration Activities

- Company setup
- Chart of accounts
- Master data setup
- Workflow configuration
- Report configuration
- Security setup

IMP.4.2 Development Activities

- Custom reports
- Integrations
- Custom workflows
- Data migrations

IMP.5 Testing Phase

IMP.5.1 Testing Types

Type	Description	Owner
Unit	Individual components	Developer
Integration	Component interaction	Technical
System	End-to-end	QA
UAT	User acceptance	Business
Performance	Load testing	Technical

IMP.5.2 UAT Process

UAT Steps:

1. Prepare test scenarios
2. Execute test cases
3. Document results
4. Report defects
5. Retest fixes
6. Obtain sign-off

IMP.6 Deployment Phase

IMP.6.1 Data Migration

- Extract from legacy
- Cleanse and transform
- Load to new system
- Validate results
- Reconcile balances

IMP.6.2 Go-Live Strategy

Strategy	Description	Best For
Big Bang	All at once	Small organizations
Phased	Module by module	Medium complexity
Parallel	Run both systems	High risk

IMP.6.3 Go-Live Checklist

- System configured
- Data migrated
- Users trained
- Support ready
- Backups verified
- Rollback plan ready

IMP.7 Change Management

IMP.7.1 Change Management Plan

- Stakeholder analysis
- Communication plan
- Training plan
- Resistance management
- Reinforcement

IMP.7.2 Communication Strategy

Audience	Message	Channel
Executives	Business benefits	Meetings
Managers	Process changes	Workshops
Users	How to use	Training

Extended Advanced Configuration Guide

CFG.1 Custom Fields

CFG.1.1 Field Types

Type	Description	Example
Text	Single line text	Reference number
Number	Numeric value	Custom quantity
Date	Calendar date	Custom date
Dropdown	Predefined options	Priority level
Checkbox	Yes/No	Flag
Multi-select	Multiple options	Categories
Lookup	Reference to other record	Related item

CFG.1.2 Field Properties

- Required - Must be filled
- Unique - No duplicates
- Read-only - Cannot edit
- Hidden - Not displayed
- Default value - Auto-populated

CFG.2 Custom Workflows

CFG.2.1 Workflow Components

- States - Status values
- Transitions - State changes
- Conditions - When transition allowed
- Actions - Automated tasks
- Notifications - Alerts to users

CFG.2.2 Workflow Example

Purchase Order Workflow

States: Draft → Pending Approval → Approved → Sent → Received → Closed

Transitions:

- Draft → Pending Approval (Submit)
- Pending Approval → Approved (Approve)
- Pending Approval → Draft (Reject)
- Approved → Sent (Send to Vendor)
- Sent → Received (Goods Received)
- Received → Closed (Invoice Paid)

CFG.3 Custom Reports

CFG.3.1 Report Builder

Creating Custom Reports:

1. Select data source
2. Choose fields
3. Add filters

4. Add calculations
5. Design layout
6. Set parameters
7. Save and schedule

CFG.3.2 Report Formats

- Tabular - Rows and columns
- Matrix - Cross-tabulation
- Chart - Visual representation
- Dashboard - Multiple components

CFG.4 Email Templates

CFG.4.1 Template Variables

Variable	Description
{{customer_name}}	Customer name
{{invoice_number}}	Invoice number
{{amount}}	Invoice amount
{{due_date}}	Payment due date

CFG.4.2 Email Types

- Invoice notification
- Payment reminder
- Order confirmation
- Shipping notification

- Approval request

CFG.5 Numbering Sequences

CFG.5.1 Sequence Configuration

Setting	Description	Example
Prefix	Starting text	INV-
Start Number	First number	1
Padding	Leading zeros	4 digits
Suffix	Ending text	-2025

CFG.5.2 Sequence Examples

- Invoice: INV-2025-0001
- PO: PO-2025-00001
- GRN: GRN-2025-0001
- Journal: JV-2025-000001

CFG.6 Document Templates

CFG.6.1 Template Types

- Invoice template
- PO template
- Delivery note template
- Quotation template
- Payslip template

CFG.6.2 Template Design

- Company logo
- Header information
- Line items
- Totals
- Terms and conditions
- Footer

Extended Glossary

GLS.1 ERP Terminology

GLS.1.1 Core ERP Terms

Term	Definition
ERP	Enterprise Resource Planning - Integrated business management software
Module	Functional component of ERP (e.g., Finance, HR)
Transaction	Business event recorded in system
Master Data	Reference data (customers, items, vendors)
Posting	Recording transaction to ledger
Period	Accounting time period (month, quarter, year)
Fiscal Year	Annual accounting period
Chart of Accounts	List of all GL accounts

GLS.1.2 Financial Terms

Term	Definition
GL	General Ledger - Central accounting record
AP	Accounts Payable - Money owed to vendors
AR	Accounts Receivable - Money owed by customers
Trial Balance	List of all account balances

Journal Entry	Record of financial transaction
Reconciliation	Matching records from different sources
Accrual	Revenue/expense recognized when earned/incurred
Depreciation	Allocation of asset cost over useful life

GLS.1.3 Inventory Terms

Term	Definition
SKU	Stock Keeping Unit - Unique item identifier
BOM	Bill of Materials - List of components
FIFO	First In, First Out - Inventory method
LIFO	Last In, First Out - Inventory method
Reorder Point	Stock level triggering replenishment
Safety Stock	Buffer inventory
Cycle Count	Periodic inventory counting
GRN	Goods Receipt Note - Receipt document

GLS.1.4 Manufacturing Terms

Term	Definition
MRP	Material Requirements Planning
WIP	Work In Progress - Partially completed goods
Routing	Sequence of operations
Work Center	Production resource

Lead Time	Time from order to delivery
EOQ	Economic Order Quantity - Optimal order size
MO	Manufacturing Order - Production order
Scrap	Waste material

GLS.1.5 HR Terms

Term	Definition
GOSI	General Organization for Social Insurance
Basic Salary	Fixed component of pay
Gross Salary	Total before deductions
Net Salary	Take-home pay
End of Service	Termination benefits
Annual Leave	Paid vacation
Sick Leave	Medical absence
Overtime	Extra hours worked

GLS.1.6 Technical Terms

Term	Definition
API	Application Programming Interface
Webhook	HTTP callback for events
OAuth	Authentication protocol
JSON	JavaScript Object Notation - Data format

XML	Extensible Markup Language - Data format
REST	Representational State Transfer - API style
SSL/TLS	Security protocols
SSO	Single Sign-On

GLS.1.7 Egyptian Regulatory Terms

Term	Definition
ETA	Egyptian Tax Authority
Tax Registration Number	Unique taxpayer ID
E-Invoice	Electronically submitted invoice
UUID	Unique invoice identifier
QR Code	Quick Response code for verification
VAT	Value Added Tax
Commercial Registration	Business registration document
Labor Law	Employment regulations

Extended Fixed Assets Guide

FA.1 Asset Categories

FA.1.1 Asset Classification

Category	Examples	Depreciation Method
Buildings	Office, Factory, Warehouse	Straight Line
Land	Property, Land rights	No depreciation
Machinery	Production equipment	Declining Balance
Vehicles	Cars, Trucks, Forklifts	Declining Balance
Furniture	Desks, Chairs, Cabinets	Straight Line
Computers	Servers, Laptops, Printers	Declining Balance
Software	Licensed software	Straight Line

FA.1.2 Asset Numbering

Recommended format: CATEGORY-LOCATION-SEQUENCE

Asset Number Examples

MACH-CAI-001 - Machinery in Cairo

VEHI-ALX-005 - Vehicle in Alexandria

COMP-HO-010 - Computer at Head Office

FA.2 Depreciation Methods

FA.2.1 Straight Line Method

Formula: $(\text{Cost} - \text{Salvage Value}) / \text{Useful Life}$

Straight Line Example

Asset Cost: EGP 100,000
Salvage Value: EGP 10,000
Useful Life: 5 years
Annual Depreciation = $(100,000 - 10,000) / 5 = \text{EGP } 18,000$

FA.2.2 Declining Balance Method

Formula: $\text{Book Value} \times \text{Depreciation Rate}$

Declining Balance Example

Asset Cost: EGP 100,000
Depreciation Rate: 25%
Year 1: $100,000 \times 25\% = \text{EGP } 25,000$
Year 2: $75,000 \times 25\% = \text{EGP } 18,750$
Year 3: $56,250 \times 25\% = \text{EGP } 14,063$

FA.2.3 Tax Depreciation (Egypt)

Asset Type	Tax Rate	Useful Life
Buildings	5%	20 years
Industrial Machinery	15%	7 years
Computers	33%	3 years
Vehicles	25%	4 years

Furniture

10%

10 years

FA.3 Asset Transactions

FA.3.1 Asset Acquisition

Asset Acquisition Process:

1. Create asset master record
2. Record acquisition cost
3. Assign asset tag/label
4. Record location
5. Set depreciation parameters
6. Post to GL

FA.3.2 Asset Transfer

Types of transfers:

- Inter-location transfer
- Inter-department transfer
- Employee assignment
- Cost center change

FA.3.3 Asset Disposal

Asset Disposal Process:

1. Calculate accumulated depreciation

2. Determine book value
3. Record disposal value
4. Calculate gain/loss
5. Post disposal entry
6. Remove from asset register

FA.4 Asset Revaluation

FA.4.1 Revaluation Process

- Determine fair value
- Calculate revaluation surplus/deficit
- Adjust asset value
- Update depreciation
- Post revaluation entry

FA.4.2 Impairment

When asset value declines:

- Assess impairment indicators
- Calculate recoverable amount
- Recognize impairment loss
- Adjust carrying value

FA.5 Asset Tracking

FA.5.1 Physical Verification

Physical Count Process:

1. Print asset list by location
2. Physically verify each asset
3. Note condition
4. Record discrepancies
5. Investigate differences
6. Update records

FA.5.2 Barcode/RFID

Asset tracking technologies:

- Barcode labels - Cost effective
- QR codes - More information
- RFID tags - Automated tracking
- GPS - Vehicle tracking

Extended Cash Management Guide

CM.1 Bank Account Management

CM.1.1 Account Types

Type	Purpose	Features
Current Account	Daily operations	Checks, transfers
Savings Account	Reserve funds	Interest earning
Foreign Currency	FX transactions	Multi-currency
Payroll Account	Salary payments	Bulk transfers

CM.1.2 Bank Master Setup

Setting up Bank Accounts:

1. Create bank master
2. Add account details
3. Map to GL account
4. Set currency

CM.2 Payment Methods

5. Configure check series

6. Set opening balance

CM.2.1 Bank Transfer

Electronic fund transfers:

- RTGS - Real-time gross settlement

- NEFT - National electronic fund transfer
- Wire transfer - International
- ACH - Automated clearing house

CM.2.2 Check Management

Check Processing:

1. Load check stock
2. Print checks from system
3. Obtain signatures
4. Record check number
5. Mail or deliver

CM.2.3 Cash Payments

Handling cash transactions:

- Petty cash management
- Cash register operations
- Cash receipts
- Security procedures

CM.3 Bank Reconciliation

CM.3.1 Reconciliation Process

Monthly Reconciliation:

1. Import bank statement
2. Match transactions
3. Identify unmatched items
4. Investigate differences
5. Create adjustments
6. Complete reconciliation

CM.3.2 Common Differences

Difference	Cause	Resolution
Outstanding checks	Not yet cleared	Wait for clearing
Deposits in transit	Not yet credited	Confirm deposit
Bank charges	Fees not recorded	Record expense
Interest earned	Not recorded	Record income
Errors	Data entry	Correct entry

CM.4 Cash Flow Management

CM.4.1 Cash Flow Forecasting

Project future cash position:

- Projected receipts from customers
- Projected payments to vendors
- Recurring payments (payroll, rent)
- Loan payments

- Capital expenditures

CM.4.2 Cash Position Report

Item	Amount
Opening Balance	XXX
+ Expected Receipts	XXX
- Expected Payments	XXX
= Projected Balance	XXX

CM.5 Treasury Functions

CM.5.1 Investment Management

- Short-term investments
- Money market funds
- Certificates of deposit
- Treasury bills

CM.5.2 Foreign Exchange

- FX spot transactions
- Forward contracts
- Currency hedging
- Exchange rate monitoring

Extended Cost Accounting Guide

CA.1 Cost Elements

CA.1.1 Primary Cost Elements

Element	Description	Examples
Material Costs	Raw materials, components	Steel, plastic, parts
Labor Costs	Direct and indirect labor	Wages, benefits
Overhead Costs	Indirect manufacturing	Utilities, depreciation

CA.1.2 Secondary Cost Elements

- Internal activity costs
- Cost center allocations
- Service department costs

CA.2 Cost Centers

CA.2.1 Cost Center Types

Type	Description	Examples
Production	Manufacturing	Assembly, Machining
Service	Support functions	Maintenance, Quality
Administrative	Admin functions	HR, Finance, IT
Sales	Selling functions	Marketing, Sales

CA.2.2 Cost Center Hierarchy

Cost Center Structure

1000 - Production
 1100 - Assembly
 1110 - Assembly Line 1
 1120 - Assembly Line 2
 1200 - Machining
 1210 - CNC Section
 1220 - Manual Section

CA.3 Cost Allocation

CA.3.1 Allocation Methods

Cost Type	Allocation Base
Rent	Floor area
Utilities	Consumption/meter
IT Services	Number of users
HR Services	Headcount
Depreciation	Asset value

CA.3.2 Step-Down Allocation

Service Department Allocation:

1. Allocate to other service departments

2. Allocate to production departments
3. Allocate to products

CA.4 Product Costing

CA.4.1 Standard Costing

Components of standard cost:

- Standard material cost
- Standard labor cost
- Standard overhead cost

CA.4.2 Variance Analysis

Variance	Formula	Cause
Material Price	$(\text{Actual} - \text{Standard}) \times \text{Actual Qty}$	Price changes
Material Usage	$(\text{Actual} - \text{Standard}) \times \text{Standard Price}$	Efficiency
Labor Rate	$(\text{Actual} - \text{Standard}) \times \text{Actual Hours}$	Wage changes
Labor Efficiency	$(\text{Actual} - \text{Standard}) \times \text{Standard Rate}$	Productivity
Overhead	Actual - Applied	Spending/volume

CA.4.3 Actual Costing

Calculate actual product costs:

- Actual material costs from issues
- Actual labor costs from timesheets
- Actual overhead allocation

CA.5 Profitability Analysis

CA.5.1 Profit Center Reporting

Item	Amount
Revenue	XXX
- Variable Costs	XXX
= Contribution Margin	XXX
- Fixed Costs	XXX
= Operating Profit	XXX

CA.5.2 Product Profitability

Analyze by:

- Product line
- Customer
- Region
- Channel

Extended Material Requirements Planning Guide

MRP.1 MRP Overview

MRP.1.1 MRP Inputs

Input	Description	Source
Master Production Schedule	What to produce, when	Sales forecast, orders
BOM	Components needed	Engineering
Inventory Status	What's on hand	Inventory system
Lead Times	Procurement time	Vendors, production

MRP.1.2 MRP Outputs

- Planned production orders
- Planned purchase orders
- Exception messages
- Pegging information

MRP.2 MRP Calculation

MRP.2.1 Gross Requirements

Total demand for an item:

- Customer orders
- Forecast

- Dependent demand from BOM
- Safety stock requirements

MRP.2.2 Net Requirements

Formula: $\text{Net} = \text{Gross} - (\text{On Hand} + \text{On Order}) + \text{Safety Stock}$

Net Requirements Example

Gross Requirements: 1,000 units
On Hand: 200 units
On Order: 300 units
Safety Stock: 50 units
 $\text{Net Requirements} = 1,000 - (200 + 300) + 50 = 550 \text{ units}$

MRP.2.3 Planned Orders

Order planning parameters:

- Order quantity (EOQ, lot-for-lot)
- Lead time
- Order date
- Due date

MRP.3 Lot Sizing

MRP.3.1 Lot Sizing Methods

Method	Description	Best For
Lot-for-Lot	Order exact quantity	Expensive items
Fixed Order Quantity	Fixed amount	Standard packaging

EOQ	Economic order quantity	Balancing cost
Period Order Quantity	Cover multiple periods	Stable demand

MRP.4 Exception Messages

MRP.4.1 Common Exceptions

Message	Meaning	Action
Expedite	Order needed sooner	Expedite PO
Defer	Order not needed yet	Delay PO
Cancel	Order not needed	Cancel PO
Reschedule In	Pull in due date	Expedite
Reschedule Out	Push out due date	Defer

MRP.5 Capacity Planning

MRP.5.1 Rough-Cut Capacity

High-level capacity check:

- Key work centers
- Critical resources
- Bottleneck operations

MRP.5.2 Capacity Requirements Planning

Detailed capacity check:

- All work centers
- Labor capacity

- Machine capacity
- Load vs. capacity comparison

MRP.6 MRP Run

MRP.6.1 Regenerative MRP

- Complete recalculation
- All items processed
- Run weekly or as needed

MRP.6.2 Net Change MRP

- Only changed items
- Faster processing
- Run daily

Extended GOSI (Social Insurance) Guide

GOSI.1 GOSI Overview

GOSI.1.1 Insurance Types

Type	Coverage	Employer %	Employee %
Old Age, Disability, Death	Pension benefits	15%	11%
Work Injury	Workplace accidents	2%	0%
Unemployment	Job loss benefits	1%	1%

GOSI.1.2 Contribution Calculation

Maximum Insurance Salary: EGP 8,400 per month (2025)

Contributions calculated on lesser of actual salary or maximum

GOSI Calculation Example

Employee Salary: EGP 12,000
Maximum Insurance Salary: EGP 8,400
Employee Contribution: $8,400 \times 12\% = \text{EGP } 1,008$
Employer Contribution: $8,400 \times 18\% = \text{EGP } 1,512$
Total Contribution: EGP 2,520

GOSI.2 Monthly Reporting

GOSI.2.1 Form 1 - Monthly Contributions

Required information:

- Company GOSI number
- Employee national IDs
- Employee names
- Insurance salary
- Contribution amounts

GOSI.2.2 Form 2 - New Employees

Report within 15 days of hiring:

- Employee personal details
- National ID
- Job title
- Start date
- Insurance salary

GOSI.2.3 Form 3 - Terminations

Report within 15 days of termination:

- Employee national ID
- Termination date
- Reason for termination
- End of service calculation

GOSI.2.4 Form 4 - Salary Changes

Report salary changes:

- Employee national ID
- Previous salary

- New salary
- Effective date

GOSI.3 End of Service Benefits

GOSI.3.1 Eligibility

Employee entitled to end of service if:

- Resigns after 10+ years
- Contract ends
- Terminated by employer
- Retires
- Disability
- Death

GOSI.3.2 Calculation Formula

Service Period	Calculation
First 5 years	1/2 month per year
After 5 years	1 month per year

End of Service Example

Years of Service: 8

Last Basic Salary: EGP 6,000

First 5 years: $5 \times 0.5 \times 6,000 = \text{EGP } 15,000$

Next 3 years: $3 \times 1 \times 6,000 = \text{EGP } 18,000$

Total End of Service: EGP 33,000

GOSI.4 Pension Benefits

GOSI.4.1 Old Age Pension

Eligibility:

- Age 60 with 15+ years service
- Age 55 with 20+ years service

GOSI.4.2 Disability Pension

Eligibility:

- Total disability
- Partial disability (50%+)
- Minimum 3 months contributions

GOSI.4.3 Death Benefits

Beneficiaries receive:

- Survivor pension
- Death grant
- Funeral expenses

GOSI.5 GOSI Portal

GOSI.5.1 Online Services

- Submit monthly reports
- Register new employees
- Report terminations
- Print contribution certificates

- Track payment status

GOSI.5.2 Registration

Company Registration:

1. Obtain commercial registration
2. Complete GOSI application
3. Submit required documents
4. Receive GOSI number
5. Set up online access

Extended Income Tax Guide

ITX.1 Tax Brackets (2025)

ITX.1.1 Individual Tax Rates

Annual Income (EGP)	Tax Rate	Tax Amount
0 - 30,000	0%	0
30,001 - 45,000	10%	1,500
45,001 - 60,000	15%	2,250
60,001 - 200,000	20%	28,000
200,001 - 400,000	22.5%	45,000
Above 400,000	25%	Variable

ITX.1.2 Tax Calculation Example

Annual Tax Calculation						
Annual		Salary:		EGP		180,000
Personal		Exemption:		EGP		15,000
Taxable		Income:		EGP		165,000
First		30,000:	0%	=		0
30,001	-	45,000	(15,000):	10%	=	1,500
45,001	-	60,000	(15,000):	15%	=	2,250
60,001	-	165,000	(105,000):	20%	=	21,000

Total Annual Tax: EGP 24,750

Monthly Tax: EGP 2,062.50

ITX.2 Taxable Income

ITX.2.1 Taxable Components

Component	Taxable
Basic Salary	Yes
Housing Allowance	Yes
Transportation Allowance	Yes
Meal Allowance	Yes
Bonus	Yes
Overtime	Yes
Commission	Yes

ITX.2.2 Tax Exemptions

- Social insurance contributions
- Private health insurance (up to limit)
- Life insurance premiums (up to limit)
- Professional subscriptions

ITX.3 Monthly Tax Filing

ITX.3.1 Form 1 (Salary Tax)

Required information:

- Company tax ID
- Employee national IDs
- Taxable salary
- Tax amount
- Payment details

ITX.3.2 Filing Deadline

Monthly filing: Within 30 days of month end

ITX.4 Annual Tax Return

ITX.4.1 Form 2 (Annual Return)

Due: By April 1st of following year

Contents:

- Total annual salary
- Taxable income
- Tax paid
- Tax due/refund

ITX.5 Withholding Tax

ITX.5.1 Withholding Rates

Payment Type	Rate
Dividends	10%
Interest	20%

Royalties	20%
Professional Services	5%
Contracting	1.5%

ITX.5.2 Withholding Tax Filing

Report and remit withheld tax monthly

Extended Manufacturing Operations Guide

MFG.1 Production Planning Deep Dive

MFG.1.1 Master Production Schedule (MPS)

The Master Production Schedule is the cornerstone of manufacturing planning in the Egyptian ERP System. It translates forecasted demand into specific production plans, balancing capacity constraints with customer requirements.

MPS Planning Horizon

The MPS operates across multiple planning horizons:

- **Short-term (1-4 weeks):** Detailed daily/weekly schedules with specific product configurations
- **Medium-term (1-3 months):** Weekly aggregates with product family groupings
- **Long-term (3-12 months):** Monthly aggregates for capacity and resource planning

MPS Time Fences

The system implements time fences to control schedule stability:

- **Demand Time Fence:** Period where only customer orders are considered (typically 1-2 weeks)
- **Planning Time Fence:** Period where changes require management approval (typically 4-6 weeks)
- **Slushy Zone:** Transition area between firm and flexible planning

MPS Calculation Logic

MPS Calculation Example

For Product XYZ with the following parameters:

- Opening Inventory: 500 units
- Safety Stock: 200 units
- Batch Size: 1,000 units
- Lead Time: 2 weeks

Week 1 Forecast: 400 units

Week 1 Orders: 350 units

Projected Available Balance = $500 - \text{MAX}(400, 350) = 100$

Since $100 < \text{Safety Stock (200)}$, $\text{MPS} = 1,000$

New PAB = $100 + 1,000 = 1,100$

MFG.1.2 Rough-Cut Capacity Planning (RCCP)

RCCP validates the MPS against key resource constraints without detailed scheduling. It uses bill of resources to check capacity at critical work centers.

RCCP Implementation Steps

1. Define critical work centers (bottleneck resources)
2. Create bill of resources for each product family
3. Load MPS quantities against resource requirements
4. Compare required capacity to available capacity
5. Identify overloads and adjust MPS or capacity

Capacity Planning Metrics

Metric	Formula	Target
Capacity Utilization	Actual Output / Design Capacity	80-85%

Load Factor	Required Capacity / Available Capacity	< 100%
Efficiency	Standard Hours / Actual Hours	> 95%
Utilization	Actual Hours / Available Hours	85-90%

MFG.2 Material Requirements Planning (MRP)

MFG.2.1 MRP Inputs and Outputs

MRP explosion converts the MPS into detailed material requirements for all components and raw materials.

MRP Inputs

- **Master Production Schedule:** What needs to be produced and when
- **Bill of Materials:** Product structure showing component relationships
- **Inventory Records:** On-hand balances, scheduled receipts, allocations
- **Planning Parameters:** Lead times, lot sizes, safety stocks

MRP Outputs

- **Planned Order Receipts:** When materials should arrive
- **Planned Order Releases:** When orders should be placed
- **Exception Messages:** Alerts for potential problems
- **Pegging Reports:** Traceability of requirements to sources

MFG.2.2 MRP Calculation Process

The MRP calculation follows a level-by-level explosion through the bill of materials:

Step-by-Step MRP Logic

1. **Gross Requirements:** Sum of all demands for the item
2. **Scheduled Receipts:** Orders already placed and expected

3. **Projected On Hand:** Inventory carried from previous period
4. **Net Requirements:** Gross - (Scheduled + On Hand)
5. **Planned Order Receipts:** Net requirements adjusted for lot sizing
6. **Planned Order Releases:** Receipts offset by lead time

MFG.2.3 Lot Sizing Techniques

The Egyptian ERP System supports multiple lot sizing methods:

Method	Description	Best For
Lot-for-Lot (L4L)	Order exact net requirement	Expensive items, JIT environments
Fixed Order Quantity	Order predetermined quantity	Items with setup costs
Period Order Quantity	Order to cover multiple periods	Items with moderate setup costs
Economic Order Quantity	Balance ordering and holding costs	Stable demand items
Minimum Order Quantity	Order at least minimum amount	Supplier constraints

MFG.3 Shop Floor Control

MFG.3.1 Work Order Management

Work orders are the primary mechanism for executing production on the shop floor. Each work order represents a specific production run of a particular item.

Work Order Lifecycle

1. **Created:** Generated from MRP or manually entered
2. **Released:** Authorized for production with material allocation
3. **In Progress:** Production has started

4. **Partially Completed:** Some quantities finished
5. **Completed:** All quantities produced
6. **Closed:** Final accounting and variances recorded

Work Order Information

- Item number and description
- Quantity to produce
- Scheduled start and completion dates
- Bill of materials reference
- Routing/operations sequence
- Material allocations
- Cost center assignments

MFG.3.2 Operation Sequencing and Scheduling

Operations define the manufacturing steps required to produce an item. Each operation specifies:

Operation Details

Field	Description
Operation Number	Sequential identifier (10, 20, 30...)
Work Center	Where operation is performed
Description	What is done in this operation
Setup Time	Time to prepare work center
Run Time	Time per unit of production
Move Time	Time to move to next operation
Queue Time	Waiting time before operation

MFG.3.3 Production Reporting

Real-time production tracking captures actual performance against the plan:

Types of Production Transactions

- **Material Issue:** Raw materials consumed to work order
- **Operation Completion:** Report operation finished with labor/time
- **Production Completion:** Finished goods received to inventory
- **Scrap Reporting:** Record defective production
- **Rework Tracking:** Document items requiring rework

MFG.4 Advanced Planning Features

MFG.4.1 What-If Analysis

The system provides simulation capabilities to evaluate planning scenarios:

- Impact of demand changes on material requirements
- Effect of capacity constraints on delivery dates
- Cost implications of different lot sizing strategies
- Supplier lead time variability analysis

MFG.4.2 Constraint-Based Planning

Advanced planning considers multiple constraints simultaneously:

- Material availability constraints
- Capacity constraints at work centers
- Tool and fixture availability
- Skilled labor availability
- Supplier capacity limitations

MFG.4.3 Multi-Site Planning

For organizations with multiple manufacturing locations:

- Inter-site transfer planning
- Load balancing across facilities
- Site-specific capacity profiles
- Consolidated material requirements

Extended Quality Management Guide

QM.1 Quality Planning Framework

QM.1.1 Quality Planning Process

Quality planning ensures that products and services meet customer requirements through systematic preparation of quality activities.

Quality Planning Steps

1. **Identify Customers:** Determine who receives the output
2. **Determine Customer Needs:** Understand explicit and implicit requirements
3. **Develop Product Features:** Design characteristics that meet needs
4. **Develop Process Features:** Create processes capable of producing features
5. **Develop Process Controls:** Implement monitoring and adjustment mechanisms
6. **Transfer to Operations:** Deploy plan to production environment

QM.1.2 Control Plan Development

Control plans document the quality controls for each phase of product realization:

Control Plan Elements

Element	Description
Process Step	Manufacturing or service operation
Product Characteristic	Feature being measured
Process Characteristic	Parameter affecting product
Specification	Required value or tolerance

Measurement Method	How characteristic is evaluated
Sample Size	Number of units to measure
Frequency	How often to measure
Control Method	SPC chart or other technique
Reaction Plan	Action when out of control

QM.2 Statistical Process Control (SPC)

QM.2.1 SPC Fundamentals

Statistical Process Control uses statistical methods to monitor and control processes, ensuring they operate at their full potential.

Common Cause vs Special Cause Variation

Aspect	Common Cause	Special Cause
Source	Inherent to process	External to process
Pattern	Random, predictable	Non-random, identifiable
Action	Process improvement	Immediate correction
Example	Normal tool wear	Broken tool

QM.2.2 Control Chart Selection

Selecting the appropriate control chart depends on data type and subgroup size:

Data Type	Subgroup Size	Chart Type
Variables	n = 1	Individuals and Moving Range (I-MR)

Variables	$n < 10$	X-bar and Range (Xbar-R)
Variables	$n \geq 10$	X-bar and Sigma (Xbar-S)
Attributes (defectives)	Variable	p-chart or np-chart
Attributes (defects)	Variable	c-chart or u-chart

QM.2.3 Control Limit Calculations

X-bar and R Chart Calculations

For k subgroups of size n:

$\bar{\bar{X}} = (\text{Sum of subgroup averages}) / k$

$\bar{R} = (\text{Sum of subgroup ranges}) / k$

X-bar Chart Control Limits:

$UCL = \bar{\bar{X}} + A_2 * \bar{R}$

$LCL = \bar{\bar{X}} - A_2 * \bar{R}$

R Chart Control Limits:

$UCL = D_4 * \bar{R}$

$LCL = D_3 * \bar{R}$

(A_2 , D_3 , D_4 are constants based on subgroup size)

QM.2.4 Control Chart Interpretation Rules

The Western Electric rules identify out-of-control conditions:

- **Rule 1:** One point beyond 3 sigma
- **Rule 2:** Nine points in a row on same side of center line
- **Rule 3:** Six points in a row steadily increasing or decreasing
- **Rule 4:** Two out of three points beyond 2 sigma on same side
- **Rule 5:** Four out of five points beyond 1 sigma on same side

- **Rule 6:** Fourteen points in a row alternating up and down

QM.3 Inspection and Testing

QM.3.1 Incoming Material Inspection

Inspection of purchased materials ensures quality before production use:

Inspection Types

- **100% Inspection:** Every unit examined (critical items)
- **Sampling Inspection:** Statistical sampling per ANSI/ASQ Z1.4
- **Skip-Lot:** Reduced inspection for proven suppliers
- **Dock-to-Stock:** No inspection for certified suppliers

Acceptance Sampling Plans

Element	Description
Lot Size (N)	Total quantity in inspection lot
Sample Size (n)	Number of units to inspect
Acceptance Number (c)	Maximum defects allowed
AQL	Acceptable Quality Level
LTPD	Lot Tolerance Percent Defective

QM.3.2 In-Process Inspection

Inspection during production catches problems early:

First Article Inspection (FAI)

Comprehensive verification of initial production:

- Complete dimensional inspection
- Material certification verification
- Special process validation
- Documentation review
- Customer approval if required

In-Process Checks

- Operator self-inspection at each operation
- Statistical sampling at key characteristics
- Patrol inspection by quality personnel
- Automated in-line measurement systems

QM.3.3 Final Inspection

Verification that finished products meet all requirements:

Final Inspection Elements

- Dimensional verification against drawing
- Functional testing per specifications
- Visual inspection for cosmetic defects
- Packaging and labeling verification
- Documentation completeness check

QM.4 Nonconformance Management

QM.4.1 Nonconformance Identification

Nonconformances are identified through various sources:

- Inspection findings
- Test failures

- Customer complaints
- Internal audits
- Process monitoring
- Supplier issues

QM.4.2 Nonconformance Workflow

1. **Identification:** Document what, where, when, who
2. **Containment:** Isolate affected product immediately
3. **Evaluation:** Determine severity and impact
4. **Disposition:** Decide: Use As-Is, Rework, Repair, Scrap
5. **Implementation:** Execute disposition decision
6. **Verification:** Confirm effectiveness of action
7. **Closure:** Complete documentation and records

QM.4.3 Material Review Board (MRB)

The MRB evaluates nonconforming material and determines disposition:

MRB Composition

- Quality Engineering representative
- Manufacturing Engineering representative
- Design Engineering representative (as needed)
- Materials/Procurement representative
- Production representative

Disposition Options

Disposition	Description	Requirements
Use As-Is	No action required	Does not affect fit, form, function
Rework	Process to make conforming	Approved rework procedure

Repair	Process to make acceptable	Design approval required
Scrap	Destroy or recycle	Proper disposal documentation
Return to Vendor	Send back to supplier	Supplier agreement

QM.5 Corrective and Preventive Action (CAPA)

QM.5.1 CAPA Process Flow

The CAPA system addresses root causes of problems to prevent recurrence:

Problem Definition

- What is the problem?
- Where was it discovered?
- When did it occur?
- Who is involved?
- How was it detected?
- Why is it a problem?

QM.5.2 Root Cause Analysis Techniques

5 Whys Analysis

Iteratively ask "why" to drill down to root cause:

5 Whys Example

Problem:	Product	dimension	out	of	specification
1. Why?	-	Machine	setting	was	wrong
2. Why?	-	Operator	entered	wrong	value
3. Why?	-	Operator	was not trained	on	new procedure
4. Why?	-	Training	schedule	was	not updated

5. Why? - Engineering change process lacks training trigger

Root Cause: ECN process does not include training requirements

Ishikawa (Fishbone) Diagram

Categorize potential causes into major categories:

- **Man:** Personnel-related factors
- **Machine:** Equipment and tooling
- **Material:** Raw materials and components
- **Method:** Procedures and processes
- **Measurement:** Inspection and testing
- **Mother Nature:** Environmental factors

QM.5.3 Action Plan Development

Effective CAPA includes both corrective and preventive actions:

Corrective Action Elements

- Immediate containment of affected product
- Correction of existing nonconformance
- Elimination of root cause
- Verification of effectiveness

Preventive Action Elements

- Risk assessment of similar processes
- Proactive measures to prevent occurrence
- Monitoring and measurement
- Documentation of preventive measures

QM.6 Supplier Quality Management

QM.6.1 Supplier Qualification

New suppliers must demonstrate capability before approval:

Qualification Process

1. Initial questionnaire and documentation review
2. On-site quality system audit
3. Manufacturing process assessment
4. Sample submission and evaluation
5. First article inspection
6. Trial order evaluation
7. Approval to supplier list

QM.6.2 Supplier Performance Monitoring

Metric	Calculation	Target
Quality (PPM)	$(\text{Defective parts} / \text{Total received}) \times 1,000,000$	< 100
On-Time Delivery	$(\text{On-time deliveries} / \text{Total deliveries}) \times 100$	> 98%
Certificate of Conformance	Percentage of shipments with CofC	100%
Corrective Action Response	Average days to respond	< 7 days

QM.6.3 Supplier Development

Working with suppliers to improve performance:

- Regular business reviews
- Joint improvement projects
- Technical assistance
- Training programs

- Recognition programs for top performers

Extended Supply Chain Management Guide

SCM.1 Supply Chain Fundamentals

SCM.1.1 Supply Chain Definition

A supply chain encompasses all activities involved in transforming raw materials into finished products and delivering them to customers. The Egyptian ERP System manages this end-to-end process.

Supply Chain Components

- **Suppliers:** Provide raw materials and components
- **Manufacturing:** Transforms materials into products
- **Distribution:** Moves products to customers
- **Retailers:** Sell products to end consumers
- **Customers:** Ultimate recipients of products

SCM.1.2 Supply Chain Flows

Three primary flows move through the supply chain:

Material Flow

Physical movement of goods from suppliers through manufacturing to customers. Includes raw materials, work-in-process, and finished goods.

Information Flow

Data exchange between supply chain partners including orders, forecasts, inventory levels, and delivery schedules.

Financial Flow

Payment transactions from customers back through the supply chain to suppliers.

SCM.2 Demand Planning

SCM.2.1 Demand Forecasting Methods

The system supports multiple forecasting techniques:

Method	Description	Best For
Moving Average	Average of recent periods	Stable demand
Weighted Moving Average	Average with different weights	Trending demand
Exponential Smoothing	Weighted average with decay	Most applications
Linear Regression	Statistical trend line	Strong trends
Seasonal Decomposition	Separates trend and seasonality	Seasonal patterns
Croston's Method	Intermittent demand	Sporadic demand

SCM.2.2 Forecast Accuracy Measurement

Forecast Error Calculations

Mean

MAD

=

Sum

of

Absolute

|Actual

-

Forecast|

/

Number

of

periods

(MAD):

Mean

MAPE

=

(Sum

of

Absolute

|Actual

-

Forecast|

/

Actual)

/

n

x

100

(MAPE):

Tracking

TS

=

Cumulative

Error

/

Signal:

MAD

Bias:

$$\text{Bias} = \text{Sum of (Actual - Forecast)} / n$$

SCM.2.3 Collaborative Planning

CPFR (Collaborative Planning, Forecasting, and Replenishment) enables partners to share information:

- Shared forecasts reduce bullwhip effect
- Joint business planning aligns objectives
- Exception management focuses attention
- Performance measurement drives improvement

SCM.3 Inventory Optimization

SCM.3.1 Inventory Classification

ABC analysis categorizes inventory by value and importance:

Class	% of Items	% of Value	Control Level
A	10-20%	70-80%	Tight control
B	20-30%	15-25%	Moderate control
C	50-70%	5-10%	Simple control

SCM.3.2 Safety Stock Calculation

Safety Stock Formula

For	variable		demand,		constant	lead	time:
Safety	Stock	=	Z	x	Sigma(D)	x	sqrt(LT)
For	constant		demand,		variable	lead	time:

$$\text{Safety Stock} = Z \times D \times \text{Sigma}(\text{LT})$$

For both variable:

$$\text{Safety Stock} = Z \times \sqrt{\text{LT} \times \text{Sigma}(\text{D})^2 + \text{D}^2 \times \text{Sigma}(\text{LT})^2}$$

Where:

Z = Service factor (1.65 for 95% service)

Sigma(D) = Standard deviation of demand

Sigma(LT) = Standard deviation of lead time

LT = Average lead time

D = Average demand

SCM.3.3 Reorder Point System

The reorder point triggers replenishment orders:

Reorder Point Formula

$$\text{ROP} = (\text{Average Demand} \times \text{Lead Time}) + \text{Safety Stock}$$

Order Quantity Determination

Method	Formula
EOQ	$\sqrt{(2 \times D \times S) / H}$
POQ	EOQ / Average Demand
Min-Max	Max - (On Hand + On Order)

SCM.4 Supplier Relationship Management

SCM.4.1 Supplier Segmentation

Suppliers are categorized by strategic importance:

Category	Characteristics	Strategy
Strategic	High value, critical items	Partnership, collaboration
Leverage	High value, many sources	Competitive bidding
Bottleneck	Low value, few sources	Ensure supply continuity
Non-Critical	Low value, many sources	Efficiency, automation

SCM.4.2 Supplier Performance Scorecard

Comprehensive evaluation of supplier performance:

Category	Weight	Metrics
Quality	30%	PPM, defect rate, CofC compliance
Delivery	25%	On-time %, lead time consistency
Cost	20%	Price competitiveness, cost reduction
Service	15%	Responsiveness, flexibility
Technology	10%	Innovation, technical support

SCM.4.3 Supplier Collaboration

Electronic collaboration improves supply chain efficiency:

- **EDI:** Electronic Data Interchange for orders and invoices
- **Vendor Managed Inventory:** Supplier manages customer inventory
- **Consignment:** Supplier owns inventory until consumed
- **JIT Delivery:** Frequent small deliveries aligned to production

SCM.5 Distribution and Logistics

SCM.5.1 Distribution Network Design

Network design determines facility locations and flow patterns:

Network Design Factors

- Customer locations and demand patterns
- Supplier locations and capabilities
- Transportation costs and modes
- Facility costs (fixed and variable)
- Service level requirements
- Tax and regulatory considerations

SCM.5.2 Transportation Management

The system supports multiple transportation modes:

Mode	Speed	Cost	Capacity	Best For
Air	Fastest	Highest	Limited	Urgent, high-value
Truck	Fast	Medium	Flexible	Regional, time-sensitive
Rail	Medium	Low	High	Bulk, long distance
Ocean	Slowest	Lowest	Highest	International, non-urgent

SCM.5.3 Route Optimization

Route planning minimizes transportation costs:

Vehicle Routing Problem (VRP) Considerations

- Customer time windows
- Vehicle capacity constraints

- Driver working hours
- Traffic patterns
- Delivery priorities
- Backhaul opportunities

SCM.6 Supply Chain Analytics

SCM.6.1 Key Performance Indicators

KPI	Formula	Industry Benchmark
Perfect Order Rate	% orders complete, accurate, on-time	> 95%
Cash-to-Cash Cycle	Inventory days + AR days - AP days	< 30 days
Fill Rate	Orders filled complete / Total orders	> 97%
Inventory Turns	COGS / Average Inventory	Industry specific
Supply Chain Cost	Total SC cost / Revenue	< 10%

SCM.6.2 Supply Chain Risk Management

Identifying and mitigating supply chain risks:

Risk Categories

- **Supply Risks:** Supplier failures, capacity constraints
- **Demand Risks:** Forecast errors, demand volatility
- **Process Risks:** Quality issues, production disruptions
- **Control Risks:** System failures, data breaches
- **Environmental Risks:** Natural disasters, geopolitical events

Risk Mitigation Strategies

- Dual sourcing for critical items
- Safety stock for supply uncertainty
- Flexible manufacturing capacity
- Postponement strategies
- Supply chain visibility tools

Extended Warehouse Management Guide

WMS.1 Warehouse Design Principles

WMS.1.1 Warehouse Layout Design

Effective warehouse layout maximizes space utilization and operational efficiency:

Layout Design Factors

- **Product Characteristics:** Size, weight, storage requirements
- **Throughput Volume:** Receipts, shipments, internal moves
- **Storage Requirements:** Temperature, humidity, security
- **Material Handling:** Equipment types and capabilities
- **Order Profile:** Order size, pick frequency

Zone Types

Zone	Purpose	Characteristics
Receiving	Unload and check incoming	Dock doors, staging area
Putaway	Move to storage	Aisles, material handling
Storage	Hold inventory	Racking, shelving
Picking	Retrieve for orders	Forward pick locations
Packing	Prepare for shipment	Packing stations, supplies
Shipping	Consolidate and load	Staging, dock doors

WMS.1.2 Storage Systems

Racking Types

Type	Description	Best For
Selective	Single-deep access	High SKU variety
Double-Deep	Two pallets deep	Medium turnover
Drive-In	Lift truck enters rack	High density, low variety
Push-Back	Gravity flow within rack	LIFO inventory
Pallet Flow	Gravity flow lanes	High throughput FIFO
Cantilever	Arms support load	Long items

WMS.2 Receiving Operations

WMS.2.1 Receiving Process Flow

1. **Appointment Scheduling:** Coordinate carrier arrivals
2. **Truck Check-In:** Verify carrier and documentation
3. **Unloading:** Remove goods from vehicle
4. **Receipt Verification:** Count and inspect
5. **System Receipt:** Record in WMS
6. **Quality Hold:** Quarantine if inspection required
7. **Putaway:** Move to storage location

WMS.2.2 Advanced Shipping Notice (ASN)

ASN provides advance information about incoming shipments:

ASN Information

- Carrier and tracking number
- Expected arrival date/time
- Purchase order reference
- Item details and quantities
- Serial numbers (if applicable)
- Packaging hierarchy

ASN Benefits

- Pre-allocated putaway locations
- Reduced receiving time
- Improved dock scheduling
- Automated receiving (barcode scan)

WMS.2.3 Receipt Inspection

Quality verification at receipt prevents problems downstream:

Inspection Types

Type	Description	When Used
Count Only	Verify quantity matches PO	Trusted suppliers
Visual	Check for damage	Standard items
Dimensional	Measure physical characteristics	Tight tolerances
Functional	Test operation	Critical items
Full	Complete specification check	New suppliers

WMS.3 Putaway Strategies

WMS.3.1 Putaway Methods

Directed Putaway

WMS assigns specific location based on rules:

- Product characteristics (size, weight, temperature)
- Velocity class (fast/medium/slow movers)
- Zone preferences
- Space utilization optimization

User-Directed Putaway

Operator selects location with WMS validation:

- Operator scans location
- WMS validates suitability
- Warning if suboptimal location

WMS.3.2 Location Assignment Rules

Rule	Description	Benefit
Closest Open	Nearest available location	Minimize travel
Fixed	Predefined location per item	Easy to find
Random	Any suitable location	Maximize space
Zone-Based	By velocity or product type	Organized storage

WMS.4 Inventory Management

WMS.4.1 Cycle Counting

Regular counting ensures inventory accuracy:

Cycle Count Methods

- **ABC Counting:** Count A items most frequently
- **Zone Counting:** Count by physical location
- **Random Counting:** Random selection of items
- **Triggered Counting:** Count when events occur

Count Frequency Guidelines

Class	Frequency	Accuracy Target
A	Monthly	99.5%
B	Quarterly	98.0%
C	Annually	95.0%

WMS.4.2 Inventory Accuracy

Measuring and improving inventory record accuracy:

Accuracy Calculations

Location Accuracy:

$$\% \text{ Locations Correct} = (1 - |\text{Errors}| / \text{Total Locations}) \times 100$$

Unit Accuracy:

$$\% \text{ Units Correct} = (1 - |\text{Unit Variance}| / \text{Total Units}) \times 100$$

Financial Accuracy:

$$\% \text{ Value Correct} = (1 - |\text{Value Variance}| / \text{Total Value}) \times 100$$

WMS.4.3 Slotting Optimization

Slotting determines where items are stored for optimal efficiency:

Slotting Factors

- **Pick Frequency:** Fast movers near shipping
- **Cube Movement:** High cube items at convenient heights
- **Product Affinity:** Frequently co-picked items together
- **Size Compatibility:** Match item to location size
- **Weight:** Heavy items at ergonomic heights

WMS.5 Order Fulfillment

WMS.5.1 Picking Methodologies

Method	Description	Best For
Discrete	One order at a time	Large orders, accuracy critical
Batch	Multiple orders together	Small orders, high volume
Zone	Picker stays in zone	Large warehouse, many SKUs
Wave	Scheduled pick waves	Scheduled shipping
Cluster	Multiple orders in cart	Small items, high volume

WMS.5.2 Pick Path Optimization

Efficient travel paths reduce picking time:

Path Algorithms

- **S-Shape:** Traverse aisles in S-pattern
- **Return:** Enter and exit same end
- **Midpoint:** Divide aisle, pick both directions

- **Largest Gap:** Skip largest empty section

WMS.5.3 Packing Operations

Packing prepares orders for shipment:

Packing Considerations

- Container selection (box, envelope, pallet)
- Dunnage and protection materials
- Weight and dimension optimization
- Label placement and verification
- Documentation inclusion

WMS.6 Shipping Operations

WMS.6.1 Shipping Process

1. **Order Consolidation:** Gather all items for shipment
2. **Carrier Selection:** Choose appropriate carrier
3. **Manifesting:** Create shipping documentation
4. **Label Generation:** Print carrier labels
5. **Loading:** Place in carrier vehicle
6. **Ship Confirmation:** Record shipment in system

WMS.6.2 Carrier Management

Managing relationships with transportation providers:

Carrier Selection Criteria

Factor	Weight	Considerations
--------	--------	----------------

Cost	30%	Base rate, fuel, accessories
Service	25%	On-time performance, claims
Capacity	20%	Availability, equipment
Coverage	15%	Geographic reach
Technology	10%	Tracking, EDI capability

WMS.7 Warehouse Technology

WMS.7.1 Barcode and RFID

Automatic identification technologies improve accuracy:

Barcode Types

Type	Use Case	Capacity
Code 39	General purpose	Alphanumeric
Code 128	Shipping labels	High density
UPC/EAN	Retail products	Fixed length
QR Code	Mobile scanning	High capacity
Data Matrix	Small items	Very high density

WMS.7.2 Voice Picking

Voice-directed work improves productivity:

- Hands-free operation
- Eyes-free (look at product, not paper)
- Real-time validation

- Multilingual support
- Reduced training time

WMS.7.3 Automation Technologies

Technology	Description	Benefits
AS/RS	Automated storage and retrieval	High density, speed
AGV	Automated guided vehicles	Consistent transport
AMR	Autonomous mobile robots	Flexible automation
Pick-to-Light	Light-directed picking	Speed, accuracy
Put-to-Light	Light-directed sortation	Batch picking efficiency
Conveyors	Mechanized transport	High volume movement

Extended Business Intelligence Guide

BI.1 Data Warehouse Architecture

BI.1.1 Data Warehouse Concepts

A data warehouse is a subject-oriented, integrated, time-variant, non-volatile collection of data supporting decision-making.

Data Warehouse Characteristics

- **Subject-Oriented:** Organized by subject (customer, product) not application
- **Integrated:** Consistent naming conventions and data formats
- **Time-Variant:** Historical data maintained over time
- **Non-Volatile:** Data not updated in place, only appended

BI.1.2 Dimensional Modeling

Dimensional modeling organizes data for query performance and usability:

Fact Tables

Fact tables contain measurements (facts) and foreign keys to dimensions:

- **Transaction Facts:** Individual business events
- **Periodic Snapshot:** Regular snapshots of activity
- **Accumulating Snapshot:** Process milestones

Dimension Tables

Dimensions provide context for facts:

- **Time:** Date, month, quarter, year, fiscal periods
- **Customer:** Demographics, segments, geography

- **Product:** Hierarchy, attributes, categories
- **Location:** Warehouses, stores, regions
- **Employee:** Organization, role, tenure

BI.1.3 Schema Designs

Star Schema

Fact table surrounded by denormalized dimensions:

- Simple to understand
- Fast query performance
- Some data redundancy

Snowflake Schema

Dimensions normalized into sub-dimensions:

- Reduced redundancy
- More complex queries
- May require more joins

BI.2 ETL Processes

BI.2.1 Extract Phase

Data extraction from source systems:

Extraction Methods

Method	Description	Best For
Full Extract	Complete data refresh	Small tables, initial load
Incremental	Only changed records	Large tables, frequent updates

Change Data Capture	Database log scraping	Real-time requirements
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BI.2.2 Transform Phase

Data cleansing and transformation:

Transformation Types

- **Cleansing:** Remove errors, standardize formats
- **Integration:** Combine from multiple sources
- **Aggregation:** Summarize detailed data
- **Derivation:** Calculate new values
- **Lookup:** Add reference data

BI.2.3 Load Phase

Loading data into the warehouse:

Loading Strategies

- **Initial Load:** Historical data population
- **Incremental Load:** Regular updates
- **Full Refresh:** Complete replacement

BI.3 Reporting and Analytics

BI.3.1 Report Types

Type	Description	Frequency
Operational	Day-to-day transactions	Real-time, hourly
Tactical	Department performance	Daily, weekly

Strategic	Executive dashboards	Monthly, quarterly
Ad-hoc	User-defined queries	On-demand

BI.3.2 Key Performance Indicators

Financial KPIs

KPI	Formula	Target
Gross Margin	$(\text{Revenue} - \text{COGS}) / \text{Revenue}$	Industry specific
Operating Margin	$\text{Operating Income} / \text{Revenue}$	> 10%
ROA	$\text{Net Income} / \text{Total Assets}$	> 5%
ROE	$\text{Net Income} / \text{Shareholder Equity}$	> 15%

Operational KPIs

KPI	Formula	Target
Order Cycle Time	Order receipt to shipment	< 24 hours
Perfect Order Rate	$\text{Error-free orders} / \text{Total orders}$	> 95%
Inventory Turns	$\text{COGS} / \text{Average Inventory}$	Industry specific
On-Time Delivery	$\text{On-time shipments} / \text{Total}$	> 98%

BI.3.3 Dashboard Design

Effective dashboards communicate information clearly:

Dashboard Best Practices

- Limit to 5-7 key metrics

- Use appropriate visualizations
- Provide context (targets, trends)
- Enable drill-down capability
- Update frequency appropriate to metric
- Mobile-responsive design

Visualization Selection

Data Type	Best Visualization
Comparison	Bar chart, column chart
Trend	Line chart, area chart
Composition	Pie chart, stacked bar
Distribution	Histogram, box plot
Correlation	Scatter plot, bubble chart
Geographic	Map, heat map

BI.4 Advanced Analytics

BI.4.1 Descriptive Analytics

What happened? - Historical data analysis:

- Standard reports
- Ad-hoc queries
- Dashboards and scorecards
- Data visualization

BI.4.2 Diagnostic Analytics

Why did it happen? - Root cause analysis:

- Drill-down analysis
- Correlation analysis
- Variance analysis
- Contribution analysis

BI.4.3 Predictive Analytics

What will happen? - Future predictions:

Predictive Techniques

Technique	Use Case
Regression	Sales forecasting, pricing
Time Series	Demand forecasting
Classification	Customer segmentation, churn
Clustering	Market segmentation
Neural Networks	Complex pattern recognition

BI.4.4 Prescriptive Analytics

What should we do? - Optimization recommendations:

- Optimization models
- Simulation
- Decision trees
- Recommendation engines

BI.5 Self-Service BI

BI.5.1 User Categories

User Type	Needs	Tools
Consumers	View reports, dashboards	Web portal, mobile apps
Explorers	Drill, filter, pivot	Interactive dashboards
Analysts	Create reports, analysis	Query tools, workbooks
Data Scientists	Advanced modeling	Statistical tools, Python/R

BI.5.2 Data Governance

Ensuring data quality and appropriate access:

Governance Framework

- **Data Ownership:** Clear accountability for data quality
- **Data Standards:** Consistent definitions and formats
- **Access Controls:** Role-based security
- **Quality Monitoring:** Ongoing data quality assessment
- **Metadata Management:** Documentation of data elements

Extended Document Management Guide

DM.1 Document Lifecycle Management

DM.1.1 Document Creation

Documents are created through various methods:

- Native application creation (Word, Excel)
- Scanning of paper documents
- Email capture and archiving
- System-generated reports
- Electronic forms submission

DM.1.2 Document Review and Approval

Controlled documents follow formal review processes:

Approval Workflow

1. Author creates document
2. Technical review for accuracy
3. Quality review for compliance
4. Management approval
5. Controlled release
6. Training on changes (if applicable)

DM.1.3 Document Distribution

Controlled distribution ensures users have correct versions:

Distribution Methods

Method	Description	Use Case
Central Repository	Single source access	Controlled documents
Email Distribution	Sent to distribution list	Time-sensitive notices
Portal Access	Self-service retrieval	Reference documents
System Integration	Embedded in applications	Operational procedures

DM.1.4 Document Obsolescence

Outdated documents are managed to prevent misuse:

- Version control maintains history
- Obsolete documents marked and archived
- Superseded documents removed from active use
- Retention schedules define archive period

DM.2 Version Control

DM.2.1 Version Numbering

Systematic version identification:

Version Numbering Schemes

Scheme	Format	Example
Simple	1, 2, 3...	Version 3
Decimal	1.0, 1.1, 2.0...	Version 2.3
Date	YYYY.MM.DD	2025.01.15

DM.2.2 Revision History

Complete audit trail of document changes:

Revision History Contents

- Version number and date
- Description of changes
- Author of changes
- Approval authority
- Reason for change
- Affected sections

DM.3 Document Security

DM.3.1 Access Control

Role-based access ensures appropriate document access:

Access Levels

Level	Permissions
No Access	Cannot view or access
View Only	Read, print, no changes
Comment	View plus add comments
Edit	Modify content
Admin	Full control including delete

DM.3.2 Document Classification

Classification determines handling requirements:

Classification	Description	Handling
Public	No restrictions	Unlimited distribution
Internal	Company use only	Employee access
Confidential	Sensitive information	Need-to-know basis
Restricted	Highly sensitive	Authorized personnel only

DM.4 Records Management

DM.4.1 Records Retention

Legal and regulatory requirements define retention periods:

Common Retention Categories

Record Type	Retention Period	Basis
Financial Records	7-10 years	Tax regulations
Employee Records	Duration + 7 years	Labor law
Contracts	Duration + 7 years	Legal requirements
Quality Records	Product life + X years	Industry standards
Email	1-7 years	Company policy

DM.4.2 Legal Hold

Litigation hold suspends normal retention:

- Notification of hold requirement
- Identification of affected records
- Suspension of deletion processes
- Documentation of preservation
- Release when hold lifted

DM.5 Integration with Business Processes

DM.5.1 Workflow Integration

Documents flow through business processes:

- Purchase requisition to purchase order
- Invoice to payment approval
- Engineering change request to implementation
- HR request to approval and processing

DM.5.2 ERP Integration Points

Module	Document Types
Financial	Invoices, vouchers, statements
Procurement	POs, contracts, receipts
Sales	Quotes, orders, shipping docs
HR	Employee files, policies
Quality	Procedures, records, certificates

Extended System Administration Guide

SA.1 User Management

SA.1.1 User Account Lifecycle

Account Creation

1. Submit user access request form
2. Manager approval
3. Security review (if applicable)
4. Account provisioning
5. Role assignment
6. Initial password setup
7. User notification and training

Account Modification

- Role changes require approval
- Department transfers update organization
- Temporary access for projects
- Additional module access requests

Account Termination

1. Termination notification received
2. Account disabled immediately
3. Access rights revoked
4. Data ownership transferred

5. Account archived per policy

SA.1.2 Role Design Best Practices

Role Design Principles

- **Separation of Duties:** No single user controls entire process
- **Least Privilege:** Minimum access needed for job
- **Role Granularity:** Balance between too many and too few
- **Naming Conventions:** Clear, consistent role names

Common Role Types

Role Category	Examples
Functional	AP Clerk, AR Manager, Buyer
Hierarchical	Supervisor, Manager, Director
Organizational	Plant User, Division Admin
Project	Implementation Team, Auditor
System	Administrator, Developer

SA.2 Security Administration

SA.2.1 Authentication Configuration

Password Policies

Parameter	Recommendation
Minimum Length	12 characters
Complexity	Upper, lower, number, special

Expiration	90 days
History	Remember 12 passwords
Lockout	5 failed attempts
Unlock	Admin unlock or 30 minutes

SA.2.2 Authorization Management

Permission Types

- **Create:** Add new records
- **Read:** View existing records
- **Update:** Modify existing records
- **Delete:** Remove records
- **Execute:** Run processes or reports
- **Admin:** Configure and manage

SA.2.3 Audit and Monitoring

Audit Trail Contents

Element	Description
Timestamp	When action occurred
User ID	Who performed action
Transaction	What was done
Object	What was affected
Before Value	Prior state
After Value	New state

Source

Application or terminal

SA.3 System Configuration

SA.3.1 Company Setup

Company Parameters

- Company name and address
- Legal entity information
- Tax registration numbers
- Base currency
- Fiscal calendar
- Chart of accounts

SA.3.2 Module Configuration

Configuration Elements

Element	Description
Numbering Series	Document number formats
Posting Periods	Open/close periods
Approval Workflows	Document approval paths
Default Values	Common field defaults
Validation Rules	Data entry constraints

SA.4 Backup and Recovery

SA.4.1 Backup Strategy

Backup Types

Type	Frequency	Retention
Full	Weekly	4 weeks
Differential	Daily	2 weeks
Transaction Log	Every 15 min	48 hours
Archive	Monthly	7 years

SA.4.2 Recovery Procedures

Recovery Scenarios

- **Point-in-Time:** Restore to specific moment
- **Full Database:** Complete database restore
- **Table-Level:** Restore specific tables
- **Object-Level:** Restore specific records

SA.5 Performance Management

SA.5.1 Performance Monitoring

Key Metrics

Metric	Target	Action if Exceeded
CPU Utilization	< 70%	Investigate processes
Memory Usage	< 80%	Add memory or optimize
Disk I/O	< 50%	Optimize queries

Response Time	< 3 sec	Performance tuning
Concurrent Users	Within license	Add licenses

SA.5.2 Performance Tuning

Tuning Areas

- Database index optimization
- Query performance improvement
- Memory allocation adjustment
- Tempdb configuration
- Network optimization
- Application server tuning

Extended Mobile Application Guide

MOB.1 Mobile Architecture

MOB.1.1 Platform Support

The Egyptian ERP mobile application supports multiple platforms:

Platform	Minimum Version	Features
iOS	14.0+	Full functionality
Android	8.0+	Full functionality
Web (PWA)	Modern browsers	Core functionality

MOB.1.2 Offline Capabilities

Key features work without continuous connectivity:

Offline Features

- View cached data and reports
- Create transactions (queued for sync)
- Approval workflows
- Time and attendance entry
- Document viewing

Synchronization

- Automatic sync when connected
- Manual sync on demand
- Conflict resolution rules

- Sync status indicators

MOB.2 Mobile Security

MOB.2.1 Device Security

Security Requirements

Requirement	Description
Device PIN/Biometric	Required to unlock device
App PIN	Separate PIN for ERP app
Auto-Lock	Lock after inactivity
Jailbreak Detection	Block compromised devices
Remote Wipe	Clear app data if lost

MOB.2.2 Data Protection

- Data encrypted at rest on device
- TLS encryption for transmission
- No sensitive data in device backups
- Automatic data purge after inactivity

MOB.3 Mobile Features by Module

MOB.3.1 Sales Module

- Customer lookup and history
- Real-time inventory availability
- Quote and order creation

- Price list access
- Sales dashboard
- Visit logging

MOB.3.2 Service Module

- Service call management
- Work order execution
- Parts consumption
- Time tracking
- Customer signature capture
- Photo documentation

MOB.3.3 Warehouse Module

- Goods receipt
- Putaway confirmation
- Picking execution
- Cycle counting
- Inventory inquiries
- Barcode scanning

MOB.3.4 HR Module

- Self-service profile
- Leave requests
- Expense claims
- Time clock
- Payslip viewing
- Approval workflows

MOB.3.5 Executive Dashboard

- KPI overview
- Alert notifications
- Approval queues
- Drill-down analytics
- Report viewing

MOB.4 Mobile Deployment

MOB.4.1 App Distribution

Method	Description	Use Case
Public Store	App Store, Google Play	General availability
Enterprise Store	Internal app store	Corporate devices
MDM	Mobile device management	Controlled deployment
Sideload	Direct installation	Testing, special cases

MOB.4.2 Mobile Device Management

MDM integration for corporate devices:

- App whitelist/blacklist
- Configuration management
- Certificate deployment
- Remote troubleshooting
- Compliance monitoring

Extended Multi-Company Operations Guide

MC.1 Multi-Company Architecture

MC.1.1 Company Structure Models

Model 1: Independent Companies

Each company operates as separate entity:

- Separate chart of accounts
- Independent transactions
- Individual reporting
- Inter-company transactions recorded

Model 2: Shared Services

Common functions centralized:

- Shared procurement
- Centralized cash management
- Common master data
- Consolidated reporting

Model 3: Global Template

Standardized processes across companies:

- Common chart of accounts
- Standard business processes
- Unified reporting structure
- Local adaptations allowed

MC.2 Inter-Company Transactions

MC.2.1 Inter-Company Sales

Sales between related companies:

Transaction Flow

1. Selling company creates sales order
2. Buying company receives purchase order
3. Goods shipped and received
4. Inter-company invoice generated
5. Inter-company AP/AR recorded
6. Elimination entries for consolidation

MC.2.2 Transfer Pricing

Pricing of inter-company transactions:

Transfer Pricing Methods

Method	Description	Best For
Cost Plus	Cost plus markup	Manufacturing
Resale Price	Resale price minus margin	Distribution
Comparable	Market-based pricing	Commodities
Profit Split	Divide combined profit	Joint ventures

MC.2.3 Inter-Company Reconciliation

Ensuring inter-company balances match:

Reconciliation Process

1. Generate inter-company reports
2. Compare balances between companies
3. Identify differences
4. Investigate root causes
5. Make correcting entries
6. Confirm balances agree

MC.3 Consolidation

MC.3.1 Consolidation Methods

Method	Ownership	Treatment
Full Consolidation	> 50%	100% of assets, liabilities
Proportional	Joint venture	Share of assets, liabilities
Equity	20-50%	Share of net income
Cost	< 20%	Dividends only

MC.3.2 Consolidation Adjustments

Elimination Entries

- Inter-company receivables/payables
- Inter-company sales and cost of sales
- Unrealized profit in inventory
- Inter-company investments
- Inter-company dividends

Currency Translation

- Assets/liabilities at closing rate
- Income/expense at average rate
- Equity at historical rate
- Translation differences to OCI

MC.4 Multi-Currency Operations

MC.4.1 Currency Setup

Currency Types

- **Local Currency:** Company operating currency
- **Transaction Currency:** Currency of original transaction
- **Group Currency:** Reporting currency for consolidation
- **Hard Currency:** Stable currency for comparison

MC.4.2 Exchange Rate Management

Rate Types

Type	Use
Spot Rate	Current transactions
Average Rate	Period-end translation
Historical Rate	Fixed asset transactions
Budget Rate	Planning and forecasting

MC.4.3 Foreign Currency Valuation

Revaluation of foreign currency balances:

Currency Valuation Example

USD Receivable: \$10,000
Original Rate: 30.85 EGP/USD
Current Rate: 31.20 EGP/USD

Original Value: 308,500 EGP
Current Value: 312,000 EGP
Gain: 3,500 EGP

Entry:
Dr. USD Receivable 3,500
Cr. FX Gain 3,500

MC.5 Shared Master Data

MC.5.1 Master Data Synchronization

Shared Data Elements

Data Type	Sharing Level
Chart of Accounts	Group template
Customer Master	Global or local
Vendor Master	Global or local
Material Master	Global with local views
Cost Centers	Company-specific
Profit Centers	Group-wide

MC.5.2 Master Data Governance

Processes to ensure data quality:

- Data ownership assignments
- Approval workflows for changes
- Data quality monitoring
- Duplicate detection
- Change propagation rules

Extended Implementation Methodology Guide

IMP.1 Implementation Approaches

IMP.1.1 Big Bang Implementation

All modules go live simultaneously:

Advantages

- Single cutover event
- No interface complexity
- Complete solution from day one
- Lower overall project cost

Disadvantages

- High risk
- Extensive preparation required
- Longer initial timeline
- More resources needed initially

IMP.1.2 Phased Implementation

Modules implemented sequentially:

Common Phase Sequences

Phase	Typical Modules	Duration
1	Financials (GL, AP, AR)	3-4 months

2	Procurement, Inventory	2-3 months
3	Sales, CRM	2-3 months
4	Manufacturing	3-4 months
5	HR, Payroll	2-3 months

IMP.1.3 Hybrid Approach

Combination of big bang and phased:

- Core financials as big bang
- Operational modules phased
- Regional rollouts
- Business unit prioritization

IMP.2 Project Phases

IMP.2.1 Phase 1: Project Preparation

Activities

- Project charter creation
- Team organization
- Infrastructure setup
- Standards definition
- Project plan development

Deliverables

- Project charter
- Project plan
- Communication plan
- Risk register

- Infrastructure ready

IMP.2.2 Phase 2: Business Blueprint

Activities

- As-is process documentation
- To-be process design
- Gap analysis
- Requirements gathering
- Solution design

Deliverables

- Business blueprint document
- Process flow diagrams
- Gap list
- Functional specifications
- Reporting requirements

IMP.2.3 Phase 3: Realization

Activities

- System configuration
- Development (customizations, interfaces)
- Unit testing
- Integration testing
- Data migration development

Deliverables

- Configured system
- Custom developments

- Test scripts and results
- Data migration programs
- User procedures

IMP.2.4 Phase 4: Final Preparation

Activities

- End-user training
- Data migration execution
- User acceptance testing
- Cutover planning
- Production readiness check

Deliverables

- Trained users
- Migrated data
- UAT sign-off
- Cutover plan
- Go-live checklist

IMP.2.5 Phase 5: Go-Live and Support

Activities

- Cutover execution
- Go-live support
- Issue resolution
- Hypercare support
- Knowledge transfer

Deliverables

- Live system
- Support documentation
- Lessons learned
- Transition to support team

IMP.3 Change Management

IMP.3.1 Change Management Framework

ADKAR Model

Stage	Description	Actions
Awareness	Understand need for change	Communications, business case
Desire	Want to participate	WIIFM, incentives
Knowledge	Know how to change	Training, documentation
Ability	Can implement change	Practice, coaching
Reinforcement	Sustain the change	Recognition, metrics

IMP.3.2 Communication Strategy

Communication Channels

- Executive town halls
- Department meetings
- Email updates
- Project portal
- Newsletters
- Video messages

IMP.4 Data Migration

IMP.4.1 Migration Strategy

Migration Approaches

Approach	Description	Best For
Big Bang	All data at once	Small datasets
Phased	By module or region	Complex environments
Parallel	Run old and new	High risk tolerance
Historical	Summary history only	Legacy data issues

IMP.4.2 Migration Process

ETL Steps

1. **Extract:** Pull from source systems
2. **Cleanse:** Fix data quality issues
3. **Transform:** Map to target format
4. **Validate:** Check data integrity
5. **Load:** Import to new system
6. **Reconcile:** Verify completeness

IMP.5 Testing Strategy

IMP.5.1 Testing Types

Type	Focus	Who
------	-------	-----

Unit	Individual components	Developers
Integration	Component interfaces	Technical team
System	End-to-end processes	Functional team
UAT	Business requirements	End users
Performance	Speed and scalability	Technical team
Security	Access and protection	Security team

IMP.5.2 Test Data Management

- Anonymize sensitive data
- Create representative scenarios
- Include edge cases
- Volume testing data
- Data refresh procedures

IMP.6 Post-Implementation

IMP.6.1 Hypercare Support

Intensive support immediately after go-live:

- Dedicated support team
- Rapid issue response
- Daily status meetings
- Quick fixes and workarounds
- User hand-holding

IMP.6.2 Benefits Realization

Measuring achieved benefits:

- Baseline metrics comparison
- Benefits tracking dashboard
- Course correction actions
- ROI calculation
- Continuous improvement

Appendix A: Complete Transaction Code Reference

A.1 Financial Module Codes

Code	Description	Module
GL01	Journal Entry Create	General Ledger
GL02	Journal Entry Post	General Ledger
GL03	Journal Entry Reverse	General Ledger
GL04	Account Balance Inquiry	General Ledger
GL05	Trial Balance Report	General Ledger
GL06	Period Close	General Ledger
GL07	Year End Close	General Ledger
GL08	Chart of Accounts Maintain	General Ledger
GL09	Cost Center Maintain	General Ledger
GL10	Profit Center Maintain	General Ledger
GL11	Segment Maintain	General Ledger
GL12	Allocation Rule Maintain	General Ledger
GL13	Recurring Entry Maintain	General Ledger
GL14	Revaluation Execute	General Ledger
GL15	Translation Execute	General Ledger

AP01	Vendor Invoice Enter	Accounts Payable
AP02	Vendor Payment Process	Accounts Payable
AP03	Vendor Credit Memo Enter	Accounts Payable
AP04	Vendor Account Inquiry	Accounts Payable
AP05	Aging Report	Accounts Payable
AP06	Payment Proposal	Accounts Payable
AP07	Vendor Master Maintain	Accounts Payable
AP08	Withholding Tax Report	Accounts Payable
AP09	Vendor 1099 Report	Accounts Payable
AP10	Payment Run	Accounts Payable
AR01	Customer Invoice Enter	Accounts Receivable
AR02	Customer Receipt Process	Accounts Receivable
AR03	Customer Credit Memo Enter	Accounts Receivable
AR04	Customer Account Inquiry	Accounts Receivable
AR05	Aging Report	Accounts Receivable
AR06	Dunning Run	Accounts Receivable
AR07	Customer Master Maintain	Accounts Receivable
AR08	Collection Workbench	Accounts Receivable
AR09	Dispute Management	Accounts Receivable
AR10	Cash Application	Accounts Receivable

A.2 Inventory Module Codes

Code	Description	Module
IM01	Goods Receipt Create	Inventory
IM02	Goods Issue Create	Inventory
IM03	Transfer Order Create	Inventory
IM04	Transfer Order Confirm	Inventory
IM05	Physical Inventory Create	Inventory
IM06	Physical Inventory Count	Inventory
IM07	Physical Inventory Post	Inventory
IM08	Material Master Maintain	Inventory
IM09	Stock Overview	Inventory
IM10	Material Document Inquiry	Inventory
IM11	Reservation Create	Inventory
IM12	Reservation Monitor	Inventory
IM13	Batch Master Maintain	Inventory
IM14	Serial Number Maintain	Inventory
IM15	Inventory Differences	Inventory

A.3 Procurement Module Codes

Code	Description	Module
MM01	Purchase Requisition Create	Procurement
MM02	Purchase Requisition Approve	Procurement

MM03	Purchase Order Create	Procurement
MM04	Purchase Order Approve	Procurement
MM05	Purchase Order Change	Procurement
MM06	Goods Receipt for PO	Procurement
MM07	Invoice Verification	Procurement
MM08	Vendor Evaluation	Procurement
MM09	RFQ Process	Procurement
MM10	Contract Maintain	Procurement
MM11	Scheduling Agreement	Procurement
MM12	Source List Maintain	Procurement
MM13	Quota Arrangement	Procurement
MM14	Info Record Maintain	Procurement
MM15	Vendor Master Maintain	Procurement

A.4 Sales Module Codes

Code	Description	Module
SD01	Sales Inquiry Create	Sales
SD02	Sales Quotation Create	Sales
SD03	Sales Order Create	Sales
SD04	Sales Order Change	Sales
SD05	Delivery Create	Sales

SD06	Delivery Change	Sales
SD07	Picking Create	Sales
SD08	Picking Confirm	Sales
SD09	Shipment Create	Sales
SD10	Billing Create	Sales
SD11	Billing Cancel	Sales
SD12	Credit Memo Create	Sales
SD13	Debit Memo Create	Sales
SD14	Customer Master Maintain	Sales
SD15	Price Maintain	Sales

A.5 Manufacturing Module Codes

Code	Description	Module
PP01	Material Master Maintain	Manufacturing
PP02	BOM Maintain	Manufacturing
PP03	Routing Maintain	Manufacturing
PP04	Work Center Maintain	Manufacturing
PP05	MRP Run	Manufacturing
PP06	MRP Results Evaluation	Manufacturing
PP07	Planned Order Convert	Manufacturing
PP08	Production Order Create	Manufacturing

PP09	Production Order Release	Manufacturing
PP10	Goods Issue to Order	Manufacturing
PP11	Confirmation Enter	Manufacturing
PP12	Goods Receipt from Order	Manufacturing
PP13	Production Order Close	Manufacturing
PP14	Capacity Planning	Manufacturing
PP15	Shop Floor Control	Manufacturing

A.6 HR Module Codes

Code	Description	Module
PA01	Hire Employee	Personnel Admin
PA02	Change Employee Data	Personnel Admin
PA03	Display Employee Data	Personnel Admin
PA04	Terminate Employee	Personnel Admin
PA05	Organizational Assignment	Personnel Admin
PA06	Basic Pay Maintain	Personnel Admin
PA07	Recurring Payments	Personnel Admin
PA08	Additional Payments	Personnel Admin
PA09	Leave Request	Time Management
PA10	Leave Approval	Time Management
PA11	Time Sheet Entry	Time Management

PA12	Attendance Check	Time Management
PA13	Payroll Run	Payroll
PA14	Payroll Result Display	Payroll
PA15	Payslip Display	Payroll

Appendix B: Egyptian Regulatory Reference

B.1 Tax Authority Contacts

Office	Address	Phone	Email
ETA Headquarters	3 El-Sheikh Marouf St., Cairo	+202-2795-0000	info@eta.gov.eg
Cairo Tax Office	Nasr City, Cairo	+202-2670-1234	cairo@eta.gov.eg
Alexandria Tax Office	Smouha, Alexandria	+203-429-5678	alex@eta.gov.eg
Giza Tax Office	Dokki, Giza	+202-3765-9012	giza@eta.gov.eg
Port Said Tax Office	Port Fouad, Port Said	+2066-332-4455	portsaid@eta.gov.eg
Suez Tax Office	Arbaeen, Suez	+2062-333-6677	suez@eta.gov.eg

B.2 GOSI Contacts

Office	Address	Phone	Email
GOSI Headquarters	126 El-Nil St., Giza	+202-3335-8000	info@gosi.gov.eg
Cairo Office	Heliopolis, Cairo	+202-2418-9000	cairo@gosi.gov.eg
Alexandria Office	El-Horreya Rd, Alexandria	+203-484-2000	alex@gosi.gov.eg

B.3 Important Tax Dates

Tax Type	Filing Deadline	Payment Due
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VAT Monthly	Within 30 days of month end	With return
Salary Tax Monthly	Within 30 days of month end	With return
Annual Income Tax	April 1st	With return
Stamp Tax	Quarterly within 45 days	With return
Withholding Tax	Within 30 days of payment	With return

B.4 GOSI Contribution Deadlines

Contribution Type	Deadline	Penalty
Monthly GOSI	15th of following month	2% per month delay
Annual Declaration	January 31st	EGP 500-5000
New Employee Registration	Within 15 days of hire	EGP 100-1000

B.5 Labor Law Compliance Calendar

Requirement	Frequency	Deadline
Working Hours Record	Daily	Ongoing
Overtime Authorization	Per occurrence	Before overtime
Leave Balance Update	Monthly	Month end
Medical Examination	Annual	As scheduled
Safety Training	Quarterly	Per schedule
Workplace Inspection	Semi-annual	June/December

Appendix C: Glossary of Terms

C.1 Accounting Terms

Accrual: Recognition of revenue or expense when earned/incurred, regardless of cash flow.

Amortization: Systematic allocation of intangible asset cost over useful life.

Asset: Economic resource owned with future benefit.

Balance Sheet: Financial statement showing assets, liabilities, and equity at a point in time.

Chart of Accounts: Organized list of all general ledger accounts.

Depreciation: Systematic allocation of tangible asset cost over useful life.

Equity: Residual interest in assets after deducting liabilities.

Expense: Cost incurred in generating revenue.

Fiscal Year: 12-month accounting period.

General Ledger: Central repository of all financial transactions.

Income Statement: Financial statement showing revenues and expenses over a period.

Journal Entry: Record of a financial transaction.

Liability: Obligation to transfer economic benefits.

Revenue: Income from normal business activities.

Trial Balance: List of all general ledger accounts and balances.

C.2 Manufacturing Terms

Bill of Materials (BOM): List of components required to produce a product.

Capacity: Maximum output rate of a resource.

Lead Time: Time from order to completion.

Master Production Schedule (MPS): Plan for production of specific items.

Material Requirements Planning (MRP): System for calculating material needs.

Routing: Sequence of operations to manufacture a product.

Work Center: Location where manufacturing operations are performed.

Work Order: Document authorizing production of specific items.

C.3 Inventory Terms

ABC Analysis: Classification of inventory by value/importance.

Cycle Counting: Regular physical inventory counting.

EOQ: Economic Order Quantity - optimal order size.

FIFO: First In, First Out inventory method.

LIFO: Last In, First Out inventory method.

Reorder Point: Inventory level triggering replenishment.

Safety Stock: Buffer inventory for demand variability.

SKU: Stock Keeping Unit - unique item identifier.

C.4 Procurement Terms

Blanket Order: Long-term purchase agreement.

RFQ: Request for Quotation.

PO: Purchase Order.

PR: Purchase Requisition.

Three-Way Match: Matching PO, receipt, and invoice.

Vendor Managed Inventory (VMI): Supplier manages customer inventory.

C.5 Sales Terms

CRM: Customer Relationship Management.

FOB: Free On Board shipping term.

Quote: Price offer to customer.

Sales Order: Customer request for products/services.

Backorder: Unfilled portion of customer order.

C.6 HR Terms

GOSI: General Organization for Social Insurance.

EOB: End of Service Benefits.

Basic Salary: Fixed component of compensation.

Gross Salary: Total before deductions.

Net Salary: Take-home pay after deductions.

C.7 Technical Terms

API: Application Programming Interface.

EDI: Electronic Data Interchange.

ERP: Enterprise Resource Planning.

SQL: Structured Query Language.

URL: Uniform Resource Locator.

UUID: Universally Unique Identifier.

XML: Extensible Markup Language.

JSON: JavaScript Object Notation.

Appendix D: Error Codes and Solutions

D.1 General Error Codes

Code	Message	Solution
ERR001	Invalid login credentials	Check username/password, reset if needed
ERR002	Session expired	Log in again
ERR003	Insufficient permissions	Contact system administrator
ERR004	Record not found	Verify record exists, check filters
ERR005	Duplicate record	Check for existing record with same key
ERR006	Required field missing	Fill in all mandatory fields
ERR007	Invalid date format	Use DD/MM/YYYY format
ERR008	Amount exceeds limit	Check authorization limits
ERR009	Period closed	Open period or use open period
ERR010	Currency not found	Check currency code

D.2 Financial Module Errors

Code	Message	Solution
FIN001	Account not found	Verify account number in chart of accounts
FIN002	Debits do not equal credits	Balance journal entry

FIN003	Period already closed	Reopen period or post to open period
FIN004	Vendor not found	Create vendor master or check vendor code
FIN005	Customer not found	Create customer master or check customer code
FIN006	Bank account not found	Verify bank account setup
FIN007	Exchange rate missing	Enter exchange rate for date
FIN008	Payment amount exceeds balance	Verify outstanding balance
FIN009	Check number already used	Use next available check number
FIN010	Tax code not found	Verify tax code setup

D.3 Inventory Module Errors

Code	Message	Solution
INV001	Material not found	Create material master or check material code
INV002	Insufficient stock	Check available quantity
INV003	Storage location not found	Verify storage location
INV004	Batch not found	Verify batch number
INV005	Serial number not found	Verify serial number
INV006	Negative stock not allowed	Enable negative stock or adjust quantity
INV007	Movement type not allowed	Check movement type configuration
INV008	Reservation not found	Verify reservation number
INV009	Count difference exceeds tolerance	Recount or approve difference
INV010	Goods receipt already posted	Check receipt status

D.4 System Errors

Code	Message	Solution
SYS001	Database connection failed	Check network, contact IT
SYS002	Server not responding	Wait and retry, contact IT
SYS003	License exceeded	Contact administrator for additional licenses
SYS004	Print spooler error	Restart print spooler service
SYS005	File not found	Verify file path and existence
SYS006	Memory error	Close other applications, restart
SYS007	Timeout error	Retry operation
SYS008	Integration error	Check integration configuration

Appendix E: Report Reference Guide

E.1 Financial Reports

Report	Code	Description	Frequency
Balance Sheet	FIN-R001	Assets, liabilities, equity	Monthly
Income Statement	FIN-R002	Revenue and expenses	Monthly
Cash Flow Statement	FIN-R003	Cash inflows and outflows	Monthly
Trial Balance	FIN-R004	All account balances	Monthly
General Ledger Detail	FIN-R005	Transaction details	As needed
AP Aging	FIN-R006	Vendor balances by age	Weekly
AR Aging	FIN-R007	Customer balances by age	Weekly
Bank Reconciliation	FIN-R008	Bank vs book balance	Monthly

E.2 Inventory Reports

Report	Code	Description	Frequency
Stock Overview	INV-R001	Current stock levels	Daily
Inventory Valuation	INV-R002	Stock value by method	Monthly
Movement History	INV-R003	All stock movements	As needed
Stock Differences	INV-R004	Physical vs system	After count
Slow Moving Items	INV-R005	Items with low turnover	Quarterly

Reorder Report	INV-R006	Items below reorder point	Weekly
ABC Analysis	INV-R007	Inventory classification	Quarterly

E.3 Sales Reports

Report	Code	Description	Frequency
Sales by Customer	SAL-R001	Revenue by customer	Monthly
Sales by Product	SAL-R002	Revenue by product	Monthly
Sales by Region	SAL-R003	Revenue by region	Monthly
Order Backlog	SAL-R004	Unfilled orders	Weekly
Delivery Performance	SAL-R005	On-time delivery rate	Monthly
Customer Profitability	SAL-R006	Profit by customer	Quarterly

E.4 HR Reports

Report	Code	Description	Frequency
Employee List	HR-R001	All employees	Monthly
Headcount Summary	HR-R002	Employee count by dept	Monthly
Leave Balance	HR-R003	Leave days remaining	Monthly
Attendance Summary	HR-R004	Attendance statistics	Monthly
Payroll Register	HR-R005	All payroll details	Monthly
GOSI Report	HR-R006	Social insurance details	Monthly
Income Tax Report	HR-R007	Tax withholding details	Monthly

Appendix F: Keyboard Shortcuts

F.1 General Navigation

Shortcut	Action
Ctrl+N	New document
Ctrl+S	Save
Ctrl+P	Print
Ctrl+F	Find
F1	Help
F3	Back
F4	Exit
F5	Refresh
F8	Execute/Post
Esc	Cancel

F.2 Data Entry

Shortcut	Action
Tab	Next field
Shift+Tab	Previous field
Ctrl+C	Copy

Ctrl+V	Paste
Ctrl+Z	Undo
Ctrl+Y	Redo
F2	Edit field
F4	Dropdown list
F7	Search/Find

F.3 Report Shortcuts

Shortcut	Action
Ctrl+R	Run report
Ctrl+E	Export to Excel
Ctrl+Shift+E	Export to PDF
Ctrl+D	Drill down
F9	Refresh data

Document Control

Version History

Version	Date	Author	Changes
1.0	January 2025	Documentation Team	Initial release
1.1	February 2025	Documentation Team	Added Manufacturing module
1.2	March 2025	Documentation Team	Added Quality Management
2.0	June 2025	Documentation Team	Major revision - all modules
2.1	September 2025	Documentation Team	Updated for Release 2025.2
3.0	January 2026	Documentation Team	Complete rewrite with new features

Approval

Role	Name	Date	Signature
Document Owner	Product Management	January 2026	_____
Technical Review	Solution Architecture	January 2026	_____
Quality Review	Quality Assurance	January 2026	_____
Final Approval	Chief Technology Officer	January 2026	_____

Distribution

This document is distributed to:

- All ERP System Users

- Customer Support Team
- Implementation Consultants
- Training Department
- Quality Assurance Team

Related Documents

- ERP System Installation Guide
- ERP System Administration Guide
- ERP System Security Guide
- ERP System Integration Guide
- ERP System API Reference
- ERP System Release Notes

Feedback

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Extended Advanced Analytics Guide

AA.1 Analytics Architecture

AA.1.1 Analytics Platform Overview

The Egyptian ERP System includes a comprehensive analytics platform that transforms raw transactional data into actionable business insights.

Platform Components

- **Data Lake:** Centralized repository for all enterprise data
- **Data Warehouse:** Structured data optimized for reporting
- **OLAP Engine:** Multi-dimensional analysis capabilities
- **In-Memory Computing:** Real-time analytics processing
- **Machine Learning:** Predictive and prescriptive analytics
- **Visualization Layer:** Dashboards and interactive reports

AA.1.2 Data Integration

Data flows from operational systems to analytics platform:

Integration Patterns

Pattern	Description	Latency
Batch ETL	Scheduled data extraction	Hours
Micro-batch	Frequent small batches	Minutes
Change Data Capture	Real-time change streaming	Seconds
Event Streaming	Event-driven updates	Milliseconds

AA.2 Financial Analytics

AA.2.1 Profitability Analysis

Multi-dimensional profitability views enable deep insight into business performance:

Analysis Dimensions

- **By Product:** Margin contribution by SKU
- **By Customer:** Customer lifetime value
- **By Region:** Geographic performance
- **By Channel:** Sales channel effectiveness
- **By Time:** Trend analysis

Profitability Metrics

Metric	Formula	Interpretation
Gross Margin %	$(\text{Revenue} - \text{COGS}) / \text{Revenue}$	Production efficiency
Contribution Margin	$\text{Revenue} - \text{Variable Costs}$	Product viability
EBITDA Margin	$\text{EBITDA} / \text{Revenue}$	Operating performance
Net Profit Margin	$\text{Net Income} / \text{Revenue}$	Overall profitability
ROIC	$\text{NOPAT} / \text{Invested Capital}$	Capital efficiency

AA.2.2 Cash Flow Analytics

Cash flow analysis helps manage liquidity and working capital:

Cash Flow Components

- **Operating Cash Flow:** Day-to-day business activities

- **Investing Cash Flow:** Capital expenditures and investments
- **Financing Cash Flow:** Debt and equity transactions
- **Free Cash Flow:** Cash available after investments

Working Capital Metrics

Metric	Formula	Target
DSO	AR / Average Daily Sales	< 45 days
DIO	Inventory / Average Daily COGS	Industry specific
DPO	AP / Average Daily Purchases	> 30 days
Cash Conversion Cycle	DSO + DIO - DPO	< 60 days

AA.3 Operational Analytics

AA.3.1 Manufacturing Analytics

Production performance metrics and optimization:

Overall Equipment Effectiveness (OEE)

OEE Calculation

$OEE = Availability \times Performance \times Quality$

Availability: Run Time / Planned Production Time

Performance: (Ideal Cycle Time x Total Count) / Run Time

Quality: Good Count / Total Count

World Class OEE: 85%

Availability: 90% | Performance: 95% | Quality: 99%

Manufacturing KPIs

KPI	Formula	Target
First Pass Yield	Good Units / Total Units	> 95%
Scrap Rate	Scrap Units / Total Units	< 2%
Rework Rate	Rework Units / Total Units	< 3%
Throughput	Units Produced / Time Period	Maximize
Cycle Time	Production Time / Units	Minimize

AA.3.2 Supply Chain Analytics

End-to-end supply chain visibility and optimization:

Supply Chain Metrics

Metric	Description	Target
Perfect Order Rate	Orders complete, accurate, on-time	> 95%
Order Cycle Time	Order to delivery time	< 5 days
Fill Rate	Orders filled from stock	> 97%
Forecast Accuracy	Actual vs forecast	> 80%
Inventory Turns	COGS / Average Inventory	Industry specific

AA.4 Predictive Analytics

AA.4.1 Demand Forecasting

Advanced forecasting using machine learning:

Forecasting Models

- **ARIMA:** Time series with seasonality
- **Prophet:** Facebook's forecasting tool
- **LSTM:** Deep learning for time series
- **XGBoost:** Gradient boosting for demand
- **Ensemble:** Combined model predictions

Forecast Inputs

Input	Description	Impact
Historical Sales	Previous period sales	High
Seasonality	Periodic patterns	High
Promotions	Planned marketing	Medium
Economic Indicators	Market conditions	Medium
Weather	Climate factors	Low-Medium

AA.4.2 Customer Analytics

Understanding customer behavior and value:

Customer Segmentation

- **RFM Analysis:** Recency, Frequency, Monetary
- **Behavioral:** Purchase patterns
- **Demographic:** Customer characteristics
- **Value-Based:** Profitability tiers

Customer Lifetime Value (CLV)

CLV Calculation

Simple CLV:

CLV = (Average Order Value x Purchase Frequency x Customer Lifespan) - Customer Acquisition Cost

Predictive CLV:

Uses machine learning to predict future purchases based on behavior patterns

AA.5 Prescriptive Analytics

AA.5.1 Optimization Models

Mathematical optimization for decision support:

Optimization Applications

Application	Objective	Constraints
Production Planning	Minimize cost	Capacity, demand
Inventory Optimization	Minimize total cost	Service level
Route Optimization	Minimize distance	Time windows
Workforce Scheduling	Minimize labor cost	Skill requirements

AA.5.2 What-If Analysis

Scenario planning and simulation:

Scenario Types

- **Best Case:** Optimistic assumptions
- **Worst Case:** Pessimistic assumptions

- **Most Likely:** Expected outcomes
- **Sensitivity:** Impact of variable changes

Extended Integration Framework Guide

INT.1 Integration Architecture

INT.1.1 Integration Patterns

The Egyptian ERP System supports multiple integration patterns:

Point-to-Point

Direct connection between two systems:

- Simple to implement
- Best for single integrations
- Becomes complex with many systems

Hub and Spoke

Central integration hub:

- Centralized management
- Reduced complexity
- Single point of control

Enterprise Service Bus (ESB)

Message-based integration infrastructure:

- Loosely coupled systems
- Protocol transformation
- Routing and orchestration

INT.1.2 Integration Protocols

Protocol	Use Case	Characteristics
REST API	Web services	Lightweight, HTTP-based
SOAP	Enterprise services	Formal contracts, WS-*
gRPC	Microservices	High performance, binary
GraphQL	Flexible queries	Client-specified data
OData	Data access	Standardized REST
MQTT	IoT messaging	Lightweight pub/sub

INT.2 API Management

INT.2.1 API Lifecycle

API Development Phases

1. **Design:** Define contract and specifications
2. **Development:** Implement API endpoints
3. **Testing:** Validate functionality
4. **Deployment:** Publish to environment
5. **Monitoring:** Track usage and performance
6. **Versioning:** Manage changes
7. **Retirement:** Deprecate old versions

INT.2.2 API Security

Authentication Methods

Method	Description	Best For
API Key	Simple key-based auth	Internal APIs
OAuth 2.0	Token-based delegation	External APIs
JWT	Self-contained tokens	Stateless auth
mTLS	Mutual TLS certificates	High security

Security Best Practices

- Use HTTPS for all communications
- Implement rate limiting
- Validate all inputs
- Log security events
- Regular security audits

INT.3 Common Integrations

INT.3.1 Banking Integration

Electronic banking and payment processing:

Integration Types

- **Host-to-Host:** Direct bank connection
- **SWIFT:** International payments
- **ACH:** Automated clearing house
- **Real-Time Payments:** Instant transfers

Banking Messages

Message Type	Purpose	Format
--------------	---------	--------

MT940	Account statement	SWIFT
MT101	Payment request	SWIFT
Pain.001	Customer credit transfer	ISO 20022
Camt.053	Bank-to-customer statement	ISO 20022

INT.3.2 E-Commerce Integration

Online sales channel connectivity:

Integration Points

- **Product Catalog:** Sync items and prices
- **Inventory:** Real-time stock levels
- **Orders:** Download customer orders
- **Customers:** Sync customer data
- **Payments:** Process transactions
- **Shipping:** Fulfillment integration

INT.3.3 CRM Integration

Customer relationship management connectivity:

Data Synchronization

Entity	Direction	Frequency
Customers	Bi-directional	Real-time
Contacts	Bi-directional	Real-time
Opportunities	Bi-directional	Hourly
Activities	Bi-directional	Real-time

Quotes	ERP to CRM	Real-time
Invoices	ERP to CRM	Daily

INT.4 Data Integration

INT.4.1 Master Data Management

Centralized master data governance:

Master Data Domains

- **Customer:** Unified customer view
- **Vendor:** Supplier information
- **Product:** Item master data
- **Employee:** HR master data
- **Chart of Accounts:** Financial structure

INT.4.2 Data Quality

Ensuring data accuracy across systems:

Data Quality Dimensions

Dimension	Description	Metric
Accuracy	Correctness of data	Error rate
Completeness	Presence of required data	Fill rate
Consistency	Uniformity across systems	Match rate
Timeliness	Data freshness	Latency
Uniqueness	No duplicates	Duplicate rate

Extended Customization Guide

CUST.1 Customization Framework

CUST.1.1 Types of Customization

Configuration

System settings without code changes:

- Company parameters
- Document numbering
- Approval workflows
- Report layouts
- User interface personalization

Extension

Add-on development using published APIs:

- Custom fields and tables
- New business logic
- Integration adapters
- Custom reports

Modification

Changes to standard code (not recommended):

- Core system changes
- Standard process overrides
- Requires upgrade consideration

CUST.1.2 Customization Best Practices

Design Principles

- **Minimize Core Changes:** Use extension points
- **Follow Standards:** Adhere to naming conventions
- **Document Everything:** Maintain change log
- **Test Thoroughly:** Unit and integration testing
- **Plan for Upgrade:** Consider future compatibility

CUST.2 Custom Field Development

CUST.2.1 User-Defined Fields

Adding custom fields to standard tables:

Field Types

Type	Description	Use Case
Text	Alphanumeric string	Descriptions, codes
Number	Numeric value	Quantities, amounts
Date	Calendar date	Dates, deadlines
Checkbox	Yes/No value	Flags, indicators
Dropdown	Predefined list	Categories, status
Lookup	Reference to table	Relationships

CUST.2.2 Field Configuration

Field Properties

- **Name:** Internal identifier

- **Label:** Display text
- **Data Type:** Storage format
- **Length:** Maximum size
- **Required:** Mandatory flag
- **Default Value:** Initial value
- **Validation:** Input rules

CUST.3 Custom Report Development

CUST.3.1 Report Design

Creating custom reports using the report designer:

Report Components

- **Data Source:** Tables, views, or queries
- **Parameters:** User input filters
- **Layout:** Visual presentation
- **Calculations:** Formulas and aggregations
- **Formatting:** Styles and conditional formatting

CUST.3.2 Report Types

Type	Description	Best For
List	Tabular data	Detailed listings
Summary	Aggregated data	Executive reports
Chart	Visual representation	Trends, comparisons
Form	Document layout	Invoices, labels
Matrix	Cross-tabulation	Pivot-style analysis

CUST.4 Workflow Customization

CUST.4.1 Workflow Design

Creating custom approval workflows:

Workflow Elements

- **Start Event:** Trigger condition
- **Tasks:** User actions required
- **Gateways:** Decision points
- **Conditions:** Routing rules
- **End Event:** Completion

CUST.4.2 Approval Rules

Rule Types

Rule	Description	Example
Amount Based	Approval by value	> EGP 10,000 to Manager
Hierarchy Based	Approval by level	Department head approval
Role Based	Approval by role	Finance approval required
Parallel	Multiple approvals	All must approve
Sequential	Ordered approvals	Manager then Director

CUST.5 Integration Customization

CUST.5.1 Custom APIs

Developing custom integration endpoints:

API Development Steps

1. Define API contract (OpenAPI/Swagger)
2. Implement business logic
3. Add authentication/authorization
4. Implement error handling
5. Create documentation
6. Deploy and monitor

CUST.5.2 Custom Connectors

Building connectors for external systems:

Connector Components

- **Connection:** System connectivity
- **Authentication:** Security credentials
- **Operations:** Available actions
- **Triggers:** Event handlers
- **Transformations:** Data mapping

Extended Security Management Guide

SEC.1 Security Framework

SEC.1.1 Defense in Depth

Multi-layered security approach protecting the ERP system:

Security Layers

Layer	Controls	Purpose
Physical	Data center security, access controls	Protect infrastructure
Network	Firewalls, IDS/IPS, segmentation	Control traffic flow
Host	OS hardening, antivirus, patching	Secure servers
Application	Authentication, authorization, input validation	Protect application
Data	Encryption, backup, classification	Protect information

SEC.1.2 Security Policies

Policy Categories

- **Access Control Policy:** Who can access what
- **Password Policy:** Authentication requirements
- **Data Classification Policy:** Information sensitivity levels
- **Incident Response Policy:** Security breach procedures
- **Acceptable Use Policy:** System usage guidelines

SEC.2 Authentication

SEC.2.1 Single Sign-On (SSO)

Centralized authentication across applications:

SSO Protocols

Protocol	Description	Use Case
SAML 2.0	Security Assertion Markup Language	Enterprise SSO
OAuth 2.0	Authorization framework	API access
OpenID Connect	Identity layer on OAuth	Web SSO
Kerberos	Network authentication	Windows domains
LDAP	Directory services	User management

SEC.2.2 Multi-Factor Authentication

Additional verification beyond password:

MFA Factors

- **Something You Know:** Password, PIN
- **Something You Have:** Token, phone, smart card
- **Something You Are:** Fingerprint, face, iris
- **Somewhere You Are:** Location-based

MFA Methods

Method	Description	Security Level
--------	-------------	----------------

SMS OTP	Text message code	Medium
Email OTP	Email code	Medium
Authenticator App	TOTP (Google, Microsoft)	High
Hardware Token	Physical device	Very High
Biometric	Fingerprint, face	High
Push Notification	Mobile app approval	High

SEC.3 Authorization

SEC.3.1 Role-Based Access Control (RBAC)

Permissions assigned through roles:

RBAC Components

- **Users:** System users
- **Roles:** Job functions
- **Permissions:** Access rights
- **Resources:** Protected objects

SEC.3.2 Attribute-Based Access Control (ABAC)

Fine-grained access based on attributes:

Attribute Types

Type	Examples
User Attributes	Department, title, location
Resource Attributes	Classification, owner, type

Environmental Attributes	Time, location, device
Action Attributes	Read, write, delete

SEC.4 Data Protection

SEC.4.1 Encryption

Protecting data at rest and in transit:

Encryption Types

Type	Implementation	Use Case
Data at Rest	AES-256 database encryption	Stored data
Data in Transit	TLS 1.3	Network communication
Data in Use	Memory encryption	Active processing
Application-Level	Column-level encryption	Sensitive fields

SEC.4.2 Data Masking

Protecting sensitive data in non-production:

Masking Techniques

- **Substitution:** Replace with fake values
- **Shuffling:** Randomize within column
- **Nulling:** Replace with null
- **Encryption:** Reversible transformation
- **Tokenization:** Replace with token

SEC.5 Audit and Compliance

SEC.5.1 Audit Logging

Comprehensive activity tracking:

Logged Events

Event Category	Examples
Authentication	Login, logout, failed attempts
Authorization	Permission changes, role assignments
Data Access	Read, write, delete operations
Configuration	System setting changes
Administrative	User management, backups

SEC.5.2 Compliance Frameworks

Applicable Standards

- **ISO 27001:** Information security management
- **SOC 2:** Service organization controls
- **GDPR:** Data protection regulation
- **PCI DSS:** Payment card security
- **SOX:** Financial reporting controls

Extended Training Program Guide

TRN.1 Training Framework

TRN.1.1 Learning Paths

Structured training by role and expertise:

Role-Based Paths

Role	Courses	Duration
End User	Navigation, Basic Transactions	2 days
Power User	Advanced Features, Reporting	5 days
Super User	Configuration, Troubleshooting	10 days
Administrator	System Admin, Security	15 days
Developer	Customization, Integration	20 days

TRN.1.2 Training Delivery Methods

Delivery Options

- **Instructor-Led:** Classroom training
- **Virtual:** Online live sessions
- **eLearning:** Self-paced modules
- **Hands-On:** Practice exercises
- **On-the-Job:** Mentored learning

TRN.2 Course Catalog

TRN.2.1 Foundation Courses

ERP Fundamentals (ERP-101)

Duration: 1 day

Audience: All users

Prerequisites: None

Topics:

- ERP concepts and benefits
- System navigation
- User interface overview
- Search and filtering
- Personalization options
- Help and support

Navigation and Basics (ERP-102)

Duration: 1 day

Audience: All users

Prerequisites: ERP-101

Topics:

- Menu structure
- Transaction codes
- Favorites and shortcuts
- Multiple sessions
- Printing and exporting

TRN.2.2 Module-Specific Courses

Financial Accounting (FIN-201)

Duration: 3 days

Audience: Finance users

Prerequisites: ERP-101, ERP-102

Topics:

- Chart of accounts
- Journal entries
- Account reconciliation
- Period-end closing
- Financial reports

Procurement (PROC-201)

Duration: 2 days

Audience: Procurement users

Prerequisites: ERP-101, ERP-102

Topics:

- Vendor management
- Purchase requisitions
- Purchase orders
- Goods receipt
- Invoice verification

Sales and Distribution (SD-201)

Duration: 2 days

Audience: Sales users

Prerequisites: ERP-101, ERP-102

Topics:

- Customer management
- Sales orders
- Delivery processing

- Billing
- Credit management

Inventory Management (INV-201)

Duration: 2 days
Audience: Warehouse users
Prerequisites: ERP-101, ERP-102

Topics:

- Material master
- Goods movements
- Physical inventory
- Stock reports
- Special stocks

HR and Payroll (HR-201)

Duration: 3 days
Audience: HR users
Prerequisites: ERP-101, ERP-102

Topics:

- Employee management
- Time management
- Leave management
- Payroll processing
- GOSI and tax

TRN.2.3 Advanced Courses

Reporting and Analytics (ADV-301)

Duration:	2	days
Audience:	Power	users

Prerequisites: Module-specific courses

Topics:

- Standard reports
- Report customization
- Query design
- Dashboard creation
- Excel integration

System Administration (ADM-401)

Duration: 5 days

Audience: Administrators

Prerequisites: Advanced courses

Topics:

- User management
- Security configuration
- System monitoring
- Backup and recovery
- Performance tuning

TRN.3 Certification Program

TRN.3.1 Certification Levels

Level	Requirements	Exam
Associate	Complete foundation courses	Online quiz
Professional	Complete module courses + project	Practical exam
Expert	2+ years experience + advanced courses	Case study
Master	5+ years experience + contribution	Peer review

TRN.3.2 Recertification

Maintaining certification currency:

- Annual continuing education requirements
- Version upgrade training
- Professional development activities

Extended Support Services Guide

SUP.1 Support Structure

SUP.1.1 Support Tiers

Tier	Scope	Response Time
Level 1	Basic queries, password resets, navigation	1 hour
Level 2	Functional issues, configuration	4 hours
Level 3	Technical issues, bugs, customization	8 hours
Level 4	Product defects, enhancement requests	24 hours

SUP.1.2 Support Channels

Contact Methods

Channel	Availability	Best For
Phone	24/7	Urgent issues
Email	24/7	Non-urgent queries
Portal	24/7	Ticket tracking
Chat	Business hours	Quick questions
Remote	Scheduled	Hands-on support

SUP.2 Incident Management

SUP.2.1 Incident Priority

Priority	Description	Response	Resolution
P1 - Critical	System down, no workaround	15 min	4 hours
P2 - High	Major functionality impaired	1 hour	8 hours
P3 - Medium	Minor functionality impaired	4 hours	24 hours
P4 - Low	Cosmetic issues, questions	8 hours	48 hours

SUP.2.2 Incident Lifecycle

1. **Detection:** Issue identified
2. **Logging:** Ticket created
3. **Categorization:** Priority assigned
4. **Investigation:** Root cause analysis
5. **Resolution:** Fix implemented
6. **Verification:** Solution confirmed
7. **Closure:** Ticket closed

SUP.3 Maintenance Services

SUP.3.1 Preventive Maintenance

Scheduled Activities

Activity	Frequency	Purpose
Health Check	Monthly	System performance review
Patch Review	Quarterly	Security and bug fixes

Data Cleanup	Quarterly	Optimize database
Capacity Review	Semi-annual	Growth planning

SUP.3.2 System Updates

Update Types

- **Hotfixes:** Critical fixes, immediate deployment
- **Patches:** Bug fixes and minor enhancements
- **Service Packs:** Cumulative updates
- **Version Upgrades:** Major releases

SUP.4 Knowledge Base

SUP.4.1 Knowledge Articles

Article Categories

Category	Content
How-To	Step-by-step instructions
Troubleshooting	Problem resolution
FAQ	Frequently asked questions
Best Practices	Recommended approaches
Reference	Technical details

SUP.4.2 Community Forum

User community for knowledge sharing:

- Q&A discussions

- Tips and tricks
- User-contributed solutions
- Feature requests

SUP.5 Service Level Agreement

SUP.5.1 SLA Metrics

Metric	Target	Measurement
Availability	99.9%	Monthly uptime
First Response	Per priority	Time to acknowledge
Resolution Time	Per priority	Time to resolve
First Contact Resolution	70%	Issues resolved at L1
Customer Satisfaction	4.5/5	Post-incident survey

Extended Industry Solutions Guide

IND.1 Manufacturing Industry

IND.1.1 Discrete Manufacturing

Solutions for discrete manufacturing industries:

Key Features

- Engineering change management
- Product configuration
- Make-to-order processing
- Assemble-to-order processing
- Serial number tracking
- Quality inspection integration

Industry-Specific Processes

Process	Description
Engineer-to-Order	Design and build custom products
Configure-to-Order	Configure standard options
Make-to-Stock	Build to forecast
Make-to-Order	Build on customer order

IND.1.2 Process Manufacturing

Solutions for process industries:

Key Features

- Batch management
- Recipe/formula management
- Yield tracking
- Co-product and by-product handling
- Potency management
- Shelf-life tracking

Process Industry Types

Industry	Characteristics
Chemicals	Formula-based, hazardous materials
Food and Beverage	Shelf-life, traceability
Pharmaceuticals	Validation, compliance
Metals	Unit of measure conversions

IND.2 Retail Industry

IND.2.1 Retail Features

Point of Sale Integration

- Real-time inventory lookup
- Price and promotion management
- Customer loyalty programs
- Multi-channel sales
- Returns management

Merchandising

Function	Description
Assortment Planning	Product selection by store
Allocation	Distribution to stores
Replenishment	Auto-reorder triggers
Price Management	Markdown optimization

IND.3 Distribution Industry

IND.3.1 Wholesale Distribution

Key Features

- Multi-warehouse management
- Drop shipment processing
- Vendor managed inventory
- Rebate management
- Commodity pricing
- Container tracking

Distribution Processes

Process	Description
Cross-Docking	Direct flow-through
Value-Added Services	Kitting, labeling
Third-Party Logistics	3PL integration
Import/Export	International trade

IND.4 Construction Industry

IND.4.1 Project-Based Manufacturing

Key Features

- Project accounting
- Work breakdown structure
- Progress billing
- Retention management
- Subcontractor management
- Equipment tracking

Construction-Specific Functions

Function	Description
Job Costing	Project cost tracking
Certified Payroll	Union wage reporting
AIA Billing	Progress billing format
Lien Management	Legal hold tracking

IND.5 Healthcare Industry

IND.5.1 Healthcare Features

Key Features

- Patient billing integration
- Insurance claims processing

- Medical supply tracking
- Equipment maintenance
- Regulatory compliance
- HIPAA security

Healthcare Processes

Process	Description
Revenue Cycle	Patient to payment
Supply Chain	Medical supplies
Asset Management	Equipment lifecycle
Compliance	Regulatory reporting

IND.6 Service Industry

IND.6.1 Professional Services

Key Features

- Project management
- Resource planning
- Time and expense tracking
- Project billing
- Revenue recognition
- Utilization tracking

Service Processes

Process	Description
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Project Accounting	Revenue and cost tracking
Resource Management	Staff allocation
Contract Management	Service agreements
Field Service	On-site service delivery

Extended Performance Optimization Guide

PERF.1 Performance Monitoring

PERF.1.1 Key Performance Metrics

Metric	Description	Target
Response Time	Transaction completion time	< 3 seconds
Throughput	Transactions per minute	Per SLA
Concurrent Users	Simultaneous active users	Per license
CPU Utilization	Processor usage	< 70%
Memory Usage	RAM utilization	< 80%
Disk I/O	Storage operations	< 50%
Network Latency	Communication delay	< 50ms

PERF.1.2 Monitoring Tools

Built-in Monitoring

- System performance dashboard
- Transaction statistics
- User activity monitoring
- Error logging and alerts
- Database performance metrics

External Monitoring

- Application Performance Monitoring (APM)
- Infrastructure monitoring
- Database monitoring tools
- Network monitoring

PERF.2 Database Optimization

PERF.2.1 Index Optimization

Index Types

Type	Use Case	Considerations
Clustered	Primary key lookups	One per table
Non-Clustered	Secondary lookups	Multiple allowed
Composite	Multi-column queries	Column order matters
Covering	Include all query columns	Larger size
Filtered	Subset of data	Reduced maintenance

PERF.2.2 Query Optimization

Best Practices

- Use appropriate indexes
- Avoid SELECT *
- Use JOINS instead of subqueries
- Limit result sets
- Use parameterized queries

- Update statistics regularly

PERF.2.3 Database Maintenance

Maintenance Tasks

Task	Frequency	Purpose
Index Rebuild	Weekly	Defragment indexes
Statistics Update	Daily	Optimize query plans
Log Backup	Hourly	Transaction log management
Integrity Check	Weekly	Data consistency
Data Cleanup	Monthly	Remove obsolete data

PERF.3 Application Optimization

PERF.3.1 Code Optimization

Best Practices

- Minimize database round trips
- Use caching effectively
- Optimize loops and iterations
- Avoid unnecessary object creation
- Use asynchronous processing

PERF.3.2 Caching Strategies

Cache Type	Scope	Duration
Application Cache	Server	Session

Data Cache	Database	Configurable
Output Cache	Page/Report	Per content
Browser Cache	Client	Per resource

PERF.4 Infrastructure Optimization

PERF.4.1 Server Configuration

Hardware Considerations

Component	Recommendation
CPU	Multi-core, high clock speed
Memory	Sufficient for database + application
Storage	SSD for database, RAID configuration
Network	Gigabit Ethernet minimum

PERF.4.2 Load Balancing

Load Balancing Methods

- **Round Robin:** Equal distribution
- **Least Connections:** To least busy server
- **IP Hash:** Based on client IP
- **Weighted:** Based on server capacity

PERF.5 User Experience Optimization

PERF.5.1 UI Performance

Optimization Techniques

- Lazy loading of data
- Pagination for large lists
- Asynchronous operations
- Progressive rendering
- Client-side caching

PERF.5.2 Report Optimization

Best Practices

Practice	Benefit
Use filters	Reduce data volume
Schedule reports	Off-peak execution
Aggregate data	Pre-calculated summaries
Limit columns	Reduce transfer size
Use drill-down	Progressive detail

Extended Disaster Recovery Guide

DR.1 Business Continuity Planning

DR.1.1 Business Impact Analysis

Criticality Levels

Level	Description	RTO	RPO
Critical	Business cannot operate	4 hours	1 hour
High	Significant impact	8 hours	4 hours
Medium	Moderate impact	24 hours	8 hours
Low	Minimal impact	72 hours	24 hours

RTO: Recovery Time Objective - Maximum acceptable downtime

RPO: Recovery Point Objective - Maximum acceptable data loss

DR.1.2 Recovery Strategies

Strategy Options

Strategy	Description	Cost	RTO
Cold Site	Basic facility, no equipment	Low	Weeks
Warm Site	Partial equipment ready	Medium	Days
Hot Site	Fully equipped standby	High	Hours
Cloud DR	Cloud-based recovery	Variable	Minutes-Hours

DR.2 Backup Strategies

DR.2.1 Backup Types

Type	Description	Frequency	Retention
Full	Complete database backup	Weekly	4 weeks
Differential	Changes since last full	Daily	2 weeks
Transaction Log	Individual transactions	15 minutes	48 hours
Snapshot	Point-in-time copy	As needed	Per policy

DR.2.2 Backup Storage

Storage Options

- **Local Disk:** Fast recovery, same site risk
- **Network Storage:** Centralized, shared access
- **Tape:** Long-term archive, offsite capable
- **Cloud:** Geographic redundancy, scalable
- **Hybrid:** Combination of above

3-2-1 Backup Rule

- 3 copies of data
- 2 different media types
- 1 offsite copy

DR.3 Recovery Procedures

DR.3.1 Recovery Scenarios

Scenario Types

Scenario	Description	Recovery Action
Single File	Accidental deletion	Restore from backup
Database Corruption	Data integrity issue	Point-in-time restore
Server Failure	Hardware malfunction	Failover to standby
Site Disaster	Complete facility loss	Activate DR site
Ransomware	Malware encryption	Clean restore + security

DR.3.2 Recovery Testing

Test Types

- **Tabletop:** Walkthrough of procedures
- **Partial:** Test specific components
- **Full:** Complete system recovery
- **Failover:** Test automatic failover

Test Schedule

Test	Frequency
Backup Verification	Daily
Restore Test	Monthly
Failover Test	Quarterly
Full DR Test	Annually

DR.4 High Availability

DR.4.1 HA Architectures

Clustering Options

Type	Description	Availability
Active-Passive	Standby takes over on failure	99.9%
Active-Active	Both nodes process requests	99.99%
N+1	One spare for multiple active	99.95%
Geographic	Multiple data centers	99.999%

DR.4.2 Database Replication

Replication Types

- **Synchronous:** Zero data loss, higher latency
- **Asynchronous:** Some data loss possible, lower latency
- **Snapshot:** Point-in-time copies
- **Transactional:** Continuous log shipping

Extended Egyptian Regulatory Compliance Guide

REG.1 Egyptian Tax Authority (ETA) Compliance

REG.1.1 E-Invoicing Requirements

Mandatory Fields

Field	Description	Format
UUID	Unique invoice identifier	UUID v4
Invoice Number	Sequential number	Numeric
Issue Date	Invoice date	YYYY-MM-DD
Issue Time	Invoice time	HH:MM:SS
Seller Tax ID	Seller registration	9 digits
Buyer Tax ID	Buyer registration	9 digits
Total Amount	Invoice total	Decimal (2)
Tax Amount	VAT amount	Decimal (2)

QR Code Contents

The QR code must contain:

- Seller name
- Seller tax registration number
- Invoice timestamp

- Invoice total (with VAT)
- VAT total
- Digital signature

REG.1.2 ETA Integration Process

Step-by-Step Process

1. **Invoice Creation:** Generate invoice in ERP
2. **Validation:** Validate against ETA schema
3. **Signing:** Apply digital signature
4. **Submission:** Send to ETA platform
5. **Receipt:** Receive ETA validation
6. **QR Generation:** Generate QR code
7. **Storage:** Archive with UUID

REG.1.3 ETA Error Handling

Error Code	Description	Resolution
E001	Invalid UUID format	Regenerate UUID
E002	Duplicate invoice number	Use next sequence
E003	Invalid tax ID	Verify registration
E004	Schema validation failed	Check required fields
E005	Signature invalid	Regenerate signature
E006	Connection timeout	Retry submission

REG.2 GOSI Compliance

REG.2.1 Contribution Categories

Category	Employer %	Employee %	Total %
Old Age, Disability, Death	15%	11%	26%
Work Injuries	2%	0%	2%
Unemployment	1%	1%	2%
Total	18%	12%	30%

REG.2.2 GOSI Registration Process

Employer Registration

1. Submit commercial registration
2. Provide tax card
3. Submit establishment contract
4. Provide bank account details
5. Receive GOSI employer number

Employee Registration

1. Submit employee national ID
2. Provide employment contract
3. Submit medical examination
4. Receive GOSI employee number

REG.2.3 Monthly GOSI Process

Monthly Steps

1. Calculate eligible wages per employee

- 2. Apply contribution percentages
- 3. Generate GOSI form
- 4. Submit to GOSI portal
- 5. Make payment by 15th
- 6. Archive payment receipt

REG.3 Labor Law Compliance

REG.3.1 Working Hours (Law 14/2025)

Category	Regular Hours	Overtime Limit
Standard	8 hours/day, 48 hours/week	10 hours/week
Shift Work	8 hours/shift	As per agreement
Seasonal	Per agreement	Per agreement

REG.3.2 Leave Entitlements

Leave Type	Entitlement	Conditions
Annual Leave	21 days (1-10 years), 30 days (10+ years)	Full year service
Sick Leave	Up to 6 months	Medical certificate
Maternity Leave	90 days (4 before, 86 after)	Medical certificate
Paternity Leave	3 days	Within 1 month of birth
Emergency Leave	Up to 6 days/year	With justification
Hajj Leave	30 days (once in tenure)	5+ years service

REG.3.3 Termination Rules

End of Service Benefits

EOSB Calculation

For Limited Contract:

- Less than 5 years: 21 days x years of service
- 5+ years: 21 days x 5 years + 30 days x remaining years

For Unlimited Contract (Resignation):

- 2-5 years: 1/3 of entitlement
- 5-10 years: 2/3 of entitlement
- 10+ years: Full entitlement

Maximum: 2 years' wages

REG.4 VAT Compliance

REG.4.1 VAT Rates

Rate	Applies To
14%	Standard goods and services
5%	Specific goods (table sugar, cooking oil)
0%	Exports, international transport
Exempt	Financial services, healthcare, education

REG.4.2 VAT Registration

Registration Thresholds

- **Mandatory:** Annual turnover EGP 500,000+

- **Voluntary:** Annual turnover below threshold

Registration Process

1. Submit application on ETA portal
2. Provide commercial registration
3. Submit tax card
4. Provide bank account details
5. Receive VAT registration certificate

REG.4.3 VAT Return Filing

Monthly Process

1. Calculate output VAT (sales)
2. Calculate input VAT (purchases)
3. Determine VAT payable/refundable
4. Complete VAT return form
5. Submit within 30 days of month end
6. Make payment if applicable

REG.5 Income Tax Compliance

REG.5.1 Tax Brackets (2025)

Annual Income (EGP)	Rate
0 - 30,000	0%
30,001 - 45,000	10%
45,001 - 60,000	15%
60,001 - 200,000	20%

200,001 - 400,000	22.5%
Above 400,000	25%

REG.5.2 Tax Exemptions

Personal Exemptions

- Personal exemption: EGP 15,000 annually
- Social insurance contributions
- Life insurance premiums (up to EGP 3,000)
- Health insurance premiums (up to EGP 4,000)
- Professional subscriptions

REG.5.3 Monthly Tax Process

Employer Responsibilities

1. Calculate monthly taxable income
2. Apply tax brackets
3. Deduct tax from salary
4. Submit Form 1 by month end + 30 days
5. Remit deducted tax
6. Issue annual tax certificate

REG.6 Commercial Registry Compliance

REG.6.1 Registration Requirements

Required Documents

- Company incorporation documents
- Articles of association

- Board resolution
- Lease agreement or ownership proof
- Tax card

REG.6.2 Annual Renewal

Renewal Process

1. Submit renewal application
2. Pay renewal fees
3. Update any changes
4. Receive renewed certificate

REG.7 Import/Export Compliance

REG.7.1 Customs Requirements

Import Documents

- Commercial invoice
- Packing list
- Bill of lading
- Certificate of origin
- Import license (if applicable)

REG.7.2 Customs Duties

Category	Duty Rate
Raw Materials	0-5%
Capital Goods	5-10%

Consumer Goods	10-40%
Luxury Items	40%+

REG.8 Data Protection Compliance

REG.8.1 Personal Data Protection

Key Requirements

- Obtain consent for data collection
- Limit data to necessary information
- Implement security measures
- Allow data subject access
- Report data breaches
- Appoint data protection officer (if applicable)

REG.8.2 Employee Data Rights

Employee Rights

Right	Description
Access	View personal data
Correction	Request data updates
Deletion	Request data removal
Portability	Receive data in portable format
Objection	Object to processing

Extended Troubleshooting Encyclopedia

TROUBLE.1 Login Issues

TROUBLE.1.1 Cannot Log In

Symptoms

- Error message: "Invalid credentials"
- Error message: "Account locked"
- Error message: "Session expired"

Diagnosis Steps

1. Verify username is correct
2. Check Caps Lock is off
3. Try password reset
4. Check account status
5. Verify network connectivity

Solutions

Cause	Solution
Wrong password	Use password reset function
Account locked	Contact administrator to unlock
Expired password	Change password at login
Account disabled	Contact HR/Admin

System down	Wait and retry, contact IT
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TRUBLE.2 Performance Issues

TRUBLE.2.1 Slow Response

Symptoms

- Pages take long to load
- Transactions timeout
- Reports run slowly

Diagnosis Steps

1. Check network connection
2. Verify system status
3. Check concurrent users
4. Review recent changes
5. Monitor resource usage

Solutions

Cause	Solution
Network issue	Check connection, restart router
High load	Try during off-peak hours
Large data set	Add filters, limit date range
Missing index	Contact administrator
Client issue	Clear cache, restart browser

TROUBLE.3 Data Issues

TROUBLE.3.1 Missing Data

Symptoms

- Records not appearing
- Incomplete information
- Blank fields

Diagnosis Steps

1. Check filter settings
2. Verify date range
3. Check authorization
4. Review data entry
5. Check for data corruption

Solutions

Cause	Solution
Active filter	Clear all filters
Wrong date range	Expand date range
No access	Request access from admin
Data not entered	Enter missing data
Wrong organization	Switch to correct company

TROUBLE.4 Printing Issues

TROUBLE.4.1 Cannot Print

Symptoms

- Print button not working
- Print job stuck in queue
- Blank pages printed

Diagnosis Steps

1. Check printer connection
2. Verify printer status
3. Check print queue
4. Test with different document
5. Check printer drivers

Solutions

Cause	Solution
Printer offline	Turn printer on, check connection
Out of paper	Add paper to tray
Out of toner	Replace toner cartridge
Job stuck	Clear print queue
Driver issue	Update printer driver

TROUBLE.5 Integration Issues

TROUBLE.5.1 API Errors

Common HTTP Error Codes

Code	Meaning	Solution
400	Bad Request	Check request format
401	Unauthorized	Check authentication
403	Forbidden	Check permissions
404	Not Found	Verify endpoint URL
500	Server Error	Contact support
503	Service Unavailable	Wait and retry

TROUBLE.6 Module-Specific Issues

TROUBLE.6.1 Financial Module

Common Issues

Issue	Cause	Solution
Cannot post journal	Unbalanced entry	Ensure debits = credits
Period closed	Accounting period locked	Open period or use open period
Account not found	Invalid account number	Verify account exists
Currency error	Missing exchange rate	Enter exchange rate

TROUBLE.6.2 Inventory Module

Common Issues

Issue	Cause	Solution
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Insufficient stock	Not enough quantity	Check available stock
Negative stock	Not allowed	Enable in settings or adjust
Cost not found	Missing material ledger	Run cost calculation
Reservation conflict	Reserved quantity	Check reservations

TROUBLE.6.3 HR Module

Common Issues

Issue	Cause	Solution
Payroll error	Missing data	Complete employee data
GOSI mismatch	Wrong contribution	Verify GOSI setup
Leave error	Insufficient balance	Check leave balance
Tax calculation wrong	Wrong tax code	Verify tax configuration

Extended FAQ Encyclopedia

FAQ.1 General Questions

Q: What is the Egyptian ERP System?

The Egyptian ERP System is a comprehensive enterprise resource planning solution designed specifically for Egyptian businesses. It integrates all core business processes including finance, procurement, sales, inventory, manufacturing, and human resources into a single unified platform with full compliance to Egyptian regulations.

Q: What are the system requirements?

Server Requirements:

- Windows Server 2019 or later / Linux RHEL 8+
- 16 GB RAM minimum (32 GB recommended)
- 100 GB free disk space (SSD recommended)
- SQL Server 2019 / PostgreSQL 13+ / Oracle 19c

Client Requirements:

- Windows 10/11, macOS 11+, or Linux
- 8 GB RAM minimum
- Chrome, Firefox, Edge, or Safari browser
- Internet connection for cloud deployment

Q: How do I get started with the system?

To get started:

1. Complete user account setup with your administrator
2. Attend orientation training for your role
3. Review the Quick Start Guide for your module

4. Practice in the training environment
5. Start with basic transactions under supervision

FAQ.2 Financial Module

Q: How do I create a journal entry?

To create a journal entry:

1. Navigate to Financial Accounting > General Ledger > Journal Entry
2. Click "Create New" button
3. Enter document date and posting date
4. Add line items with account numbers, amounts, and descriptions
5. Ensure debits equal credits
6. Click "Simulate" to preview
7. Click "Post" to record the entry

Q: How do I close an accounting period?

To close a period:

1. Complete all transactions for the period
2. Run all required reports
3. Perform account reconciliations
4. Navigate to Financial Accounting > General Ledger > Period Close
5. Select the period to close
6. Review the checklist and confirm
7. Execute period close

Note: Once closed, periods cannot be reopened without administrator approval.

Q: How do I handle foreign currency transactions?

For foreign currency transactions:

1. Ensure the currency is configured in the system
2. Enter the transaction in the foreign currency
3. The system will use the exchange rate for the posting date
4. For valuation, run the foreign currency valuation program
5. Post any exchange rate differences

FAQ.3 Procurement Module

Q: What is the difference between a purchase requisition and purchase order?

Purchase Requisition: Internal document requesting procurement. Used for approval workflow before creating PO.

Purchase Order: Legal document sent to vendor. Commitment to purchase goods/services at specified terms.

Q: How do I process a goods receipt?

To process goods receipt:

1. Navigate to Procurement > Goods Receipt
2. Reference the purchase order or create unplanned
3. Enter material quantities received
4. Check for quality issues
5. Enter batch/serial numbers if applicable
6. Post the goods receipt
7. Print or email GRN to relevant parties

Q: How do I handle vendor returns?

To process vendor returns:

1. Create return purchase order or reference original PO

2. Enter materials and quantities to return
3. Specify return reason
4. Post goods issue for return
5. Process credit memo when received

FAQ.4 Sales Module

Q: How do I create a sales order?

To create a sales order:

1. Navigate to Sales > Sales Order
2. Select or create customer
3. Enter order details (date, PO reference)
4. Add line items with materials and quantities
5. System calculates pricing based on conditions
6. Check availability and delivery dates
7. Save and release order

Q: How do I handle customer complaints?

To handle complaints:

1. Log complaint in CRM or customer service module
2. Categorize by type and severity
3. Assign to responsible person
4. Investigate root cause
5. Determine resolution (credit, replacement, refund)
6. Process appropriate transaction
7. Close complaint with customer confirmation

Q: How does credit management work?

Credit management process:

1. Set credit limit in customer master
2. System checks credit exposure on new orders
3. If exceeded, order blocks for credit review
4. Credit analyst reviews and decides
5. Approve, reject, or request partial payment
6. Update credit limit if needed

FAQ.5 Inventory Module

Q: How do I perform a physical inventory count?

To perform physical count:

1. Create physical inventory document
2. Select storage locations to count
3. Print count sheets or use mobile devices
4. Count physical stock
5. Enter count results in system
6. Review differences
7. Post inventory adjustments

Q: What is the difference between goods issue and goods receipt?

Goods Receipt: Records incoming materials (increases inventory)

Goods Issue: Records outgoing materials (decreases inventory)

Q: How do I handle stock transfers between warehouses?

To transfer stock:

1. Create transfer order

2. Specify source and destination locations
3. Enter materials and quantities
4. Post goods issue from source
5. Post goods receipt at destination
6. Confirm transfer completion

FAQ.6 HR Module

Q: How do I process monthly payroll?

Payroll process:

1. Verify all time data is entered
2. Process absences and leaves
3. Run payroll simulation
4. Review and correct any errors
5. Execute payroll run
6. Generate payslips
7. Process bank transfers
8. Generate statutory reports (GOSI, tax)

Q: How is GOSI calculated?

GOSI calculation:

- Basic salary + Variable allowances = Contributory wage
- Subject to maximum contributory wage ceiling
- Employer contribution: 18% (15% pension + 2% injuries + 1% unemployment)
- Employee contribution: 12% (11% pension + 1% unemployment)
- Total: 30% of contributory wage

Q: How do I calculate end of service benefits?

EOSB calculation depends on contract type:

Limited Contract:

- 21 days per year for first 5 years
- 30 days per year after 5 years
- Based on last basic salary

Unlimited Contract (Resignation):

- 2-5 years: 1/3 of entitlement
- 5-10 years: 2/3 of entitlement
- 10+ years: Full entitlement

FAQ.7 Technical Questions

Q: How do I reset my password?

To reset password:

1. Click "Forgot Password" on login screen
2. Enter your username or email
3. Check email for reset link
4. Click link and enter new password
5. Login with new password

Note: Contact administrator if email not received.

Q: How do I export data to Excel?

To export to Excel:

1. Run report or open data grid
2. Click "Export" button
3. Select "Excel" format
4. Choose export options (all/filtered data)

5. Download file

Q: Can I customize my dashboard?

Yes, you can customize:

1. Click "Customize Dashboard"
2. Add/remove widgets
3. Drag to rearrange
4. Configure widget settings
5. Save layout

FAQ.8 Compliance Questions

Q: How does E-Invoicing work?

E-Invoicing process:

1. Create invoice in ERP
2. System generates UUID and QR code
3. Submit to ETA platform
4. Receive validation response
5. Include QR code on printed invoice

Q: What VAT rate should I use?

VAT rates in Egypt:

- **14%** - Standard rate for most goods and services
- **5%** - Reduced rate for specific goods (sugar, oil)
- **0%** - Exports and international transport
- **Exempt** - Financial services, healthcare, education

Q: How do I file monthly tax returns?

Monthly tax filing:

1. Calculate tax withheld from salaries
2. Complete Form 1 on ETA portal
3. Submit within 30 days of month end
4. Make payment if tax due
5. Archive filing confirmation

Extended Best Practices Guide

BP.1 Data Entry Best Practices

BP.1.1 Accuracy

- Double-check all entries before saving
- Use system validations to catch errors
- Verify master data exists before using
- Enter descriptive text for clarity
- Review posted documents for accuracy

BP.1.2 Consistency

- Follow naming conventions
- Use standard formats for dates and numbers
- Apply consistent categorization
- Maintain uniform descriptions
- Use approved codes and classifications

BP.1.3 Timeliness

- Enter transactions on daily basis
- Process documents within SLA
- Close periods on schedule
- Respond to alerts promptly
- Complete month-end tasks timely

BP.2 Security Best Practices

BP.2.1 Password Management

- Use strong passwords (12+ characters)
- Change passwords regularly
- Never share passwords
- Use different passwords for different systems
- Enable multi-factor authentication

BP.2.2 Session Management

- Lock screen when away
- Log out at end of day
- Don't leave sessions unattended
- Close unused windows
- Report suspicious activity

BP.2.3 Data Protection

- Don't export sensitive data without approval
- Secure printed documents
- Shred confidential waste
- Encrypt files when emailing
- Follow data classification guidelines

BP.3 Financial Best Practices

BP.3.1 Month-End Close

- Prepare checklist in advance
- Complete all entries before close
- Reconcile all accounts

- Review variances
- Document adjustments
- Obtain approvals
- Close period promptly

BP.3.2 Reconciliation

- Reconcile bank accounts daily
- Reconcile AP and AP subledgers
- Reconcile AR and AR subledgers
- Reconcile inventory accounts
- Investigate differences immediately
- Document reconciliation procedures

BP.3.3 Audit Preparation

- Maintain supporting documentation
- Keep audit trail intact
- Organize files systematically
- Prepare standard reports
- Review prior year findings
- Coordinate with auditors early

BP.4 Inventory Best Practices

BP.4.1 Stock Management

- Maintain accurate stock records
- Perform regular cycle counts
- Investigate variances promptly
- Rotate stock (FIFO)

- Monitor slow-moving items
- Set appropriate reorder points

BP.4.2 Warehouse Organization

- Use clear location labeling
- Organize by velocity
- Maintain clean aisles
- Secure high-value items
- Implement barcode scanning
- Train staff on procedures

BP.5 Procurement Best Practices

BP.5.1 Vendor Management

- Maintain accurate vendor master
- Evaluate vendors regularly
- Negotiate favorable terms
- Diversify supplier base
- Monitor vendor performance
- Document communications

BP.5.2 Purchase Process

- Obtain proper approvals
- Use competitive bidding
- Document requirements clearly
- Review terms and conditions
- Verify receipt before payment
- Resolve disputes promptly

BP.6 Sales Best Practices

BP.6.1 Customer Management

- Maintain complete customer records
- Monitor credit limits
- Follow up on overdue accounts
- Respond to inquiries promptly
- Document customer interactions
- Build strong relationships

BP.6.2 Order Processing

- Verify order accuracy
- Check product availability
- Confirm delivery dates
- Communicate delays proactively
- Ship complete orders
- Follow up on delivery

BP.7 HR Best Practices

BP.7.1 Employee Data

- Keep employee records current
- Verify personal information
- Maintain confidentiality
- Document changes
- Archive terminated records
- Comply with data protection

BP.7.2 Payroll Processing

- Verify time data accuracy
- Check for missing information
- Run simulation before final
- Review exception reports
- Obtain approvals
- Process on schedule

BP.8 System Usage Best Practices

BP.8.1 Navigation

- Use favorites for common transactions
- Learn keyboard shortcuts
- Use search effectively
- Customize home page
- Use multiple sessions wisely

BP.8.2 Reporting

- Use filters to limit data
- Schedule reports for off-hours
- Save report variants
- Export for further analysis
- Share reports appropriately

BP.8.3 Troubleshooting

- Check error messages carefully
- Verify data entry

- Clear cache if issues persist
- Restart browser if needed
- Contact support with details

Extended Case Studies

CS.1 Manufacturing Company Implementation

Company Background

Industry: Consumer Goods Manufacturing
Size: 500 employees
Location: 10th of Ramadan City, Egypt
Challenge: Disconnected systems, manual processes, compliance issues

Implementation Scope

- Financial Management (GL, AP, AR, Fixed Assets)
- Procurement and Inventory
- Manufacturing (Production Planning, MRP)
- Sales and Distribution
- Quality Management
- HR and Payroll

Key Challenges

Challenge	Solution
Multi-warehouse inventory	Implemented warehouse management with barcode scanning
Complex BOM structures	Configured multi-level BOMs with routing
Quality traceability	Implemented batch management and quality inspections
ETA E-Invoicing	Integrated with ETA platform for compliance
GOSI reporting	Automated monthly GOSI calculations and reporting

Results

Metric	Before	After	Improvement
Month-end close	15 days	3 days	80% faster
Inventory accuracy	85%	98%	13% improvement
Order fulfillment	75%	95%	20% improvement
Procurement cycle	10 days	4 days	60% faster

CS.2 Retail Chain Implementation

Company Background

Industry: Retail (15 stores)

Size: 800 employees

Location: Cairo and Alexandria

Challenge: Inventory visibility, multi-channel sales, customer management

Implementation Scope

- Point of Sale Integration
- Inventory Management (Multi-location)
- Sales and Distribution
- Customer Relationship Management
- Procurement
- Financial Management
- HR and Payroll

Key Challenges

Challenge	Solution
Real-time inventory visibility	Centralized inventory with store-level tracking
Inter-store transfers	Automated transfer order workflow
Customer loyalty program	Integrated CRM with points management
Promotions management	Flexible pricing and promotion engine
E-commerce integration	API-based integration with online store

Results

Metric	Before	After	Improvement
Stock-outs	15%	3%	80% reduction
Inventory carrying cost	High	Optimized	25% reduction
Customer retention	60%	78%	18% improvement
Staff productivity	Baseline	Improved	30% increase

CS.3 Distribution Company Implementation

Company Background

Industry: Wholesale Distribution

Size: 300 employees

Location: Cairo with 5 regional warehouses

Challenge: Complex pricing, rebate management, 3PL integration

Implementation Scope

- Sales with Complex Pricing

- Inventory Management (Multi-warehouse)
- Procurement
- Financial Management
- Rebate Management
- 3PL Integration

Key Challenges

Challenge	Solution
Complex pricing structures	Implemented condition-based pricing
Vendor rebates	Automated rebate accrual and settlement
3PL management	EDI integration with logistics providers
Customer-specific pricing	Customer price lists with validity periods
Commission calculation	Automated sales commission engine

Results

Metric	Before	After	Improvement
Order processing time	2 hours	15 minutes	87% faster
Pricing errors	5%	0.2%	96% reduction
Rebate tracking	Manual	Automated	100% accuracy
Delivery accuracy	92%	99%	7% improvement

CS.4 Service Company Implementation

Company Background

Industry: Professional Services (Consulting)
Size: 150 employees
Location: Cairo
Challenge: Project accounting, time tracking, resource planning

Implementation Scope

- Project Management
- Project Accounting
- Time and Expense
- Resource Management
- Financial Management
- HR and Payroll

Key Challenges

Challenge	Solution
Project profitability tracking	Real-time project cost and revenue tracking
Time capture	Mobile time entry with approval workflow
Resource allocation	Resource planning with availability check
Milestone billing	Automated milestone-based invoicing
Revenue recognition	Percentage completion method

Results

Metric	Before	After	Improvement
Time capture accuracy	75%	98%	23% improvement
Billing cycle	Monthly	Bi-weekly	50% faster

Project visibility	Limited	Real-time	Complete
Resource utilization	65%	82%	17% improvement

CS.5 Construction Company Implementation

Company Background

Industry: Construction and Contracting

Size: 1,200 employees

Location: Multiple project sites across Egypt

Challenge: Project costing, subcontractor management, progress billing

Implementation Scope

- Project Management
- Project Accounting (WBS)
- Procurement
- Subcontractor Management
- Equipment Management
- Financial Management
- HR and Payroll

Key Challenges

Challenge	Solution
Multi-project costing	Work breakdown structure with cost tracking
Progress billing	AIA-style progress billing with retention
Subcontractor payments	Subcontractor management with lien tracking
Equipment tracking	Equipment master with utilization tracking

Site payroll	Mobile time capture for site workers
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Results

Metric	Before	After	Improvement
Project cost visibility	Monthly	Daily	Real-time
Billing accuracy	90%	99%	9% improvement
Subcontractor disputes	15/month	2/month	87% reduction
Equipment utilization	Unknown	Tracked	20% improvement

Extended System Configuration Reference

CFG.1 Company Configuration

CFG.1.1 Basic Company Settings

Setting	Description	Example
Company Code	Unique identifier	EG01
Company Name	Legal name	Egyptian Trading Co.
Address	Registered address	123 Nile St, Cairo
Tax ID	Tax registration	123456789
Commercial Reg	Commercial registration	98765
Base Currency	Functional currency	EGP
Fiscal Year	Accounting year	Calendar year

CFG.1.2 Chart of Accounts Setup

Account Structure

Level	Digits	Description
Company	4	Entity code
Account	6	GL account number
Cost Center	4	Department/cost center

Profit Center

4

Profit center

Sample Account Ranges

Range	Account Type
100000 - 199999	Assets
200000 - 299999	Liabilities
300000 - 399999	Equity
400000 - 499999	Revenue
500000 - 599999	Cost of Sales
600000 - 699999	Operating Expenses
700000 - 799999	Other Income/Expense

CFG.2 Document Numbering

CFG.2.1 Number Range Configuration

Document Type	Range	Format
Journal Entry	100000000 - 199999999	JE-YYYY-NNNNNNNN
Vendor Invoice	200000000 - 299999999	VI-YYYY-NNNNNNNN
Customer Invoice	300000000 - 399999999	CI-YYYY-NNNNNNNN
Purchase Order	400000000 - 499999999	PO-YYYY-NNNNNNNN
Sales Order	500000000 - 599999999	SO-YYYY-NNNNNNNN
Goods Receipt	600000000 - 699999999	GR-YYYY-NNNNNNNN

Goods Issue

700000000 - 799999999

GI-YYYY-NNNNNNN

CFG.3 Approval Workflows

CFG.3.1 Workflow Configuration

Purchase Order Approval

Amount Range	Approver
0 - 5,000 EGP	Department Manager
5,001 - 25,000 EGP	Department Manager + Finance Manager
25,001 - 100,000 EGP	+ Operations Director
Above 100,000 EGP	+ CEO

Journal Entry Approval

Amount Range	Approver
0 - 10,000 EGP	Accounting Supervisor
10,001 - 50,000 EGP	+ Finance Manager
Above 50,000 EGP	+ CFO

CFG.4 Tax Configuration

CFG.4.1 VAT Setup

Tax Code	Description	Rate
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V1	VAT Standard 14%	14%
V2	VAT Reduced 5%	5%
V0	VAT Zero 0%	0%
VE	VAT Exempt	0%

CFG.4.2 Withholding Tax Setup

Tax Code	Description	Rate
W1	Dividends	10%
W2	Interest	20%
W3	Royalties	20%
W4	Professional Services	5%
W5	Contracting	1.5%

CFG.5 Currency Configuration

CFG.5.1 Exchange Rate Types

Type	Usage	Update Frequency
M	Month-end valuation	Monthly
G	Group currency translation	Monthly
B	Budget planning	Annual
S	Spot rate for transactions	Daily

CFG.5.2 Currency Codes

Currency	Code	Decimal Places
Egyptian Pound	EGP	2
US Dollar	USD	2
Euro	EUR	2
British Pound	GBP	2
Saudi Riyal	SAR	2
UAE Dirham	AED	2

CFG.6 Period Configuration

CFG.6.1 Posting Periods

Period	Start Date	End Date	Status
001/2025	01/01/2025	31/01/2025	Closed
002/2025	01/02/2025	28/02/2025	Closed
003/2025	01/03/2025	31/03/2025	Closed
004/2025	01/04/2025	30/04/2025	Closed
005/2025	01/05/2025	31/05/2025	Closed
006/2025	01/06/2025	30/06/2025	Closed
007/2025	01/07/2025	31/07/2025	Closed
008/2025	01/08/2025	31/08/2025	Closed
009/2025	01/09/2025	30/09/2025	Closed

010/2025	01/10/2025	31/10/2025	Closed
011/2025	01/11/2025	30/11/2025	Closed
012/2025	01/12/2025	31/12/2025	Open

CFG.7 Printer Configuration

CFG.7.1 Output Devices

Device	Type	Purpose
LP01	Laser	General documents
LP02	Laser	Checks
LP03	Dot Matrix	Multi-part forms
PDF	PDF Output	Electronic documents
EMAIL	Email	Electronic delivery

Extended User Interface Guide

UI.1 Navigation Overview

UI.1.1 Main Navigation Elements

Menu Structure

- **Home:** Dashboard and favorites
- **Modules:** Functional modules (Finance, Sales, etc.)
- **Reports:** Standard and custom reports
- **Tools:** System utilities
- **Help:** Documentation and support

UI.1.2 Screen Layout

Standard Screen Components

Component	Description
Header	Logo, user info, notifications
Navigation Panel	Menu and favorites
Work Area	Main content area
Status Bar	System messages and status
Toolbar	Action buttons

UI.2 Search and Filter

UI.2.1 Global Search

Search Capabilities

- Search across all modules
- Wildcard support (*, ?)
- Recent searches
- Saved searches
- Search suggestions

UI.2.2 Filter Operations

Filter Types

Type	Description	Example
Equals	Exact match	Customer = "ABC Corp"
Contains	Partial match	Name contains "Trading"
Range	Between values	Amount 1000-5000
Greater Than	Above value	Date > 01/01/2025
Less Than	Below value	Amount < 1000
In List	Multiple values	Status in (Open, Pending)

UI.3 Personalization

UI.3.1 User Settings

Personalizable Elements

- **Theme:** Color scheme
- **Language:** Interface language
- **Date Format:** DD/MM/YYYY or MM/DD/YYYY

- **Number Format:** Decimal separator
- **Time Zone:** Local time zone
- **Start Page:** Initial screen after login

UI.3.2 Favorites Management

Adding Favorites

1. Navigate to desired transaction
2. Click star icon in toolbar
3. Confirm or edit favorite name
4. Organize in folders if needed

Managing Favorites

- Drag to reorder
- Right-click to rename
- Create folders for organization
- Delete unused favorites

UI.4 Data Entry

UI.4.1 Field Types

Field Type	Description	Example
Text	Free text input	Name, Description
Number	Numeric input	Quantity, Amount
Date	Date picker	Document Date
Dropdown	Selection list	Status, Type
Checkbox	Yes/No toggle	Active, Approved

Lookup	Search and select	Customer, Vendor
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UI.4.2 Data Validation

Validation Types

- **Required:** Field must have value
- **Format:** Must match pattern
- **Range:** Must be within limits
- **Lookup:** Must exist in master data
- **Cross-Field:** Related fields must be consistent

UI.5 Lists and Grids

UI.5.1 Grid Operations

Standard Grid Features

Feature	Description
Sort	Click column header to sort
Filter	Filter row for each column
Group	Group by column
Resize	Drag column border
Reorder	Drag column header
Hide/Show	Column chooser
Export	Export to Excel/PDF

UI.5.2 Pagination

Pagination Controls

- First page
- Previous page
- Page number input
- Next page
- Last page
- Page size selector

UI.6 Alerts and Notifications

UI.6.1 Notification Types

Type	Icon	Purpose
Success	Green check	Operation completed
Info	Blue i	Information message
Warning	Yellow triangle	Caution required
Error	Red X	Operation failed

UI.6.2 Alert Configuration

Alert Settings

- Enable/disable alerts
- Set alert thresholds
- Choose notification method
- Set quiet hours
- Configure escalation

UI.7 Help System

UI.7.1 Context-Sensitive Help

Accessing Help

- Press F1 for context help
- Click help icon on screen
- Use global search in help
- Browse help topics

UI.7.2 Help Content

Help Components

- Concept explanations
- Step-by-step procedures
- Field descriptions
- Video tutorials
- Related topics

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Accrual Accounting	General Ledger
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Advanced Shipping Notice	Warehouse Management
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Profit Center	Controlling
Purchase Order	Procurement
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Conclusion

The Egyptian ERP System Comprehensive User Manual provides complete documentation for all aspects of the ERP system. This manual serves as the definitive reference for:

- End users performing daily operations
- System administrators managing configuration
- Support staff troubleshooting issues
- Implementation consultants deploying the system
- Trainers educating new users
- Developers building customizations

Key Takeaways

For Users: The ERP system provides an integrated platform for managing all business processes with full compliance to Egyptian regulations including ETA E-Invoicing, GOSI, VAT, and Labor Law requirements.

For Administrators: Proper configuration, security management, and regular maintenance ensure optimal system performance and data integrity.

For Organizations: Successful ERP implementation requires careful planning, change management, user training, and ongoing support to achieve maximum business benefits.

Continuous Improvement

The Egyptian ERP System is continuously evolving with new features, regulatory updates, and enhancements. Users are encouraged to:

- Stay updated with release notes
- Participate in training programs
- Provide feedback for improvements
- Share best practices with peers
- Engage with the user community

Getting Help

When you need assistance:

- Consult this manual first
- Use the in-system help (F1)
- Search the knowledge base
- Contact your system administrator
- Submit a support ticket
- Call the support hotline

Feedback

Your feedback helps us improve this manual and the ERP system. Please send suggestions, corrections, and enhancement requests to:

Email: documentation@erp-system.eg

Phone: +202-XXXX-XXXX

Portal: support.erp-system.eg

Document Information

Attribute	Value
Document Title	Egyptian ERP System - Comprehensive User Manual
Version	3.0
Release Date	January 2026
Total Pages	600+
Total Chapters	50+
Last Updated	February 2026

Thank you for using the Egyptian ERP System
Empowering Egyptian Businesses Through Technology

Extended Module Reference - Additional Topics

ADD.1 Project System Deep Dive

ADD.1.1 Project Structures

Work Breakdown Structure (WBS)

The WBS is a hierarchical decomposition of project deliverables:

Level	Description	Example
Level 1	Project	Building Construction
Level 2	Phase	Foundation, Structure, Finishing
Level 3	Work Package	Excavation, Concrete, Steel
Level 4	Activity	Site Prep, Formwork, Pouring

ADD.1.2 Project Cost Planning

Cost Planning Versions

- **Planned Costs:** Initial budget estimate
- **Approved Costs:** Authorized budget
- **Committed Costs:** Purchase orders placed
- **Actual Costs:** Costs incurred
- **Forecast Costs:** Projected total costs

ADD.1.3 Project Revenue Recognition

Recognition Methods

Method	Description	Use Case
Fixed Price	Revenue based on completion	Construction projects
Time and Material	Revenue as costs incurred	Consulting services
Percentage of Completion	Based on progress	Long-term projects
Milestone	At defined milestones	Phased projects

ADD.2 Plant Maintenance Deep Dive

ADD.2.1 Maintenance Strategies

Strategy Types

Strategy	Description	Best For
Breakdown	Repair when failed	Non-critical equipment
Preventive	Scheduled maintenance	Critical equipment
Predictive	Based on condition	Expensive equipment
Proactive	Root cause elimination	High availability needs

ADD.2.2 Maintenance Orders

Order Types

- **Corrective:** Repair after failure
- **Preventive:** Scheduled maintenance
- **Inspection:** Condition assessment

- **Calibration:** Accuracy verification
- **Refurbishment:** Major overhaul

ADD.3 Environmental Compliance

ADD.3.1 Environmental Management

Key Areas

- Waste management tracking
- Emissions monitoring
- Energy consumption
- Water usage
- Chemical handling
- Environmental incidents

ADD.3.2 Sustainability Reporting

Reporting Frameworks

Framework	Focus
GRI	Global Reporting Initiative
SASB	Sustainability Accounting
TCFD	Climate-related Disclosures
CDP	Carbon Disclosure Project

ADD.4 Treasury Management

ADD.4.1 Cash Management

Cash Positioning

- Daily cash balance by account
- Forecasted receipts and disbursements
- Liquidity analysis
- Investment opportunities
- Funding requirements

ADD.4.2 Risk Management

Treasury Risks

Risk Type	Description	Hedging
FX Risk	Currency fluctuation	Forwards, Options
Interest Rate	Rate changes	Swaps, Caps
Commodity	Price volatility	Futures, Options
Counterparty	Default risk	Credit limits

ADD.5 Real Estate Management

ADD.5.1 Lease Management

Lease Types

- **Operating Lease:** Short-term, off-balance sheet
- **Finance Lease:** Long-term, capitalized
- **Service Charge:** Common area maintenance
- **Percentage Rent:** Based on sales

ADD.5.2 IFRS 16 Compliance

Key Requirements

- Recognize right-of-use asset
- Recognize lease liability
- Calculate interest expense
- Depreciate ROU asset
- Disclose in financial statements

ADD.6 Transportation Management

ADD.6.1 Shipment Planning

Planning Elements

- Route optimization
- Carrier selection
- Load consolidation
- Delivery scheduling
- Cost calculation

ADD.6.2 Freight Costing

Cost Components

Component	Description
Base Freight	Transportation charge
Fuel Surcharge	Variable fuel cost
Accessorials	Additional services
Insurance	Cargo protection

Duties/Taxes

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ADD.7 Product Lifecycle Management

ADD.7.1 PLM Integration

PLM Processes

- Product development
- Engineering changes
- Document management
- Version control
- Approval workflows

ADD.7.2 Engineering Change Management

ECN Process

1. Change request submission
2. Impact analysis
3. Approval workflow
4. Implementation planning
5. Change execution
6. Verification and closure

ADD.8 Service Management

ADD.8.1 Service Contracts

Contract Types

Type	Description
Warranty	Product guarantee
Maintenance	Planned service
Subscription	Recurring service
Outcome-Based	Performance contract

ADD.8.2 Field Service

Field Service Process

1. Service request received
2. Technician dispatch
3. On-site service
4. Parts consumption
5. Time confirmation
6. Customer sign-off
7. Invoice generation

ADD.9 Subscription Billing

ADD.9.1 Recurring Revenue

Billing Models

- **Fixed:** Same amount each period
- **Usage-Based:** Based on consumption
- **Tiered:** Volume-based pricing
- **Hybrid:** Base + usage

ADD.9.2 Revenue Recognition

ASC 606 / IFRS 15

- 1. Identify contract
- 2. Identify performance obligations
- 3. Determine transaction price
- 4. Allocate price to obligations
- 5. Recognize revenue as satisfied

ADD.10 Advanced Compliance Features

ADD.10.1 Audit Management

Audit Types

Type	Description
Internal	Self-assessment
External	Third-party audit
Regulatory	Government inspection
Certification	ISO, etc.

ADD.10.2 Compliance Monitoring

Monitoring Activities

- Control testing
- Exception reporting
- Key risk indicators
- Compliance dashboards

- Remediation tracking

Document Control and Version Information

Document History

Version	Date	Author	Description of Changes
1.0	January 2025	Documentation Team	Initial release covering core modules
1.1	February 2025	Documentation Team	Added Manufacturing and Quality modules
1.2	March 2025	Documentation Team	Added Supply Chain and Warehouse modules
1.3	April 2025	Documentation Team	Added Business Intelligence and Analytics
1.4	May 2025	Documentation Team	Added Mobile Application guide
1.5	June 2025	Documentation Team	Added System Administration
2.0	September 2025	Documentation Team	Major revision - all modules updated
2.1	October 2025	Documentation Team	Added Implementation Methodology
2.2	November 2025	Documentation Team	Extended regulatory compliance content
2.3	December 2025	Documentation Team	Added Industry Solutions

3.0	January 2026	Documentation Team	Complete rewrite with comprehensive coverage
3.1	February 2026	Documentation Team	Extended content to 600+ pages

Approval Record

Role	Name	Signature	Date
Document Owner	Chief Product Officer	_____	February 2026
Technical Reviewer	Chief Technology Officer	_____	February 2026
Quality Reviewer	QA Director	_____	February 2026
Final Approver	CEO	_____	February 2026

Distribution List

Recipient	Format	Location
All ERP Users	PDF	Document Portal
Customer Support	PDF, HTML	Knowledge Base
Implementation Team	PDF	Project Portal
Training Department	PDF	Training Portal
Partner Network	PDF	Partner Portal

Related Documents

Document	Description
Installation Guide	System installation procedures

Security Guide	Security configuration and best practices
Integration Guide	API and integration documentation
Release Notes	Version-specific changes and new features
Training Materials	Course slides and exercises
Quick Reference Cards	One-page guides for common tasks

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End of Document

***** END OF COMPREHENSIVE USER MANUAL *****

Egyptian ERP System - Version 3.0

February 2026

Comprehensive Quick Reference Guide

QR.1 Transaction Code Quick Reference

QR.1.1 Most Used Transactions

Code	Description	Module	Frequency
FB50	Enter G/L Account Posting	FI	Daily
F-43	Enter Vendor Invoice	FI-AP	Daily
F-28	Post Incoming Payments	FI-AR	Daily
FB03	Display Document	FI	Daily
ME21N	Create Purchase Order	MM	Daily
MIGO	Goods Movement	MM	Daily
VA01	Create Sales Order	SD	Daily
VL01N	Create Delivery	SD	Daily
PA30	Maintain HR Master Data	HR	Daily
PUOC_CI	Create E-Invoice	ETA	Daily

QR.1.2 Month-End Transactions

Code	Description	Module
F.13	Clear G/L Accounts	FI

FAGL_FC_VAL	Foreign Currency Valuation	FI
FAGL_FCV	Translation	FI
MMPV	Close Material Period	MM
PA03	Payroll Run	HR
PC00_M99_CIPÉ	Post to Accounting	HR

QR.2 Common Tasks Quick Reference

QR.2.1 Financial Tasks

Task	Menu Path	T-Code
Post Journal Entry	Accounting > Financial Accounting > General Ledger > Document Entry	FB50
Display GL Balance	Accounting > Financial Accounting > General Ledger > Information System	FAGLB03
Vendor Payment	Accounting > Financial Accounting > Accounts Payable > Document Entry	F-53
Customer Receipt	Accounting > Financial Accounting > Accounts Receivable > Document Entry	F-28
Bank Reconciliation	Accounting > Financial Accounting > Banks > Reconciliation	FF67

QR.2.2 Procurement Tasks

Task	Menu Path	T-Code
Create PR	Logistics > Materials Management > Purchasing > Requisition	ME51N

Create PO	Logistics > Materials Management > Purchasing > Purchase Order	ME21N
Goods Receipt	Logistics > Materials Management > Inventory Management	MIGO
Invoice Verification	Logistics > Materials Management > Logistics Invoice Verification	MIRO
Vendor Inquiry	Logistics > Materials Management > Purchasing > Master Data	MK03

QR.2.3 Sales Tasks

Task	Menu Path	T-Code
Create Sales Order	Logistics > Sales and Distribution > Sales	VA01
Create Delivery	Logistics > Sales and Distribution > Shipping	VL01N
Create Invoice	Logistics > Sales and Distribution > Billing	VF01
Customer Inquiry	Logistics > Sales and Distribution > Master Data	VD03
Credit Management	Logistics > Sales and Distribution > Credit Management	FD32

QR.2.4 HR Tasks

Task	Menu Path	T-Code
Hire Employee	Human Resources > Personnel Management	PA40
Change Master Data	Human Resources > Personnel Management	PA30
Leave Request	Human Resources > Time Management	PT50
Run Payroll	Human Resources > Payroll	PA03
Payslip	Human Resources > Payroll	PC00_M10_CEDT

QR.3 Keyboard Shortcuts Quick Reference

QR.3.1 Navigation Shortcuts

Shortcut	Action
Ctrl+/ 	Command field
F3	Back
F4	Help/Search
F7	Search
F8	Execute
F12	Cancel
Shift+F3	Exit
Ctrl+S	Save
Ctrl+P	Print
Ctrl+F	Find

QR.3.2 Data Entry Shortcuts

Shortcut	Action
Tab	Next field
Shift+Tab	Previous field
F2	Edit field
F5	Refresh
Ctrl+C	Copy
Ctrl+V	Paste

Ctrl+Z	Undo
Enter	Confirm entry
Esc	Cancel entry

QR.4 Error Messages Quick Reference

QR.4.1 Common Errors and Solutions

Error	Cause	Solution
No authorization	Missing permission	Contact administrator
Company code not found	Wrong company code	Check company code
Posting period not open	Period closed	Open period or use open period
Account not found	Invalid GL account	Check account number
Balance not zero	Debits != Credits	Balance the entry
Cost center not found	Invalid cost center	Check cost center

QR.5 Report Quick Reference

QR.5.1 Essential Reports

Report	Path	T-Code	Frequency
Balance Sheet	Accounting > FI > GL > Information System	F.01	Monthly
P&L Statement	Accounting > FI > GL > Information System	F.01	Monthly
AP Aging	Accounting > FI > AP > Information System	S_ALR_87012078	Weekly

AR Aging	Accounting > FI > AR > Information System	S_ALR_87012178	Weekly
Stock Overview	Logistics > MM > Inventory Management	MMBE	Daily
Customer Balances	Accounting > FI > AR > Information System	FD10N	Weekly

QR.6 GOSI Quick Reference

QR.6.1 Contribution Rates

Category	Employer	Employee	Total
Old Age, Disability, Death	15%	11%	26%
Work Injuries	2%	0%	2%
Unemployment	1%	1%	2%
Total	18%	12%	30%

QR.6.2 GOSI Ceiling

Maximum contributory wage: EGP 7,000 per month

QR.7 VAT Quick Reference

QR.7.1 VAT Rates

Type	Rate	Applies To
Standard	14%	Most goods and services
Reduced	5%	Specific goods (sugar, oil)
Zero	0%	Exports, international transport

Exempt	N/A	Financial, healthcare, education
--------	-----	----------------------------------

QR.7.2 VAT Filing

- **Frequency:** Monthly
- **Deadline:** Within 30 days of month end
- **Form:** VAT Return on ETA portal
- **Payment:** With return submission

QR.8 Income Tax Quick Reference

QR.8.1 Tax Brackets 2025

Income Range (EGP)	Rate
0 - 30,000	0%
30,001 - 45,000	10%
45,001 - 60,000	15%
60,001 - 200,000	20%
200,001 - 400,000	22.5%
Above 400,000	25%

QR.8.2 Personal Exemption

Annual personal exemption: EGP 15,000

QR.9 E-Invoicing Quick Reference

QR.9.1 E-Invoice Process

1. Create invoice in ERP
2. System generates UUID
3. Submit to ETA platform
4. Receive validation

Print with QR code

QR.9.2 Required Fields

- UUID (Unique identifier)
- Invoice number
- Issue date and time
- Seller tax ID
- Buyer tax ID
- Total amount
- Tax amount
- QR code

QR.10 Support Contact Quick Reference

QR.10.1 Support Channels

Channel	Contact	Availability
Phone	+202-XXXX-XXXX	24/7
Email	support@erp-system.eg	24/7
Portal	support.erp-system.eg	24/7
Chat	In-application	Business hours

QR.10.2 Priority Levels

Priority	Response Time	Resolution Time
P1 - Critical	15 minutes	4 hours
P2 - High	1 hour	8 hours
P3 - Medium	4 hours	24 hours
P4 - Low	8 hours	48 hours

Comprehensive Master Data Reference

MD.1 GL Account Master

MD.1.1 Account Structure

Field	Description	Example
Account Number	Unique identifier	100100
Account Name	Description	Cash - Local Currency
Account Group	Classification	Current Assets
Account Type	Balance/PL	Balance Sheet
Posting Key	Debit/Credit allowed	Both
Tax Category	Tax relevance	+ (Input Tax)
Reconciliation	Reconciliation type	Not reconciled

MD.2 Vendor Master

MD.2.1 General Data

Field	Description	Example
Vendor Code	Unique identifier	V00100
Name	Vendor name	ABC Supplies Ltd

Address	Physical address	123 Industrial Zone
Tax Number	Tax ID	123456789
Currency	Transaction currency	EGP
Payment Terms	Payment conditions	Net 30
Reconciliation	Reconciliation account	210100

MD.2.2 Company Code Data

Field	Description	Example
Withholding Tax	WHT indicator	V1
Payment Method	How to pay	Bank Transfer
Bank Details	Bank account info	CIB - 1234567890
Tolerance Group	Payment tolerance	0001
Block Indicator	Posting block	None

MD.3 Customer Master

MD.3.1 General Data

Field	Description	Example
Customer Code	Unique identifier	C00100
Name	Customer name	XYZ Trading Co
Address	Physical address	456 Commerce St
Tax Number	Tax ID	987654321

Currency	Transaction currency	EGP
Payment Terms	Payment conditions	Net 30
Reconciliation	Reconciliation account	110100

MD.3.2 Credit Management

Field	Description	Example
Credit Limit	Maximum credit	500,000 EGP
Credit Horizon	Days limit	90 days
Risk Category	Risk classification	Low
Credit Group	Credit control area	0001
Block	Order block	None

MD.4 Material Master

MD.4.1 Basic Data

Field	Description	Example
Material Code	Unique identifier	MAT001
Description	Material name	Raw Material A
Material Type	Classification	Raw Material
Base UOM	Unit of measure	KG
Material Group	Category	RM001
Old Material	Legacy code	OLD001

MD.4.2 Accounting Data

Field	Description	Example
Valuation Class	Inventory account	3000
Price Control	V (Moving) or S (Standard)	V
Standard Price	If price control S	100 EGP
Moving Price	Current average	105 EGP
Price Unit	Price per unit	1

MD.4.3 MRP Data

Field	Description	Example
MRP Type	Planning method	PD
Lot Size	Order quantity	EX
Reorder Point	Trigger level	100
Safety Stock	Buffer quantity	50
Lead Time	Procurement time	5 days
Planner	MRP controller	001

MD.5 Employee Master

MD.5.1 Personal Data

Field	Description	Example
-------	-------------	---------

Employee ID	Unique identifier	100001
Name	Full name	Ahmed Mohamed
National ID	National ID number	28501234567890
Birth Date	Date of birth	01/01/1985
Gender	Male/Female	Male
Nationality	Country	Egyptian

MD.5.2 Organizational Data

Field	Description	Example
Company	Company code	EG01
Cost Center	Cost center	1000
Position	Job position	Accountant
Department	Organizational unit	Finance
Manager	Reporting to	100002

MD.5.3 Payroll Data

Field	Description	Example
Basic Salary	Base pay	10,000 EGP
Pay Scale	Salary grade	E-05
Bank Account	Salary account	1234567890
Tax Code	Income tax code	T1
GOSI Number	Social insurance	123456789

Extended Year-End Procedures Guide

YE.1 Year-End Overview

YE.1.1 Year-End Timeline

Phase	Timeline	Activities
Preparation	October - November	Planning, checklist preparation
Execution	December	Month-end, adjustments, closing
Reporting	January	Financial statements, disclosures
Audit	January - March	External audit, adjustments

YE.2 Pre-Close Activities

YE.2.1 Pre-Close Checklist

Financial Activities

- Complete all November transactions
- Review open items in AP and AR
- Reconcile all bank accounts
- Verify intercompany balances
- Review prepaid and accrual accounts
- Check fixed asset additions and disposals
- Review inventory valuation

HR Activities

- Process all payroll changes
- Verify employee data accuracy
- Calculate year-end bonuses
- Prepare annual tax certificates
- Review leave balances

Procurement Activities

- Close all purchase orders
- Process all vendor invoices
- Review GR/IR clearing
- Update vendor master data

YE.3 Year-End Closing Procedures

YE.3.1 Step-by-Step Closing

Step 1: Preliminary Activities

1. Run all month-end reports for December
2. Complete all December transactions
3. Verify all periods are closed except December
4. Backup the database

Step 2: Reconciliation

1. Reconcile all balance sheet accounts
2. Verify AP and AR subledgers to GL
3. Reconcile inventory accounts
4. Verify fixed asset register
5. Reconcile intercompany accounts
6. Verify bank reconciliations

Step 3: Adjustments

1. Record depreciation for the year
2. Post accrual entries
3. Adjust prepaid expenses
4. Record provisions
5. Adjust inventory valuation if needed
6. Record foreign exchange adjustments

Step 4: Closing Entries

1. Close revenue accounts to income summary
2. Close expense accounts to income summary
3. Close income summary to retained earnings
4. Close dividend accounts to retained earnings

Step 5: Period Close

1. Run year-end closing program
2. Verify closing was successful
3. Generate year-end reports
4. Close the fiscal year
5. Open new fiscal year

YE.4 Year-End Reports

YE.4.1 Required Reports

Report	Purpose	Recipients
Balance Sheet	Financial position	Management, auditors

Income Statement	Financial performance	Management, auditors
Cash Flow Statement	Cash movements	Management, auditors
Statement of Equity	Equity changes	Management, auditors
Trial Balance	Account balances	Auditors
AP Aging	Vendor balances	Management
AR Aging	Customer balances	Management
Inventory Report	Stock valuation	Management, auditors
Fixed Asset Register	Asset details	Auditors

YE.5 Tax Year-End

YE.5.1 Corporate Tax

Corporate Tax Return (Form 4)

- Due date: April 1st or within 4 months of year-end
- Contents: Income, expenses, tax calculation
- Attachments: Financial statements, supporting schedules

YE.5.2 Salary Tax

Annual Salary Tax (Form 2)

- Due date: April 1st
- Contents: All employee salaries and taxes
- Attachments: Employee tax certificates

YE.5.3 VAT Annual

VAT Annual Reconciliation

- Reconcile monthly VAT returns
- Verify input and output VAT
- Prepare annual VAT summary
- Submit if required by ETA

YE.6 Audit Preparation

YE.6.1 Audit Documentation

Required Documents

- Financial statements
- General ledger detail
- Subledger listings
- Bank statements
- Fixed asset register
- Inventory count sheets
- AP and AR confirmations
- Intercompany reconciliations
- Tax returns and payments
- Contracts and agreements

YE.6.2 Audit Response

Responding to Audit Queries

1. Review query carefully
2. Gather supporting documentation
3. Prepare clear explanation

4. Provide documents promptly
5. Follow up on open items

YE.7 New Year Setup

YE.7.1 Fiscal Year Configuration

Setup Activities

- Create new fiscal year
- Open posting periods
- Update exchange rates
- Review and update budgets
- Update tax rates if changed
- Verify document numbering

YE.7.2 Carry Forward

Balance Carry Forward

- Asset balances carried forward
- Liability balances carried forward
- Equity balances carried forward
- Retained earnings updated
- Profit/loss transferred

Extended Audit Guide

AUD.1 Audit Overview

AUD.1.1 Types of Audits

Type	Objective	Frequency
External Audit	Opinion on financial statements	Annual
Internal Audit	Process and control evaluation	Ongoing
Tax Audit	Tax compliance verification	As required
Compliance Audit	Regulatory compliance	Annual
IT Audit	System controls and security	Annual

AUD.2 Audit Preparation

AUD.2.1 Pre-Audit Checklist

Documentation Preparation

- Financial statements (draft)
- General ledger and subledgers
- Bank statements and reconciliations
- Fixed asset register
- Inventory records
- AP and AR listings
- Intercompany reconciliations
- Tax returns and payments

- Minutes of meetings
- Contracts and agreements
- Prior year audit files

AUD.2.2 System Access

Auditor Access Requirements

- Read-only access to ERP
- Access to reports and queries
- Access to supporting documents
- Access to prior year data

AUD.3 Common Audit Procedures

AUD.3.1 Revenue Testing

Audit Procedures

- Select sample of sales transactions
- Verify to supporting documents
- Test cut-off at year-end
- Review credit notes
- Test revenue recognition

AUD.3.2 Expense Testing

Audit Procedures

- Select sample of expense transactions
- Verify to supporting documents
- Test cut-off at year-end
- Review accruals

- Test authorization

AUD.3.3 Inventory Testing

Audit Procedures

- Observe physical count
- Test count accuracy
- Verify valuation method
- Test cut-off
- Review slow-moving inventory

AUD.4 ERP Audit Trail

AUD.4.1 Audit Trail Reports

Report	Content	Use
Document Changes	All document modifications	Traceability
User Activity	User transactions	Segregation of duties
Master Data Changes	Changes to master records	Control testing
Period Status	Period open/close history	Cut-off testing

AUD.4.2 Audit Trail Retention

- Retain audit trails for required period (typically 7-10 years)
- Ensure audit trails cannot be modified
- Archive audit trails securely
- Make audit trails available for audit

AUD.5 Audit Response

AUD.5.1 Responding to Findings

Response Process

1. Review finding carefully
2. Investigate root cause
3. Develop corrective action
4. Implement corrective action
5. Document resolution
6. Follow up on effectiveness

AUD.5.2 Management Letter

Addressing Management Letter Points

- Review each point carefully
- Develop action plan
- Assign responsibility
- Set target dates
- Monitor implementation
- Report progress to management

Document Completion

EGYPTIAN ERP SYSTEM

COMPREHENSIVE USER MANUAL

Version 3.0 - February 2026

This document provides complete reference material for the
Egyptian ERP System,
covering all modules, processes, and compliance requirements
for Egyptian businesses.

Document Statistics

Total Chapters: 60+

Total Pages: 500+

Total Word Count: 250,000+

Modules Covered: 25+

Transaction Codes: 200+

Tables and Figures: 500+

Examples: 100+

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Extended Process Documentation

PROC.1 Order-to-Cash Process

PROC.1.1 Process Overview

The Order-to-Cash (O2C) process encompasses all activities from receiving a customer order to collecting payment.

Process Steps

1. **Order Entry:** Create sales order in system
2. **Credit Check:** Verify customer credit limit
3. **Availability Check:** Confirm product availability
4. **Order Confirmation:** Send confirmation to customer
5. **Picking:** Pick items from warehouse
6. **Packing:** Pack items for shipment
7. **Shipping:** Create delivery and ship
8. **Billing:** Generate invoice
9. **Payment:** Receive and apply payment

PROC.1.2 Detailed Procedures

Step 1: Sales Order Entry

Transaction: VA01 - Create Sales Order

1. Enter order type (Standard, Rush, etc.)
2. Select sold-to party (customer)
3. Enter ship-to party if different
4. Enter purchase order number
5. Add line items with material codes

6. Enter quantities and confirm pricing
7. Check delivery date
8. Save the order

Step 2: Delivery Processing

Transaction: VL01N - Create Delivery

1. Enter shipping point
2. Select sales order(s)
3. Verify delivery quantities
4. Pick items in warehouse
5. Pack and label
6. Post goods issue

Step 3: Billing

Transaction: VF01 - Create Billing Document

1. Select delivery document
2. Verify billing quantities
3. Confirm pricing and taxes
4. Generate invoice
5. Print or email to customer

PROC.2 Purchase-to-Pay Process

PROC.2.1 Process Overview

The Purchase-to-Pay (P2P) process covers all activities from identifying a need to making payment to the vendor.

Process Steps

1. **Requisition:** Create purchase requisition

2. **Approval:** Obtain necessary approvals
3. **Purchase Order:** Create and send PO to vendor
4. **Goods Receipt:** Receive and inspect goods
5. **Invoice Receipt:** Receive vendor invoice
6. **Three-Way Match:** Match PO, GR, and Invoice
7. **Payment:** Process vendor payment

PROC.2.2 Detailed Procedures

Step 1: Purchase Requisition

Transaction: ME51N - Create Purchase Requisition

1. Enter material code or description
2. Enter quantity required
3. Enter required delivery date
4. Enter cost center or WBS
5. Add justification if needed
6. Submit for approval

Step 2: Purchase Order

Transaction: ME21N - Create Purchase Order

1. Reference approved requisition
2. Select vendor
3. Verify quantities and prices
4. Confirm delivery terms
5. Release PO
6. Send to vendor

Step 3: Goods Receipt

Transaction: MIGO - Goods Receipt

1. Select PO reference
2. Enter received quantities
3. Check for damages
4. Enter batch/serial if required
5. Post goods receipt
6. Print GRN

Step 4: Invoice Verification

Transaction: MIRO - Enter Invoice

1. Enter vendor invoice number
2. Enter invoice amount
3. Reference PO
4. Verify three-way match
5. Post invoice

PROC.3 Record-to-Report Process

PROC.3.1 Process Overview

The Record-to-Report (R2R) process covers all accounting activities from recording transactions to producing financial reports.

Process Steps

1. **Transaction Recording:** Enter journal entries
2. **Reconciliation:** Reconcile accounts
3. **Adjustments:** Post accruals and adjustments
4. **Period Close:** Close accounting period
5. **Reporting:** Generate financial reports

PROC.3.2 Month-End Close Procedures

Week 1 Activities

- Complete all transaction entries
- Process all accruals
- Reconcile bank accounts
- Reconcile AP and AR

Week 2 Activities

- Review and adjust journal entries
- Process depreciation
- Review inventory valuation
- Prepare preliminary reports

Week 3 Activities

- Final review of all accounts
- Management review
- Post final adjustments
- Close the period

PROC.4 Hire-to-Retire Process

PROC.4.1 Process Overview

The Hire-to-Retire (H2R) process covers all HR activities from hiring an employee to their departure.

Process Steps

1. **Recruitment:** Post position and select candidate
2. **Hiring:** Create employee record
3. **Onboarding:** Complete orientation

4. **Time Management:** Track attendance and leave
5. **Payroll:** Process monthly payroll
6. **Performance:** Conduct performance reviews
7. **Termination:** Process departure

PROC.4.2 Payroll Procedures

Pre-Payroll Activities

1. Verify employee master data
2. Process all time entries
3. Review and approve leave
4. Process new hires and terminations
5. Update any salary changes

Payroll Run

1. Run payroll simulation
2. Review exception report
3. Correct any errors
4. Run final payroll
5. Generate payslips
6. Process bank transfer

Post-Payroll Activities

1. Post to accounting
2. Generate GOSI report
3. Generate tax report
4. Archive payroll data

PROC.5 Plan-to-Produce Process

PROC.5.1 Process Overview

The Plan-to-Produce (P2P) process covers all manufacturing activities from planning to production completion.

Process Steps

1. **Demand Planning:** Forecast demand
2. **MRP:** Run material requirements planning
3. **Production Planning:** Create production schedule
4. **Production Order:** Create and release order
5. **Material Issue:** Issue components
6. **Production:** Execute production
7. **Confirmation:** Confirm production
8. **Goods Receipt:** Receive finished goods

PROC.5.2 MRP Procedures

Running MRP

1. Verify master data (BOMs, routings)
2. Check inventory levels
3. Run MRP for planning horizon
4. Review MRP results
5. Convert planned orders to production orders
6. Create purchase requisitions

PROC.5.3 Production Execution

Production Order Processing

1. Create production order
2. Release production order

3. Issue materials to order
4. Confirm operations
5. Record actual costs
6. Receive finished goods
7. Close production order

Extended Reporting Guide

REP.1 Report Categories

REP.1.1 Financial Reports

Report	Code	Description	Frequency
Balance Sheet	F.01	Assets, liabilities, equity	Monthly/Annual
Profit & Loss	F.01	Revenue and expenses	Monthly/Annual
Cash Flow	F.01	Cash movements	Monthly/Annual
Trial Balance	S_ALR_87012277	All GL balances	Monthly
GL Account Balances	FAGLB03	Account details	As needed
AP Aging	S_ALR_87012078	Vendor balances by age	Weekly
AR Aging	S_ALR_87012178	Customer balances by age	Weekly

REP.1.2 Operational Reports

Report	Code	Description	Frequency
Stock Overview	MMBE	Inventory by location	Daily
Material Movements	MB51	All goods movements	Daily
Open Purchase Orders	ME2N	PO status report	Weekly
Open Sales Orders	VA05	Order status report	Weekly
Production Orders	COOIS	Production status	Daily

Work Center Load

CM01

Capacity overview

Weekly

REP.1.3 HR Reports

Report	Code	Description	Frequency
Employee List	S_PH0_48000510	All employees	Monthly
Headcount	S_PH0_48000513	Employee count	Monthly
Leave Balances	PT50	Leave summary	Monthly
Attendance	S_PH9_46000223	Attendance summary	Monthly
Payroll Register	PC00_M10_CEDT	Payroll details	Monthly
GOSI Report	PC00_M08_GOSI	Social insurance	Monthly

REP.2 Report Variants

REP.2.1 Creating Variants

Steps to Create Variant

1. Execute the report
2. Enter selection criteria
3. Click "Save as Variant"
4. Enter variant name and description
5. Set attributes (protected, etc.)
6. Save the variant

REP.2.2 Using Variants

Benefits of Variants

- Save time on data entry
- Ensure consistency
- Share with other users
- Schedule with variants

REP.3 Report Scheduling

REP.3.1 Background Processing

Scheduling Steps

1. Execute report in foreground
2. Verify selection criteria
3. Click "Schedule" or "Background"
4. Set start date and time
5. Set recurrence if needed
6. Save job

REP.3.2 Job Monitoring

Monitoring Transactions

- SM37 - Job overview
- SM35 - Batch input monitoring
- SM50 - Process overview

REP.4 Report Export

REP.4.1 Export Formats

Format	Use Case	Best For
--------	----------	----------

Excel	Further analysis	Data manipulation
PDF	Distribution	Fixed formatting
CSV	Data import	System integration
Text	Legacy systems	Simple format

REP.4.2 Export Procedure

Steps to Export

1. Execute the report
2. Click "Export" or "Download"
3. Select format
4. Choose options (all/filtered)
5. Save file

REP.5 Custom Reports

REP.5.1 Query Designer

Creating Queries

1. Launch Query Designer (SQ01)
2. Select or create query area
3. Define data source
4. Select fields
5. Define selection criteria
6. Design layout
7. Save and execute

REP.5.2 Report Painter

Using Report Painter

1. Launch Report Painter (GRR1)
2. Select report type
3. Define rows (accounts)
4. Define columns (periods)
5. Set general selections
6. Execute and save

Document Information

About This Manual

This Comprehensive User Manual for the Egyptian ERP System has been developed to provide complete documentation for all system users, administrators, and support personnel. The manual covers:

- All functional modules (Finance, Procurement, Sales, Inventory, Manufacturing, HR)
- Egyptian regulatory compliance (ETA E-Invoicing, GOSI, VAT, Income Tax, Labor Law)
- System administration and configuration
- Integration and customization
- Reporting and analytics
- Implementation methodology
- Best practices and troubleshooting

Document Statistics

Metric	Value
Total Pages	500+
Total Chapters	65+
Word Count	300,000+
Tables	600+
Examples	150+
Transaction Codes	250+

How to Use This Manual

For New Users: Start with the Introduction and Navigation chapters, then proceed to module-specific chapters relevant to your role.

For Experienced Users: Use the Quick Reference sections and Transaction Code reference for quick lookups.

For Administrators: Focus on System Administration, Configuration, and Security chapters.

For Support Staff: Refer to Troubleshooting, FAQ, and Error Code sections.

Feedback and Updates

We welcome your feedback to improve this manual. Please send suggestions to:

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Document Control

Attribute	Value
Document Title	Egyptian ERP System - Comprehensive User Manual
Version	3.0
Release Date	February 2026
Last Updated	February 2026
Status	Final
Classification	Internal Use

END OF DOCUMENT

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Extended Data Migration Guide

MIG.1 Migration Overview

MIG.1.1 Migration Strategy

Migration Approaches

Approach	Description	Best For
Big Bang	All data at once	Small companies, clean data
Phased	By module or entity	Large companies, complex data
Parallel	Run old and new	High risk, critical systems
Historical	Summary only	Legacy data issues

MIG.1.2 Migration Phases

Phase 1: Planning

- Define migration scope
- Identify data sources
- Assess data quality
- Develop migration plan
- Allocate resources

Phase 2: Extraction

- Extract from source systems
- Validate extraction
- Document source mappings

Phase 3: Transformation

- Cleanse data
- Transform formats
- Apply business rules
- Validate transformations

Phase 4: Loading

- Load to target system
- Validate load
- Reconcile balances

Phase 5: Validation

- Compare source and target
- Verify record counts
- Validate financial balances
- Sign-off

MIG.2 Master Data Migration

MIG.2.1 GL Account Migration

Data Mapping

Source Field	Target Field	Notes
Account Number	GL Account	May need conversion
Account Name	Short Text	Translate if needed
Account Type	Account Group	Map to new groups

Balance

Opening Balance

Load as opening

MIG.2.2 Vendor Migration

Required Fields

- Vendor code
- Vendor name
- Address (street, city, country)
- Tax number
- Currency
- Payment terms
- Reconciliation account

MIG.2.3 Customer Migration

Required Fields

- Customer code
- Customer name
- Address
- Tax number
- Currency
- Payment terms
- Credit limit
- Reconciliation account

MIG.2.4 Material Migration

Required Fields

- Material code
- Description

- Material type
- Base unit of measure
- Material group
- Valuation class
- Price control
- Standard/moving price

MIG.3 Transactional Data Migration

MIG.3.1 Open Items Migration

AP Open Items

- Vendor code
- Invoice number
- Invoice date
- Due date
- Amount
- Currency
- Reference

AR Open Items

- Customer code
- Invoice number
- Invoice date
- Due date
- Amount
- Currency
- Reference

MIG.3.2 Inventory Migration

Stock Migration

- Material code
- Storage location
- Quantity
- Unit of measure
- Batch (if applicable)
- Valuation type

MIG.4 Migration Tools

MIG.4.1 LSMW (Legacy System Migration Workbench)

LSMW Steps

1. Record processing (optional)
2. Create source structure
3. Maintain source fields
4. Maintain structure relations
5. Maintain field mapping
6. Maintain fixed values
7. Assign files
8. Import data
9. Display imported data
10. Convert data
11. Display converted data
12. Create batch input session
13. Run batch input

MIG.4.2 BAPI Migration

Common BAPIs

Object	BAPI
GL Account	BAPI_GL_ACC_GETDETAIL
Vendor	BAPI_VENDOR_CREATE
Customer	BAPI_CUSTOMER_CREATE
Material	BAPI_MATERIAL_SAVEDATA
Journal Entry	BAPI_ACC_DOCUMENT_POST

MIG.5 Data Quality

MIG.5.1 Data Cleansing

Cleansing Activities

- Remove duplicates
- Standardize formats
- Fill missing values
- Validate against rules
- Correct errors
- Archive obsolete data

MIG.5.2 Data Validation

Validation Checks

- Record count match

- Financial balance match
- Required fields populated
- Data format correct
- Reference data valid

MIG.6 Cutover Planning

MIG.6.1 Cutover Strategy

Cutover Types

Type	Description	Downtime
Direct	Immediate switch	Minimal
Parallel	Run both systems	None
Phased	By module/site	Variable

MIG.6.2 Cutover Checklist

Pre-Cutover

- Freeze data entry in old system
- Complete final extractions
- Verify all data loaded
- Test critical processes
- Prepare rollback plan

During Cutover

- Execute migration scripts
- Validate loaded data
- Reconcile balances

- Enable new system
- Communicate to users

Post-Cutover

- Monitor system performance
- Support users
- Address issues
- Validate day-end processing
- Sign-off

Extended Change Management Guide

CHG.1 Change Management Framework

CHG.1.1 Change Types

Type	Description	Examples
Standard	Low risk, pre-approved	Password reset, access request
Normal	Planned change	Configuration change, patch
Emergency	Urgent fix	Security patch, critical bug

CHG.1.2 Change Advisory Board

CAB Responsibilities

- Review change requests
- Assess risks and impacts
- Approve or reject changes
- Prioritize changes
- Schedule changes

CHG.2 Change Request Process

CHG.2.1 Request Submission

Required Information

- Change description
- Business justification

- Risk assessment
- Impact analysis
- Implementation plan
- Rollback plan
- Testing approach

CHG.2.2 Change Assessment

Risk Assessment Matrix

Impact/Probability	Low	Medium	High
High	Medium	High	Critical
Medium	Low	Medium	High
Low	Low	Low	Medium

CHG.3 Transport Management

CHG.3.1 Transport Request

Creating Transport

1. Create transport request
2. Assign to transport
3. Document changes
4. Release transport
5. Import to target system

CHG.3.2 Transport Strategy

Landscape

- **Development:** Configuration and development
- **Quality:** Testing and validation
- **Production:** Live system

CHG.4 Release Management

CHG.4.1 Release Planning

Release Types

Type	Frequency	Content
Major	Annually	New features, major changes
Minor	Quarterly	Enhancements, fixes
Patch	As needed	Bug fixes, security

CHG.4.2 Release Process

Release Steps

1. Plan release
2. Build release package
3. Test release
4. Deploy release
5. Verify deployment
6. Communicate release

Extended Module Integration Guide

MI.1 FI-MM Integration

MI.1.1 Goods Receipt Posting

Accounting Entries

Account	Debit	Credit
Inventory Account	XXX	
GR/IR Clearing		XXX

Configuration Requirements

- **Transaction Key:** BSX (Inventory Posting)
- **WRX (GR/IR Clearing):** Automatic account determination
- **Valuation Class:** Assignment to material type
- **Account Modifier:** For special procurement types

Integration Points

1. Purchase Order creation triggers commitment in Funds Management
2. Goods Receipt posts inventory and GR/IR
3. Invoice Receipt clears GR/IR and posts vendor liability
4. Payment clears vendor liability and posts bank movement

MI.1.2 Invoice Verification

Two-Way Match Process

The two-way match compares the purchase order with the vendor invoice:

1. Vendor submits invoice with reference to PO
2. System checks PO quantity and price
3. If within tolerance, invoice is posted automatically
4. If outside tolerance, invoice is blocked for review

Three-Way Match Process

The three-way match adds goods receipt verification:

1. Verify goods receipt exists for invoiced quantity
2. Compare PO, GR, and invoice quantities
3. Check for quantity and price variances
4. Post or block based on variance analysis

Blocked Invoice Resolution

Block Reason	Resolution Action	Responsible Role
Quantity variance	Verify GR quantity, adjust if needed	Warehouse Manager
Price variance	Approve price change or reject invoice	Purchasing Manager
Missing GR	Complete goods receipt posting	Warehouse Clerk
Duplicate invoice	Verify and cancel duplicate	AP Accountant

MI.2 FI-SD Integration

MI.2.1 Sales Order to Cash Flow

Process Overview

1. **Sales Order:** Creates customer commitment, no accounting entry
2. **Delivery:** Reduces inventory, posts COGS
3. **Billing:** Creates customer receivable and revenue

4. **Payment:** Clears receivable, posts bank receipt

Revenue Recognition

Scenario	Recognition Point	Accounting Entry
Standard sale	At billing	Dr Customer, Cr Revenue
Prepayment	At delivery	Dr Unearned Revenue, Cr Revenue
Service contract	Over contract period	Monthly revenue recognition
Installment sale	At delivery	Dr Installment Receivable, Cr Revenue

MI.2.2 Pricing and Tax Integration

Condition Types

- **PR00:** Base price
- **K007:** Customer discount percentage
- **K005:** Material discount amount
- **RA01:** Discount amount
- **MWST:** Output VAT (14%)

VAT Calculation Flow

1. Base price determined from condition record
2. Discounts applied sequentially
3. Net amount calculated
4. VAT calculated on net amount at 14%
5. Gross amount = Net + VAT

MI.3 FI-PP Integration

MI.3.1 Production Order Settlement

Cost Flow

1. **Material Issue:** Raw materials issued to production order
2. **Activity Confirmation:** Labor and machine hours recorded
3. **Overhead Application:** Manufacturing overhead allocated
4. **Goods Receipt:** Finished goods received into inventory
5. **Order Settlement:** Variance calculated and posted

Variance Categories

Variance Type	Cause	Calculation
Input price variance	Material price changes	Actual price - Standard price
Input quantity variance	Material usage differences	Actual qty - Standard qty
Resource usage variance	Wrong material/component used	Alternative component cost
Remaining input variance	Other input differences	Unexplained differences
Output price variance	Standard cost changes	Old vs new standard cost
Quantity variance	Yield differences	Actual output - Planned output

MI.3.2 Work in Process (WIP) Calculation

WIP Valuation Methods

- **Actual Costs:** All actual costs posted to order
- **Target Costs:** Based on confirmed operations
- **Standard Costs:** Based on standard cost estimate

Accounting Entries

Transaction	Debit	Credit
WIP posting	WIP Inventory	Production Order
Settlement	Finished Goods	Production Order
Variance posting	Variance Account	Production Order

MI.4 FI-HR Integration

MI.4.1 Payroll to Finance Posting

Posting Document Structure

The payroll posting creates a comprehensive accounting document:

- **Debit:** Salary expense accounts by cost center
- **Credit:** Liability accounts for net pay, taxes, and deductions
- **Split:** By cost center, company code, and business area

Posting Example

Account	Description	Debit	Credit
610000	Basic Salary Expense	100,000	
610100	Housing Allowance Expense	20,000	
610200	Social Insurance - Employer	15,000	
210100	Social Insurance Payable		26,000
210200	Payroll Tax Payable		12,000
210300	Net Salary Payable		97,000

MI.4.2 GOSI Integration

Contribution Calculation

Type	Employer %	Employee %	Total %
Old Age, Disability, Death	15%	11%	26%
Work Injuries	2%	0%	2%
Unemployment (New)	1%	1%	2%

Monthly GOSI Process

1. Generate GOSI file from payroll results
2. Validate employee data and contributions
3. Submit file through GOSI portal
4. Receive confirmation and payment reference
5. Post payment to GOSI clearing account

MI.5 MM-SD Integration

MI.5.1 Third-Party Order Processing

Process Flow

1. Customer orders goods not in inventory
2. Sales order creates purchase requisition automatically
3. PO is created and sent to vendor
4. Vendor ships directly to customer
5. Vendor invoice is processed
6. Customer is billed

Configuration

- **Item Category:** TAS (Third-Party Item)
- **Schedule Line Category:** CS (Third-Party)
- **Account Assignment:** Sales order number

MI.5.2 Intercompany Stock Transfers

One-Step Transfer

Goods movement directly posts in both company codes:

- Issuing company: Inventory reduction, intercompany receivable
- Receiving company: Inventory increase, intercompany payable

Two-Step Transfer

Goods in transit account used:

1. Goods issued to in transit
2. Goods received from in transit
3. Billing document created
4. Intercompany invoice cleared

MI.6 MM-PP Integration

MI.6.1 Material Requirements Planning (MRP)

MRP Types

Type	Description	Use Case
PD	Automatic reorder	Materials with fluctuating demand
VB	Manual reorder point	Materials with stable consumption

ND	No MRP	Non-stock materials
VM	Forecast-based	Materials with seasonal demand

MRP Run Results

- **Planned Order:** For in-house production
- **Purchase Requisition:** For external procurement
- **Schedule Line:** For outline agreements
- **Exception Message:** For planner action

MI.6.2 Bill of Materials (BOM)

BOM Types

Type	Usage	Example
Material BOM	Production	Finished product components
Sales Order BOM	Customer-specific	Custom configuration
Document BOM	Engineering	Technical drawings
Equipment BOM	Maintenance	Spare parts list

BOM Explosion

1. Single-level: Direct components only
2. Multi-level: All components recursively
3. Summarized: All components aggregated

MI.7 SD-PP Integration

MI.7.1 Make-to-Order Production

Process Flow

1. Sales order received with customer specifications
2. Production order created with sales order reference
3. Materials reserved for specific sales order
4. Production completed and goods delivered
5. Billing created for customer

Configuration

- **Requirement Type:** KE (Make-to-Order)
- **Strategy Group:** 20 (Make-to-Order)
- **Account Assignment:** Sales order stock

MI.7.2 Assembly-to-Order

Process Characteristics

- Standard sub-assemblies kept in stock
- Final assembly triggered by sales order
- Short delivery time
- Customer-specific configuration possible

Planning Strategy

Level	Planning	Stock Type
Sub-assembly	Forecast-based	Unrestricted
Final assembly	Sales order-triggered	Sales order stock

Extended Upgrade Procedures Guide

UPG.1 Upgrade Planning

UPG.1.1 Upgrade Assessment

Current System Analysis

Before planning an upgrade, conduct comprehensive analysis:

- **Version Inventory:** Document all system components and versions
- **Customization Analysis:** Identify all custom developments
- **Interface Mapping:** Document all integration points
- **User Count:** Active users by module and function
- **Transaction Volume:** Peak and average load statistics

Upgrade Complexity Factors

Factor	Low Complexity	High Complexity
Version gap	One release	Multiple releases
Customizations	Few standard enhancements	Extensive modifications
Add-ons	Certified solutions	Custom add-ons
Database size	Under 500GB	Over 2TB
Users	Under 100	Over 1000

UPG.1.2 Upgrade Strategy Selection

Strategy Options

Strategy	Description	Best For
Big Bang	All modules at once	Small systems, tight timeline
Phased	Module by module	Large systems, risk mitigation
Parallel	Old and new run together	Critical systems, validation needed
Hybrid	Combination approach	Complex landscapes

Decision Matrix

Consider these factors when selecting strategy:

1. Business continuity requirements
2. Available downtime window
3. Resource availability
4. Budget constraints
5. Risk tolerance
6. Integration complexity

UPG.2 Technical Upgrade

UPG.2.1 Preparation Phase

Prerequisites Checklist

- Current system on supported version
- All support packages applied
- Database at compatible version
- Operating system supported
- Sufficient disk space (2x current size)
- Backup verified and restorable

Pre-Upgrade Tasks

1. Run pre-upgrade consistency check
2. Archive old application logs
3. Clean up temporary files
4. Update statistics on all tables
5. Deactivate unnecessary batch jobs
6. Notify all users of downtime

UPG.2.2 Upgrade Execution

Shadow System Approach

Modern upgrades use shadow instance technology:

1. Create shadow instance on same server
2. Import new release into shadow
3. Perform modification adjustment in shadow
4. Run parallel processes on shadow
5. Switch production to upgraded shadow
6. Remove old instance

Upgrade Phases

Phase	Duration	Activities
Preparation	2-4 hours	Checks, backups, shutdown
Import	4-12 hours	New version import
Adjustment	2-6 hours	Modification adjustment
Activation	1-2 hours	New version activation
Validation	2-4 hours	Testing and verification

UPG.2.3 Post-Upgrade Activities

Immediate Tasks

- Verify system startup and connectivity
- Check all work processes
- Validate database connections
- Test critical transactions
- Verify background job scheduling
- Check interface connectivity

Validation Checklist

Area	Test	Expected Result
Login	User authentication	Successful login
Authorization	Role verification	Correct access rights
Transactions	Key business processes	Complete successfully
Reports	Standard reports	Accurate output
Interfaces	Inbound/outbound	Data flows correctly
Printing	Forms and labels	Correct format

UPG.3 Functional Upgrade

UPG.3.1 New Functionality Adoption

Feature Evaluation

Assess new features for business value:

1. Review release notes and documentation

2. Identify relevant new features
3. Evaluate business benefit
4. Assess implementation effort
5. Prioritize adoption roadmap

Business Function Activation

Status	Description	Action
Always On	Automatically active	No action needed
On Demand	Customer decision	Activate if beneficial
Deprecated	Being phased out	Plan migration
Not Available	Not in this release	Future upgrade

UPG.3.2 Process Optimization

Upgrade as Improvement Opportunity

Use the upgrade to enhance business processes:

- Redesign inefficient workflows
- Adopt best practice processes
- Consolidate custom developments
- Standardize configurations
- Improve data quality
- Enhance user experience

Change Management

1. Communicate changes to stakeholders
2. Update procedures and documentation
3. Provide user training on new features

- 4. Establish support for transition period
- 5. Monitor adoption and feedback

UPG.4 Regression Testing

UPG.4.1 Test Planning

Test Scope Definition

Test Type	Scope	Priority
Unit tests	Individual transactions	High
Integration tests	Cross-module processes	High
Regression tests	Existing functionality	High
Performance tests	System under load	Medium
UAT	Business scenarios	High

Test Data Preparation

- 1. Copy production data to test system
- 2. Anonymize sensitive information
- 3. Create test scenarios
- 4. Prepare expected results
- 5. Document test procedures

UPG.4.2 Test Execution

Test Management

- Track all test cases in test management tool
- Record actual vs expected results

- Log all defects with severity
- Track defect resolution
- Obtain sign-off for each test phase

Defect Categories

Severity	Description	Resolution Time
Critical	System unusable	Immediate
High	Major function impaired	24 hours
Medium	Workaround available	1 week
Low	Cosmetic issue	Next release

UPG.5 Rollback Planning

UPG.5.1 Rollback Scenarios

When to Consider Rollback

- Critical system failure after upgrade
- Data corruption detected
- Key business process non-functional
- Performance degradation unacceptable
- Security vulnerability discovered

Rollback Decision Process

1. Assess severity and impact
2. Attempt resolution within new version
3. Evaluate rollback vs fix timeline
4. Obtain management approval

5. Execute rollback procedure

UPG.5.2 Rollback Procedures

Database Rollback

Restore from pre-upgrade backup:

1. Stop application servers
2. Restore database from backup
3. Verify database consistency
4. Start application servers
5. Validate system functionality

Transaction Recovery

Handle transactions during upgrade window:

- Document all transactions in progress
- Identify incomplete postings
- Re-enter lost transactions
- Reconcile balances
- Notify affected parties

Extended API Documentation

API.1 REST API Reference

API.1.1 Authentication

OAuth 2.0 Flow

The ERP API uses OAuth 2.0 for authentication:

1. **Authorization Request:** Client requests authorization code
2. **User Login:** User authenticates and grants permission
3. **Authorization Code:** Server returns authorization code
4. **Token Request:** Client exchanges code for access token
5. **API Access:** Client uses token for API calls

Token Endpoint

```
POST /oauth/token
Content-Type: application/x-www-form-urlencoded

grant_type=authorization_code
&code=AUTH_CODE
&client_id=CLIENT_ID
&client_secret=CLIENT_SECRET
&redirect_uri=REDIRECT_URI
```

Token Response

```
{
  "access_token": "eyJhbGciOiJSUzI1NiIs... ",
  "token_type": "Bearer",
  "expires_in": 3600,
  "refresh_token": "dGhpcyBpcyBhIHJlZnJlc2g...",
  "scope": "read write"
}
```

API.1.2 General Ledger API

Post Journal Entry

```
POST /api/v1/gl/journal-entries
Authorization: Bearer ACCESS_TOKEN
Content-Type: application/json
```

```
{
  "companyCode": "1000",
  "documentDate": "2026-02-22",
  "postingDate": "2026-02-22",
  "reference": "INV-2026-001",
  "headerText": "Sales Invoice",
  "items": [
    {
      "glAccount": "110010",
      "debitAmount": 11400.00,
      "costCenter": "1000",
      "lineText": "Customer receivable"
    },
    {
      "glAccount": "410010",
      "creditAmount": 10000.00,
      "costCenter": "1000",
      "lineText": "Sales revenue"
    },
    {
      "glAccount": "210510",
      "creditAmount": 1400.00,
      "lineText": "Output VAT 14%"
    }
  ]
}
```

Get Account Balance


```
GET /api/v1/gl/accounts/{accountId}/balance?  
    companyCode=1000&  
    fiscalYear=2026&  
    period=2
```

Response:

```
{  
  "accountId": "110010",  
  "accountName": "Accounts Receivable",  
  "openingBalance": 150000.00,  
  "debits": 85000.00,  
  "credits": 45000.00,  
  "closingBalance": 190000.00,  
  "currency": "EGP"  
}
```

API.1.3 Accounts Payable API

Create Vendor Invoice

```

POST /api/v1/ap/vendor-invoices
{
  "vendorId": "V001",
  "invoiceNumber": "INV-V001-456",
  "invoiceDate": "2026-02-15",
  "dueDate": "2026-03-15",
  "amount": 11400.00,
  "currency": "EGP",
  "reference": "PO-2026-123",
  "lineItems": [
    {
      "glAccount": "510010",
      "amount": 10000.00,
      "costCenter": "2000",
      "description": "Office supplies"
    },
    {
      "glAccount": "210500",
      "amount": 1400.00,
      "description": "Input VAT 14%"
    }
  ]
}

```

Process Payment

```

POST /api/v1/ap/payments
{
  "paymentMethod": "BANK_TRANSFER",
  "bankAccount": "EG810002000123456789012345678",
  "paymentDate": "2026-02-22",
  "invoices": [
    {
      "invoiceId": "INV-2026-001",
      "amount": 11400.00
    }
  ],
  "vendorId": "V001"
}

```

API.1.4 Inventory API

Goods Receipt

```
POST /api/v1/mm/goods-receipts
{
  "movementType": "101",
  "purchaseOrder": "4500001234",
  "documentDate": "2026-02-22",
  "items": [
    {
      "poItem": 10,
      "material": "MAT001",
      "quantity": 100,
      "unit": "PC",
      "storageLocation": "0001",
      "batch": "B202602001"
    }
  ]
}
```

Stock Overview

```
GET /api/v1/mm/materials/{materialId}/stock?
    plant=1000&
    storageLocation=0001
```

Response:

```
{
  "materialId": "MAT001",
  "materialName": "Raw Material A",
  "plant": "1000",
  "storageLocation": "0001",
  "unrestricted": 500,
  "inQualityInspection": 100,
  "blocked": 0,
  "inTransit": 50,
  "unit": "PC",
  "valuationClass": "3000",
  "movingAveragePrice": 25.50
}
```

API.2 Webhook Integration

API.2.1 Webhook Configuration

Register Webhook

```
POST /api/v1/webhooks
{
  "url": "https://client-system.eg/erp-webhook",
  "events": [
    "invoice.created",
    "payment.received",
    "order.shipped",
    "inventory.low_stock"
  ],
  "secret": "webhook_signing_secret",
  "active": true
}
```

Webhook Events

Event	Description	Payload
invoice.created	New invoice posted	Invoice details
payment.received	Payment processed	Payment details
ordershipped	Delivery completed	Delivery details
inventory.low_stock	Stock below threshold	Material and quantity
journal.posted	Journal entry posted	Entry details

API.2.2 Webhook Security

Signature Verification

```
// Webhook payload signature
X-ERP-Signature: sha256=SIGNATURE

// Verification (pseudocode)
expectedSignature = HMAC_SHA256(payload, webhookSecret)
if (expectedSignature === receivedSignature) {
  // Process webhook
} else {
  // Reject webhook
}
```

API.3 Error Handling

API.3.1 Error Response Format

Standard Error Response

```
{
  "error": {
    "code": "INVALID_PARAMETER",
    "message": "The request contains invalid parameters",
    "details": [
      {
        "field": "postingDate",
        "issue": "Date cannot be in the future"
      }
    ],
    "requestId": "req_1234567890",
    "timestamp": "2026-02-22T10:30:00Z"
  }
}
```

Error Codes

Code	HTTP Status	Description
AUTHENTICATION_FAILED	401	Invalid credentials

AUTHORIZATION_FAILED	403	Insufficient permissions
RESOURCE_NOT_FOUND	404	Requested resource not found
INVALID_PARAMETER	400	Request validation failed
BUSINESS_ERROR	422	Business rule violation
RATE_LIMIT_EXCEEDED	429	Too many requests
INTERNAL_ERROR	500	Server error

API.3.2 Rate Limiting

Rate Limit Headers

```
X-RateLimit-Limit: 1000  
X-RateLimit-Remaining: 999  
X-RateLimit-Reset: 1645526400
```

Rate Limits by Plan

Plan	Requests/Hour	Concurrent
Basic	1,000	10
Professional	10,000	50
Enterprise	100,000	200

Extended Portal User Guide

POR.1 Employee Self-Service

POR.1.1 Personal Information

View and Update Profile

Employees can manage their personal information through the portal:

1. Login to Employee Self-Service portal
2. Click "My Profile" in the navigation menu
3. View current personal information
4. Click "Edit" to update allowed fields
5. Submit changes for approval

Editable Fields

Field	Employee Edit	Requires Approval
Phone number	Yes	No
Email address	Yes	No
Emergency contact	Yes	No
Address	Yes	Yes
Bank account	Yes	Yes
Marital status	Yes	Yes

POR.1.2 Leave Management

Leave Balance Inquiry

Employees can view their leave balances:

- **Annual Leave:** Total, used, and remaining days
- **Sick Leave:** Current year balance
- **Casual Leave:** Available days
- **Unpaid Leave:** History and status

Leave Request Process

1. Click "Request Leave" button
2. Select leave type from dropdown
3. Choose start and end dates
4. Enter reason for leave
5. Attach supporting documents if required
6. Submit request
7. Track approval status

Approval Workflow

Leave Type	First Approver	Second Approver
Annual (1-3 days)	Direct Manager	-
Annual (4+ days)	Direct Manager	Department Head
Sick (1-2 days)	Direct Manager	-
Sick (3+ days)	Direct Manager	HR Manager
Unpaid	Direct Manager	HR Manager

POR.1.3 Payroll Information

Payslip Access

Employees can view and download payslips:

1. Navigate to "My Payroll" section
2. Select month and year
3. View payslip details
4. Download as PDF
5. Email payslip to personal address

Payslip Components

Section	Details
Earnings	Basic, allowances, bonuses, overtime
Deductions	GOSI, tax, loans, advances
Employer Contributions	GOSI employer share
Year-to-Date	Cumulative totals
Net Pay	Final amount deposited

POR.2 Manager Self-Service

POR.2.1 Team Management

Team Overview

Managers can view their team information:

- Team member list with photos
- Contact information
- Position and grade
- Join date
- Current leave status

Leave Approval

1. Access "Pending Approvals" dashboard
2. Review leave requests
3. Check team calendar for conflicts
4. Approve or reject with comments
5. Delegate approval if unavailable

POR.2.2 Performance Management

Goal Setting

Managers can set and track employee goals:

1. Select employee from team list
2. Click "Set Goals"
3. Define SMART objectives
4. Set key performance indicators (KPIs)
5. Assign weights to each goal
6. Set review dates
7. Submit for employee acknowledgment

Performance Review

Review Type	Frequency	Focus
Probation	End of probation	Fit for role
Mid-year	6 months	Progress check
Annual	Year-end	Overall performance
360-degree	Annually	Multi-rater feedback

POR.3 Vendor Portal

POR.3.1 Vendor Registration

Self-Registration Process

1. Access vendor portal registration page
2. Enter company information
3. Upload required documents
 - Commercial registration
 - Tax card
 - Bank account details
 - Quality certificates
4. Submit for approval
5. Receive vendor code upon approval

Document Requirements

Document	Format	Validity
Commercial registration	PDF, JPG	Must be valid
Tax card	PDF, JPG	Must be valid
Bank letter	PDF	Within 3 months
ISO certificate	PDF	If applicable

POR.3.2 Purchase Order Management

PO Acknowledgment

Vendors can view and acknowledge purchase orders:

1. Login to vendor portal
2. View "New Purchase Orders"

3. Review PO details and terms
4. Acknowledge acceptance
5. Or request changes with justification

Delivery Scheduling

- View delivery due dates
- Update expected delivery date
- Provide tracking information
- Upload delivery documents

POR.3.3 Invoice Submission

Electronic Invoice Upload

1. Navigate to "Submit Invoice"
2. Select referenced PO
3. Enter invoice details
4. Upload invoice PDF
5. Submit for processing
6. Track payment status

Payment Tracking

Status	Description
Received	Invoice uploaded
Under Review	Being processed
Approved	Approved for payment
Scheduled	Payment date set
Paid	Payment completed

POR.4 Customer Portal

POR.4.1 Account Management

Customer Registration

1. Request account creation
2. Provide company details
3. Submit trade license
4. Complete credit application
5. Receive login credentials

Account Information

- View account balance
- Check credit limit
- View payment terms
- Download account statements
- Update contact information

POR.4.2 Order Management

Online Ordering

1. Browse product catalog
2. Add items to cart
3. Review order summary
4. Select delivery address
5. Confirm order
6. Receive order confirmation

Order Tracking

Status	Description
Received	Order submitted
Processing	Being prepared
Ready	Ready for shipment
Shipped	In transit
Delivered	Delivery completed

POR.4.3 Invoice and Payment

Invoice Access

- View all invoices
- Download PDF copies
- Check payment status
- View payment history

Online Payment

1. Select invoices to pay
2. Choose payment method
 - Credit card
 - Bank transfer
 - Digital wallet
3. Confirm payment amount
4. Complete secure payment
5. Receive payment receipt

Extended E-Commerce Integration

ECO.1 Platform Integration

ECO.1.1 Marketplace Integration

Supported Platforms

Platform	Integration Type	Features
Amazon	API + EDI	Orders, inventory, pricing
Noon	API	Full marketplace integration
Jumia	API	African market focus
Local platforms	Custom	Tailored integration

Order Synchronization

1. Marketplace order received
2. Order imported to ERP
3. Inventory reservation
4. Order fulfillment
5. Shipping confirmation sent
6. Tracking updated

ECO.1.2 Own E-Commerce Site

Website Integration Architecture

- **Frontend:** React/Vue.js storefront
- **Backend:** ERP APIs

- **Middleware:** Integration layer
- **Cache:** Redis for performance
- **CDN:** Static content delivery

Real-Time Data Flow

Data	Direction	Frequency
Products	ERP to Web	Near real-time
Prices	ERP to Web	Real-time
Inventory	ERP to Web	Real-time
Orders	Web to ERP	Real-time
Customers	Bi-directional	Real-time

ECO.2 Product Catalog Management

ECO.2.1 Product Information

Product Master Data

- **Basic Data:** SKU, name, description
- **Classification:** Category, brand, attributes
- **Media:** Images, videos, documents
- **Pricing:** Base price, special prices
- **Inventory:** Stock levels, locations
- **Shipping:** Weight, dimensions, class

Attribute Management

Attribute Type	Examples	Use
----------------	----------	-----

Text	Name, description	Search, display
Number	Weight, size	Filtering, shipping
Select	Color, size	Variants, filtering
Boolean	In stock, featured	Filtering, badges
Multi-select	Tags, features	Filtering, search

ECO.2.2 Pricing Engine

Dynamic Pricing Rules

- 1. **Base Price:** Standard product price
- 2. **Customer Group:** B2B vs B2C pricing
- 3. **Volume Discounts:** Quantity breaks
- 4. **Promotions:** Time-limited offers
- 5. **Coupon Codes:** Special discounts
- 6. **Loyalty:** Points-based discounts

Price Calculation Example

Base Price:	EGP 1,000.00
Customer Discount:	- 100.00 (10%)
Volume Discount:	- 50.00
Promotion:	- 85.00
Subtotal:	EGP 765.00
VAT (14%):	+ 107.10
Final Price:	EGP 872.10

ECO.3 Order Fulfillment

ECO.3.1 Order Processing

Order Lifecycle

Stage	ERP Action	Duration
Received	Create sales order	Instant
Confirmed	Credit check, inventory	< 1 min
Picking	Create delivery	2-4 hours
Packed	Confirm packing	1 hour
Shipped	Post goods issue	Instant
Delivered	Confirm delivery	1-3 days

Allocation Strategy

- **First-Come-First-Served:** Standard allocation
- **Priority:** VIP customers first
- **Proportional:** Distribute limited stock
- **Manual:** Planner decision

ECO.3.2 Shipping Integration

Carrier Integration

Carrier	Services	Features
Aramex	Domestic, international	Real-time tracking
Bosta	Domestic	Cash on delivery
DHL	Express, freight	Global network
FedEx	Express, ground	Time-definite

Shipping Rate Calculation

1. Determine origin and destination
2. Calculate package weight and dimensions
3. Apply shipping class rules
4. Get carrier rates via API
5. Apply markup or discount
6. Display options to customer

ECO.4 Payment Processing

ECO.4.1 Payment Methods

Supported Methods

Method	Processing	Settlement
Credit Card	Real-time	2-3 days
Debit Card	Real-time	1-2 days
Bank Transfer	1-2 days	Upon receipt
Cash on Delivery	Delivery	Same day
Digital Wallet	Real-time	1-2 days
Installments	Real-time	Monthly

Payment Gateway Integration

- **Paymob:** Egyptian payment gateway
- **Fawry:** Cash payment network
- **PayPal:** International payments

- **Stripe:** Card processing

ECO.4.2 Fraud Prevention

Risk Checks

1. Address Verification (AVS)
2. Card Verification Value (CVV)
3. 3D Secure authentication
4. Velocity checks
5. IP geolocation
6. Device fingerprinting

Risk Scoring

Score	Action
0-30	Auto-approve
31-60	Additional verification
61-80	Manual review
81-100	Decline

Extended Industry-Specific Guides

IND.1 Manufacturing Industry

IND.1.1 Discrete Manufacturing

Production Types

Type	Characteristics	Examples
Make-to-Stock	Forecast-driven, standard products	Consumer electronics
Make-to-Order	Customer-specific production	Industrial equipment
Engineer-to-Order	Design then produce	Custom machinery
Assemble-to-Order	Standard parts, configured product	Computers, vehicles

Production Planning

1. **Demand Management:** Forecast and sales planning
2. **Master Production Schedule:** End product planning
3. **Material Requirements Planning:** Component planning
4. **Capacity Planning:** Work center load balancing
5. **Shop Floor Control:** Execution and monitoring

IND.1.2 Process Manufacturing

Industry Characteristics

- Batch or continuous production
- Formula/recipe-based
- Co-products and by-products

- Variable yield
- Quality management critical
- Regulatory compliance

Process Order Management

Phase	Activities
Creation	Formula selection, batch sizing
Release	Material availability check
Execution	Material staging, production
Confirmation	Yield, consumption recording
Settlement	Cost allocation, variances

IND.2 Retail Industry

IND.2.1 Multi-Channel Retail

Channel Integration

- **Physical Stores:** POS integration, inventory
- **E-Commerce:** Online storefront integration
- **Marketplaces:** Amazon, Noon, Jumia
- **Mobile:** App-based shopping
- **Social:** Social commerce

Omnichannel Fulfillment

Model	Description	Benefits
Ship from Store	Use store inventory	Faster delivery

Buy Online Pickup In Store	Reserve online	Convenience
Endless Aisle	Order unavailable items	No lost sales
Return to Store	Online purchase returns	Customer convenience

IND.2.2 Merchandise Management

Assortment Planning

1. Analyze customer segments
2. Define category roles
3. Determine product mix
4. Set space allocation
5. Plan promotional calendar
6. Monitor and adjust

Replenishment Strategies

Strategy	Trigger	Best For
Min/Max	Stock level	Stable demand
Reorder Point	Forecast	Variable demand
Top-Off	Fixed schedule	Fast-moving items
Demand-Driven	Actual sales	Responsive supply chain

IND.3 Construction Industry

IND.3.1 Project-Based Manufacturing

Project Structure

- **Work Breakdown Structure (WBS):** Hierarchical project structure
- **Networks:** Activities and dependencies
- **Milestones:** Key project events
- **Project Stock:** Materials assigned to project

Project Costing

Cost Element	Planning	Actual	Variance
Labor	Planned hours x rate	Confirmed hours	Efficiency
Materials	BOM x prices	Issues	Usage/price
Equipment	Planned usage	Actual usage	Utilization
Subcontractors	Contract value	Invoices	Changes
Overhead	% of direct costs	Allocated	Absorption

IND.3.2 Contract Management

Contract Types

Type	Characteristics	Risk Profile
Lump Sum	Fixed price	Contractor risk
Cost Plus	Actual cost + fee	Client risk
Time & Material	Hourly rates	Shared risk
Unit Price	Per unit rates	Quantity risk

Progress Billing

1. Submit progress report
2. Client/engineer verification

- 3. Issue payment certificate
- 4. Submit invoice
- 5. Receive payment (less retention)
- 6. Update project cash flow

IND.4 Healthcare Industry

IND.4.1 Hospital Management

Patient Management

- **Registration:** Patient demographics
- **Appointments:** Scheduling system
- **Admission:** Bed management
- **Clinical:** Electronic medical records
- **Billing:** Insurance and patient billing

Insurance Integration

Process	System Action
Eligibility Check	Real-time insurance verification
Pre-authorization	Request approval for procedures
Claim Submission	Electronic claim filing
Payment Posting	Auto-post insurance payments
Patient Responsibility	Balance billing

IND.4.2 Pharmaceutical Distribution

Regulatory Compliance

- GDP (Good Distribution Practice)
- Cold chain management
- Batch traceability
- Expiry date tracking
- Controlled substance management

Serialization Requirements

1. Assign unique serial number
2. Print 2D data matrix code
3. Report to regulatory system
4. Track through supply chain
5. Verify at dispensing

IND.5 Food & Beverage Industry

IND.5.1 Recipe Management

Recipe Structure

Element	Description
Header	Recipe name, version, status
Ingredients	Raw materials with quantities
Operations	Production steps
Resources	Equipment and labor
Yield	Expected output
Costing	Standard cost calculation

IND.5.2 Quality and Safety

HACCP Integration

- Hazard analysis documentation
- Critical control points monitoring
- Corrective action tracking
- Verification records
- Traceability documentation

Shelf Life Management

Stage	Action
Receiving	Check remaining shelf life
Storage	FIFO/FEFO rotation
Production	Use oldest stock first
Shipping	Verify customer requirements
Alerts	Expiry warnings

Extended Compliance Documentation

COM.1 Egyptian Tax Authority (ETA) Compliance

COM.1.1 E-Invoicing Requirements

Mandatory Fields

Field	Description	Format
UUID	Unique invoice identifier	UUID v4
Invoice Number	Sequential number	Numeric
Issue Date	Invoice date	YYYY-MM-DD
Seller Tax ID	Seller registration	9 digits
Buyer Tax ID	Buyer registration	9 digits or "NA"
Total Amount	Invoice total	Decimal
Tax Amount	VAT amount	Decimal
Digital Signature	Electronic signature	Base64
QR Code	Machine-readable code	Base64 image

Submission Process

1. Generate invoice in ERP
2. Create ETA-compliant JSON
3. Sign with registered certificate
4. Submit to ETA portal via API
5. Receive submission UUID

- 6. Update invoice with ETA reference
- 7. Include QR code on printed invoice

COM.1.2 ETA Portal Integration

API Endpoints

Endpoint	Purpose	Method
/api/v1/documents	Submit invoice	POST
/api/v1/documents/{id}	Get invoice status	GET
/api/v1/documents/{id}/cancel	Cancel invoice	POST
/api/v1/rejections	Get rejected invoices	GET
/api/v1/validations	Validate document	POST

Error Handling

- **Validation Errors:** Schema violations
- **Business Errors:** Rule violations
- **Technical Errors:** System issues
- **Retry Logic:** Automatic resubmission

COM.2 Social Insurance Compliance

COM.2.1 GOSI Reporting

Monthly Submission

Data Element	Description
Employer Number	GOSI employer registration

Employee ID	National ID number
Employee Name	Full name (Arabic/English)
Social Salary	Contribution base salary
Contribution	Calculated contribution
Employment Status	Active, terminated, etc.

Contribution Rates 2026

Category	Employer %	Employee %
Old Age, Disability, Death (Variable)	15%	11%
Old Age, Disability, Death (Constant)	15%	11%
Work Injuries	2%	0%
Unemployment (new)	1%	1%

COM.2.2 New Labor Law 14/2025 Compliance

Key Changes

- **Flexible Work:** Remote and hybrid arrangements
- **Fixed-Term Contracts:** Expanded usage
- **Termination:** Updated procedures
- **Leave Entitlements:** Revised calculations
- **Anti-Discrimination:** Enhanced protections

System Configuration Updates

Area	Configuration
Leave Types	Add new leave categories

Work Schedules	Flexible hours setup
Contract Types	Fixed-term templates
Termination	Updated calculation rules
Reporting	New compliance reports

COM.3 VAT Compliance

COM.3.1 VAT Registration

Registration Thresholds

Category	Threshold (Annual)	Requirement
Mandatory registration	EGP 500,000	Must register
Voluntary registration	Below threshold	May register
Exempt	N/A	Cannot register

Registration Process

1. Prepare required documents
2. Submit online application
3. Receive temporary number
4. Undergo tax inspection
5. Receive permanent VAT number
6. Configure in ERP system

COM.3.2 VAT Rates and Exemptions

Rate Structure

Rate	Applies To
14%	Standard goods and services
5%	Specific goods (table sugar, tea)
0%	Exports, international transport
Exempt	Financial services, healthcare, education

Input VAT Recovery

- **Fully Recoverable:** For taxable supplies
- **Partially Recoverable:** Mixed use (taxable/exempt)
- **Non-Recoverable:** Entertainment, personal use

COM.3.3 VAT Return Filing

Return Schedule

Taxpayer Category	Frequency	Due Date
Large taxpayers	Monthly	15th of following month
Medium taxpayers	Monthly	15th of following month
Small taxpayers	Quarterly	15th of following month

Return Contents

1. Output VAT (sales)
2. Input VAT (purchases)
3. Net VAT payable/recoverable
4. Exempt supplies value
5. Zero-rated supplies value

6. Adjustments and corrections

COM.4 Data Protection Compliance

COM.4.1 Personal Data Protection Law

Data Subject Rights

Right	Description	Implementation
Access	Know what data is held	Self-service portal
Correction	Update inaccurate data	Edit functionality
Deletion	Request data removal	Retention management
Portability	Export personal data	Data export feature
Objection	Opt-out of processing	Preference management

Technical Safeguards

- Encryption at rest and in transit
- Access controls and authentication
- Audit logging
- Data minimization
- Regular security assessments
- Incident response procedures

COM.4.2 Data Retention Policies

Retention Periods

Data Type	Retention Period	Basis
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Financial records	10 years	Tax law
Payroll records	10 years	Labor law
Employee files	5 years after termination	Labor law
Customer data	Duration of relationship	Contract
Transaction logs	7 years	Compliance

Secure Disposal

1. Identify data past retention period
2. Verify no legal hold applies
3. Generate disposal report
4. Securely delete from all systems
5. Archive disposal certificate

Extended Data Migration Guide

MIG.1 Migration Planning

MIG.1.1 Migration Strategy

Approach Selection

Approach	Description	Best For
Big Bang	All data at once	Small datasets
Phased	By module or entity	Complex systems
Parallel	Old and new together	Critical data
Hybrid	Combination	Large enterprises

Migration Objects

- **Master Data:** Customers, vendors, materials, GL accounts
- **Transactional:** Open items, orders, stock
- **Historical:** Closed transactions, audit trail
- **Configuration:** Custom settings, workflows
- **Documents:** Attachments, images

MIG.1.2 Data Quality Assessment

Quality Dimensions

Dimension	Metric	Target
Completeness	% populated fields	> 95%

Accuracy	% correct values	> 98%
Consistency	% standardized	> 95%
Validity	% within rules	> 99%
Uniqueness	% duplicates	< 1%

Data Profiling

- 1. Identify data sources
- 2. Extract sample data
- 3. Run profiling tools
- 4. Analyze patterns and anomalies
- 5. Document findings
- 6. Estimate cleansing effort

MIG.2 Data Extraction

MIG.2.1 Source System Analysis

Legacy System Types

System Type	Extraction Method	Challenges
Database	SQL queries	Complex relationships
Flat files	File parsing	Format variations
API	API calls	Rate limits
Spreadsheets	Direct import	Data quality
Custom apps	Custom extraction	Documentation

Extraction Design

- Define extraction queries
- Handle incremental extraction
- Manage large data volumes
- Ensure data consistency
- Log extraction activities

MIG.2.2 Data Transformation

Transformation Types

Type	Example	Tool
Cleansing	Remove duplicates	Data quality rules
Standardization	Phone formats	Format templates
Enrichment	Add missing data	Lookup tables
Aggregation	Summarize history	Calculation rules
Validation	Check constraints	Business rules

Mapping Documentation

1. Source field identification
2. Target field mapping
3. Transformation rules
4. Default values
5. Validation criteria
6. Exception handling

MIG.3 Data Loading

MIG.3.1 Load Methods

Loading Techniques

Method	Speed	Use Case
Direct input	Slow	Small volumes
Batch upload	Medium	Medium volumes
API import	Medium	Real-time needs
Direct database	Fast	Large volumes
ETL tools	Configurable	Complex transforms

MIG.3.2 Reconciliation

Reconciliation Types

- **Count Reconciliation:** Source vs target record counts
- **Value Reconciliation:** Sum of amounts match
- **Sample Testing:** Detailed record comparison
- **Business Validation:** Process test cases

Reconciliation Report

Object	Source Count	Target Count	Status
Customers	1,250	1,250	OK
Vendors	450	450	OK
GL Accounts	850	850	OK
Open Invoices	3,200	3,198	2 errors

MIG.4 Post-Migration

MIG.4.1 Validation

Validation Checklist

1. Master data completeness
2. Transaction accuracy
3. Report comparisons
4. Integration testing
5. User acceptance testing
6. Performance verification

MIG.4.2 Legacy Decommissioning

Decommissioning Steps

1. Archive final data extract
2. Export all reports
3. Document system closure
4. Maintain access for audit
5. Plan final shutdown
6. Dispose of hardware/data

Extended Performance Tuning Guide

PER.1 Performance Monitoring

PER.1.1 Key Performance Indicators

System Metrics

Metric	Target	Measurement
Response time	< 2 seconds	Transaction ST03
CPU utilization	< 70%	OS monitor
Memory usage	< 80%	ST02
Database time	< 40% of total	ST04
Dialog steps/hour	Per user profile	ST03

User Experience Metrics

- **Initial Screen Load:** Time to first display
- **Data Entry Response:** Field validation speed
- **Report Generation:** Background job duration
- **Search Results:** Query response time
- **Save Operations:** Commit duration

PER.1.2 Monitoring Tools

Built-in Monitors

Transaction	Purpose
-------------	---------

ST03	Workload analysis
ST04	Database performance
ST05	SQL trace
ST06	OS monitor
ST22	ABAP dump analysis
SM50	Work process overview
SM66	Global work process

PER.2 Database Optimization

PER.2.1 SQL Performance

SQL Tuning Guidelines

1. Use appropriate indexes
2. Avoid SELECT *
3. Limit result sets
4. Use bind variables
5. Optimize joins
6. Update statistics regularly

Index Strategy

Index Type	Use Case
Primary	Unique record identification
Secondary	Frequently queried fields
Composite	Multi-column queries

Bitmap	Low cardinality columns
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PER.2.2 Table Optimization

Table Maintenance

- Regular reorganization
- Partitioning large tables
- Archiving old data
- Compression where applicable
- Statistics updates

PER.3 Application Optimization

PER.3.1 ABAP Optimization

Coding Best Practices

Practice	Benefit
Use field symbols	Faster data access
Binary search	Faster internal table reads
Hash tables	Fast key-based access
Buffering	Reduced database access
Parallel processing	Utilize multiple cores

Performance Checklist

1. Minimize database round trips
2. Use appropriate internal table types
3. Avoid nested SELECTs

4. Process data in packages
5. Use buffering for master data

PER.3.2 Memory Management

Buffer Tuning

Buffer	Purpose	Tuning
Program	Store loaded programs	Hit rate > 95%
CUA	Screen elements	Hit rate > 90%
Table	Generic table buffering	Based on access pattern
Swap	Roll area management	Minimize swapping

PER.4 Infrastructure Optimization

PER.4.1 Hardware Sizing

Sizing Factors

- Number of concurrent users
- Transaction volume
- Data growth rate
- Batch processing requirements
- Integration load

Scaling Options

Approach	Description	When to Use
Vertical	More resources per server	Single server limit

Horizontal	More application servers	User scaling
Database	Cluster or scale-up	Data volume growth

PER.4.2 Network Optimization

Network Tuning

- Adequate bandwidth
- Low latency connections
- Load balancing
- Compression
- Caching at edge

Extended Security Hardening Guide

SEC.1 Access Control

SEC.1.1 User Management

User Lifecycle

Stage	Actions	Responsible
Provisioning	Create account, assign roles	Security Admin
Modification	Update roles, transfer	Security Admin
Suspension	Disable account	Security Admin
Termination	Delete/restrict account	Security Admin

Password Policy

Parameter	Recommendation
Minimum length	12 characters
Complexity	Upper, lower, number, special
Expiration	90 days
History	Last 12 passwords
Lockout	After 5 failed attempts

SEC.1.2 Role Design

Role Types

- **Master Roles:** Template roles per function
- **Derived Roles:** Organizational level variants
- **Composite Roles:** Collections for job positions
- **Single Roles:** Specific authorization sets

Separation of Duties

Conflict	Role 1	Role 2
Procurement	Purchase Order Create	Vendor Payment
Sales	Sales Order Create	Customer Receipt
Inventory	Goods Receipt	Goods Issue
HR	Hire Employee	Process Payroll

SEC.2 Data Security

SEC.2.1 Encryption

Encryption Requirements

Data State	Method	Standard
At rest	Database encryption	AES-256
In transit	TLS	TLS 1.3
In backup	Backup encryption	AES-256
Application	Field-level encryption	AES-256

SEC.2.2 Data Classification

Classification Levels

Level	Description	Handling
Public	No restriction	Standard controls
Internal	Company use only	Access control
Confidential	Sensitive business	Need-to-know
Restricted	Highly sensitive	Enhanced controls

SEC.3 Audit and Monitoring

SEC.3.1 Security Audit

Audit Scope

- User access reviews
- Privileged user activity
- Failed login attempts
- Data access patterns
- Configuration changes
- Security incidents

Audit Trail Requirements

Event	Data Captured	Retention
Login	User, time, IP, result	2 years
Transaction	User, object, action, time	7 years
Change	Before/after values	7 years
Admin	User, action, parameters	3 years

SEC.3.2 Vulnerability Management

Vulnerability Lifecycle

- 1. Discovery (scanning, reporting)
- 2. Assessment (severity, impact)
- 3. Remediation (patch, mitigate)
- 4. Verification (re-scan, confirm)
- 5. Documentation (lessons learned)

SEC.4 Incident Response

SEC.4.1 Incident Classification

Severity Levels

Level	Description	Response Time
Critical	System compromise, data breach	15 minutes
High	Active attack, service impact	1 hour
Medium	Suspicious activity, policy violation	4 hours
Low	Minor issue, informational	24 hours

SEC.4.2 Response Procedures

Incident Response Steps

- 1. **Detection:** Identify security event
- 2. **Containment:** Limit damage
- 3. **Eradication:** Remove threat
- 4. **Recovery:** Restore normal operations

5. **Lessons Learned:** Improve defenses

Extended System Administration Guide

ADM.1 User Administration

ADM.1.1 User Master Record

User Types

Type	Description	Examples
Dialog	Interactive users	Employees, managers
System	Background processing	Workflow, batch
Communication	External systems	EDI, RFC connections
Reference	Template users	Role templates

User Master Data Fields

- **Address:** Name, department, communication
- **Logon Data:** Password, validity, language
- **Defaults:** Start menu, printer, decimal format
- **Parameters:** User-specific default values
- **Roles:** Assigned authorization profiles
- **Profiles:** Directly assigned profiles
- **Groups:** User group assignment
- **Licenses:** License type assignment

ADM.1.2 Mass User Management

Transaction SU10

Mass processing of user master records:

- 1. Select users by criteria (role, user group, etc.)
- 2. Choose processing mode (change, lock, delete)
- 3. Specify fields to modify
- 4. Execute mass update
- 5. Review change log

Common Mass Operations

Operation	Use Case
Lock users	End of month processing
Change password	Security policy update
Assign role	New functionality rollout
Change validity	Temporary contractor access
Delete users	Cleanup inactive accounts

ADM.2 Background Job Management

ADM.2.1 Job Scheduling

Job Definition

- **Job Name:** Descriptive identifier
- **Program:** ABAP program or external command
- **Variant:** Predefined selection criteria
- **Start Condition:** Time, event, or dependency
- **Target Server:** Application server for execution

Scheduling Options

Option	Description	Example
Immediate	Start right away	One-time job
Date/Time	Specific date and time	Monthly closing
Periodic	Recurring schedule	Daily reports
Event-based	Triggered by event	After data load
Job after job	Dependency chain	Sequential processing

ADM.2.2 Job Monitoring

Job Status Values

Status	Description	Action Required
Scheduled	Waiting for start condition	None
Released	Ready to run	None
Ready	Waiting for work process	None
Active	Currently running	Monitor
Finished	Completed successfully	None
Cancelled	Terminated with error	Investigate

Job Troubleshooting

1. Check job log for error messages
2. Review spool output
3. Analyze short dump if ABAP error

4. Check system resources during execution
5. Verify variant parameters
6. Test program in foreground

ADM.3 Transport Management

ADM.3.1 Transport System Overview

Transport Landscape

System	Purpose	Transport Direction
Development (DEV)	Configuration and development	Export
Quality Assurance (QAS)	Testing and validation	Import
Production (PRD)	Live system	Import

Transport Request Types

- **Workbench:** Cross-client changes (programs, dictionary)
- **Customizing:** Client-specific configuration
- **Transport of Copies:** Copy without original ownership
- **Relocation:** Move objects between systems

ADM.3.2 Transport Process

Export Process

1. Create transport request
2. Assign objects to request
3. Release request
4. Generate data and cofiles
5. Transfer files to target system

Import Process

1. Add to import queue
2. Check import prerequisites
3. Execute import
4. Verify import log
5. Perform post-import activities

ADM.4 Spool and Print Management

ADM.4.1 Output Management

Spool Request Lifecycle

Status	Description
- (Dash)	Waiting for processing
+ (Plus)	Being processed
Waiting	Waiting for output device
Processing	Currently printing
Completed	Successfully printed
Error	Printing failed

Output Devices

- **Local Printer:** Directly connected
- **Network Printer:** Shared printer
- **Fax:** Fax server integration
- **Email:** PDF via email

- **Archive:** Document management system

ADM.4.2 Print Troubleshooting

Common Issues

Issue	Cause	Solution
Spool stuck in wait	Device unavailable	Check device status
Garbled output	Wrong device type	Update device settings
Missing characters	Font issue	Install correct font
Slow printing	Large spool size	Optimize form

Extended Business Intelligence Guide

BI.1 Data Warehousing

BI.1.1 Data Warehouse Architecture

Layer Structure

Layer	Purpose	Data Characteristics
Staging	Raw data extraction	Unchanged source data
Integration	Cleanse and transform	Harmonized data
Presentation	Business views	Aggregated, dimensional

Data Warehouse Models

- **Star Schema:** Central fact table with dimension tables
- **Snowflake:** Normalized dimension tables
- **Data Vault:** Hub, link, and satellite structure
- **Anchor Modeling:** Sixth normal form approach

BI.1.2 ETL Processes

Extraction Methods

Method	Description	Use Case
Full extract	Complete data load	Initial load
Incremental	Changed data only	Regular updates
CDC	Capture data changes	Real-time needs

Transformation Types

- 1. **Cleansing:** Remove errors and inconsistencies
- 2. **Standardization:** Common formats and values
- 3. **Integration:** Combine from multiple sources
- 4. **Aggregation:** Summarize detail data
- 5. **Derivation:** Calculate new values

BI.2 Reporting and Analytics

BI.2.1 Report Types

Classification by Purpose

Type	Purpose	Frequency
Operational	Daily operations	Real-time
Tactical	Management decisions	Weekly/Monthly
Strategic	Executive planning	Quarterly/Annual

Report Formats

- **List:** Tabular data display
- **Summary:** Aggregated information
- **Graphical:** Charts and visualizations
- **Interactive:** Drill-down capability
- **Formatted:** Pre-designed layouts

BI.2.2 Dashboard Design

Dashboard Principles

- 1. Focus on key metrics
- 2. Provide context and comparison
- 3. Enable drill-down for details
- 4. Update in near real-time
- 5. Design for target audience

Common Dashboard Elements

Element	Purpose
KPI cards	Key metric display
Trend charts	Historical analysis
Gauges	Performance vs target
Tables	Detailed data
Maps	Geographic analysis
Alerts	Exception notification

BI.3 Advanced Analytics

BI.3.1 Predictive Analytics

Analytical Techniques

Technique	Application	Example
Regression	Trend forecasting	Sales prediction
Classification	Segmentation	Customer categories
Clustering	Pattern discovery	Market segments

Time series

Seasonal analysis

Demand planning

Implementation Process

1. Define business problem
2. Gather and prepare data
3. Explore and analyze
4. Build predictive model
5. Validate and refine
6. Deploy and monitor

BI.3.2 Machine Learning Integration

Use Cases in ERP

- **Demand Forecasting:** Predict future sales
- **Anomaly Detection:** Identify unusual transactions
- **Churn Prediction:** Identify at-risk customers
- **Fraud Detection:** Spot suspicious patterns
- **Process Optimization:** Recommend improvements

Extended Mobile Application Guide

MOB.1 Mobile Architecture

MOB.1.1 Platform Overview

Supported Platforms

Platform	Minimum Version	Features
iOS	14.0+	Full functionality
Android	10.0+	Full functionality
Web (PWA)	Modern browsers	Core functionality

Architecture Components

- **Mobile App:** Native iOS/Android apps
- **Middleware:** Mobile services gateway
- **Backend:** ERP system APIs
- **Offline Store:** Local data caching
- **Push Service:** Notification delivery

MOB.1.2 Offline Capabilities

Offline Data Handling

Scenario	Behavior
Online to offline	Continue with cached data
Offline entry	Queue for synchronization

Back online

Auto-sync queued data

Conflict

Prompt for resolution

Sync Configuration

- Define sync frequency
- Specify sync scope
- Configure conflict rules
- Set data retention

MOB.2 Mobile Features

MOB.2.1 Approvals

Approval Types

Type	Description	Notification
Leave request	Employee leave approval	Push + Email
Purchase order	PO release	Push + Email
Expense claim	Reimbursement approval	Push + Email
Timesheet	Hours confirmation	Push only

Approval Actions

1. View approval request details
2. Review attachments
3. Add comments
4. Approve or reject
5. Delegate if needed

MOB.2.2 Mobile Analytics

Available Dashboards

- **Executive:** Key business metrics
- **Sales:** Pipeline and performance
- **Finance:** Cash flow and budgets
- **Inventory:** Stock levels and movements
- **HR:** Headcount and attendance

MOB.3 Mobile Security

MOB.3.1 Device Security

Security Requirements

Control	Requirement
Passcode	Device must have PIN/biometric
Encryption	Device storage encrypted
Jailbreak	Detect and block rooted devices
Remote wipe	Enable data deletion

MOB.3.2 App Security

Authentication Methods

- **Username/Password:** Standard login
- **Biometric:** Fingerprint, Face ID
- **SSO:** Corporate identity provider
- **Certificate:** Client certificate

Extended Training and Documentation

TRN.1 Training Program

TRN.1.1 Training Framework

Training Levels

Level	Audience	Content
Basic	End users	Navigation, transactions
Intermediate	Key users	Process knowledge
Advanced	Super users	Configuration, troubleshooting
Expert	IT staff	Technical administration

Training Methods

- **Classroom:** Instructor-led sessions
- **E-Learning:** Self-paced online courses
- **Workshops:** Hands-on practice
- **Coaching:** One-on-one guidance
- **Documentation:** Reference materials

TRN.1.2 Role-Based Training

Training by Role

Role	Training Topics	Duration
AP Accountant	Vendor, invoice, payment	3 days
AR Accountant	Customer, billing, collection	3 days
Warehouse Clerk	GR, GI, inventory	2 days
Purchasing	Requisition, PO, vendor	3 days
Sales	Order, delivery, billing	3 days
HR Specialist	PA, OM, payroll	5 days

TRN.2 Documentation Standards

TRN.2.1 Documentation Types

Document Categories

Type	Purpose	Audience
User Guide	Step-by-step instructions	End users
Quick Reference	Key information summary	All users
Process Guide	Business process flow	Process owners
Configuration	Setup instructions	Consultants
Technical	Development details	Developers

TRN.2.2 Document Templates

Standard Structure

1. Document header (title, version, date)

- 2. Table of contents
- 3. Introduction and scope
- 4. Prerequisites
- 5. Main content
- 6. Troubleshooting
- 7. Related documents
- 8. Change history

TRN.3 Knowledge Management

TRN.3.1 Knowledge Base

Knowledge Categories

- **FAQs:** Common questions and answers
- **How-To:** Step-by-step guides
- **Best Practices:** Recommended approaches
- **Known Issues:** Problems and workarounds
- **Tips & Tricks:** Productivity enhancers

TRN.3.2 Community Support

Support Channels

Channel	Purpose	Response Time
Help Desk	Issue reporting	4 hours
Forum	Peer support	Community
Chat	Quick questions	Immediate
Email	Detailed queries	24 hours

Extended Project Implementation Guide

IMP.1 Implementation Methodology

IMP.1.1 Project Phases

ASAP Methodology

Phase	Duration	Key Deliverables
Project Preparation	2-4 weeks	Charter, plan, team
Business Blueprint	4-8 weeks	Requirements, To-Be
Realization	12-20 weeks	Configuration, development
Final Preparation	4-6 weeks	Testing, training
Go-Live	1-2 weeks	Production cutover
Run & Support	Ongoing	Support, optimization

IMP.1.2 Project Organization

Project Roles

- **Executive Sponsor:** Strategic oversight
- **Project Manager:** Day-to-day management
- **Functional Lead:** Business process expertise
- **Technical Lead:** Technical implementation
- **Change Manager:** Organizational change

- **Test Manager:** Quality assurance

IMP.2 Business Blueprint

IMP.2.1 Requirements Gathering

Requirement Types

Type	Description	Priority
Business	Process requirements	Must have
Functional	System functionality	Must have
Technical	Infrastructure needs	Must have
Regulatory	Compliance requirements	Must have
Nice-to-have	Desirable features	Should have

IMP.2.2 Fit-Gap Analysis

Gap Categories

- **Fit:** Standard functionality meets need
- **Gap - Configuration:** Solved by parameter changes
- **Gap - Enhancement:** Requires custom development
- **Gap - Workaround:** Manual process or alternative

IMP.3 Testing Strategy

IMP.3.1 Test Levels

Testing Pyramid

Level	Scope	Participants
Unit Test	Single component	Developer
Integration Test	Multiple components	Technical team
System Test	End-to-end process	Functional team
UAT	Business scenarios	End users
Performance Test	Load and stress	Technical team

IMP.3.2 Test Data Management

Data Requirements

- Representative of production
- Cover all scenarios
- Anonymized for privacy
- Consistent across systems
- Refreshable for retesting

IMP.4 Go-Live Preparation

IMP.4.1 Cutover Planning

Cutover Activities

Activity	Timing	Responsible
Lock legacy system	T-0	IT
Final data extract	T+2h	IT
Data migration	T+4h	IT

Validation	T+8h	Business
Enable new system	T+12h	IT
User access	T+16h	IT

IMP.4.2 Post Go-Live Support

Support Model

- **Hypercare:** Intensive support first 2 weeks
- **Stabilization:** Regular support weeks 3-4
- **Normal Operations:** Standard support after week 4

Document Summary

EGYPTIAN ERP SYSTEM COMPREHENSIVE USER MANUAL

Version 3.0

This manual provides comprehensive documentation for the Egyptian ERP System, covering all modules, processes, and compliance requirements.

Total Chapters: 85+

Total Pages: 600+

Release Date: February 2026

For the latest updates, visit:
www.erp-system.eg/documentation

END OF MANUAL

Thank you for using the Egyptian ERP System

Extended API Integration Guide

API.1 REST API Architecture

API.1.1 API Design Principles

RESTful Design Standards

The Egyptian ERP System follows REST architectural principles for all API implementations. The API design emphasizes resource-based URLs, standard HTTP methods, and consistent response formats across all endpoints.

Resource Naming Conventions

Resource	URL Pattern	Example
Customers	/api/v1/customers	GET /api/v1/customers
Single Customer	/api/v1/customers/{id}	GET /api/v1/customers/12345
Customer Invoices	/api/v1/customers/{id}/invoices	GET /api/v1/customers/12345/invoices
Products	/api/v1/products	GET /api/v1/products
Inventory	/api/v1/inventory	GET /api/v1/inventory

API.1.2 HTTP Methods and Status Codes

Standard HTTP Methods

- **GET:** Retrieve resources without modification
- **POST:** Create new resources
- **PUT:** Update existing resources completely
- **PATCH:** Partial update of existing resources

- **DELETE:** Remove resources

HTTP Status Code Reference

Code	Meaning	Usage Context
200	OK	Successful GET, PUT, PATCH, DELETE
201	Created	Successful POST with new resource
204	No Content	Successful DELETE
400	Bad Request	Invalid request format or parameters
401	Unauthorized	Authentication required
403	Forbidden	Insufficient permissions
404	Not Found	Resource does not exist
409	Conflict	Resource conflict (duplicate, etc.)
422	Unprocessable Entity	Validation errors
500	Internal Server Error	Unexpected server error

API.2 Authentication and Security

API.2.1 OAuth 2.0 Implementation

OAuth Flow Types

The Egyptian ERP System supports multiple OAuth 2.0 flows depending on the integration scenario:

Authorization Code Flow (Recommended)

Used for server-side applications where the client secret can be securely stored. This flow provides the highest security level by exchanging an authorization code for an access token.

Client Credentials Flow

Used for machine-to-machine authentication where no user interaction is required. Common for backend integrations and scheduled processes.

Password Credentials Flow

Used for trusted applications where the user directly provides credentials. This flow is restricted to first-party applications only.

API.2.2 JWT Token Management

Token Structure

JSON Web Tokens (JWT) contain three parts separated by dots:

- **Header:** Algorithm and token type
- **Payload:** Claims including user ID, roles, expiration
- **Signature:** Verification signature

Token Lifecycle

Token Type	Expiration	Usage
Access Token	15 minutes	API requests
Refresh Token	7 days	Obtain new access token
ID Token	15 minutes	User identity information

API.3 Rate Limiting and Throttling

API.3.1 Rate Limit Tiers

Tier-Based Limits

Tier	Requests/Minute	Requests/Hour	Requests/Day
Basic	60	1,000	10,000
Standard	300	10,000	100,000
Premium	1,000	50,000	500,000
Enterprise	5,000	200,000	2,000,000

API.3.2 Rate Limit Headers

Response Headers

- **X-RateLimit-Limit:** Maximum requests allowed
- **X-RateLimit-Remaining:** Remaining requests in window
- **X-RateLimit-Reset:** Unix timestamp when limit resets
- **X-RateLimit-Retry-After:** Seconds until retry allowed

API.4 Webhook Integration

API.4.1 Webhook Events

Available Event Types

Event Category	Event Type	Description
Customer	customercreated	New customer created
Customer	customerupdated	Customer information changed
Invoice	invoice.created	New invoice generated
Invoice	invoice.paid	Invoice payment received

Payment	payment.received	Payment recorded
Inventory	stock.low	Low stock alert triggered
Order	order.placed	New order received
Order	order.shipped	Order shipped

API.4.2 Webhook Security

Signature Verification

All webhook payloads include a signature header for verification. The signature is computed using HMAC-SHA256 with your webhook secret.

Verification Steps

1. Extract signature from X-Webhook-Signature header
2. Compute HMAC-SHA256 of payload using webhook secret
3. Compare computed signature with received signature
4. Reject payload if signatures do not match

API.5 API Versioning

API.5.1 Version Strategy

URL Versioning

The Egyptian ERP System uses URL path versioning (e.g., /api/v1/, /api/v2/). This approach provides clear visibility of the API version being used and simplifies caching and routing.

Version Lifecycle

Status	Description	Support Period
Current	Latest stable version	Active development

Supported	Previous stable version	12 months after new version
Deprecated	End-of-life version	6 months warning before removal
Sunset	Removed version	No longer available

API.6 Error Handling

API.6.1 Error Response Format

Standard Error Structure

All API errors follow a consistent JSON structure:

```
{
  "error": {
    "code": "INVALID_PARAMETER",
    "message": "The provided parameter is invalid",
    "details": [
      {
        "field": "email",
        "issue": "Invalid email format"
      }
    ],
    "request_id": "req_1234567890",
    "documentation_url": "https://docs.erp-system.eg/errors/INVALID_PARAMETER"
  }
}
```

API.6.2 Error Categories

Common Error Codes

Error Code	Description	HTTP Status
INVALID_REQUEST	Malformed request	400
INVALID_PARAMETER	Invalid parameter value	400

MISSING_PARAMETER	Required parameter missing	400
UNAUTHORIZED	Authentication failed	401
FORBIDDEN	Insufficient permissions	403
NOT_FOUND	Resource not found	404
CONFLICT	Resource conflict	409
VALIDATION_ERROR	Business rule violation	422
RATE_LIMITED	Rate limit exceeded	429
INTERNAL_ERROR	Server error	500

API.7 SDK and Client Libraries

API.7.1 Official SDKs

Available SDKs

Language	Package Name	Installation
JavaScript/Node.js	@egypterp/sdk	npm install @egypterp/sdk
Python	egypterp-sdk	pip install egypterp-sdk
Java	com.egypterp:sdk	Maven/Gradle dependency
C#	EgyptErp.Sdk	NuGet package
PHP	egypterp/sdk	Composer require

API.7.2 SDK Features

Common Capabilities

- **Authentication Management:** Automatic token refresh and session handling
- **Request Retry:** Configurable retry logic with exponential backoff
- **Rate Limit Handling:** Automatic rate limit compliance
- **Error Handling:** Structured exception types for all error codes
- **Logging:** Built-in request/response logging
- **Type Safety:** Strongly typed models and responses

Extended Data Migration Guide

MIG.1 Migration Strategy Framework

MIG.1.1 Migration Approaches

Big Bang Migration

In a big bang migration, all data is migrated at once during a cutover window. This approach is suitable for smaller datasets or when downtime is acceptable.

Advantages

- Single cutover event
- Simpler rollback if issues occur
- Lower ongoing maintenance of parallel systems
- Complete transition at once

Disadvantages

- Requires significant downtime
- Higher risk if issues discovered post-migration
- Intensive resource requirements during cutover
- Less time for user adaptation

Phased Migration

Data is migrated in phases, typically by module, department, or time period. This approach reduces risk and allows for iterative learning.

Advantages

- Reduced risk through incremental approach
- Less downtime per phase

- Opportunity to learn and adjust between phases
- Easier resource management

Disadvantages

- Extended overall migration timeline
- Complex integration between old and new systems
- Potential data synchronization challenges
- Longer period of dual system maintenance

MIG.1.2 Migration Methodology

Phase 1: Discovery and Assessment

The discovery phase involves comprehensive analysis of source data to understand its structure, quality, and relationships.

Discovery Activities

Activity	Description	Output
Data Inventory	Catalog all data sources	Data source list
Schema Analysis	Document source structures	Schema documentation
Data Profiling	Analyze data quality	Quality report
Volume Assessment	Estimate data sizes	Volume estimates
Relationship Mapping	Identify dependencies	Relationship diagram

Phase 2: Design and Planning

Based on discovery findings, design the target data model and migration approach.

Design Deliverables

- **Target Data Model:** Designed for Egyptian ERP structure

- **Mapping Specifications:** Source to target field mappings
- **Transformation Rules:** Data conversion and cleansing rules
- **Migration Sequence:** Order of data migration
- **Validation Criteria:** Success metrics and checkpoints

Phase 3: Development and Testing

Build migration scripts, procedures, and validation routines.

Development Components

Component	Description	Technology
Extract Scripts	Pull data from source	SQL, Python, ETL tools
Transform Logic	Convert and cleanse data	Python, Java, SQL
Load Procedures	Insert into target	SQL, API calls
Validation Scripts	Verify data integrity	SQL, Python
Reconciliation	Compare source and target	SQL, Python

Phase 4: Execution

Execute the migration according to the planned sequence.

Execution Steps

1. Pre-migration backup of source and target systems
2. Execute extract procedures
3. Apply transformation rules
4. Load data into target system
5. Run validation checks
6. Perform reconciliation
7. Address any identified issues

8. Obtain sign-off from business stakeholders

MIG.2 Data Quality Management

MIG.2.1 Data Quality Dimensions

Quality Assessment Framework

Dimension	Description	Measurement
Completeness	Presence of required values	% of non-null required fields
Accuracy	Correctness of values	% of values matching reference
Consistency	Uniformity across sources	% of consistent values
Timeliness	Data freshness	Age of data in days
Validity	Conformity to format rules	% of valid format values
Uniqueness	Absence of duplicates	% of unique records

MIG.2.2 Data Cleansing Techniques

Standardization

Convert data to consistent formats and values:

- **Date Formats:** Convert to ISO 8601 (YYYY-MM-DD)
- **Phone Numbers:** Standardize to international format (+20 XX XXX XXXX)
- **Addresses:** Normalize street names, governorate names
- **Names:** Standardize capitalization and spacing
- **Currencies:** Convert to base currency (EGP)

Deduplication

Identify and merge duplicate records:

- **Exact Match:** Identical values in all key fields
- **Fuzzy Match:** Similar values using algorithms (Levenshtein, Soundex)
- **Rule-Based:** Business rules for identifying duplicates
- **Survivorship:** Rules for selecting values from merged records

MIG.3 Master Data Migration

MIG.3.1 Chart of Accounts Migration

Account Structure Mapping

Map legacy account codes to Egyptian ERP chart of accounts structure:

Mapping Considerations

Element	Legacy System	Egyptian ERP
Account Code	Variable length	Standardized segments
Account Type	Custom types	Standard types
Currency	Single/limited	Multi-currency support
Cost Centers	Optional	Integrated structure
Tax Codes	Manual entry	Linked to tax tables

MIG.3.2 Customer Migration

Customer Data Elements

- **Basic Information:** Name, address, contact details
- **Tax Information:** Tax ID, VAT registration
- **Commercial Registration:** Registration number, date
- **Payment Terms:** Credit limit, payment conditions

- **Price Lists:** Assigned pricing tiers
- **Sales Territory:** Geographic assignment

Customer Classification

Category	Criteria	Treatment
Active	Transaction within 12 months	Full migration
Inactive	No transaction 12-24 months	Summary migration
Dormant	No transaction 24+ months	Archive only
Prospect	No transactions	Selective migration

MIG.3.3 Vendor Migration

Vendor Data Requirements

Essential vendor information for Egyptian ERP:

- **Legal Name:** Registered company name
- **Trade Name:** Operating name if different
- **Tax ID:** Egyptian Tax Authority registration
- **Commercial Registration:** Ministry of Commerce number
- **Bank Details:** Account information for payments
- **Contact Information:** Primary and secondary contacts
- **Product Categories:** Goods/services supplied
- **Payment Terms:** Agreed payment conditions

MIG.4 Transactional Data Migration

MIG.4.1 Historical Data Strategy

Data Retention Approach

Data Age	Migration Strategy	Justification
Current year	Full detail migration	Active transactions
Previous year	Full detail migration	Comparative analysis
2-3 years	Summary migration	Trend analysis
3-5 years	Archive reference	Audit requirements
5+ years	Archive only	Legal compliance

MIG.4.2 Open Transaction Migration

Open Items Categories

- **Open Invoices:** Customer and vendor invoices not fully paid
- **Open Orders:** Sales and purchase orders not completed
- **Open Payments:** Payments in transit or pending
- **Open Inventory:** Goods receipts not invoiced
- **Open Projects:** Work in progress projects

Migration Requirements

All open transactions must be migrated with:

- Original document numbers for reference
- Original dates and due dates
- Complete line item details
- Applied payments and allocations
- Original currency and amounts
- Aging information

MIG.5 Migration Validation

MIG.5.1 Validation Levels

Level 1: Record Count Validation

Compare record counts between source and target:

- Total records migrated
- Records by type/category
- Records by time period
- Records by organizational unit

Level 2: Sum Validation

Compare aggregated values:

- Total amounts by account
- Total quantities by product
- Total balances by customer/vendor
- Summary by period

Level 3: Sample Validation

Detailed verification of sample records:

- Random sample (statistically significant)
- High-value sample (top 10% by value)
- Exception sample (edge cases)
- Stratified sample (by category)

MIG.5.2 Reconciliation Process

Reconciliation Steps

1. Extract source system totals
2. Extract target system totals

3. Compare and identify variances
4. Investigate variance causes
5. Document explanations
6. Obtain business sign-off

Reconciliation Report

Category	Source Total	Target Total	Variance	Status
Customers	5,234	5,234	0	Passed
Vendors	1,876	1,876	0	Passed
Products	12,456	12,456	0	Passed
GL Balance	EGP 45,678,901	EGP 45,678,901	0	Passed
AR Balance	EGP 12,345,678	EGP 12,345,678	0	Passed
AP Balance	EGP 8,765,432	EGP 8,765,432	0	Passed

MIG.6 Cutover Planning

MIG.6.1 Cutover Timeline

Sample Cutover Schedule

Time	Activity	Duration	Owner
T-48h	Freeze master data changes	Ongoing	Data Team
T-24h	Begin transaction freeze	Ongoing	Business
T-0	Final transaction cutoff	Instant	Project Manager
T+2h	Final data extraction	2 hours	Technical Team

T+4h	Data transformation	2 hours	Technical Team
T+8h	Data loading	4 hours	Technical Team
T+12h	Validation and reconciliation	4 hours	Business + IT
T+16h	Go/No-Go decision	1 hour	Steering Committee
T+18h	Enable new system	2 hours	Technical Team
T+24h	Resume operations	Ongoing	Business

MIG.6.2 Rollback Plan

Rollback Triggers

- Critical validation failures
- System performance issues
- Data integrity problems
- Security concerns
- Business stakeholder decision

Rollback Procedure

1. Stop new system access
2. Notify all stakeholders
3. Restore legacy system from backup
4. Verify legacy system functionality
5. Resume legacy system operations
6. Document issues for resolution
7. Plan remediation and retry

Extended Security Compliance Guide

SEC.1 Information Security Framework

SEC.1.1 Security Policies

Policy Hierarchy

Level	Document Type	Scope
1	Security Policy	Organization-wide principles
2	Security Standards	Mandatory requirements
3	Security Procedures	Step-by-step instructions
4	Security Guidelines	Recommended practices
5	Security Baselines	Technical configurations

SEC.1.2 Security Governance

Security Organization

- **Information Security Officer (ISO):** Overall security responsibility
- **Security Committee:** Cross-functional security oversight
- **Data Owners:** Responsible for data classification
- **System Owners:** Responsible for system security
- **Security Administrators:** Day-to-day security operations

SEC.2 Access Control

SEC.2.1 Authentication Mechanisms

Multi-Factor Authentication (MFA)

All administrative access and privileged operations require MFA. Supported factors include:

Authentication Factors

Factor Type	Examples	Use Case
Knowledge	Password, PIN	Primary authentication
Possession	Hardware token, Mobile app	Second factor
Inherence	Fingerprint, Face recognition	Biometric verification
Location	IP address, GPS	Risk-based authentication

SEC.2.2 Role-Based Access Control (RBAC)

RBAC Principles

- **Least Privilege:** Users receive minimum necessary access
- **Separation of Duties:** Critical functions split among roles
- **Need-to-Know:** Access granted only for required information
- **Role Hierarchy:** Inheritance of permissions

Standard Roles

Role	Description	Typical Permissions
System Administrator	Full system access	All functions
Security Administrator	Security management	User management, audit
Application Administrator	Module configuration	Module setup, workflows
Power User	Advanced operations	Reporting, batch processing

Standard User	Regular operations	Transaction entry, inquiry
Read-Only User	View access only	Reports, inquiries only

SEC.3 Data Protection

SEC.3.1 Data Classification

Classification Levels

Level	Description	Examples	Handling
Public	No restriction	Marketing materials	Standard
Internal	Organization use	Policies, procedures	Internal sharing
Confidential	Limited distribution	Financial data, HR records	Need-to-know
Restricted	Strict control	Customer PII, credentials	Authorized only

SEC.3.2 Encryption Standards

Encryption Requirements

Data State	Encryption Method	Key Length
Data at Rest	AES-256	256 bits
Data in Transit	TLS 1.3	2048-bit RSA
Database	TDE (Transparent Data Encryption)	256 bits
Backups	AES-256	256 bits
Passwords	bcrypt/Argon2	Adaptive

SEC.3.3 Key Management

Key Management Lifecycle

- **Generation:** Cryptographically secure random generation
- **Distribution:** Secure channels only
- **Storage:** Hardware Security Module (HSM) or KMS
- **Rotation:** Regular scheduled rotation
- **Revocation:** Immediate upon compromise
- **Destruction:** Secure deletion when no longer needed

SEC.4 Network Security

SEC.4.1 Network Segmentation

Security Zones

Zone	Description	Access Control
DMZ	Public-facing services	Firewall, WAF
Application	Application servers	Internal firewall
Database	Database servers	Strict ACLs
Management	Administrative systems	Jump server, VPN
Backup	Backup infrastructure	Isolated network

SEC.4.2 Firewall Configuration

Firewall Rules

- **Default Deny:** All traffic blocked by default

- **Explicit Allow:** Only required traffic permitted
- **Least Privilege:** Minimum necessary access
- **Regular Review:** Quarterly rule review
- **Change Control:** Approved changes only

SEC.5 Application Security

SEC.5.1 Secure Development

Secure SDLC

Phase	Security Activity
Requirements	Security requirements definition
Design	Threat modeling, security architecture
Development	Secure coding standards, code review
Testing	SAST, DAST, penetration testing
Deployment	Security configuration, hardening
Operations	Monitoring, patching, incident response

SEC.5.2 OWASP Top 10 Mitigation

Security Controls

Vulnerability	Mitigation Control
Injection	Parameterized queries, input validation
Broken Authentication	MFA, secure session management
Sensitive Data Exposure	Encryption, secure protocols

XXE	Disable XML external entities
Broken Access Control	RBAC, authorization checks
Security Misconfiguration	Hardening, configuration management
XSS	Output encoding, CSP headers
Insecure Deserialization	Input validation, integrity checks
Known Vulnerabilities	Patch management, dependency scanning
Insufficient Logging	Comprehensive audit logging

SEC.6 Security Monitoring

SEC.6.1 Security Information and Event Management (SIEM)

SIEM Capabilities

- **Log Aggregation:** Centralized log collection
- **Correlation:** Event correlation and analysis
- **Alerting:** Real-time threat detection
- **Reporting:** Security dashboards and reports
- **Investigation:** Forensic analysis tools

SEC.6.2 Security Metrics

Key Performance Indicators

Metric	Target	Measurement
Mean Time to Detect (MTTD)	< 24 hours	Average detection time
Mean Time to Respond (MTTR)	< 4 hours	Average response time

Mean Time to Contain (MTTC)	< 8 hours	Average containment time
Patch Compliance	> 95%	Critical patches applied
Vulnerability Scan Coverage	100%	Assets scanned
Security Training Completion	100%	Employees trained

SEC.7 Compliance and Audit

SEC.7.1 Regulatory Compliance

Egyptian Regulatory Requirements

Regulation	Requirement	ERP Compliance
Data Protection Law	Personal data protection	Encryption, access controls
ETA E-Invoicing	Electronic invoice security	Digital signatures, audit trails
Central Bank	Financial data security	Segregation, monitoring
Telecom Law	Communication security	Encrypted transmissions

SEC.7.2 Audit Trail Requirements

Audit Log Content

- **User Identification:** Who performed the action
- **Timestamp:** When the action occurred
- **Action Type:** What action was performed
- **Object:** What resource was affected
- **Result:** Success or failure
- **Source:** Where the action originated
- **Before/After:** Data changes (for modifications)

Retention Requirements

Log Type	Retention Period	Storage
Authentication Logs	2 years	Tamper-proof
Transaction Logs	7 years	Archive
Administrative Logs	5 years	Secure storage
Security Events	3 years	SIEM

Extended Performance Tuning Guide

PERF.1 Performance Fundamentals

PERF.1.1 Performance Metrics

Key Performance Indicators

Metric	Description	Target
Response Time	Time to complete request	< 2 seconds
Throughput	Transactions per second	> 100 TPS
Concurrent Users	Simultaneous active users	> 1000
CPU Utilization	CPU usage percentage	< 70%
Memory Usage	RAM utilization	< 80%
Disk I/O	Disk operations per second	< 80% capacity
Network Latency	Network response time	< 50ms
Error Rate	Failed transaction percentage	< 0.1%

PERF.1.2 Performance Testing Types

Testing Categories

- **Load Testing:** System behavior under expected load
- **Stress Testing:** System behavior under extreme load
- **Endurance Testing:** System stability over extended periods
- **Spike Testing:** System response to sudden load increases
- **Volume Testing:** System behavior with large data volumes

- **Scalability Testing:** Ability to handle growing load

PERF.2 Database Performance

PERF.2.1 Query Optimization

Query Performance Best Practices

- **Select Only Needed Columns:** Avoid SELECT *
- **Use Indexes Effectively:** Proper index design and maintenance
- **Optimize Joins:** Use appropriate join types
- **Limit Result Sets:** Use pagination for large results
- **Avoid Cursors:** Use set-based operations
- **Parameterize Queries:** Enable plan reuse

Execution Plan Analysis

Operation	Cost Indicator	Optimization
Table Scan	High	Add appropriate index
Index Scan	Medium	Consider covering index
Index Seek	Low	Optimal access method
Key Lookup	Medium	Include columns in index
Sort	Medium-High	Use index for ordering
Hash Match	High	Optimize join conditions

PERF.2.2 Index Optimization

Index Types and Usage

Index Type	Use Case	Example
B-Tree	Equality and range queries	Primary keys, foreign keys
Composite	Multi-column queries	(CustomerID, OrderDate)
Covering	Frequently accessed columns	Include all query columns
Filtered	Subset of data	Active records only
Full-Text	Text search	Product descriptions

Index Maintenance

- **Rebuild:** When fragmentation > 30%
- **Reorganize:** When fragmentation 5-30%
- **Update Statistics:** Weekly or after bulk operations
- **Monitor Usage:** Identify unused indexes

PERF.3 Application Performance

PERF.3.1 Code Optimization

Performance Best Practices

- **Minimize Database Calls:** Batch operations when possible
- **Use Caching:** Cache frequently accessed data
- **Async Operations:** Use asynchronous processing
- **Connection Pooling:** Reuse database connections
- **Lazy Loading:** Load data on demand
- **Object Pooling:** Reuse expensive objects

PERF.3.2 Caching Strategies

Cache Types

Cache Level	Scope	Duration	Use Case
Application	Process	Session	User session data
Distributed	Cluster	Configurable	Shared reference data
Database	Server	Query-based	Query results
CDN	Global	Long-term	Static assets

Cache Invalidation Strategies

- **Time-Based:** Expire after specified duration
- **Event-Based:** Invalidate on data change
- **Manual:** Explicit cache clearing
- **LRU:** Least Recently Used eviction

PERF.4 Infrastructure Performance

PERF.4.1 Server Optimization

Application Server Tuning

Parameter	Description	Recommendation
Heap Size	JVM memory allocation	4-8GB based on load
Thread Pool	Concurrent request handling	100-500 threads
Connection Pool	Database connections	50-200 connections
GC Tuning	Garbage collection	G1GC for large heaps
Session Timeout	Inactive session duration	30 minutes

PERF.4.2 Network Optimization

Network Performance Factors

- **Bandwidth:** Sufficient capacity for peak load
- **Latency:** Minimize round-trip time
- **Packet Loss:** Maintain < 0.1%
- **Jitter:** Consistent latency
- **Compression:** Enable HTTP compression
- **Keep-Alive:** Reuse TCP connections

PERF.5 Monitoring and Diagnostics

PERF.5.1 Performance Monitoring Tools

Monitoring Categories

Category	Tools	Metrics
Application	APM tools	Response time, throughput
Database	Query monitors	Query time, locks, waits
Infrastructure	System monitors	CPU, memory, disk, network
User Experience	RUM tools	Page load, interaction time

PERF.5.2 Performance Baselines

Baseline Metrics

- **Normal Load:** Average daily performance
- **Peak Load:** Maximum observed performance
- **Off-Peak:** Minimum load performance

- **Growth Trend:** Performance over time

Baseline Uses

- Detect performance degradation
- Plan capacity upgrades
- Validate optimization efforts
- Support troubleshooting

Extended Disaster Recovery Guide

DR.1 Business Continuity Planning

DR.1.1 Business Impact Analysis

Criticality Classification

Criticality	RTO	RPO	Examples
Critical	4 hours	1 hour	Core ERP, E-Invoicing
High	24 hours	4 hours	Financial reporting
Medium	72 hours	24 hours	Analytics, BI
Low	1 week	1 week	Archival systems

Key Definitions

- **RTO (Recovery Time Objective):** Maximum acceptable downtime
- **RPO (Recovery Point Objective):** Maximum acceptable data loss
- **MTD (Maximum Tolerable Downtime):** Longest acceptable outage
- **WRT (Work Recovery Time):** Time to restore full operations

DR.1.2 Risk Assessment

Risk Categories

Risk Type	Examples	Mitigation
Natural	Earthquake, flood, fire	Geographic redundancy
Technical	Hardware failure, software bug	Redundancy, backups

Human	Error, sabotage	Training, access controls
External	Cyber attack, power outage	Security, UPS, generators

DR.2 Backup Strategies

DR.2.1 Backup Types

Backup Categories

Type	Description	Frequency	Retention
Full	Complete data backup	Weekly	4 weeks
Incremental	Changes since last backup	Daily	2 weeks
Differential	Changes since last full	Every 6 hours	1 week
Transaction Log	Database transactions	Every 15 min	48 hours
Snapshot	Point-in-time copy	Every hour	24 hours

DR.2.2 Backup Storage

Storage Tiers

- **Primary:** Local storage for quick recovery
- **Secondary:** Offsite storage for disaster recovery
- **Archive:** Long-term storage for compliance
- **Cloud:** Geographic redundancy

3-2-1 Backup Rule

- **3:** Keep 3 copies of data
- **2:** Store on 2 different media types

- **1:** Keep 1 copy offsite

DR.3 Recovery Strategies

DR.3.1 Recovery Options

Recovery Models

Strategy	RTO	Cost	Description
Cold Site	Weeks	Low	Facility only, no equipment
Warm Site	Days	Medium	Partial equipment ready
Hot Site	Hours	High	Fully equipped, ready to run
Active-Active	Minutes	Very High	Continuous operation

DR.3.2 Failover Procedures

Failover Types

- **Manual Failover:** Administrator-initiated switchover
- **Automatic Failover:** System-detected and executed
- **Planned Failover:** Scheduled maintenance switch
- **Unplanned Failover:** Emergency response to failure

Failover Steps

1. Detect failure or initiate planned switch
2. Assess impact and confirm failover decision
3. Update DNS or load balancer configuration
4. Activate standby systems
5. Verify system functionality

6. Notify users of status change

7. Begin monitoring and support

DR.4 Disaster Recovery Testing

DR.4.1 Test Types

Testing Methods

Test Type	Description	Frequency
Tabletop	Discussion-based walkthrough	Quarterly
Simulation	Scenario-based role play	Semi-annually
Parallel	Run systems in parallel	Annually
Full Interruption	Complete failover test	Annually

DR.4.2 Test Documentation

Test Report Contents

- Test objectives and scope
- Test scenario and timeline
- Participants and roles
- Test execution results
- Issues identified
- Lessons learned
- Remediation actions
- Plan updates required

DR.5 Crisis Management

DR.5.1 Crisis Response Team

Team Structure

Role	Responsibility	Primary Contact
Crisis Manager	Overall coordination	IT Director
Technical Lead	Technical recovery	CTO
Communications	Stakeholder updates	PR Manager
Business Lead	Business continuity	COO
Legal	Regulatory compliance	Legal Counsel

DR.5.2 Communication Plan

Communication Channels

- **Internal:** Email, SMS, intranet, phone tree
- **External:** Website, social media, press release
- **Customers:** Direct notification, status page
- **Vendors:** Direct contact, partner portal
- **Regulators:** Formal notification as required

Communication Templates

Pre-defined templates for common scenarios:

- Initial incident notification
- Status update
- Resolution notification
- Post-incident summary

Extended User Experience Guide

UX.1 User Interface Design Principles

UX.1.1 Design Standards

Core Principles

- **Consistency:** Uniform design patterns across modules
- **Clarity:** Clear and understandable interface elements
- **Efficiency:** Minimize steps to complete tasks
- **Forgiveness:** Allow easy error recovery
- **Accessibility:** Usable by people with disabilities

UX.1.2 Visual Design System

Color Palette

Color	Hex Code	Usage
Primary	#1E3A5F	Headers, primary actions
Secondary	#4A90A4	Secondary actions, links
Success	#28A745	Success messages, approvals
Warning	#FFC107	Warnings, attention needed
Danger	#DC3545	Errors, deletions
Neutral	#6C757D	Disabled states, secondary text

Typography

Element	Font	Size	Weight
H1	Noto Sans	28px	700
H2	Noto Sans	24px	600
H3	Noto Sans	20px	600
Body	Noto Sans	14px	400
Small	Noto Sans	12px	400

UX.2 Navigation Design

UX.2.1 Navigation Patterns

Primary Navigation

- **Sidebar Menu:** Module-level navigation
- **Top Bar:** Global actions and search
- **Breadcrumbs:** Location and hierarchy indicator
- **Context Menu:** Object-specific actions

UX.2.2 Search and Filter

Search Capabilities

Search Type	Description	Example
Global	Search across all modules	Customer name, invoice number
Module	Search within current module	Products in inventory
Advanced	Multi-criteria search	Date range, amount, status

Saved	Reusable search criteria	Monthly sales report filter
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UX.3 Form Design

UX.3.1 Form Layout

Layout Best Practices

- **Single Column:** Primary layout for forms
- **Logical Grouping:** Related fields together
- **Clear Labels:** Descriptive and concise
- **Inline Validation:** Real-time error checking
- **Progress Indicators:** For multi-step forms

UX.3.2 Input Types

Input Components

Input Type	Use Case	Validation
Text	Free-form text	Length, format
Number	Numeric values	Range, precision
Date	Date selection	Date format, range
Dropdown	Predefined options	Required selection
Checkbox	Multiple selection	At least one
Radio	Single selection	Required selection
Autocomplete	Large option sets	Valid selection

UX.4 Feedback and Notifications

UX.4.1 Message Types

Notification Categories

Type	Purpose	Duration
Success	Confirm action completion	3 seconds
Error	Report failure	Persistent
Warning	Caution about action	5 seconds
Info	General information	4 seconds

UX.4.2 Progress Indicators

Indicator Types

- **Spinner:** Indeterminate progress
- **Progress Bar:** Determinate progress with percentage
- **Step Indicator:** Multi-step process progress
- **Skeleton:**Content loading placeholder

UX.5 Personalization

UX.5.1 User Preferences

Customizable Settings

Setting	Options	Default
Language	Arabic, English	English
Date Format	Gregorian, Hijri	Gregorian

Number Format	Western, Arabic	Western
Currency Display	Symbol, Code	Symbol
Theme	Light, Dark	Light
Page Size	10, 25, 50, 100	25

UX.5.2 Dashboard Customization

Dashboard Widgets

- **KPI Cards:** Key metrics at a glance
- **Charts:** Visual data representation
- **Lists:** Recent items or tasks
- **Calendars:** Event and deadline view
- **Feeds:** Activity streams

Customization Options

- Add/remove widgets
- Rearrange layout
- Configure widget settings
- Save multiple layouts
- Share layouts with team

Extended Reporting Framework Guide

RPT.1 Reporting Architecture

RPT.1.1 Report Types

Report Categories

Category	Description	Examples
Operational	Day-to-day operations	Daily sales, inventory status
Financial	Financial statements	Balance sheet, P&L
Management	Executive summaries	KPI dashboards
Compliance	Regulatory reports	Tax returns, audit reports
Analytical	Deep analysis	Trend analysis, forecasting

RPT.1.2 Report Delivery

Delivery Methods

Method	Description	Best For
On-Demand	User-initiated generation	Ad-hoc queries
Scheduled	Automated generation	Regular reports
Event-Driven	Triggered by events	Alerts, notifications
Streaming	Real-time updates	Live dashboards

RPT.2 Report Design

RPT.2.1 Design Principles

Effective Report Design

- **Clear Purpose:** Define report objective
- **Target Audience:** Design for specific users
- **Appropriate Detail:** Right level of granularity
- **Visual Hierarchy:** Guide reader attention
- **Consistent Formatting:** Uniform appearance

RPT.2.2 Report Components

Standard Elements

Element	Description	Required
Header	Report title and metadata	Yes
Parameters	Filter criteria applied	Yes
Summary	Key highlights	No
Detail	Main report content	Yes
Charts	Visual representations	No
Footer	Page numbers, timestamps	Yes

RPT.3 Report Builder

RPT.3.1 Builder Interface

Builder Components

- **Data Source:** Select data source
- **Fields:** Choose report fields
- **Filters:** Define selection criteria
- **Groups:** Organize data hierarchically
- **Sort:** Define sort order
- **Layout:** Arrange report elements
- **Format:** Apply styling

RPT.3.2 Advanced Features

Advanced Capabilities

Feature	Description
Calculated Fields	Custom formula-based fields
Subreports	Nested report within report
Cross-Tabs	Pivot table style layout
Drill-Down	Expand for more detail
Parameters	User input at runtime
Conditional Formatting	Format based on values

RPT.4 Standard Reports

RPT.4.1 Financial Reports

Standard Financial Reports

Report	Description	Frequency
Trial Balance	Account balances summary	Monthly
Balance Sheet	Assets, liabilities, equity	Monthly/Quarterly
Income Statement	Revenue and expenses	Monthly
Cash Flow	Cash movements	Monthly
AR Aging	Receivables by age	Weekly
AP Aging	Payables by age	Weekly

RPT.4.2 Operational Reports

Standard Operational Reports

Report	Description	Frequency
Sales Summary	Sales by period/product	Daily
Inventory Status	Stock levels and valuation	Daily
Purchase Orders	PO status and tracking	Daily
Production Status	Work order progress	Daily
Employee Attendance	Time and attendance	Daily

RPT.5 Report Distribution

RPT.5.1 Distribution Options

Output Formats

Format	Extension	Best For
PDF	.pdf	Sharing, printing
Excel	.xlsx	Further analysis
CSV	.csv	Data import/export
Word	.docx	Documentation
HTML	.html	Web viewing
XML	.xml	System integration
JSON	.json	API consumption

RPT.5.2 Scheduled Distribution

Schedule Configuration

- **Frequency:** Hourly, Daily, Weekly, Monthly
- **Recipients:** Individual users or groups
- **Format:** Preferred output format
- **Delivery:** Email, portal, file share
- **Retention:** How long to keep generated reports

Extended Workflow Engine Guide

WF.1 Workflow Concepts

WF.1.1 Workflow Fundamentals

Key Concepts

- **Workflow:** Automated business process
- **Process:** Sequence of activities
- **Activity:** Individual step in workflow
- **Transition:** Path between activities
- **Instance:** Running workflow execution
- **Participant:** User or system involved

WF.1.2 Workflow Types

Type Classification

Type	Description	Example
Sequential	Linear step-by-step flow	Expense approval
Parallel	Multiple simultaneous paths	Document review
Conditional	Branch based on conditions	Credit approval
Ad-hoc	Flexible, user-defined flow	Case management

WF.2 Workflow Design

WF.2.1 Design Principles

Best Practices

- **Simplicity:** Keep workflows as simple as possible
- **Clarity:** Clear naming and documentation
- **Flexibility:** Allow for exceptions
- **Visibility:** Track progress and status
- **Efficiency:** Minimize wait times

WF.2.2 Activity Types

Standard Activities

Activity	Description	Use Case
User Task	Human action required	Approval, review
Service Task	Automated system action	Data update, notification
Script Task	Custom code execution	Complex calculation
Decision	Conditional branch	Route based on value
Parallel Gateway	Split/join parallel paths	Simultaneous tasks
Timer	Time-based trigger	Escalation, reminder

WF.3 Workflow Configuration

WF.3.1 Workflow Builder

Builder Interface

- **Canvas:** Visual workflow designer
- **Toolbox:** Available activities
- **Properties:** Activity configuration

- **Validation:** Error checking
- **Simulation:** Test execution

WF.3.2 Participant Configuration

Assignment Options

Method	Description
Direct	Specific user
Role	Any user with role
Group	Members of group
Rule	Dynamic based on data
Round Robin	Distribute among group
Load Balanced	Based on workload

WF.4 Standard Workflows

WF.4.1 Approval Workflows

Purchase Order Approval

1. PO created by requester
2. Manager approval (if > EGP 10,000)
3. Finance approval (if > EGP 50,000)
4. Director approval (if > EGP 100,000)
5. PO issued to vendor

Expense Claim Approval

1. Employee submits claim

2. Receipt verification
3. Manager approval
4. Finance verification
5. Payment processing

WF.4.2 Document Workflows

Document Review Process

1. Document uploaded
2. Initial review assigned
3. Parallel department reviews
4. Consolidation of feedback
5. Final approval
6. Document publication

WF.5 Workflow Monitoring

WF.5.1 Instance Tracking

Tracking Information

- **Status:** Current state of workflow
- **History:** Completed activities
- **Current Task:** Pending action
- **Assignee:** Responsible user
- **Timeline:** Start and due dates

WF.5.2 Performance Metrics

Key Metrics

Metric	Description	Target
Cycle Time	Total workflow duration	Per SLA
Wait Time	Time waiting for action	< 20% of cycle
Completion Rate	Successfully completed	> 95%
Escalation Rate	Requires escalation	< 5%

Extended Document Management Guide

DOC.1 Document Lifecycle

DOC.1.1 Lifecycle Stages

Document Phases

Stage	Description	Actions
Create	Document creation	Draft, template use
Review	Content review	Edit, comment, approve
Approve	Formal approval	Sign-off, versioning
Publish	Make available	Distribute, notify
Use	Active usage	Access, reference
Archive	Long-term storage	Compress, index
Dispose	Secure deletion	Shred, audit

DOC.1.2 Version Control

Versioning Scheme

- **Major:** Significant changes (1.0, 2.0)
- **Minor:** Small updates (1.1, 1.2)
- **Revision:** Corrections (1.1.1, 1.1.2)

Version History

Attribute	Description
Version Number	Sequential identifier
Author	Who made changes
Date	When changed
Changes	What was modified
Comments	Change description

DOC.2 Document Types

DOC.2.1 Type Classification

Standard Document Types

Type	Description	Retention
Policy	Organizational policies	Permanent
Procedure	Operational procedures	Until superseded
Form	Data entry forms	7 years
Report	Generated reports	Per regulation
Contract	Legal agreements	Contract term + 7 years
Invoice	Financial documents	7 years
Correspondence	Emails, letters	3 years

DOC.2.2 Metadata

Standard Metadata Fields

- **Title:** Document name
- **Author:** Creator
- **Date Created:** Creation date
- **Document Type:** Classification
- **Department:** Owning department
- **Confidentiality:** Security level
- **Keywords:** Search terms
- **Retention:** Retention period

DOC.3 Document Storage

DOC.3.1 Storage Structure

Folder Hierarchy

- **Level 1:** Department
- **Level 2:** Document Type
- **Level 3:** Year
- **Level 4:** Month/Project

DOC.3.2 Storage Quotas

Quota Allocation

User Type	Quota	Max File Size
Standard User	1 GB	50 MB
Power User	5 GB	100 MB
Department	50 GB	500 MB
System	Unlimited	2 GB

DOC.4 Document Security

DOC.4.1 Access Control

Permission Levels

Permission	Description
View	Read-only access
Download	Save local copy
Edit	Modify content
Delete	Remove document
Share	Distribute to others
Manage	Full control including permissions

DOC.4.2 Digital Signatures

Signature Types

- **Simple:** Click-to-sign
- **Advanced:** Password protected
- **Qualified:** Certificate-based (ETA compliant)

Signature Workflow

1. Document prepared for signing
2. Signers notified
3. Each signer reviews and signs
4. Signature validated
5. Signed document finalized

6. Audit trail maintained

DOC.5 Document Search

DOC.5.1 Search Capabilities

Search Types

Search	Description
Basic	Keyword search
Advanced	Multiple criteria
Full-Text	Content within documents
Metadata	Search by properties
Saved	Reusable searches

DOC.5.2 Search Filters

Available Filters

- Document type
- Date range
- Author
- Department
- Confidentiality level
- File format
- Size range

Extended Multi-Company Operations Guide

MCO.1 Multi-Company Architecture

MCO.1.1 Company Structure

Hierarchy Levels

Level	Description	Example
Enterprise	Top-level organization	Holding Company
Company	Legal entity	Subsidiary A
Branch	Operating location	Cairo Branch
Department	Functional unit	Sales Department
Cost Center	Budget responsibility	Marketing CC

MCO.1.2 Data Isolation

Isolation Levels

- **Complete:** No data sharing between companies
- **Shared Master:** Common master data
- **Shared Services:** Shared transactional services
- **Consolidated:** Full visibility with consolidation

MCO.2 Inter-Company Transactions

MCO.2.1 Transaction Types

Common Inter-Company Transactions

Type	Description	Example
Sales	Sale between companies	Goods transfer
Services	Service provision	Management fees
Loans	Financial advances	Working capital
Cost Allocation	Shared cost distribution	Overhead allocation
Dividends	Profit distribution	Parent to subsidiary

MCO.2.2 Transfer Pricing

Transfer Pricing Methods

- **Comparable Uncontrolled Price:** Market-based pricing
- **Resale Price:** Margin-based approach
- **Cost Plus:** Cost-based with markup
- **Transactional Net Margin:** Profit-based method

Documentation Requirements

- Transfer pricing policy
- Functional analysis
- Comparability analysis
- Method selection rationale
- Annual documentation

MCO.3 Consolidation

MCO.3.1 Consolidation Process

Consolidation Steps

1. Collect subsidiary financials
2. Currency translation
3. Eliminate inter-company transactions
4. Adjust for minority interests
5. Apply consolidation entries
6. Generate consolidated statements

MCO.3.2 Elimination Entries

Elimination Types

Elimination	Description
Investment	Investment vs equity
Receivable/Payable	Inter-company balances
Revenue/Cost	Inter-company sales
Profit in Inventory	Unrealized profit
Dividends	Internal dividends

MCO.4 Multi-Currency Operations

MCO.4.1 Currency Management

Currency Types

- **Local Currency:** Company operating currency

- **Group Currency:** Consolidation currency (EGP)
- **Hard Currency:** USD, EUR for international
- **Index Currency:** Inflation-adjusted

MCO.4.2 Exchange Rate Handling

Rate Types

Rate Type	Usage	Source
Spot	Current transactions	CBE daily rate
Period	Monthly average	CBE monthly rate
Historical	Transaction date rate	Stored rate
Budget	Planning	User defined

MCO.5 Shared Services

MCO.5.1 Shared Service Centers

Common Shared Services

Service	Description	Benefit
Finance	Accounting, AP, AR	Economies of scale
HR	Payroll, recruitment	Standardization
IT	Infrastructure, support	Cost reduction
Procurement	Purchasing	Bulk discounts

MCO.5.2 Cost Allocation

Allocation Methods

- **Direct:** Traceable costs
- **Step-Down:** Sequential allocation
- **Reciprocal:** Mutual service recognition
- **Activity-Based:** Cost driver based

Allocation Bases

Service	Allocation Base
Finance	Transaction count
HR	Headcount
IT	User count
Facilities	Square meters

Document Summary

EGYPTIAN ERP SYSTEM COMPREHENSIVE USER MANUAL

Version 3.0 - Extended Edition

This manual provides comprehensive documentation for the Egyptian ERP System, covering all modules, processes, compliance requirements, and extended guides.

Total Chapters: 95+

Total Pages: 600+

Release Date: February 2026

Extended Guides Included:

- API Integration Guide
- Data Migration Guide
- Security Compliance Guide
- Performance Tuning Guide
- Disaster Recovery Guide
- User Experience Guide
- Reporting Framework Guide
- Workflow Engine Guide
- Document Management Guide
- Multi-Company Operations Guide

For the latest updates, visit:

www.erp-system.eg/documentation

END OF MANUAL

Thank you for using the Egyptian ERP System